Fpt University

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# CAPSTONE PROJECT REPORT

## Report 3 – Software Requirement Specification

– Hanoi, Sept 2026 –

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## Change Log

| Date | A* M, D | In charge | Change Description |
|---|---|---|---|
| 18/09/2026 | A | Vu Long Nhat | Initial version created |
| 01/10/2026 | A | Nguyen Viet Hoang | Added use case diagrams, Non-Functional Requirements |
| 02/10/2026 | M | Nguyen Viet Hoang | Modified use case diagrams, Non-Functional Requirements |
| 03/10/2026 | A | Duong Duc Loc | Added context diagram, user requirement |
| 03/10/2026 | A | Nguyen Tuan Anh | Added Functional Requirements |
| 04/10/2026 | A | Vu Long Nhat | Added ERD diagram, entity descriptions |
| 05/10/2026 | A | Nguyen Tuan Anh | Added Main business flow |
| 06/10/2026 | A | Duong Duc Loc | Added Use Case, Use Case Diagram, Use Case Specification. |
| 06/10/2026 | M | Duong Duc Loc | Modified Main Business Flow |
| 08/10/2026 | M | Nguyen Tuan Anh | Modified Main Business Flow |
| 08/10/2026 | M | Duong Duc Loc | Modified Actor, Use Case, Use Case Diagram, Permission matrix |
| | | | |
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| | | | |
| | | | |

*A - Added M - Modified D - Deleted

# Definition and Acronyms

| Acronym | Definition |
|---|---|
| COS | Cafeteria Ordering System |
| SaaS | Software as a Service |

# I. Overall Requirements

## 1. Context Diagram

The ConTrack Contract Management System is a multi-tenant SaaS platform that helps businesses manage the full lifecycle of their contracts. Each business that subscribes to ConTrack gets its own isolated workspace (tenant), with separate data and user accounts. The system supports the whole process, from customer management and contract drafting to multi-level approval, e-signing, payment, archiving, and renewal tracking. It replaces scattered manual work, such as paper contracts, spreadsheets, email approvals, and missed expiry dates, with one structured, automated, and secure solution.

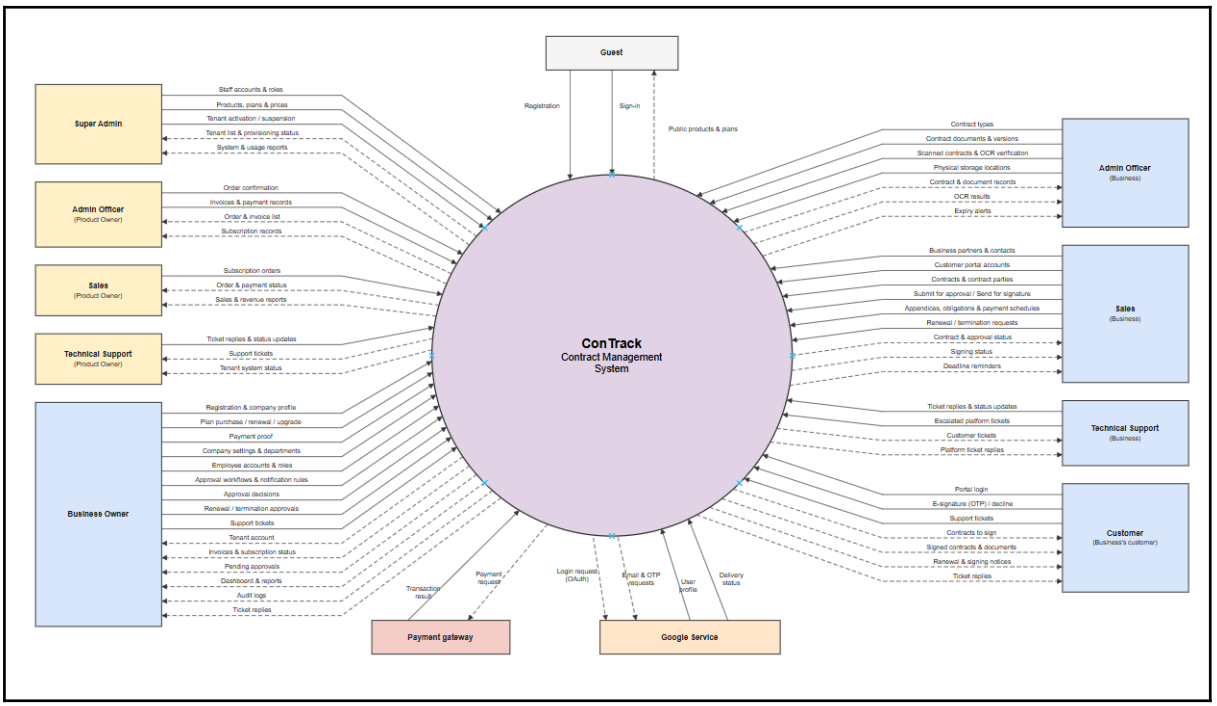



Figure 1. Context Diagram

## 2. Main Business Flow

## 2.1 Product Subscription (performed by Business Owner)

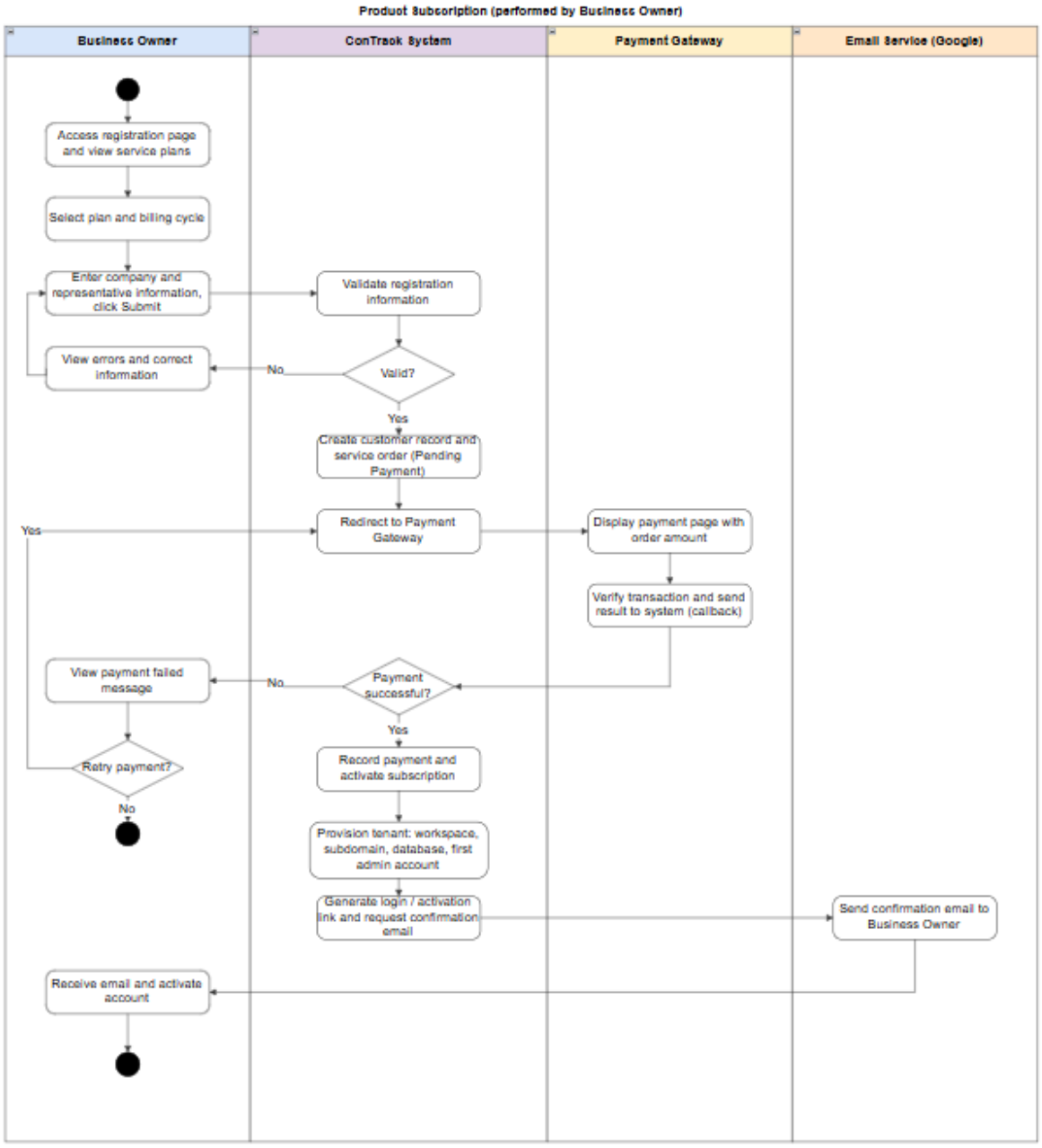



Figure 2.1. Swimlane for Product Subscription

## 2.2 Online Contract Management (Create a New Contract with a Customer)

**Online Contract Management (Create a New Contract with a Customer)**

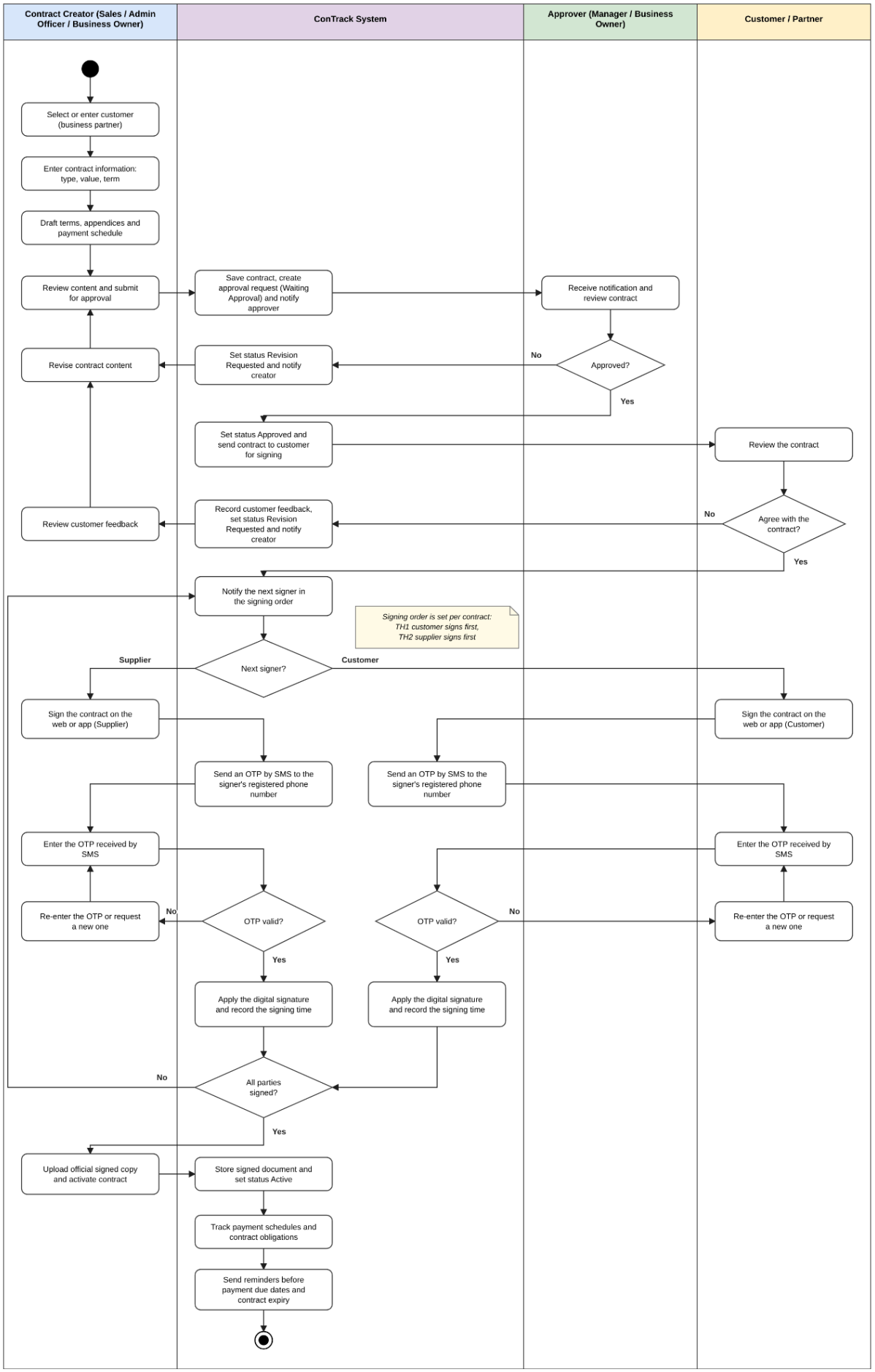


Figure 2.2 Online Contract Management

## 2.3 Upload Contracts

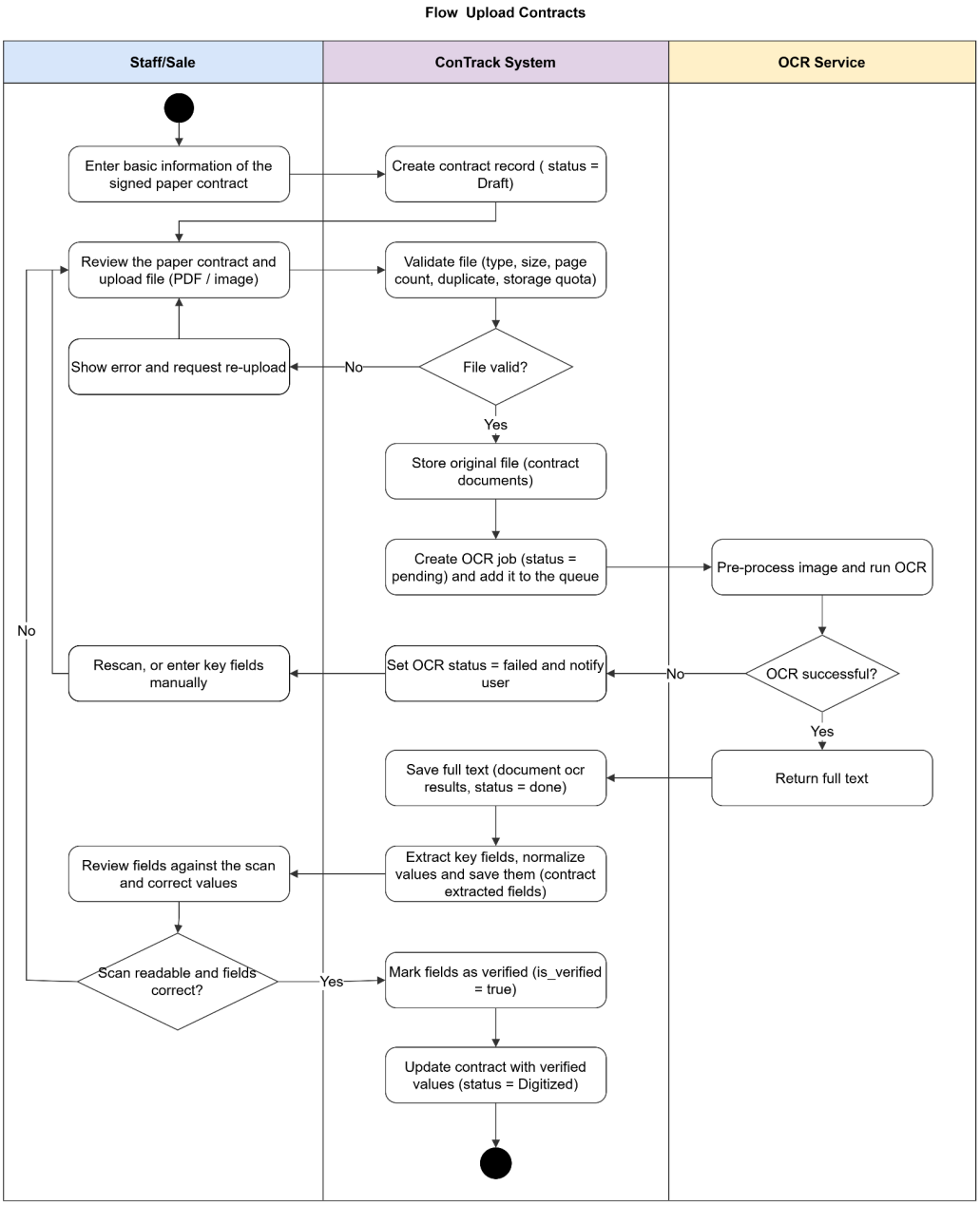



Figure 2.3. Upload Contracts

## 2.4 Business Owner Support

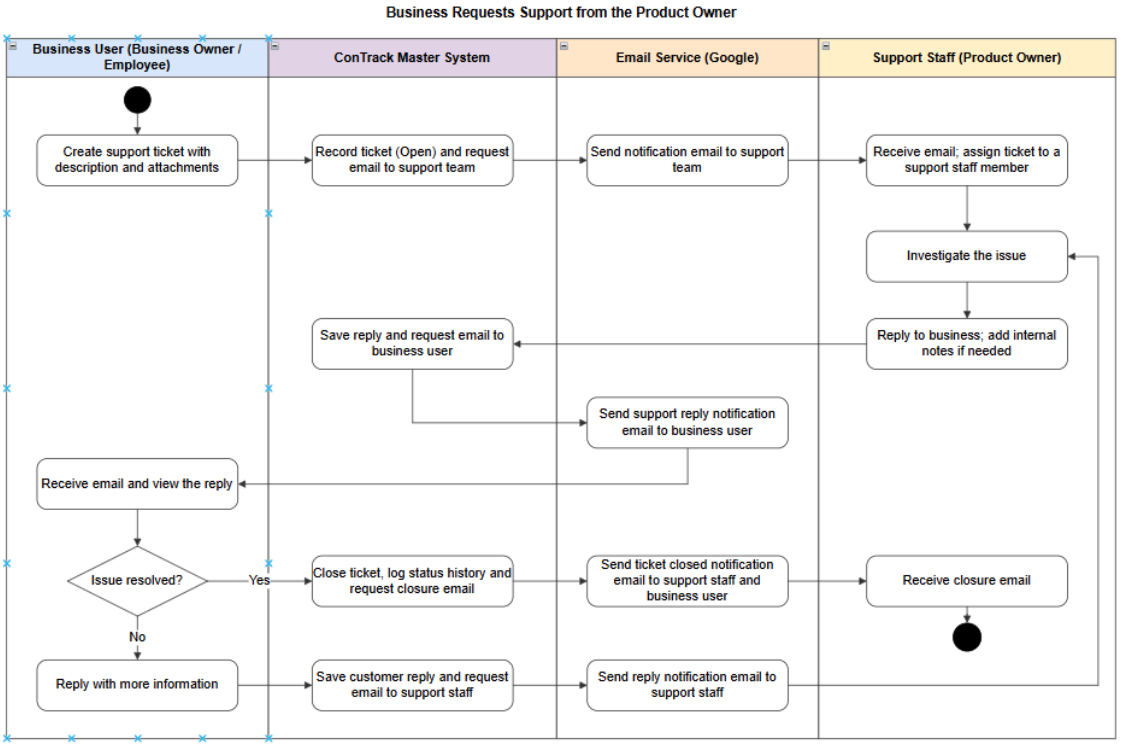



Figure 2.4. Business Owner Support

## 2.5 Search Contracts

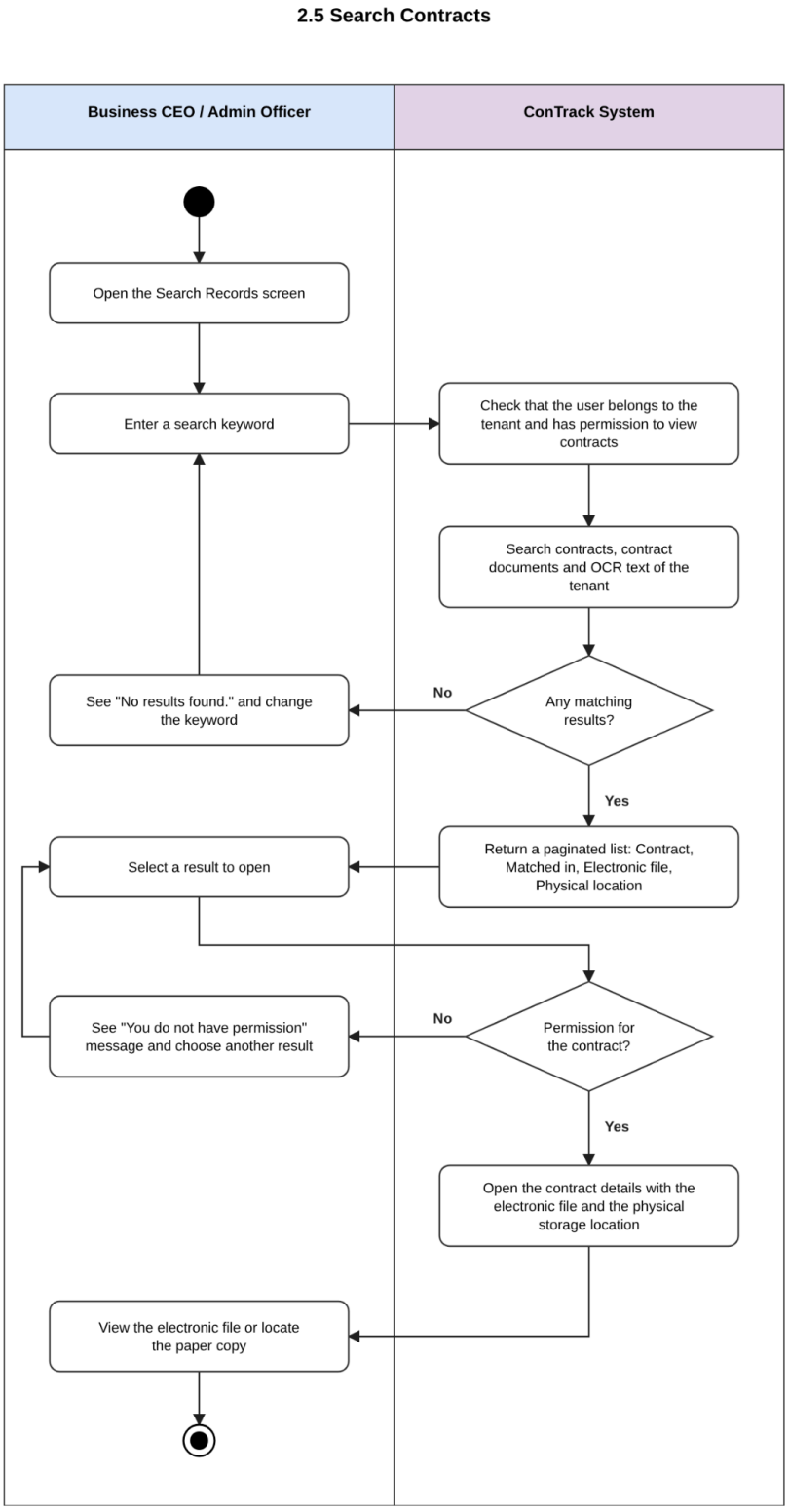



Figure 2.5. Search Contracts

# 3. Conceptual Data Model

## 3.1 Entity Relationship Diagram

### 3.1.1 MASTER DATABASE ENTITIES (Product Owner View)

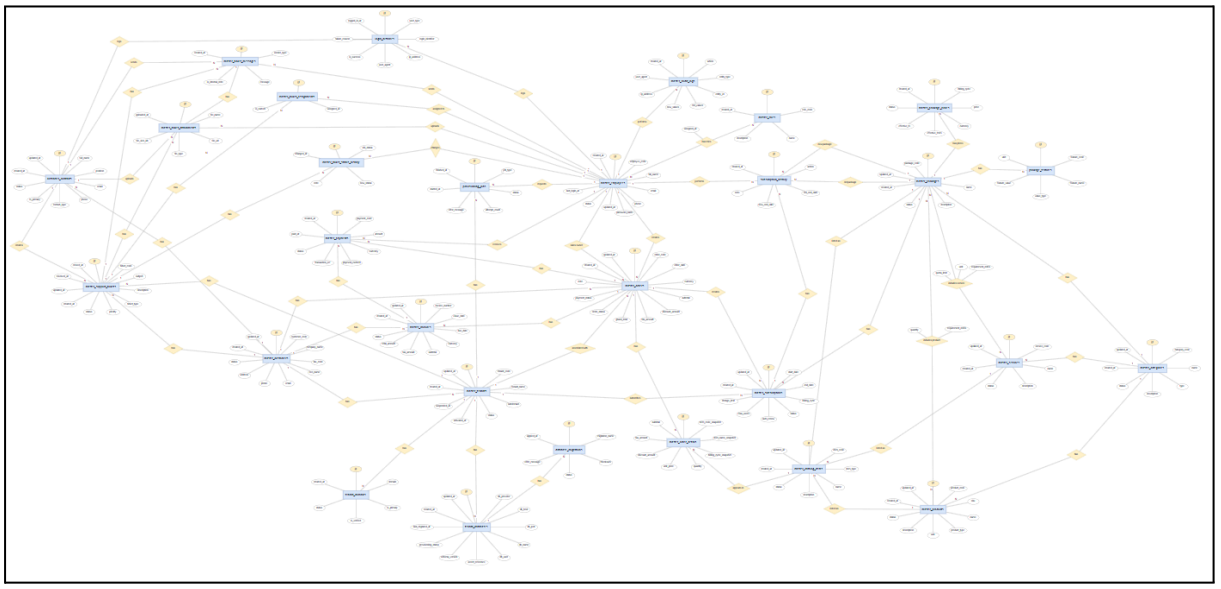


Figure 3.1. Entity diagram for master database

| # | Entity | Description |
|---|---|---|
| 1 | Employee | Internal staff of the product owner (Super Admin, Admin Officer, Sales, Technical Support), with login credentials and account status. |
| 2 | Role | A platform role that groups permissions, e.g. SUPER_ADMIN, ADMIN_OFFICER, SALES, TECHNICAL_SUPPORT. |
| 3 | Employee Role | Assigns one or more platform roles to an employee. |
| 4 | Customer | A business (legal entity) that buys ConTrack: company name, tax code, legal representative and contact information. |
| 5 | Customer Contact | A contact person of a SaaS customer who can sign in to the platform portal, place orders and raise support tickets. |
| 6 | Tenant | The isolated workspace granted to a customer, identified by tenant code and subdomain, with its status (Provisioning, Active, Suspended, Terminated). |
| 7 | Tenant Domain | A domain or subdomain used to access a tenant; one domain is marked as primary. |

| | | |
|---|---|---|
| 8 | Tenant Database | Connection information of the tenant's dedicated database (host, port, name, user, secret reference), its schema version and provisioning status. |
| 9 | Provisioning Job | A background job that creates or changes a tenant's resources (create database, run migration, create admin, suspend, terminate) and its result. |
| 10 | Database Migration | A schema migration applied to a tenant database, with version, status and time, so all tenants stay on the same schema version. |
| 11 | Category | Groups packages, products and services in the catalog. |
| 12 | Product | A sellable product of the product owner (software or hardware), with SKU and unit. |
| 13 | Service | A value-added service of the product owner, such as training or data migration. |
| 14 | Package | A subscription plan sold to customers, with its description and status. |
| 15 | Package Price | The price of a package for one billing cycle (monthly, quarterly, yearly) within an effective period. |
| 16 | Package Feature | A feature or limit included in a package, such as maximum users, storage or number of contracts. |
| 17 | Package Product | A product bundled in a package, with quantity. |
| 18 | Package Service | A service bundled in a package, with its quota. |
| 19 | Catalog Item | A unified sales item that points to a package, product or service, so order details reference a single catalog. |
| 20 | Order | A purchase made by a customer for a tenant, with amounts (subtotal, discount, tax, grand total), order status and payment status. |
| 21 | Order Detail | A line of an order: the catalog item, quantity, unit price and a snapshot of the item code and name at the time of sale. |
| 22 | Subscription | The tenant's active use of a package for a period (start and end date, billing cycle, limits, auto-renew, status). |
| 23 | Subscription History | A record of every change to a subscription: created, renewed, upgraded, downgraded, suspended or cancelled. |
| 24 | Invoice | A bill issued to a customer for an order, with issue date, due date, amounts and status. |

| | | |
|---|---|---|
| 25 | Payment | A payment received against an invoice, with amount, method, transaction reference and the employee who recorded it. |
| 26 | Support Ticket | A support request raised by a customer to the product owner, with category, priority and status. |
| 27 | Ticket Message | A message in a support ticket, sent by a customer contact or an employee; internal notes are hidden from customers. |
| 28 | Ticket Attachment | A file attached to a support ticket or message, stored by storage key. |
| 29 | Ticket Assignment | Records which employee is assigned to handle a support ticket and when. |
| 30 | Ticket Status History | Records every status change of a support ticket, who made it and when. |
| 31 | Audit Log | Records actions performed by platform employees, with old and new values. |
| 32 | Login History | Records every sign-in attempt by an employee or customer contact, with result, IP address and time. |

### 3.1.2 TENANT DATABASE ENTITIES (Business Tenant Operations)

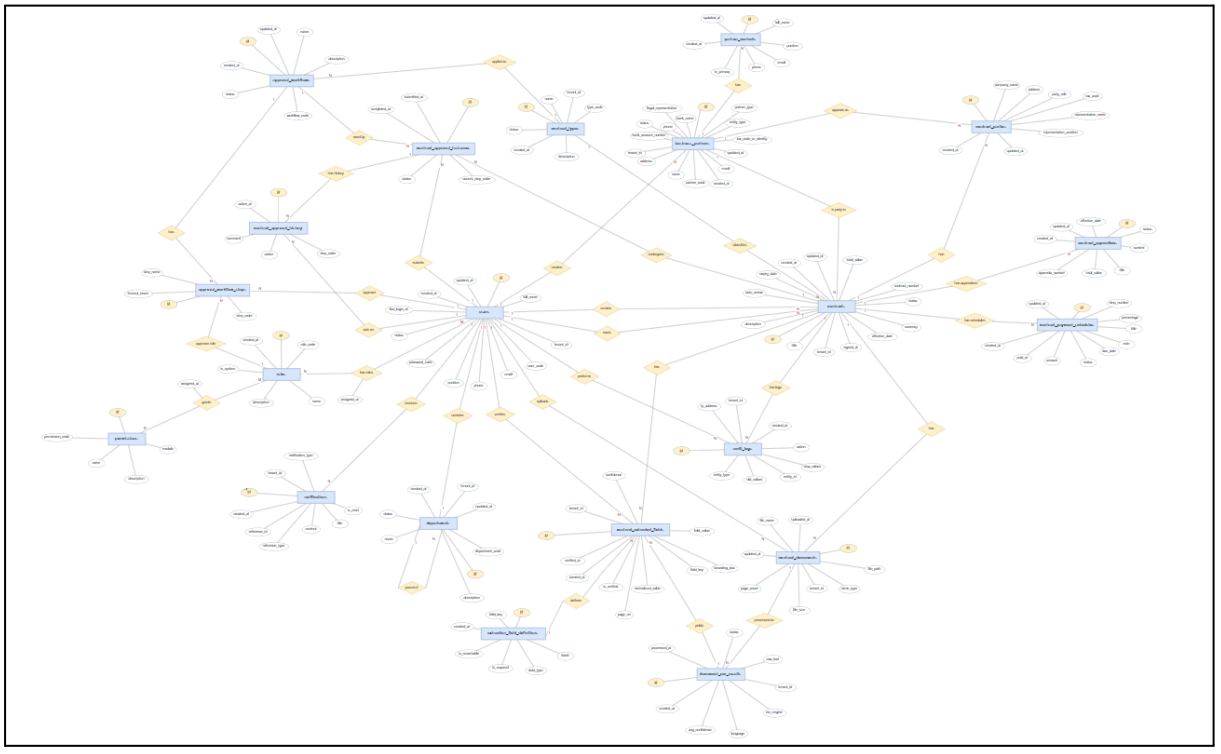


Figure 3.2. Entity diagram for tenant database

| # | Entity | Description |
|---|---|---|
| 1 | Department | An organizational unit of the enterprise within a tenant, with a parent-child hierarchy and an active/inactive status. |
| 2 | Role | A tenant-level role that groups permissions; system roles are flagged and cannot be removed. |
| 3 | Permission | A fine-grained permission, grouped by module, that can be granted to roles. |
| 4 | Role Permission | Assigns one or more permissions to a role. |
| 5 | User | A staff member of the enterprise who uses the contract system, with user code, department, position, login credentials, account status (Active, Inactive, Locked) and last sign-in time. |
| 6 | User Role | Assigns one or more roles to a user. |
| 7 | Business Partner | A counterparty of the enterprise (customer, supplier or both; company or individual), with tax code or ID number, legal representative, contact and bank information. |
| 8 | Partner Contact | A contact person of a business partner; one contact can be marked as primary. |
| 9 | Contract Document | An uploaded or scanned contract file (file name, storage path, type, size, page count), recorded with the user who uploaded it and linked to a contract once confirmed. |
| 10 | Document OCR Result | The outcome of OCR processing on a document: engine used, status (pending, processing, done, failed), raw text, language and average confidence. |
| 11 | Extraction Field Definition | A configurable definition of a data field to extract from contracts (key, label, data type, required, searchable). |
| 12 | Contract Extracted Field | A value extracted from an OCR result for a field definition, with normalized value, confidence, page and position, and whether a user has verified it. |
| 13 | Contract Type | A classification of contracts (e.g. sales, service, lease) used for filtering and for choosing the approval workflow. |
| 14 | Contract | The master record of a confirmed contract: number, title, type, partner, owner, value, currency, effective, expiry and signed dates, auto-renew flag and status (Draft, PendingApproval, Approved, Rejected, Signed, Active, Expired, Terminated, Cancelled). |

| | | |
|---|---|---|
| 15 | Contract Appendix | An amendment or addendum to a contract, with its number, content, effective date, value and status. |
| 16 | Contract Party | A party to a contract with its role (Party A, Party B, third party), keeping a snapshot of company name, tax code, representative and address at signing time. |
| 17 | Contract Payment Schedule | A payment installment of a contract: step number, title, amount, percentage, due date and status (Pending, Paid, Overdue, Cancelled), with the paid time. |
| 18 | Approval Workflow | A template approval process, optionally tied to a contract type, with its status. |
| 19 | Approval Workflow Step | One ordered step of a workflow, assigned to an approver role or a specific user, with a timeout in hours. |
| 20 | Contract Approval Instance | An actual approval run for a contract: the workflow used, current step, status, who submitted it, and submission and completion times. |
| 21 | Contract Approval History | The decision recorded at each approval step: approver, action (Approve, Reject, Request Change), comment and time. |
| 22 | Notification | An in-app message to a user, such as an expiry warning, payment due or approval request, with read status and a reference to the related record. |
| 23 | Audit Log | Records user actions on tenant data (create, update, delete, OCR verify, approve), with the affected entity, old and new values and IP address. |

## 3.3 Data Business Rules

| Rule | Description | Enforced On |
|---|---|---|
| Unique Master Employee Credentials | No two Master Employees may share the same email address or employee code. | DB: UQ_master_employees_code, UQ_master_employees_email. Employee creation and profile updates |
| Master Employee Account Status | An employee account must be Active, Inactive or Locked. Only Active accounts may sign in to the platform. | DB: CK_master_employees_status. Authentication service |
| Unique Role Code and Single Assignment | A platform role code is unique, and the same role cannot be assigned twice to the same employee. | DB: UQ_master_roles_code, UQ_master_employee_roles. Role management |

| | | |
|---|---|---|
| Unique Master Customer Identity | Each subscribing business must have a unique customer code, Tax Code and primary email address. | DB: UQ_master_customers_code, UQ_master_customers_tax, UQ_master_customers_email. Customer registration and profile updates |
| Customer Status | A customer must be Active, Inactive or Blocked. Inactive or Blocked customers cannot place new orders or open new tickets. | DB: CK_master_customers_status. Order and ticket creation APIs |
| Unique Customer Contact Email | Within one customer, each contact must have a distinct email address. A contact type must be General, Technical or Billing. | DB: UQ_customer_contacts_email, CK_customer_contacts_type. Contact management |
| Single Primary Contact | A customer has at most one primary contact at any time. | App: Contact management API |
| Accounts and Entities Soft Deletion | Employees, customers, contacts, products, services, packages and tenants that are referenced by orders, subscriptions, tickets or logs cannot be physically deleted. They must be deactivated (status = Inactive, Cancelled or Terminated). | App: Entity deletion handlers |
| Single Tenant per Customer | Each customer is linked to exactly one tenant (1 Customer = 1 Tenant). The restriction can be lifted later by dropping the unique constraint. | DB: UQ_master_tenants_customer. Tenant provisioning service |
| Unique Tenant Code and Subdomain | A tenant code is unique. A subdomain, when provided, is unique across all tenants. | DB: UQ_master_tenants_code, UX_master_tenants_subdomain. Tenant provisioning |
| Tenant Lifecycle Status | A tenant status must be Provisioning, Active, Suspended or Terminated. Only Active tenants can be accessed by users. | DB: CK_master_tenants_status. Request middleware |
| Unique Tenant Domain | A domain name can belong to only one tenant. At most one domain per tenant is marked primary, and only verified domains are used for request routing. | DB: UQ_tenant_domains_domain. App: Domain management and routing |

| | | |
|---|---|---|
| Dedicated Database per Tenant | Each tenant database is identified by a unique combination of host, port and database name, so two tenants can never share a database. | DB: UQ_tenant_databases_server. Tenant provisioning |
| Tenant Database Isolation | Cross-tenant database queries are strictly prohibited. Connection routing must resolve the connection parameters from tenant_databases for the tenant of the current request, and each request is served by that tenant's database only. | App: API Gateway / Dynamic Connection Router |
| No Plaintext Database Credentials | The Master Database stores only a secret reference (secret_reference) pointing to the secret vault. Database passwords must never be stored in any table. | App: Provisioning service / Secret manager |
| Tenant Database Provisioning Status | A tenant database status must be Pending, Provisioning, Ready, Failed or Decommissioned. Only Ready databases can receive tenant traffic. | DB: CK_tenant_databases_status. Connection router |
| Provisioning Job Lifecycle | Every create, migrate, suspend or terminate action on tenant resources runs as a provisioning job with status Pending, Running, Succeeded, Failed or Cancelled, and records the employee who requested it. | DB: CK_provisioning_jobs_status. Provisioning service |

# 4. User Requirements

## 4.1 Actors

| # | Actor | Description |
|---|---|---|
| 1 | Guest | An unauthenticated visitor to the ConTrack website. A Guest can browse public products and service packages, view package details and pricing, and register a new account (directly or with Google). Once registered and verified, the Guest becomes a Business Owner. |
| 2 | Business Owner | The owner and top administrator of a business that buys ConTrack from the product owner. On the platform side, the Business Owner creates and maintains the company profile, purchases, renews or upgrades a service package, submits payment proof, and views invoices and subscription status. The Business Owner can also send support tickets to the product owner, for example to report a software error or request a package upgrade. When the subscription is paid, the system provides a tenant (a separate workspace with its own database) for the business, and the Business Owner becomes the first administrator of that tenant. Inside the tenant, the Business Owner maintains company settings (company information, logo, contract code format) and the department structure. The Business Owner also creates employee accounts and assigns roles and permissions, configures approval workflows and notification rules, and approves or rejects contracts at the required approval step. Contract performance is monitored through the dashboard, reports and audit logs. |

| | | |
|---|---|---|
| 3 | Super Admin | The highest-level administrator on the product owner's side. The Super Admin manages internal staff accounts and roles, and maintains the catalog of packages, products and services, including package prices by billing cycle and package features. The Super Admin also oversees tenant provisioning, activates or suspends tenants, and monitors the tenant list, database migration status and system usage reports. |
| 4 | Admin Officer (Product Owner) | An administrative employee of the product owner. The Admin Officer processes subscription orders, issues invoices, records payments received from Business Owners, and maintains Business Owner accounts and subscription records. |
| 5 | Sales (Product Owner) | A sales employee of the product owner who sells ConTrack to businesses. Sales creates subscription orders for Business Owners and tracks order and payment status until the subscription is activated. Sales also views sales and revenue reports. |
| 6 | Technical Support (Product Owner) | A support employee of the product owner. Technical Support receives support tickets from Business Owners and tickets escalated by tenant businesses. Technical Support takes assignment of tickets, replies to requesters, adds internal notes, and updates ticket status until the issue is resolved and closed. It can also view a tenant's system status to diagnose problems. |
| 7 | Admin Officer (Business) | An administrative employee of the tenant business who manages contract documents and records. The Admin Officer manages contract types, uploads contract documents and new document versions, and digitizes legacy paper contracts by scanning them, checking the OCR results and confirming each scan. The Admin Officer also records where each paper original is physically stored (room, cabinet, shelf, box), so that both the electronic file and the paper copy can be located. |
| 8 | Sales (Business) | An employee of the tenant business who manages business partners and their contracts. Sales maintains business partners (customers, suppliers, partners, individuals), their contacts and bank accounts, and creates customer portal accounts. Sales also creates contracts with their parties, appendices, obligations and payment schedules. Sales then submits contracts for approval, sends approved contracts to customers for e-signature, and follows each contract through signing, renewal and termination. Sales receives reminders about expiring contracts, payment due dates and pending obligations. |
| 9 | Technical Support (Business) | A support employee of the tenant business who handles requests from the business's own customers. Technical Support receives support tickets submitted through the customer portal, replies to customers, adds internal notes that customers cannot see, and updates ticket status until the ticket is closed. Issues that come from the platform itself are escalated to the product owner's Technical Support. |

| | | |
|---|---|---|
| 10 | Customer (Business's customer) | A contact person of a business partner who has been given a customer portal account by Sales (Business). A Customer can log in to view related contracts and documents, review contracts sent for signing, and electronically sign them (confirmed by an OTP sent to email) or decline them with feedback. A Customer can also submit support tickets, follow their replies, and receive renewal and expiry notices. |
| 11 | Google Service | An external system that provides Google sign-in (OAuth) for registration and login, and email delivery. ConTrack uses it to send account information, password reset links, approval requests, one-time passwords (OTP) for e-signature, and contract and ticket notifications. |
| 12 | Payment Gateway | An external payment system that processes online payments for subscription orders. ConTrack sends a payment request when the Business Owner pays for a package, and the Payment Gateway returns the transaction result (success or failure). ConTrack uses that result to update the invoice and activate the subscription. |

## 4.2 Use Cases (UC)

| ID | Use Case | Feature | Use Case Description |
|---|---|---|---|
| Guest | | | |
| 1 | View Products | Product Management | Guest views the products provided by ConTrack. |
| 2 | View Plans | Subscription Management | Guest views available plans, pricing, duration and included limits. |
| 3 | Register Account | Registration | Guest registers an organization account to use ConTrack; the account becomes the Business Owner of a new tenant. |
| 4 | Verify Email | Registration | System sends a verification link through Google Service; the guest confirms the email before the account is activated (included in Register Account). |
| 5 | Login Account | Authentication | Guest logs into the system using account credentials. |
| Common – Super Admin, Admin Officer, Sales, Technical Support (Product Owner & Business), Business Owner, Customer | | | |

| | | | |
|---|---|---|---|
| 6 | Login | Authentication | Signed-in actor logs into ConTrack with email and password; the system routes the user to the platform or tenant workspace according to the account. |
| 7 | Login with Google | Authentication | Actor logs in with a Google account through OAuth instead of a password (extends Login; not available to Customer). |
| 8 | Forgot Password | Authentication | Actor requests a password-reset link sent by email and sets a new password (extends Login). |
| 9 | Logout | Authentication | Actor ends the current session and is returned to the login page. |
| 10 | View/Update Profile | Account Management | Actor views and updates personal information such as name, phone number and avatar. |
| 11 | Change Password | Account Management | Actor changes the current password after confirming the old one (extends View/Update Profile). |
| 12 | View Notifications | Notification | Actor views in-app notifications such as approval requests, expiry alerts and ticket replies, and marks them as read. |
| Business Owner | | | |
| 13 | Manage Company Profile | Tenant Management | Business Owner views and updates the company profile: company name, tax code, address, logo and contact information. |
| 14 | Purchase Plan | Subscription Management | Business Owner selects a plan and billing cycle and places a subscription order. |
| 15 | View Subscription Status | Subscription Management | Business Owner views the current plan, start and end dates, usage against limits and subscription status. |
| 16 | Renew Subscription | Subscription Management | Business Owner renews the subscription before or after it expires (extends View Subscription Status). |

| | | | |
|---|---|---|---|
| 17 | Upgrade/Downgrade Plan | Subscription Management | Business Owner changes to a higher or lower plan; the system calculates the price difference (extends View Subscription Status). |
| 18 | View Orders & Invoices | Billing & Payment | Business Owner views subscription orders, invoices and their payment status. |
| 19 | Submit Payment Proof | Billing & Payment | Business Owner uploads the bank-transfer receipt for an unpaid invoice (extends View Orders & Invoices). |
| 20 | View Tenant Information | Tenant Management | Business Owner views tenant information: tenant code, workspace URL, provisioning status and storage usage. |
| 21 | Manage Company Settings | Company Configuration | Business Owner configures company-wide settings such as time zone, currency, contract code format and default reminder days. |
| 22 | Manage Departments | Company Configuration | Business Owner creates, updates and deactivates departments and assigns department heads. |
| 23 | Manage Employee Accounts | User & Role Management | Business Owner creates, updates, locks and unlocks employee accounts; the system emails login information through Google Service. |
| 24 | Manage Roles & Permissions | User & Role Management | Business Owner creates roles and assigns permissions to each role (extends Manage Employee Accounts). |
| 25 | Configure Approval Workflows | Contract Approval | Business Owner defines approval workflows: steps, approver roles and conditions such as contract type or value. |
| 26 | Approve/Reject Contract | Contract Approval | Business Owner reviews a contract waiting at their approval step and approves or rejects it with a comment. |

| | | | |
|---|---|---|---|
| 27 | Configure Notification Rules | Notification | Business Owner sets rules for alerts: which events, how many days before, and who receives them. |
| 28 | View Dashboard | Reporting | Business Owner views the dashboard with contract counts by status, total value, contracts expiring soon and pending approvals. |
| 29 | View Contract Reports | Reporting | Business Owner views and exports detailed contract reports filtered by time, department, partner or type (extends View Dashboard). |
| 30 | View Audit Logs | Audit | Business Owner views the history of user actions in the tenant: who did what, when, and on which record. |
| 31 | Create Support Ticket | Platform Support | Business Owner sends a support ticket about the platform to the Product Owner's Technical Support. |
| 32 | Track Support Ticket | Platform Support | Business Owner tracks ticket status and replies, and adds further information to the ticket. |
| Super Admin | | | |
| 33 | Manage Staff Accounts | User & Role Management | Super Admin creates, updates and deactivates Product Owner staff accounts; login information is emailed through Google Service. |
| 34 | Manage Roles | User & Role Management | Super Admin defines platform roles and their permissions (extends Manage Staff Accounts). |
| 35 | Manage Categories | Product Management | Super Admin manages categories used to group products and plans. |
| 36 | Manage Catalog Items | Product Management | Super Admin creates, updates and hides products and plans shown to guests. |
| 37 | Manage Plan Prices | Subscription Management | Super Admin sets prices for each plan and billing cycle (extends Manage Catalog Items). |

| | | | |
|---|---|---|---|
| 38 | Manage Plan Features | Subscription Management | Super Admin sets the features and limits of each plan, such as number of users, contracts and storage (extends Manage Catalog Items). |
| 39 | View Tenant List | Tenant Management | Super Admin views all tenants with their plan, status and creation date. |
| 40 | Provision Tenant | Tenant Management | Super Admin triggers or retries the creation of a tenant database for a paid subscription (extends View Tenant List). |
| 41 | Activate/Suspend Tenant | Tenant Management | Super Admin activates or suspends a tenant, for example when payment is overdue (extends View Tenant List). |
| 42 | Monitor Provisioning & Migrations | Tenant Management | Super Admin monitors the provisioning status and database migration version of each tenant. |
| 43 | View System & Usage Reports | Reporting | Super Admin views platform reports: number of tenants, active users, storage usage and revenue. |
| 44 | View Audit Logs | Audit | Super Admin views the history of actions performed on the platform (Master DB). |
| Admin Officer (Product Owner) | | | |
| 45 | Manage Business Owner Accounts | Tenant Management | Admin Officer views and updates Business Owner (platform customer) accounts and their company information. |
| 46 | View Orders & Invoices | Billing & Payment | Admin Officer views all subscription orders and invoices of every tenant. |
| 47 | Confirm Order | Billing & Payment | Admin Officer confirms a valid subscription order (extends View Orders & Invoices). |
| 48 | Issue Invoice | Billing & Payment | Admin Officer issues an invoice for a confirmed order and sends it by email through Google Service (extends View Orders & Invoices). |

| | | | |
|---|---|---|---|
| 49 | Record Payment | Billing & Payment | Admin Officer checks the payment proof and records the payment; the subscription is activated (extends View Orders & Invoices). |
| 50 | Manage Subscriptions | Subscription Management | Admin Officer views and updates tenant subscriptions: dates, plan and status. |
| Sales (Product Owner) | | | |
| 51 | Create Subscription Order | Subscription Management | Sales creates a subscription order on behalf of a Business Owner after consultation. |
| 52 | Track Order Status | Subscription Management | Sales tracks the status of orders they created: pending, confirmed, paid or cancelled. |
| 53 | Update Order | Subscription Management | Sales edits the plan, cycle or note of an order that is not yet confirmed (extends Track Order Status). |
| 54 | View Sales Report | Reporting | Sales views reports on orders, conversions and revenue by period. |
| Technical Support (Product Owner) | | | |
| 55 | View Support Tickets | Platform Support | Technical Support views platform support tickets sent by Business Owners and escalated by tenants. |
| 56 | Assign Ticket | Platform Support | Technical Support assigns a ticket to themselves or to another staff member (extends View Support Tickets). |
| 57 | Reply to Ticket | Platform Support | Technical Support replies to a ticket; the requester is notified by email through Google Service (extends View Support Tickets). |
| 58 | Add Internal Note | Platform Support | Technical Support adds a note visible only to staff (extends View Support Tickets). |

| | | | |
|---|---|---|---|
| 59 | Update Ticket Status | Platform Support | Technical Support changes the ticket status: open, in progress, resolved or closed (extends View Support Tickets). |
| 60 | View Tenant Status | Tenant Management | Technical Support views a tenant's system status to diagnose problems: provisioning, database version and storage. |
| Admin Officer (Business) | | | |
| 61 | Manage Contract Types | Contract Configuration | Admin Officer creates and updates contract types and their required fields. |
| 62 | Upload Contract Document | Document Management | Admin Officer uploads contract files to the internal server and links them to a contract. |
| 63 | Upload New Document Version | Document Management | Admin Officer uploads a new version of a document while keeping the previous versions (extends Upload Contract Document). |
| 64 | Scan Paper Contract | Contract Digitization | Admin Officer uploads a scanned paper contract; the system runs OCR and extracts key fields. |
| 65 | Verify OCR Result | Contract Digitization | Admin Officer reviews and corrects extracted fields against the scan and confirms them (extends Scan Paper Contract). |
| 66 | Manage Storage Locations | Physical Archive | Admin Officer manages physical storage locations: cabinet, shelf and box. |
| 67 | Assign Physical Location | Physical Archive | Admin Officer records where the original paper contract is stored. |
| 68 | Search Contract Records | Document Management | Admin Officer searches contracts by code, partner, date, status or text inside documents (full-text search). |
| Sales (Business) | | | |

| | | | |
|---|---|---|---|
| 69 | Manage Business Partners | Partner Management | Sales creates and updates business partners (the business's customers and suppliers). |
| 70 | Manage Partner Contacts | Partner Management | Sales manages the contact persons of a partner (extends Manage Business Partners). |
| 71 | Manage Partner Bank Accounts | Partner Management | Sales manages a partner's bank accounts used in contracts (extends Manage Business Partners). |
| 72 | Create Portal Account | Partner Management | Sales creates a customer portal account for a partner contact; login information is emailed through Google Service (extends Manage Business Partners). |
| 73 | Create Contract | Contract Management | Sales creates a contract draft with type, partner, value, dates and terms. |
| 74 | Manage Contract Parties | Contract Management | Sales adds the parties and their representatives to the contract (included in Create Contract). |
| 75 | Submit Contract for Approval | Contract Approval | Sales submits a draft contract into the configured approval workflow (extends Create Contract). |
| 76 | Track Contracts | Contract Management | Sales tracks their contracts with approval status, signing status and important dates. |
| 77 | Update Contract Status | Contract Management | Sales updates the status of a contract, for example sends it for signature or marks it active (extends Track Contracts). |
| 78 | Request Contract Renewal | Contract Lifecycle | Sales creates a renewal request for a contract that is about to expire (extends Track Contracts). |
| 79 | Request Contract Termination | Contract Lifecycle | Sales creates a request to terminate a contract early, with a reason (extends Track Contracts). |

| | | | |
|---|---|---|---|
| 80 | Manage Appendices | Contract Lifecycle | Sales creates and updates contract appendices that change terms, value or dates. |
| 81 | Manage Obligations | Obligation Tracking | Sales records the obligations of each party with due dates and tracks their completion. |
| 82 | Manage Payment Schedules | Obligation Tracking | Sales defines payment installments of a contract and records paid amounts. |
| Technical Support (Business) | | | |
| 83 | View Customer Tickets | Customer Support | Technical Support views support tickets sent by the business's customers through the portal. |
| 84 | Assign Ticket | Customer Support | Technical Support assigns a customer ticket to a staff member (extends View Customer Tickets). |
| 85 | Reply to Ticket | Customer Support | Technical Support replies to a customer ticket; the customer is notified by email through Google Service (extends View Customer Tickets). |
| 86 | Add Internal Note | Customer Support | Technical Support adds a note visible only to the business's staff (extends View Customer Tickets). |
| 87 | Update Ticket Status | Customer Support | Technical Support changes the ticket status: open, in progress, resolved or closed (extends View Customer Tickets). |
| 88 | Manage Ticket Categories | Customer Support | Technical Support manages categories used to classify customer tickets. |
| Customer (Business's customer) | | | |
| 89 | View Contracts | Customer Portal | Customer views the contracts in which they are a party, with status and key dates. |
| 90 | View/Download Contract Documents | Customer Portal | Customer views and downloads contract files and signed copies (extends View Contracts). |

| | | | |
|---|---|---|---|
| 91 | Review Contract for Signing | E-signature | Customer reviews the content of a contract sent for signature. |
| 92 | Decline Contract | E-signature | Customer declines the contract with feedback; the contract returns to the creator for revision (extends Review Contract for Signing). |
| 93 | E-sign Contract | E-signature | Customer signs the contract electronically; the signature is recorded with time and IP address. |
| 94 | Verify Signing OTP | E-signature | System sends an OTP to the customer's email through Google Service; the customer enters it to confirm the signature (included in E-sign Contract). |
| 95 | Create Support Ticket | Customer Support | Customer sends a support ticket about a contract to the business. |
| 96 | Track Support Ticket | Customer Support | Customer tracks ticket status and replies from the business. |
| 97 | View Renewal Notices | Customer Portal | Customer views notices about contracts that are about to expire or be renewed. |
| System Scheduler (Timer) | | | |
| 98 | Send Expiry Alerts | Notification | System Scheduler runs daily and sends alerts for contracts about to expire, based on notification rules, by email through Google Service. |
| 99 | Send Payment Due Reminders | Notification | System Scheduler sends reminders for contract payment installments that are due or overdue. |
| 100 | Send Obligation Reminders | Notification | System Scheduler sends reminders for contract obligations that are due or overdue. |
| 101 | Expire Subscriptions | Subscription Management | System Scheduler marks tenant subscriptions past their end date as expired and notifies the Business Owner. |

| | | | |
|---|---|---|---|
| 102 | Update Contract Status | Contract Lifecycle | System Scheduler automatically changes contract status to Active on the effective date and to Expired on the end date. |

## 4.3 Use Case Diagrams

### 4.3.1 UCs for Guest

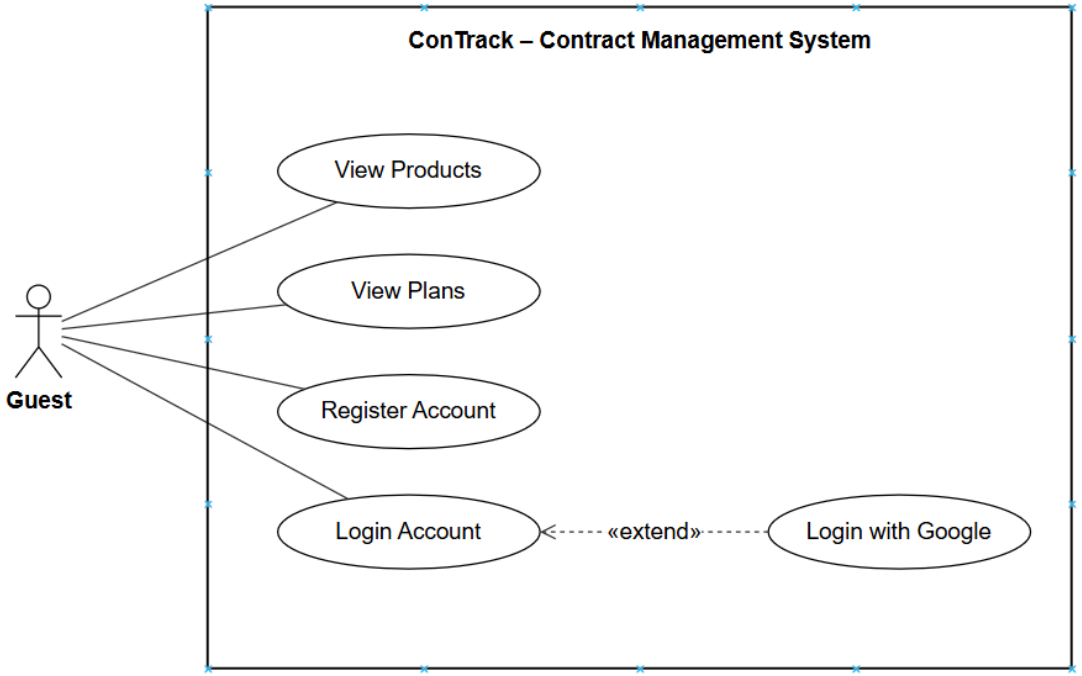



Figure 4.3.1. UCs for Guest

### 4.3.2 UCs for Business Owner

#### 4.3.2.1 UCs for Business Customer

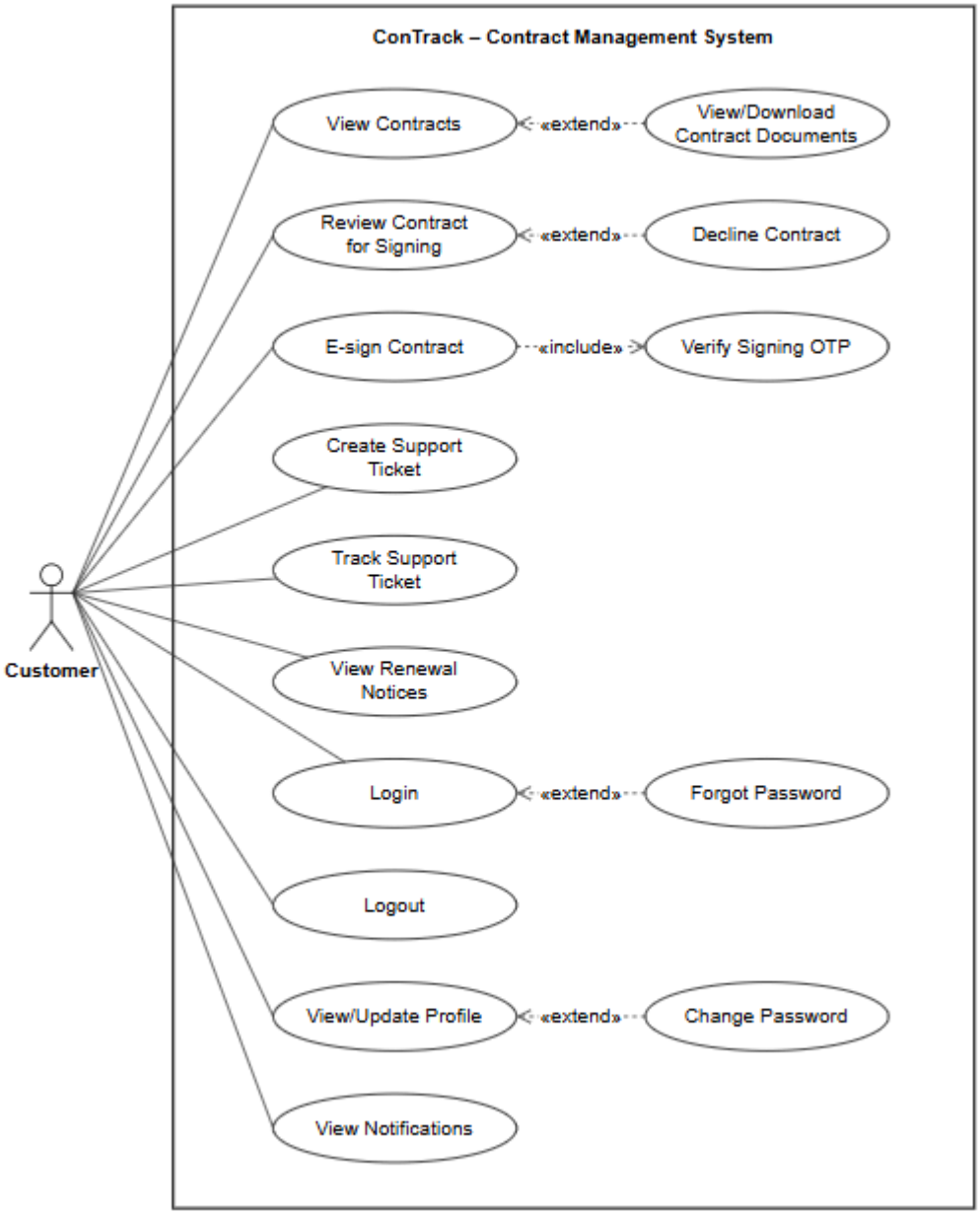



Figure 4.3.3.1 UCs for Business Customer

#### 4.3.2.2 UCs for Business Owner




Figure 4.3.3.2. UCs for Business Owner

#### 4.3.2.3 UCs for Business Admin Officer

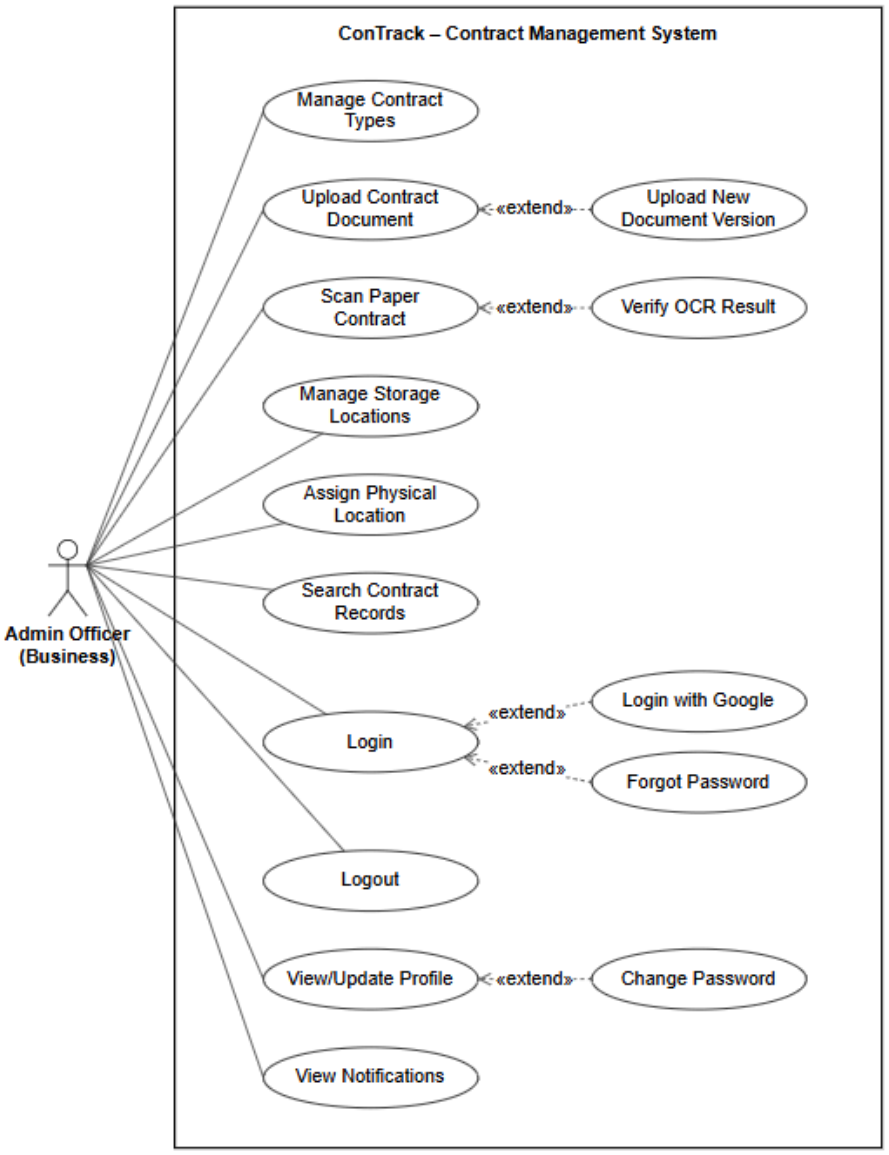



Figure 4.3.3.3. UCs for Business Admin Officer

#### 4.3.2.4 UCs for Business Sales

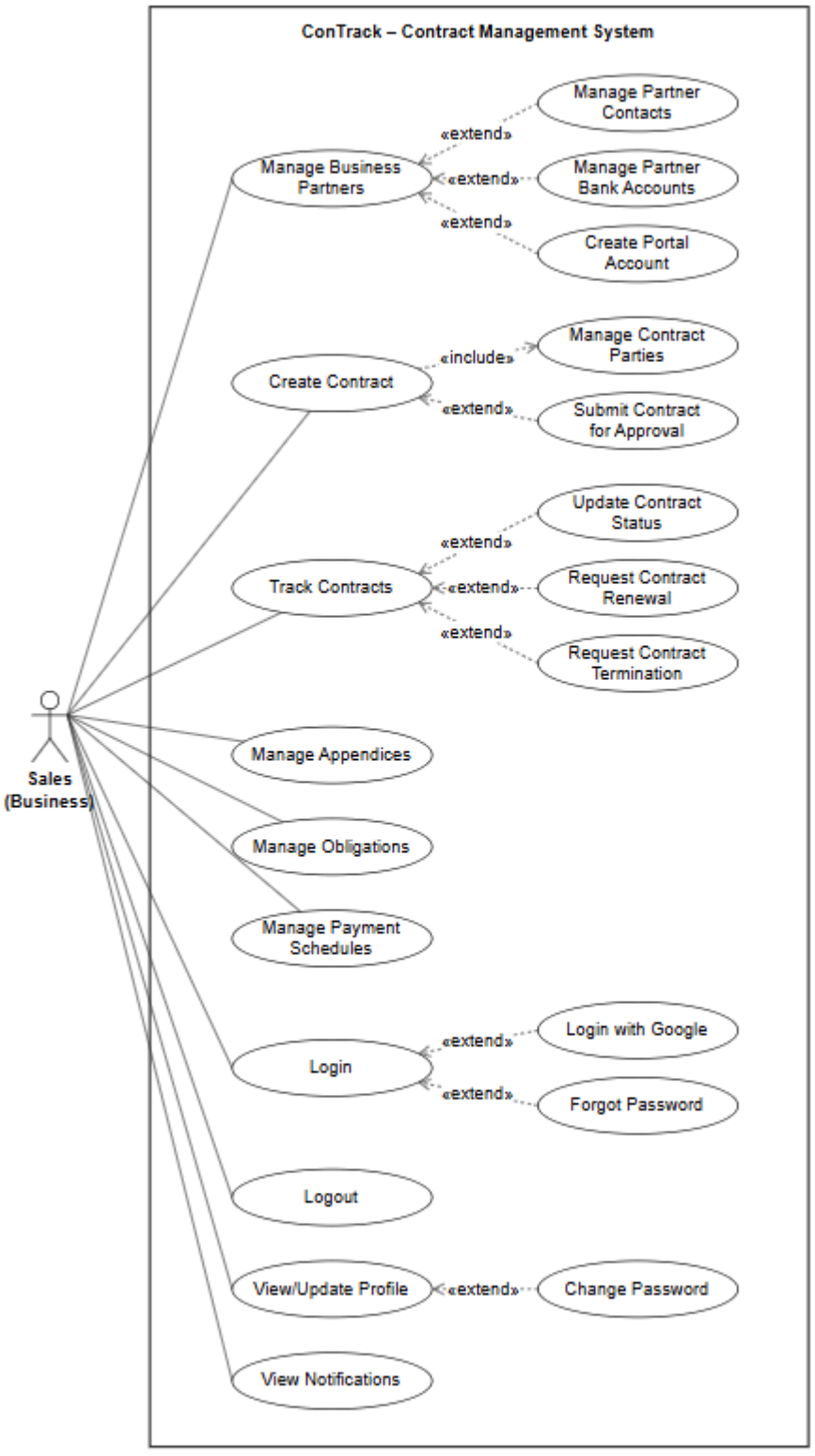



Figure 4.3.3.4. UCs for Business Sales

#### 4.3.2.5 UCs for Business Technical Support

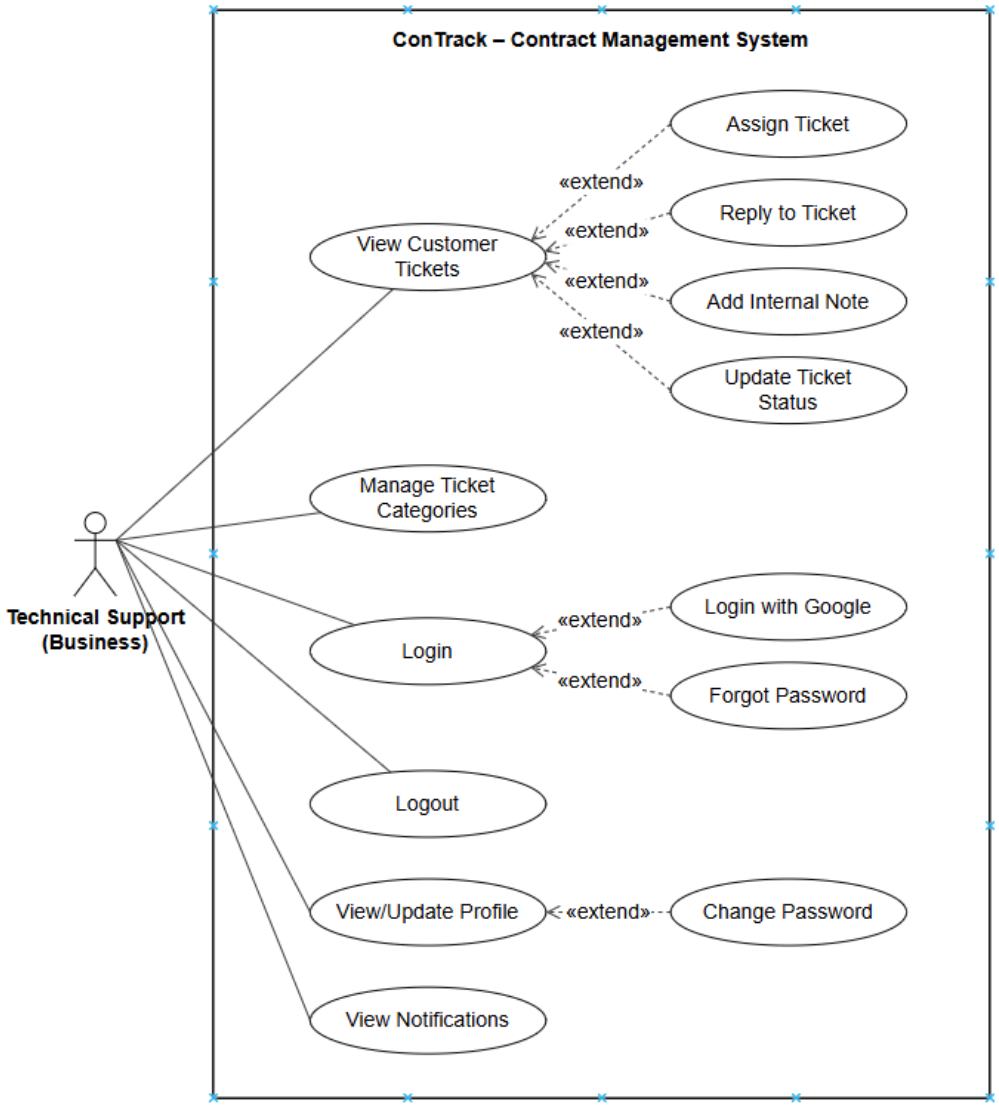



Figure 4.3.3.5. UCs for Business Technical Support

### 4.4.3 UCs for Product Owner

#### 4.4.3.1 UCs for Super Admin

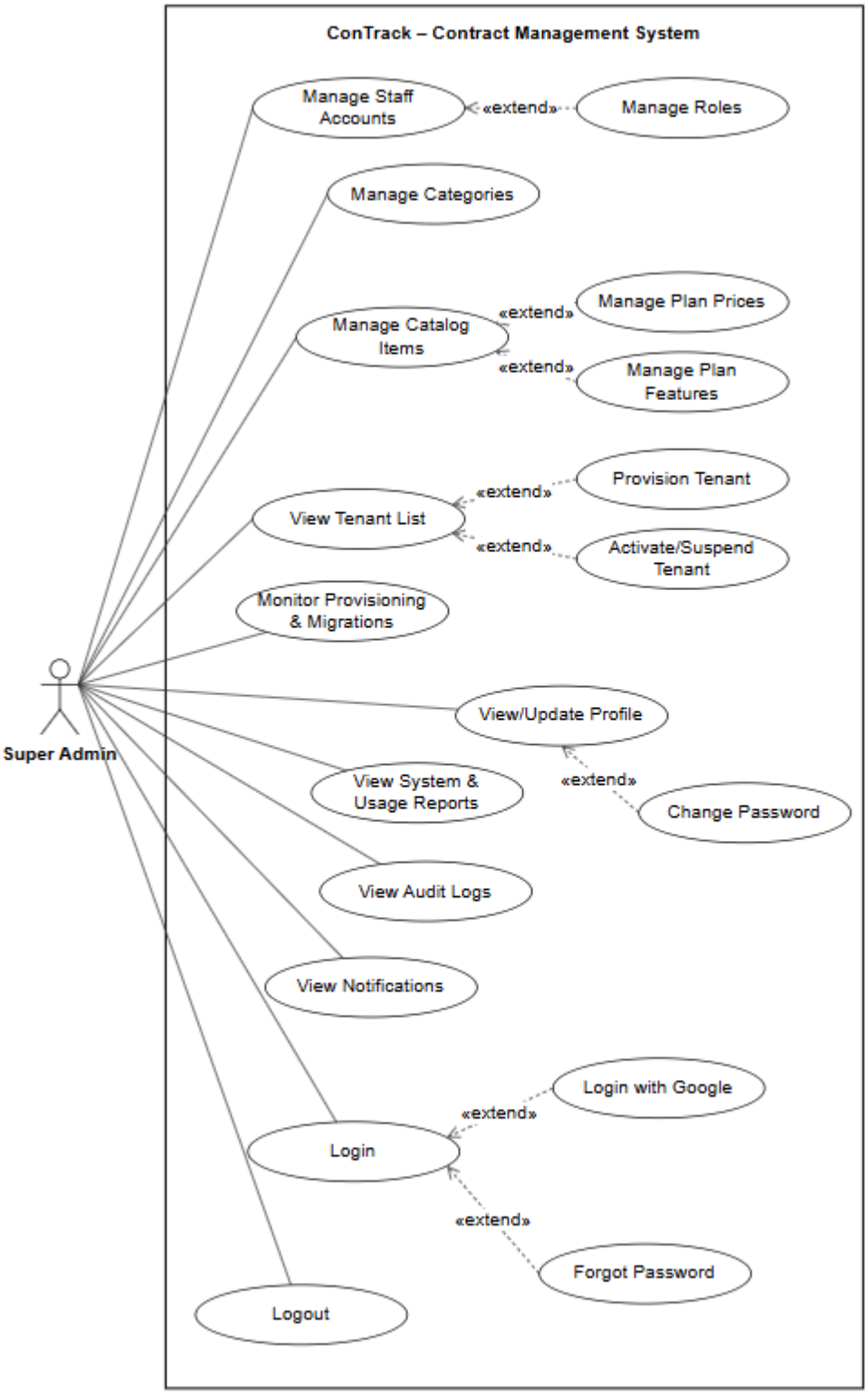



Figure 4.4.4.1. UCs for Super Admin

#### 4.4.3.2 UCs for Product Admin Officer

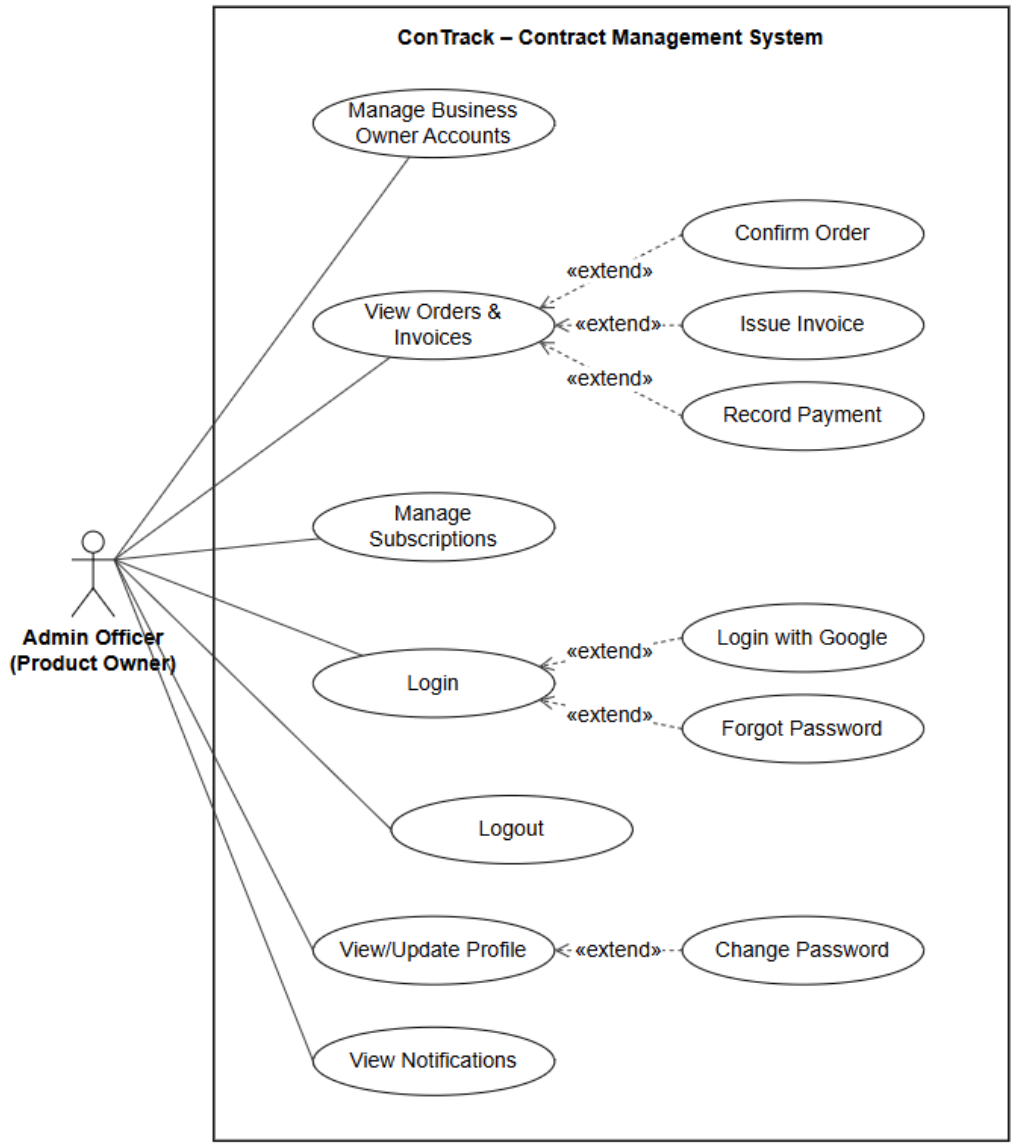



Figure 4.4.4.2. UCs for Product Admin Officer

#### 4.4.3.3 UCs for Product Sale

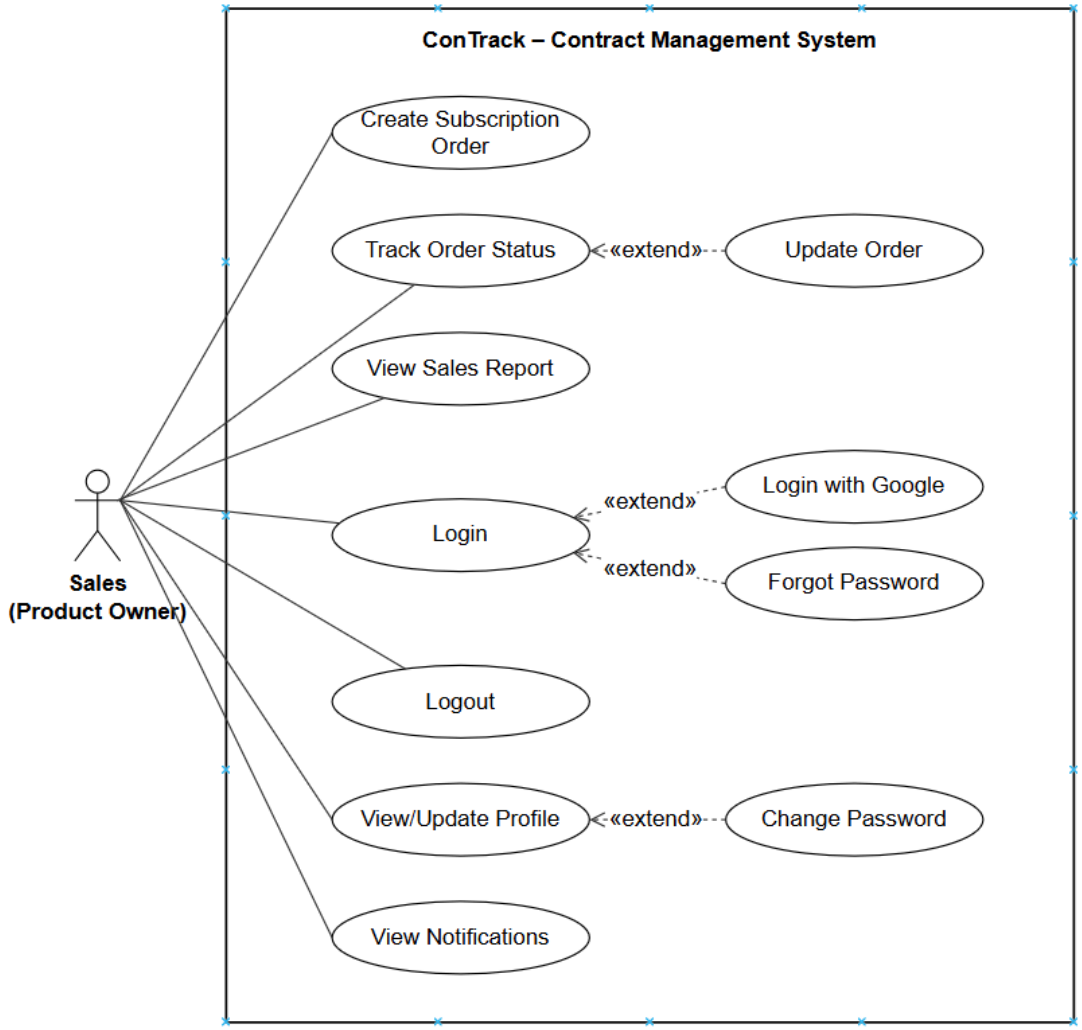



Figure 4.4.4.3. UCs for Product Sale

#### 4.4.3.4 UCs for Product Technical Support

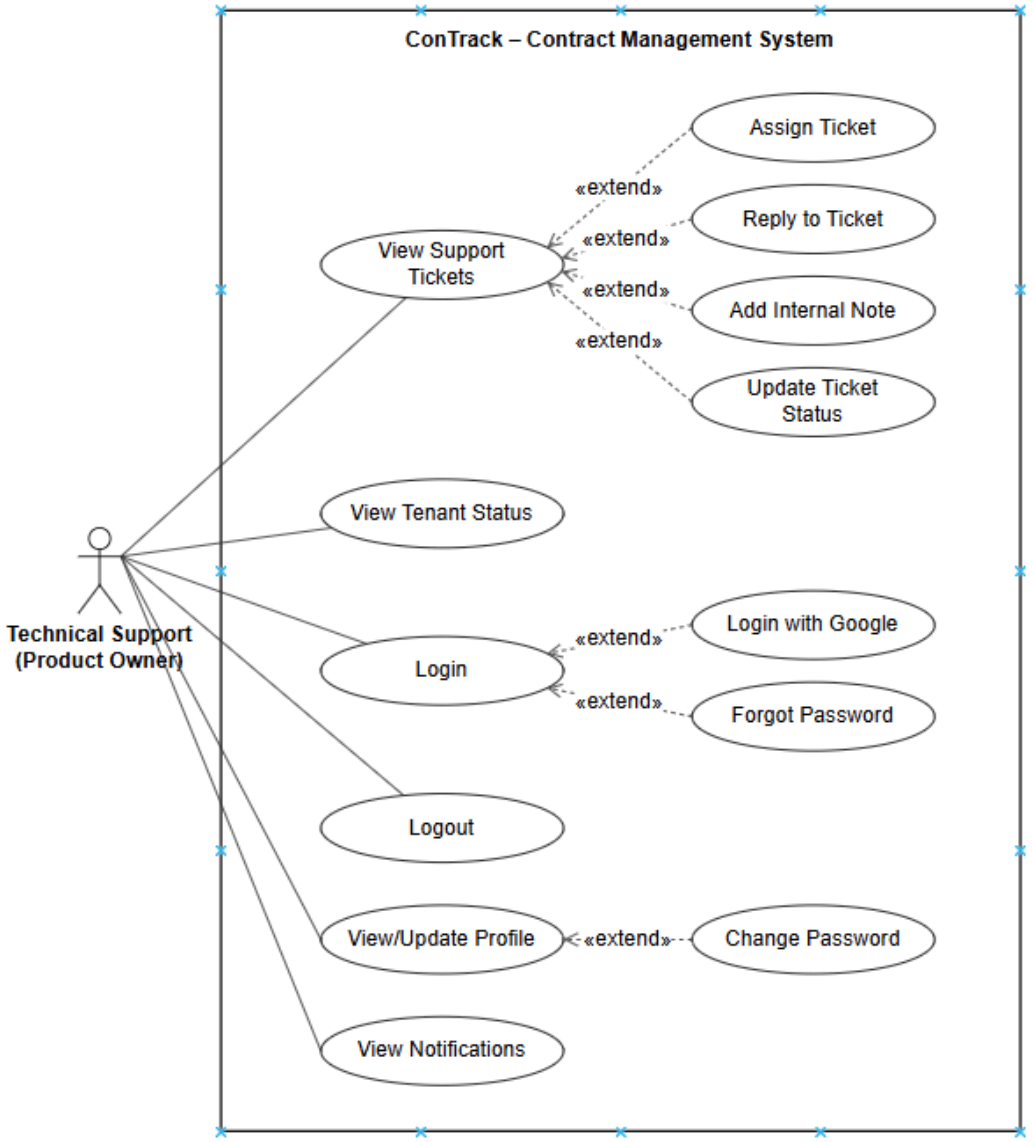



Figure 4.4.4.4. UCs for Product Technical Support

## 4.4 Permission Matrix

### 4.4.1 Product Owner

*Full = unrestricted · Restricted = conditions apply (see footnotes) · No = denied*

*SA = Super Admin · PAO = Admin Officer (Product Owner) · PS = Product Sales (Product Owner) · PTS = Product Technical Support (Product Owner) · BO = Business Owner*

| Entity | Action | SA | PAO | PS | PTS | BO | Guest |
|---|---|---|---|---|---|---|---|
| Staff account | Create | Full | No | No | No | No | No |
| Staff account | Edit / Lock / Deactivate | Full | No | No | No | No | No |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Staff account | Assign roles | Full | No | No | No | No | No |
| Category | Create / Edit / Deactivate | Full | No | No | No | No | No |
| Catalog item (package, product, service) | Create / Edit | Full | No | No | No | No | No |
| Catalog item | Discontinue | Full | No | No | No | No | No |
| Catalog item | View (public, active only) | Full | Full | Full | Full | Full | Full |
| Plan price & features | Create / Edit | Full | No | No | No | No | No |
| Business Owner account | Register | No | No | No | No | No | Full |
| Business Owner account | Create / Edit | Full | Full | Restricted[1] | No | Restricted[2] | No |
| Business Owner account | View | Full | Full | Restricted[1] | Full | Restricted[2] | No |
| Order | Create | Full | Full | Full | No | Full | No |
| Order | Update (draft only) | Full | Full | Restricted[3] | No | Restricted[4] | No |
| Order | Confirm / Cancel | Full | Full | No | No | No | No |
| Order | View | Full | Full | Restricted[3] | No | Restricted[4] | No |
| Invoice | Issue | No | Full | No | No | No | No |
| Invoice | View | Full | Full | Restricted[3] | No | Restricted[4] | No |
| Payment | Submit payment proof | No | No | No | No | Restricted[5] | No |
| Payment | Record / Confirm | No | Full | No | No | No | No |
| Subscription | Purchase / Renew / Request upgrade | No | No | No | No | Restricted[6] | No |
| Subscription | Activate / Renew / Cancel | Full | Full | No | No | No | No |
| Subscription | View | Full | Full | Restricted[1] | Full | Restricted[6] | No |
| Tenant | Provision | Full | No | No | No | No | No |
| Tenant | Activate / Suspend / Terminate | Full | No | No | No | No | No |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Tenant | View | Full | Full | No | Restricted[7] | Restricted[6] | No |
| Provisioning job & migration | View / Retry | Full | No | No | Restricted[8] | No | No |
| Support ticket | Create | No | No | No | No | Full | No |
| Support ticket | View | Full | No | No | Full | Restricted[9] | No |
| Support ticket | Assign | Full | No | No | Full | No | No |
| Support ticket | Reply / Update status | Full | No | No | Restricted[10] | Restricted[9] | No |
| Support ticket | Internal note (create / view) | Full | No | No | Full | No | No |
| Report | System & usage report | Full | No | No | No | No | No |
| Report | Sales & revenue report | Full | Full | Restricted[11] | No | No | No |
| Audit log & login history | View | Full | No | No | No | No | No |
| Notification | View own | Full | Full | Full | Full | Full | No |

**Footnotes**

1. PS: only Business Owners they consult or created orders for.
2. BO: only their own account and company profile.
3. PS: only orders they created and the invoices of those orders; updates allowed only while the order is a draft.
4. BO: only their own orders and invoices; updates allowed only while the order is a draft.
5. BO: only for their own unpaid invoices.
6. BO: only their own subscription and their own tenant.
7. PTS: technical information only (status, database version, storage); no billing data.
8. PTS: can view jobs and retry a failed job; cannot change tenant configuration.
9. BO: only tickets they created; can reply and close their own ticket, but cannot change other statuses.
10. PTS: only tickets assigned to them.
11. PS: only their own sales figures.

### 4.4.2 Business Owner

*Full = unrestricted · Restricted = conditions apply (see footnotes) · No = denied*

*BO = Business Owner · BAO = Admin Officer (Business) · BS = Sales (Business) · BTS = Technical Support (Business) · CU = Customer (Business's customer)*

| Entity | Action | BO | BAO | BS | BTS | CU |
|---|---|---|---|---|---|---|
| Company settings | View | Full | Full | Full | Full | No |
| Company settings | Edit | Full | No | No | No | No |

| | | | | | | |
|---|---|---|---|---|---|---|
| Department | Create / Edit / Deactivate | Full | No | No | No | No |
| Employee account | Create | Full | No | No | No | No |
| Employee account | Edit / Lock / Deactivate | Restricted[1] | No | No | No | No |
| Role & permission | Create / Edit / Assign | Full | No | No | No | No |
| Approval workflow | Configure | Full | No | No | No | No |
| Notification rule | Configure | Full | No | No | No | No |
| Business partner | Create / Edit | Full | No | Full | No | No |
| Business partner | View | Full | Full | Full | Full | No |
| Business partner | Deactivate | Full | No | Restricted[2] | No | No |
| Partner contact & bank account | Create / Edit | Full | No | Full | No | No |
| Customer portal account | Create / Lock | Full | No | Full | No | No |
| Contract type | Create / Edit / Deactivate | Full | Full | No | No | No |
| Contract | Create | Full | No | Full | No | No |
| Contract | Edit (incl. parties) | Full | No | Restricted[3] | No | No |
| Contract | View | Full | Full | Restricted[4] | Restricted[5] | Restricted[6] |
| Contract | Submit for approval | Full | No | Restricted[4] | No | No |
| Contract | Approve / Reject / Request revision | Full | Restricted[7] | Restricted[7] | Restricted[7] | No |
| Contract | Send for e-signature | Full | No | Restricted[8] | No | No |
| Contract | E-sign (with OTP verification) | No | No | No | No | Restricted[9] |
| Contract | Decline signing | No | No | No | No | Restricted[9] |
| Contract | View signing status | Full | Full | Restricted[4] | No | Restricted[6] |
| Contract | Update lifecycle status | Full | No | Restricted[4] | No | No |
| Contract | Cancel | Full | No | Restricted[10] | No | No |
| Appendix, obligation, payment schedule | Create / Edit | Full | No | Restricted[4] | No | No |

| | | | | | | |
|---|---|---|---|---|---|---|
| Renewal / Termination | Request | Full | No | Restricted[4] | No | No |
| Renewal / Termination | Approve | Full | No | No | No | No |
| Contract document | Upload / New version | Full | Full | Restricted[4] | No | No |
| Contract document | View / Download | Full | Full | Restricted[4] | No | Restricted[6] |
| Contract document | Delete | Full | Restricted[11] | Restricted[11] | No | No |
| Scan record | Scan / Upload scan | Full | Full | No | No | No |
| Scan record | Verify OCR / Confirm | Full | Full | No | No | No |
| Physical storage location | Create / Edit / Assign | Full | Full | No | No | No |
| Support ticket | Create | No | No | No | Full | Full |
| Support ticket | View | Full | No | No | Full | Restricted[12] |
| Support ticket | Assign | Full | No | No | Full | No |
| Support ticket | Reply / Update status | Full | No | No | Restricted[13] | Restricted[12] |
| Support ticket | Internal note (create / view) | Full | No | No | Full | No |
| Ticket category | Create / Edit | Full | No | No | Full | No |
| Dashboard & report | View | Full | Restricted[14] | Restricted[14] | No | No |
| Audit log | View | Full | No | No | No | No |
| Notification | View own | Full | Full | Full | Full | Full |

**Footnotes**

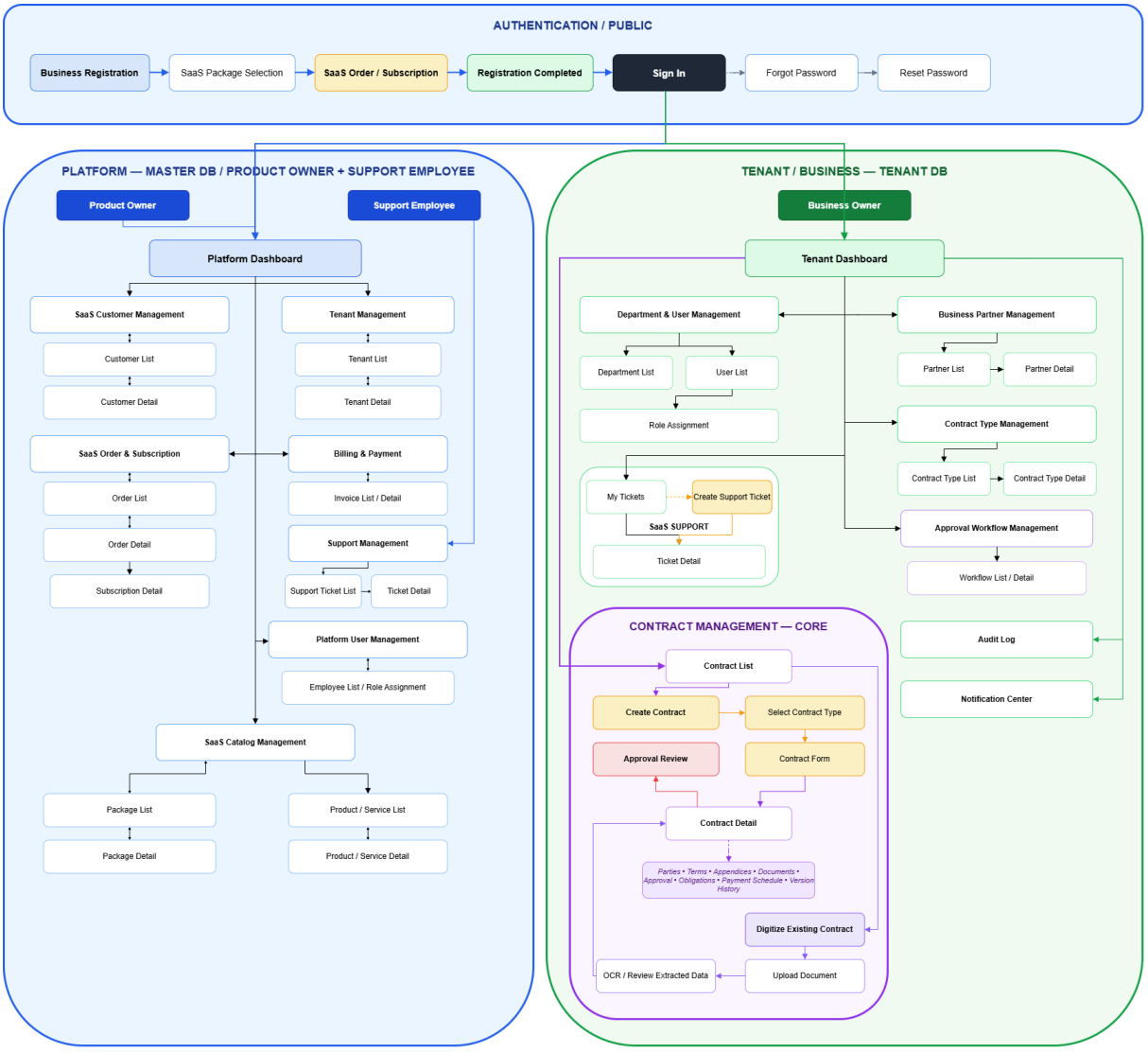
1. BO: can edit, lock or deactivate any employee account except their own Business Owner account.
2. BS: only partners they created that have no active contract.
3. BS: only contracts they created, while the status is Draft or Revision Requested.
4. BS: only contracts they created or are assigned to.
5. BTS: read-only, and only contracts linked to a ticket they handle.
6. CU: only contracts where their partner company is a party, after the contract has been sent for signature.
7. BAO / BS / BTS: only when their role is the approver of the current step in the approval workflow, and never for a contract they created.
8. BS: only their own contracts that have been fully approved.
9. CU: only contracts sent to them in Pending Signature status; e-sign requires a valid OTP sent to their email.
10. BS: only their own contracts that have not been signed yet.

11. BAO / BS: only documents they uploaded, and only while the contract is not signed or active; signed documents cannot be deleted.
12. CU: only tickets they created; can reply and close their own ticket.
13. BTS: only tickets assigned to them.
14. BAO / BS: only data within their own scope (contracts they manage or their department).

# 5. System Functionalities

## 5.1 Screen Inventory

### a. Screen Flow




### b. Screen List

[Fill details of each screen into the table with the columns and sample data as below]

| # | Screen Name | Feature | Description |
|---|---|---|---|

| | | | |
|---|---|---|---|
| 1 | **Sign In** | Authentication | Allows users to authenticate and access the system according to their role and tenant. |
| 2 | **User Register** | Authentication | Allows a business owner to create a SaaS account and register for ConTrack. |
| 3 | **SaaS Package Selection** | SaaS Subscription | Allows a business owner to view and select an available SaaS package. |
| 4 | **SaaS Order / Subscription** | SaaS Subscription | Allows a business owner to review the selected package and submit a SaaS subscription order. |
| 5 | **Registration Completed** | Authentication | Confirms successful registration and completion of the SaaS registration process. |
| 6 | **Forgot Password** | Authentication | Allows users to request a password reset when they cannot sign in. |
| 7 | **Reset Password** | Authentication | Allows users to set a new password after a password reset request. |
| 8 | **Platform Dashboard** | Platform Management | Provides the Product Owner with an overview of the ConTrack platform and platform activities. |
| 9 | **SaaS Customer List** | SaaS Customer Management | Displays businesses registered for or using ConTrack. |
| 10 | **SaaS Customer Detail** | SaaS Customer Management | Displays detailed information about a selected SaaS customer and its related tenant information. |
| 11 | **Tenant List** | Tenant Management | Displays registered tenants and allows the Product Owner to select a tenant for management. |
| 12 | **Tenant Detail** | Tenant Management | Displays tenant information, status, domain, subscription, and tenant database status. |
| 13 | **SaaS Catalog List** | SaaS Catalog Management | Displays SaaS packages, products, and services offered by the ConTrack platform. |

| | | | |
|---|---|---|---|
| 14 | **SaaS Catalog Detail** | SaaS Catalog Management | Displays detailed information about a selected SaaS package, product, or service. |
| 15 | **SaaS Order List** | SaaS Order Management | Displays SaaS orders placed by business customers. |
| 16 | **SaaS Order Detail** | SaaS Order Management | Displays detailed information about a selected SaaS order and its order items. |
| 17 | **Subscription List** | Subscription Management | Displays active and historical SaaS subscriptions of business tenants. |
| 18 | **Subscription Detail** | Subscription Management | Displays package, billing cycle, validity period, status, and subscription history. |
| 19 | **Invoice List** | Billing & Payment | Displays invoices generated for SaaS orders and subscriptions. |
| 20 | **Invoice Detail** | Billing & Payment | Displays detailed information about a selected invoice and its payment status. |
| 21 | **Payment List** | Billing & Payment | Displays payment records associated with SaaS orders and invoices. |
| 22 | **Payment Detail** | Billing & Payment | Displays detailed information about a selected payment transaction. |
| 23 | **Platform User Management** | Platform User Management | Allows the Product Owner to manage platform employees and assign platform roles. |
| 24 | **Support Ticket List** | Support Management | Allows Support Employees to view and manage support tickets submitted by SaaS customers. |
| 25 | **Support Ticket Detail** | Support Management | Displays ticket information and allows Support Employees to assign, respond to, update, and track tickets. |
| 26 | **Tenant Dashboard** | Tenant Management | Provides the Business Owner with an overview of the business tenant and its activities. |
| 27 | **User & Role Management** | Access Control | Allows the Business Owner to manage tenant users, departments, roles, and permissions. |

| | | | |
|---|---|---|---|
| 28 | **Business Partner List** | Business Partner Management | Displays customers, suppliers, and other business partners managed by the tenant. |
| 29 | **Business Partner Detail** | Business Partner Management | Displays detailed information about a selected business partner and its contacts. |
| 30 | **Contract Type Management** | Contract Management | Allows tenant users to manage contract types used to classify contracts. |
| 31 | **Approval Workflow List** | Approval Management | Displays approval workflows configured for the tenant. |
| 32 | **Approval Workflow Detail** | Approval Management | Allows authorized users to configure workflow steps and approvers for contract approval. |
| 33 | **Contract List** | Contract Management | Displays contracts managed by the tenant and provides access to contract operations. |
| 34 | **Create Contract** | Contract Management | Allows the Business Owner or authorized tenant user to start creating a contract. |
| 35 | **Select Contract Type** | Contract Management | Allows the user to select the contract type for a new contract. |
| 36 | **Contract Form** | Contract Management | Allows the user to enter contract information, parties, terms, dates, and related details. |
| 37 | **Contract Detail** | Contract Management | Displays contract information, parties, terms, documents, approval status, obligations, and payment schedules. |
| 38 | **Contract Version History** | Contract Version Management | Displays previous versions of contract documents and related change information. |
| 39 | **Digitize Existing Contract** | Contract Digitization | Allows the user to start digitizing an existing contract document. |
| 40 | **Upload Document** | Contract Digitization | Allows the user to upload a scanned or digital contract document for OCR processing. |
| 41 | **OCR / Review Extracted Data** | Contract Digitization & OCR | Allows the user to review and correct information extracted from the uploaded contract document. |

| | | | |
|---|---|---|---|
| 42 | **Approval Review** | Contract Approval | Allows an authorized approver to review a submitted contract and approve, reject, or return it for revision. |
| 43 | **Contract Obligation List** | Contract Obligation Management | Displays contractual obligations, responsible parties, due dates, and completion status. |
| 44 | **Payment Schedule List** | Contract Payment Management | Displays contract payment schedules, amounts, due dates, and payment status. |
| 45 | **Notification Center** | Notification Management | Displays notifications related to contract approvals, expiry dates, payment due dates, and support activities. |
| 46 | **Audit Log** | Audit Management | Allows authorized users to review recorded tenant activities and important data changes. |
| 47 | **My Support Tickets** | Support Management | Allows tenant users to view submitted support tickets and track their status. |
| 48 | **Create Support Ticket** | Support Management | Allows tenant users to submit a support request with its subject, description, priority, and related information. |
| 49 | **Support Ticket Detail** | Support Management | Displays a tenant support ticket and allows the user to view messages, attachments, and ticket status. |

## 5.2 External API Inventory

| # | API Name | Calling Path | Description |
|---|---|---|---|
| 1 | Google Sign-In API | User → ConTrack → Google | allows users to authenticate using their Google account and sign in to the system without creating a separate password |

## 5.3 Background Job Inventory

| # | Job Name | Group | Description |
|---|---|---|---|
| 1 | Contract Expiration Reminder | Scheduled | Periodically checks contracts approaching their expiration date and sends reminders to the Business Owner |
| 2 | Contract Status Update | Scheduled | Periodically checks contract dates and updates contract status when a contract expires |
| 3 | Support Ticket Reminder | Scheduled | Periodically checks pending support tickets and reminds Support Employees about tickets requiring attention |

# II. Use Case Specifications

# 1. Authentication & Registration

## 1.1 Register Account

| Primary Actors | Guest | Secondary Actors | Google Service |
|---|---|---|---|
| Description | As a Guest, I want to register an organization account so that I can purchase a plan and use ConTrack for my business. | | |
| Preconditions | The Guest is not logged in. | | |
| Normal Flow | 1. Guest clicks the Register button in the page header or on a plan's detail page.<br>2. System shows the Registration screen.<br>3. Guest enters company name, tax code, legal representative, email, phone and password, then accepts the terms of service.<br>4. Guest clicks Register.<br>5. System validates the input (BR-01, BR-02).<br>6. System creates a SaaS customer record and a customer account with status Pending Verification.<br>7. System sends a verification email through Google Service and shows MSG04.<br>8. Guest opens the email and clicks the verification link.<br>9. System activates the account and redirects the user to the Login screen. | | |
| Alternative Flows | **Step 3.1_Register with Google**<br>1. Guest clicks Sign up with Google.<br>2. System redirects to Google's login screen; Guest chooses a Google account.<br>3. Google returns the verified email and name.<br>4. Guest completes the company information. The email is already verified, so the flow continues at step 9.<br>**Step 5_Invalid input**<br>A required field is empty (MSG01).<br>Email or tax code has an invalid format (MSG02).<br>Email or tax code is already registered (MSG03).<br>The password does not meet the password policy (MSG02).<br>**Step 8_Verification link expired or already used**<br>System shows MSG06 and offers Resend verification email. | | |
| Postconditions | A SaaS customer and an active customer account exist in the Master DB.<br>The user can log in and purchase a plan. | | |

## 1.2 Login System

| Primary Actors | All registered users | Secondary Actors | Google Service |
|---|---|---|---|
| Description | As a user, I want to log into the system so that I can use the authenticated features for my role. | | |

| | |
|---|---|
| Preconditions | The user account has been created and activated. |
| Normal Flow | 1. The user clicks Login in the page header, or opens an authenticated page directly (link or URL).<br>2. The system shows the Login screen. On a tenant domain, it shows the tenant's logo and login page.<br>3. The user enters an email and password.<br>4. The user clicks Login.<br>5. System validates the login details (BR-03).<br>6. The system identifies the account type and context: platform staff or SaaS Customer → Master DB; employee or Business Customer → the database of the tenant identified by the domain.<br>7. The system checks that the tenant is Active (tenant users only).<br>8. The system grants access according to the user's roles and permissions.<br>9. The system records the successful login in the login history.<br>10. The system redirects the user to their home page (dashboard or customer portal), or back to the page they originally requested. |
| Alternative Flows | **Step 3.1_Google Login**<br>1. The user clicks Login with Google.<br>2. System redirects to Google's login screen.<br>3. User signs in with Google; Google validates the account and redirects back to the system.<br>4. Flow returns to step 6.<br>**Step 5_System cannot authenticate the user**<br>Email and/or password is blank (MSG10).<br>Email or password is incorrect (MSG09).<br>Email has not been verified (MSG11).<br>The account is locked or inactive (MSG12).<br>After 5 consecutive failed attempts, the account is locked for 30 minutes (MSG13, BR-03).<br>**Step 7_Tenant suspended or terminated**<br>The system denies access and shows MSG12 with a contact link for the product owner. |
| Postconditions | ● The user is logged in.<br>● The login attempt is recorded in the login history. |

# 2. Subscription & Tenant Management

## 2.1 Purchase Plan

| | | | |
|---|---|---|---|
| Primary Actors | SaaS Customer | Secondary Actors | Admin Officer (Product Owner), Google Service |

| | |
|---|---|
| Description | As a SaaS Customer, I want to choose a plan and billing cycle and place an order so that my company can get its own ConTrack workspace. |
| Preconditions | The SaaS Customer is logged in.<br>The plan is Active with a valid price for the chosen billing cycle. |
| Normal Flow | 1. SaaS Customer opens Plans and selects a plan.<br>2. System shows plan details: features, limits, and prices per billing cycle.<br>3. SaaS Customer selects a billing cycle (monthly / quarterly / yearly) and clicks Subscribe.<br>4. System shows the order summary: subtotal, discount, tax and grand total.<br>5. SaaS Customer enters the preferred subdomain and clicks Place Order.<br>6. System validates the subdomain (unique, valid format) and checks BR-04.<br>7. System creates an order with status Confirmed and payment status Unpaid, copying the item code, name and price into the order details (BR-17).<br>8. System issues an invoice and shows payment instructions (bank account, amount, transfer reference).<br>9. System emails the invoice to the customer through Google Service. |
| Alternative Flows | **Step 5.1_Order created by platform Sales**<br>Sales (PO) creates the order for the customer through UC Create Subscription Order. The customer receives the invoice by email and the flow continues at step 8.<br>**Step 6_Subdomain taken or invalid**<br>System shows MSG03 and suggests available alternatives.<br>**Step 6_Customer already has a tenant (BR-04)**<br>System redirects to Renew / Upgrade Subscription. |
| Postconditions | An order and an unpaid invoice exist. The order is waiting for payment. |

## 2.2 Record Payment & Activate Subscription

| | | | |
|---|---|---|---|
| Primary Actors | Admin Officer (Product Owner) | Secondary Actors | SaaS Customer, Google Service |
| Description | As an Admin Officer, I want to record a customer's payment so that the subscription is activated and the tenant is provisioned. | | |
| Preconditions | An invoice exists with status Issued or Overdue. | | |
| Normal Flow | 1. SaaS Customer submits payment proof (transfer reference, receipt image) on the invoice (optional).<br>2. Admin Officer opens Invoices and selects the invoice.<br>3. System shows the invoice, the amount paid so far, and any payment proof.<br>4. Admin Officer enters the payment amount, method, transaction reference and paid date, then clicks Record Payment.<br>5. System validates the payment and updates the invoice and the order's payment status.<br>6. Once the invoice is fully paid (BR-05), the system creates a subscription with status Pending and a start and end date based on the billing cycle. | | |

| | |
|---|---|
| | 7. System records the action in the subscription history and starts UC Provision Tenant.<br>8. System notifies the SaaS Customer by email that payment has been confirmed. |
| Alternative Flows | **Step 5_Partial payment**<br>Payment status becomes Partially Paid; no subscription is created and the flow ends.<br>**Step 5_Amount exceeds the outstanding balance**<br>System shows MSG07 and asks for confirmation.<br>**Step 4.1_Invalid payment proof**<br>Admin Officer rejects the proof with a reason, and the customer is notified by email. |
| Postconditions | The payment is recorded.<br>If the invoice is fully paid, a subscription exists and tenant provisioning has started. |

## 2.3 Provision Tenant

| | | | |
|---|---|---|---|
| Primary Actors | Super Admin (or triggered automatically by the system) | Secondary Actors | Google Service |
| Description | As a Super Admin, I want the system to create an isolated workspace and database for a paid customer so that the business can start using ConTrack. | | |
| Preconditions | A subscription exists in status Pending.<br>The customer has no active tenant (BR-04). | | |
| Normal Flow | 1. System creates a tenant record with status Provisioning, plus the tenant code and subdomain.<br>2. System creates provisioning job CREATE_DATABASE.<br>3. System creates a new tenant database from the latest schema version and stores its connection info with a secret reference only (BR-18).<br>4. System runs job RUN_MIGRATION and records the applied schema version.<br>5. System runs job CREATE_ADMIN: creates the company settings from the customer's profile, creates the default roles (Business Owner, Admin Officer, Sales, Technical Support), and creates the first Business Owner (CEO) account.<br>6. System sets the tenant status to Active and the subscription status to Active.<br>7. System emails the CEO the tenant URL and an account activation link through Google Service. | | |
| Alternative Flows | **Step 3–5_A provisioning job fails**<br>System marks the job Failed, stores the error, and keeps the tenant in Provisioning.<br>System notifies the Super Admin (MSG14).<br>Super Admin opens Provisioning Jobs and clicks Retry; the flow resumes from the failed job.<br>**Step 1.1_Manual provisioning**<br>Super Admin opens the tenant list, selects a paid customer without a tenant and clicks Provision Tenant; the flow continues at step 1. | | |
| Postconditions | The tenant is Active with its own database at the current schema version. | | |

| | |
|---|---|
| | The CEO has received login access. |

### 2.4 Renew / Upgrade Subscription

| | | | |
|---|---|---|---|
| Primary Actors | SaaS Customer | Secondary Actors | Admin Officer (Product Owner), System Scheduler |
| Description | As a SaaS Customer, I want to renew or change my plan so that my company keeps using ConTrack with suitable limits. | | |
| Preconditions | The SaaS Customer has a subscription with status Active or Expired. | | |
| Normal Flow | 1. SaaS Customer opens My Subscription.<br>2. System shows the current plan, end date, usage (users, storage, contracts) and the available actions.<br>3. SaaS Customer clicks Renew or Change Plan and selects a billing cycle or a new plan.<br>4. System calculates the amount (with a prorated difference for upgrades) and shows the order summary.<br>5. SaaS Customer confirms; the system creates the order and invoice (as in Purchase Plan, steps 7–9).<br>6. After payment (Record Payment & Activate Subscription), the system extends the end date or changes the plan, updates the limits, and records Renewed or Upgraded in the subscription history. | | |
| Alternative Flows | **Step 3_Downgrade exceeds limits**<br>Current usage is higher than the new plan's limits (BR-06). System shows MSG08 and blocks the downgrade until usage is reduced.<br>**Step 1.1_Subscription expires without renewal**<br>System Scheduler sets the subscription to Expired and notifies the customer.<br>After the grace period, the tenant is set to Suspended: users can log in read-only and cannot create contracts. | | |
| Postconditions | The subscription has the new end date or plan, and the history records the change. | | |

# 3. Platform Support

### 3.1 Create & Handle Platform Support Ticket

| | | | |
|---|---|---|---|
| Primary Actors | SaaS Customer (create), Technical Support – Product Owner (handle) | Secondary Actors | Google Service |
| Description | As a SaaS Customer, I want to send an issue or request to the product owner so that it is tracked until it is resolved. | | |
| Preconditions | The SaaS Customer is logged in. | | |
| Normal Flow | 1. SaaS Customer opens Support and clicks New Ticket. | | |

| | |
|---|---|
| | 2. SaaS Customer selects a category (Bug / Account / Billing / Upgrade / Other) and priority, enters a subject and description, and attaches files if needed.<br>3. System validates the input (MSG01, BR-13) and creates the ticket with status Open.<br>4. System notifies the platform Technical Support team.<br>5. Technical Support opens the ticket and clicks Assign to me (or assigns it to a colleague).<br>6. Technical Support replies to the customer and adds internal notes if needed.<br>7. Technical Support changes the status to In Progress, Waiting Customer or Resolved.<br>8. System records every status change in the ticket status history and emails the customer about each reply or status change.<br>9. SaaS Customer confirms the resolution and closes the ticket. |
| Alternative Flows | **Step 9_Customer replies instead of closing**<br>The status returns to In Progress and the flow repeats from step 6.<br>**Step 9_No customer response**<br>A ticket left Resolved for 7 days is closed automatically by the System Scheduler. |
| Postconditions | The ticket is Closed, with its full message and status history stored in the Master DB. |

# 4. Organization & User Management

## 4.1 Create Employee Account & Assign Roles

| | | | |
|---|---|---|---|
| Primary Actors | Business CEO | Secondary Actors | Google Service |
| Description | As a CEO, I want to create employee accounts and assign roles so that each employee can work with the right permissions. | | |
| Preconditions | The CEO is logged in.<br>The tenant is Active. | | |
| Normal Flow | 1. CEO opens Employees and clicks Add Employee.<br>2. CEO enters employee code, full name, email, phone, department and job title.<br>3. CEO selects one or more roles.<br>4. CEO clicks Save.<br>5. System validates the input (BR-01) and checks the plan's user limit (BR-06).<br>6. System creates the account with status Pending Activation and assigns the selected roles.<br>7. System emails the employee an activation link to set a password.<br>8. System writes the action to the audit log and shows MSG05. | | |
| Alternative Flows | **Step 5_User limit reached**<br>System shows MSG08 with a link to upgrade the plan.<br>**Step 5_Email already exists in this tenant**<br>System shows MSG03.<br>**Step 1.1_Edit roles of an existing employee**<br>1. CEO selects an employee and changes their roles. | | |

| | |
|---|---|
| | 2. System prevents removing the Business Owner role from the last Business Owner (MSG16).<br>3. The new permissions apply from the employee's next request. |
| Postconditions | The employee account exists with its assigned roles, and the employee has received the activation email. |

### 4.2 Configure Approval Workflow

| Primary Actors | Business CEO | Secondary Actors | None |
|---|---|---|---|
| Description | As a CEO, I want to define who approves which contracts so that every contract goes through the right approval steps. | | |
| Preconditions | The CEO is logged in.<br>Contract types, departments and roles exist. | | |
| Normal Flow | 1. CEO opens Approval Workflows and clicks New Workflow.<br>2. CEO enters the workflow code and name, and selects the contract type and/or department it applies to.<br>3. CEO adds steps in order. For each step: the step name, the approver type (specific user / role / department manager), the approver, and whether the step is required.<br>4. CEO clicks Save.<br>5. System validates the workflow: at least one required step; every approver exists and is active; no other active workflow for the same contract type and department.<br>6. System saves the workflow as Active and shows MSG05. | | |
| Alternative Flows | **Step 5_Validation fails**<br>System highlights the invalid step (MSG01 / MSG02).<br>**Step 3.1_Value-based rule**<br>CEO adds a condition such as "contract value > 500,000,000 VND requires CEO approval". The system inserts this step only for contracts that match the condition.<br>**Step 1.1_Edit an active workflow**<br>Changes apply only to contracts submitted afterwards. Contracts already in approval keep the original steps. | | |
| Postconditions | An active approval workflow is available for contract submission. | | |

# 5. Partner Management

### 5.1 Create Customer Portal Account

| Primary Actors | Business Sales | Secondary Actors | Google Service |
|---|---|---|---|
| Description | As a Sales employee, I want to create a portal account for a partner's contact so that the customer can review, sign and track their contracts online. | | |
| Preconditions | The business partner and its contact person exist. | | |

| | |
|---|---|
| | The contact has a valid email. |
| Normal Flow | 1. Sales opens a business partner and selects a contact person.<br>2. Sales clicks Create Portal Account.<br>3. System checks that the contact has no existing portal account and that the email is unique (BR-01).<br>4. System creates the portal account with status Pending Activation.<br>5. System emails the contact an invitation link to set a password and verify the email.<br>6. Business Customer opens the link and sets a password.<br>7. System activates the account. |
| Alternative Flows | **Step 3_Contact already has an account**<br>System shows MSG03 and offers Resend invitation.<br>**Step 6_Invitation link expired**<br>System shows MSG06; Sales can resend the invitation. |
| Postconditions | The partner contact has an active customer portal account linked to the partner. |

# 6. Contract Management & E-Signature

## 6.1 Create Contract

| | | | |
|---|---|---|---|
| Primary Actors | Business Sales | Secondary Actors | None |
| Description | As a Sales employee, I want to create a contract with its parties, value and terms so that it can go through approval and signing. | | |
| Preconditions | Sales is logged in.<br>The business partner and contract type exist. | | |
| Normal Flow | 1. Sales opens Contracts and clicks New Contract.<br>2. Sales selects the contract type, the main business partner, and the source (Online or Scanned).<br>3. Sales enters the contract name, value, currency, effective date, expiration date, payment terms and description.<br>4. Sales adds the contract parties. For each party: role (Party A / Party B / Guarantor…), representative, position and signing order.<br>5. Sales uploads the draft document (PDF/DOCX).<br>6. Sales clicks Save Draft.<br>7. System validates the input (MSG01, MSG02, BR-13): expiration date ≥ effective date; at least two parties, one of them the company itself.<br>8. System generates the contract code from the company's code format (BR-07).<br>9. System saves the contract with status Draft and stores the document as version 1 with its SHA-256 hash.<br>10. System writes the action to the audit log and shows MSG05. | | |
| Alternative Flows | **Step 5_File invalid** | | |

| | |
|---|---|
| | File type or size is not allowed (MSG19).<br>A file with the same hash already exists for this contract (MSG20).<br>**Step 2.1_Use default duration**<br>If the expiration date is left empty, the system fills it from the contract type's default duration. |
| Postconditions | A contract in Draft exists with its parties and its first document version. |

## 6.2 Submit & Approve Contract

| | | | |
|---|---|---|---|
| Primary Actors | Business Sales (submit), approvers defined by the workflow (default: Business CEO) | Secondary Actors | Google Service |
| Description | As a Sales employee, I want to submit a contract for approval; as an approver, I want to review and decide on it, so that only approved contracts are sent to customers. | | |
| Preconditions | The contract is in Draft or Revision Requested (BR-08).<br>An active approval workflow matches its type and department. | | |
| Normal Flow | 1. Sales opens the contract and clicks Submit for Approval.<br>2. System finds the matching workflow and creates an approval instance.<br>3. System records the action Submitted and sets the contract status to Waiting Approval.<br>4. System notifies the approver of the first step (in-app and email).<br>5. Approver opens Pending Approvals and selects the contract.<br>6. System shows the contract details, the parties, the current document version, and the approval history.<br>7. Approver clicks Approve and may add a comment.<br>8. System records the action (approver, time, comment) and moves to the next step (BR-09).<br>9. Steps 4–8 repeat until every required step is approved.<br>10. System sets the contract status to Approved, completes the approval instance, and notifies Sales. | | |
| Alternative Flows | **Step 2_No matching workflow**<br>System shows MSG15 and asks Sales to contact the CEO.<br>**Step 7.1_Reject / Request revision**<br>1. Approver clicks Reject or Request Revision and must enter a comment.<br>2. System records the action and sets the contract status to Revision Requested (BR-10).<br>3. System notifies Sales with the comment.<br>4. Sales edits the contract and uploads a new document version, then resubmits; the flow restarts at step 1.<br>**Step 1.1_Cancel submission** | | |

| | |
|---|---|
| | Sales withdraws the contract before the first approval. The action Cancelled is recorded and the status returns to Draft. |
| Postconditions | The contract is Approved (or Revision Requested).<br>Every approval action is stored with its approver, time and comment. |

## 6.3 Send Contract for Signature

| | | | |
|---|---|---|---|
| Primary Actors | Business Sales | Secondary Actors | Business Customer, Google Service |
| Description | As a Sales employee, I want to send an approved contract to the customer so that they can sign it electronically. | | |
| Preconditions | The contract is Approved.<br>Each customer party has a contact with an active portal account. | | |
| Normal Flow | 1. Sales opens the contract and clicks Send for Signature.<br>2. System shows the parties in signing order with their signers, and a default signing deadline (BR-11).<br>3. Sales confirms the signers and the deadline, then clicks Send.<br>4. System locks the current document version as the signing version, so no further edits are possible.<br>5. System creates a signing request for each signer and sets the contract status to Pending Signature.<br>6. System notifies the first signer by email and in the customer portal. | | |
| Alternative Flows | **Step 2_A party has no portal account**<br>System shows MSG01 and offers to open UC Create Customer Portal Account.<br>**Step 3.1_Company signs first**<br>If the company is first in the signing order, the company's representative signs first (same steps as UC E-sign Contract), then the customer is notified. | | |
| Postconditions | The contract is Pending Signature, with signing requests created and the first signer notified. | | |

## 6.4 E-sign Contract

| | | | |
|---|---|---|---|
| Primary Actors | Business Customer | Secondary Actors | Google Service |
| Description | As a Business Customer, I want to review and electronically sign a contract so that the agreement is completed without paper. | | |
| Preconditions | The Business Customer is logged into the customer portal.<br>A signing request is pending for them, not expired, and it is their turn in the signing order. | | |
| Normal Flow | 1. Business Customer opens the notification or Contracts to Sign. | | |

| | |
|---|---|
| | 2. System shows the contract summary and the document version to be signed.<br>3. Business Customer reviews all pages and ticks "I have read and agree to the contract terms".<br>4. Business Customer clicks Sign.<br>5. System sends a 6-digit OTP to the customer's email through Google Service (BR-11).<br>6. Business Customer enters the OTP and clicks Confirm.<br>7. System verifies the OTP.<br>8. System applies the signature to the document and creates a new signed document version.<br>9. System stores the signer, time, IP address and file hash of the signed version.<br>10. System marks the signing request Signed and notifies the next signer, if any.<br>11. When all parties have signed, the system sets the contract status to Signed (then Active on the effective date), stores the final document as type Signed Contract, emails a signed copy to all parties and notifies Sales. |
| Alternative Flows | **Step 4.1_Decline**<br>1. Business Customer clicks Decline and must enter a reason.<br>2. System marks the signing request Declined and sets the contract status to Revision Requested.<br>3. System notifies Sales with the reason.<br>**Step 7_OTP incorrect or expired**<br>System shows MSG17 and allows a resend.<br>After 3 wrong attempts, signing is blocked for 15 minutes (BR-11).<br>**Step 1_Signing request expired**<br>System shows MSG18. Sales must send the contract again (UC Send Contract for Signature).<br>**Step 3_Checkbox not ticked**<br>The Sign button remains disabled (MSG21). |
| Postconditions | The customer's signature is recorded with verification data.<br>When all parties have signed, the contract is Signed, can no longer be changed (BR-12), and the signed copy has been distributed. |

# 7. Document & Digitization

## 7.1 Upload New Document Version

| | | | |
|---|---|---|---|
| Primary Actors | Business Admin Officer, Business Sales | Secondary Actors | None |
| Description | As an Admin Officer, I want to upload a new version of a contract document so that the latest file is used while earlier versions are kept. | | |
| Preconditions | The contract document exists.<br>The user has upload permission for this contract. | | |

| | |
|---|---|
| Normal Flow | 1. User opens the contract's Documents tab and selects a document.<br>2. User clicks Upload New Version, chooses a file and enters change notes.<br>3. System validates the file type and size (BR-13).<br>4. System calculates the SHA-256 hash and stores the file on the file server using a storage key.<br>5. System creates a document version (version number + 1) and updates the document's current version.<br>6. System writes the action to the audit log and shows MSG05. |
| Alternative Flows | **Step 3_Invalid file**<br>System shows MSG19.<br>**Step 4_Same hash as the current version**<br>System shows MSG20; no version is created.<br>**Step 1_Contract already signed (BR-12)**<br>Only document types Appendix, Acceptance Record or Other can be added. The signed contract file cannot be replaced (MSG16). |
| Postconditions | A new document version is stored and set as current.<br>All previous versions remain available. |

## 7.2 Digitize Paper Contract

| | | | |
|---|---|---|---|
| Primary Actors | Business Admin Officer | Secondary Actors | None |
| Description | As an Admin Officer, I want to scan a legacy paper contract, check the OCR result and record where the original is stored, so that the contract can be found both electronically and physically. | | |
| Preconditions | The contract record exists with source Scanned.<br>Storage locations are defined. | | |
| Normal Flow | 1. Admin Officer opens the contract and clicks Upload Scan.<br>2. Admin Officer uploads the scanned file and enters the page count and scan quality.<br>3. System stores the file as a new document version (type Scanned Contract) and creates a scan record with status Uploaded.<br>4. System runs OCR and sets the status to OCR Processing.<br>5. When OCR finishes, the system stores the OCR text and confidence and sets the status to Waiting Verification.<br>6. Admin Officer opens the scan record and compares the OCR text with the scan.<br>7. Admin Officer corrects the key fields (contract code, parties, dates, value) if needed and clicks Confirm.<br>8. System sets the scan status to Completed and records the verifier and verification time.<br>9. Admin Officer selects the physical storage location (room / cabinet / shelf / box) for the paper original.<br>10. System links the location to the contract and shows MSG05. | | |

| | |
|---|---|
| Alternative Flows | **Step 5_Low OCR confidence (below the threshold, BR-14)**<br>System flags the record for careful review.<br>**Step 7.1_Rescan**<br>Admin Officer clicks Request Rescan, the status returns to Uploaded, and the flow restarts at step 2.<br>**Step 4_OCR fails**<br>System sets the status to Waiting Verification with empty OCR text; the Admin Officer enters the key fields manually. |
| Postconditions | The scanned document is verified, searchable by its OCR text, and linked to the physical location of the original. |

# 8. Contract Execution

## 8.1 Request & Approve Contract Renewal

| | | | |
|---|---|---|---|
| Primary Actors | Business Sales (request), Business CEO (approve) | Secondary Actors | Business Customer, Google Service |
| Description | As a Sales employee, I want to request a renewal of an expiring contract so that the business relationship continues without interruption. | | |
| Preconditions | The contract is Active or Expired. | | |
| Normal Flow | 1. Sales opens the contract and clicks Request Renewal.<br>2. Sales enters the new expiration date, any value change and a reason.<br>3. System creates a renewal request with status Pending and notifies the CEO.<br>4. CEO reviews the request and clicks Approve.<br>5. System records the approver and approval time.<br>6. System creates a contract appendix for the renewal, which is then sent for signature (UC Send Contract for Signature and UC E-sign Contract).<br>7. After the appendix is signed, the system updates the contract's expiration date and value and records the old and new values.<br>8. System notifies Sales and the Business Customer. | | |
| Alternative Flows | **Step 4.1_CEO rejects**<br>CEO must enter a reason; the request becomes Rejected and Sales is notified.<br>**Step 1.1_Termination**<br>1. Sales clicks Request Termination and enters the termination date, reason and settlement amount.<br>2. After the CEO approves, the contract status becomes Terminated (or Liquidated when the settlement is completed). | | |
| Postconditions | The contract has a new expiration date backed by a signed appendix, or the request is rejected with a reason. | | |

# 9. Customer Support

## 9.1 Create & Handle Customer Support Ticket

| Primary Actors | Business Customer (create), Business Technical Support (handle) | Secondary Actors | Google Service |
|---|---|---|---|
| Description | As a Business Customer, I want to send a request about my contract or service so that the business can respond and resolve it. | | |
| Preconditions | The Business Customer is logged into the customer portal. | | |
| Normal Flow | 1. Business Customer clicks New Ticket and selects a category and priority.<br>2. Business Customer selects a related contract (optional), enters a subject and description, and attaches files if needed.<br>3. System creates the ticket with status Open and notifies Business Technical Support.<br>4. Technical Support assigns the ticket to themself or a colleague.<br>5. Technical Support replies to the customer; internal notes are hidden from the customer.<br>6. System records every status change and emails the customer about each reply.<br>7. Technical Support sets the ticket to Resolved; the customer confirms and the ticket becomes Closed. | | |
| Alternative Flows | **Step 5.1_System issue that needs the product owner**<br>Technical Support creates a platform ticket (UC Create & Handle Platform Support Ticket) on behalf of the business and links its code to the customer ticket.<br>**Step 7_Customer reopens**<br>The customer replies and the status returns to In Progress. | | |
| Postconditions | The ticket is Closed, with its message and status history stored in the Tenant DB. | | |

# 10. Notification

## 10.1 Send Scheduled Reminders

| Primary Actors | System Scheduler | Secondary Actors | Google Service |
|---|---|---|---|
| Description | The system automatically reminds users and customers of upcoming deadlines so that no expiry, payment or signature is missed. | | |
| Preconditions | Notification rules are configured (default: 30, 15 and 7 days before). | | |
| Normal Flow | 1. The scheduler runs daily at 00:00 (tenant timezone).<br>2. For each active tenant, the system reads the notification rules (BR-15).<br>3. System finds contracts reaching an expiration threshold, payment schedules and obligations nearing their due date, and signing requests that are still pending.<br>4. System creates in-app notifications for the responsible users (contract owner, responsible user, approver). | | |

| | |
|---|---|
| | 5. System sends emails to the responsible users and to Business Customers (renewal notices, signing reminders) through Google Service.<br>6. System sets contracts past their expiration date to Expired, and signing requests past their deadline to Expired. |
| Alternative Flows | **Step 5_Email delivery fails**<br>System marks the notification Failed and retries up to 3 times.<br>**Step 2_Tenant suspended**<br>The tenant is skipped. |
| Postconditions | Notifications are created and sent.<br>Expired contracts and signing requests are updated. |

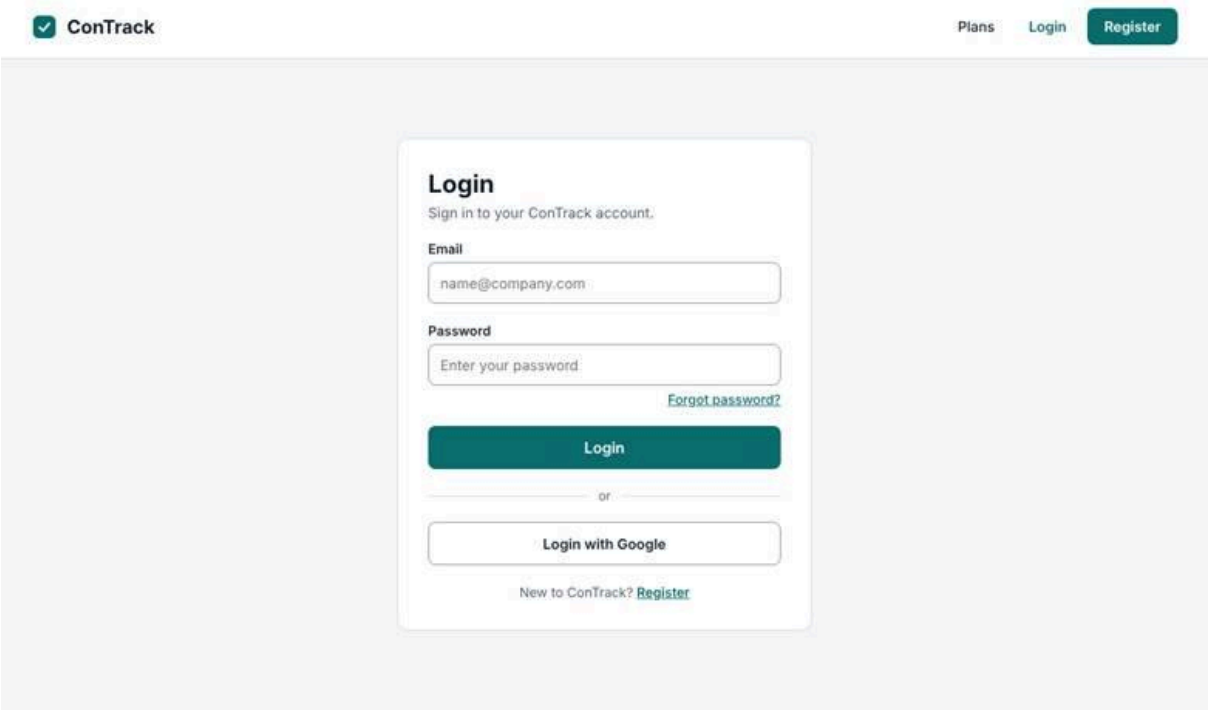
# III. Functional Requirements

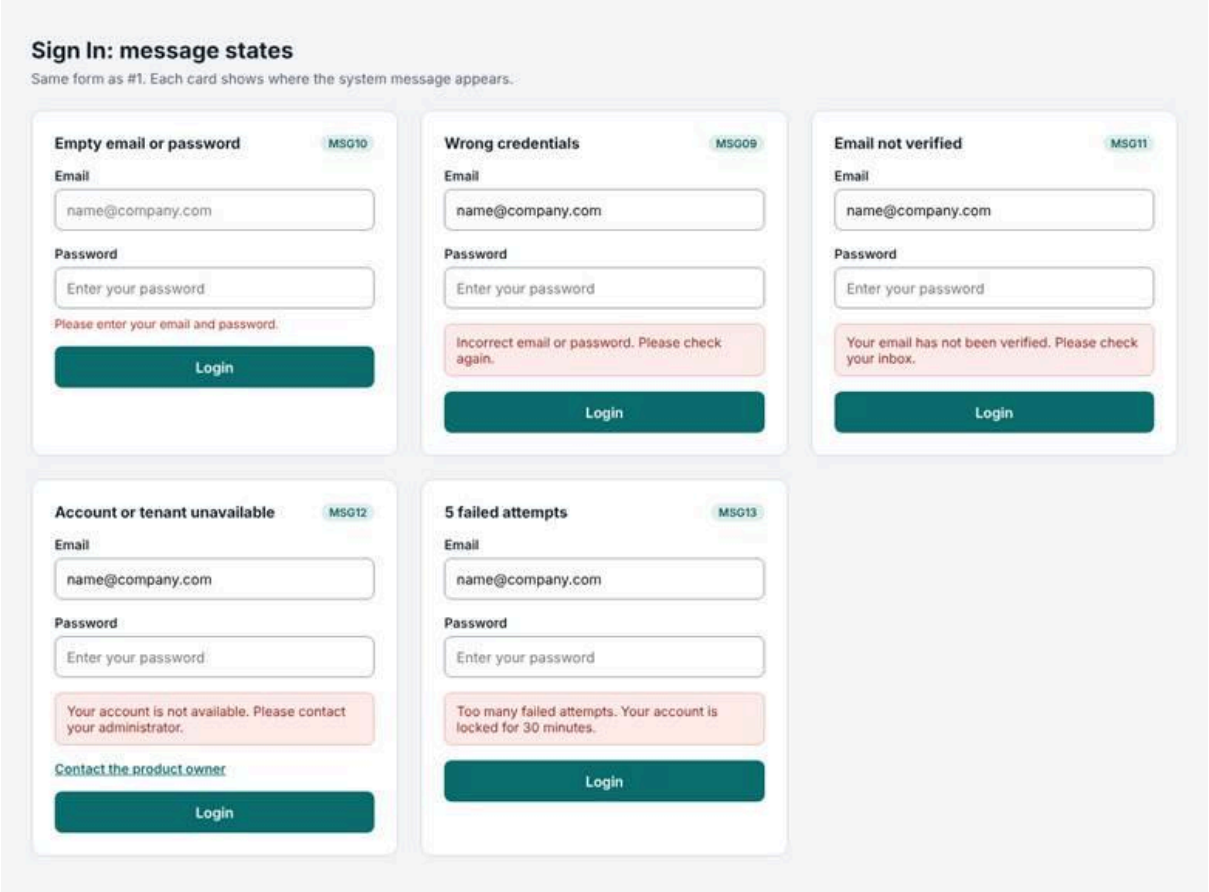
This section details the functional specifications and screen layouts of the **ConTrack Contract Management System**. The specifications are organized hierarchically by functional modules and screens, followed by the **Background Job Inventory** and the **External API Inventory**.

# 1. Authentication and Registration

## 1.1 Sign In

**Content #1: UI Layout (Wireframe Mockup)**






**Content #2: Brief Description and Use Case Mapping**

- Description: Authenticates a user by email and password and routes the user to the area of their role and tenant. Shows the generic and locked-account error states.
- Mapped use case: UC 1.2
- Actors: All registered users (Guest before sign-in)
- Entities: users, employees, customer_contacts, tenants, login_histories
- Business rules and messages: BR-03, BR-21, BR-22, BR-23; MSG09–MSG13

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|
| Email | Text input | Yes | Placeholder: name@company.com. Empty value shows MSG10 |
| Password | Password | Yes | Placeholder: Enter your password. Empty value shows MSG10 |

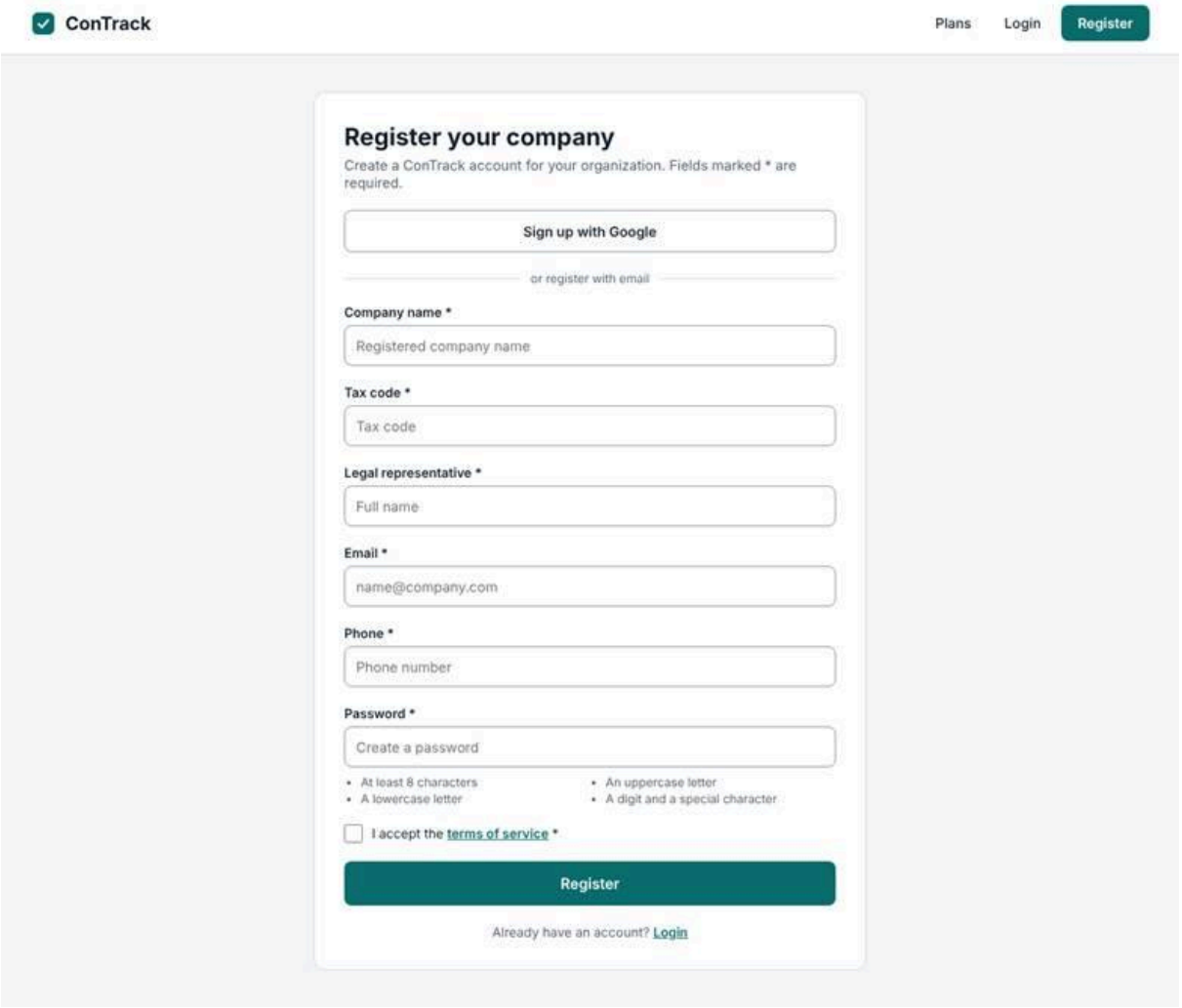
- Actions: Login, Login with Google

**Content #4: Permission and State**

- Permission: Open to Guest. After sign-in the user lands on the home screen of their role (Platform, Tenant or Customer Portal).
- State behavior: Error states: wrong credentials (MSG09), empty fields (MSG10), unverified email (MSG11), unavailable account or suspended tenant (MSG12), locked after 5 failed attempts (MSG13).

## 1.2 User Register

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: Lets a Guest register a SaaS customer account: company legal information, contact person and password.
- Mapped use case: UC 1.1

- Actors: Guest
- Entities: customers, customer_contacts
- Business rules and messages: BR-01, BR-02, BR-04; MSG01–MSG04

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|
| Company name | Text input | Yes | Placeholder: Registered company name |
| Tax code | Text input | Yes | Placeholder: Tax code. Unique (BR-01, MSG03) |
| Legal representative | Text input | Yes | Placeholder: Full name |
| Email | Text input | Yes | Placeholder: name@company.com. Unique (BR-01, MSG03) |
| Phone | Text input | Yes | Placeholder: Phone number |
| Password | Password | Yes | Placeholder: Create a password. Password policy (BR-02) |
| I accept the terms of service | Checkbox | Yes | |

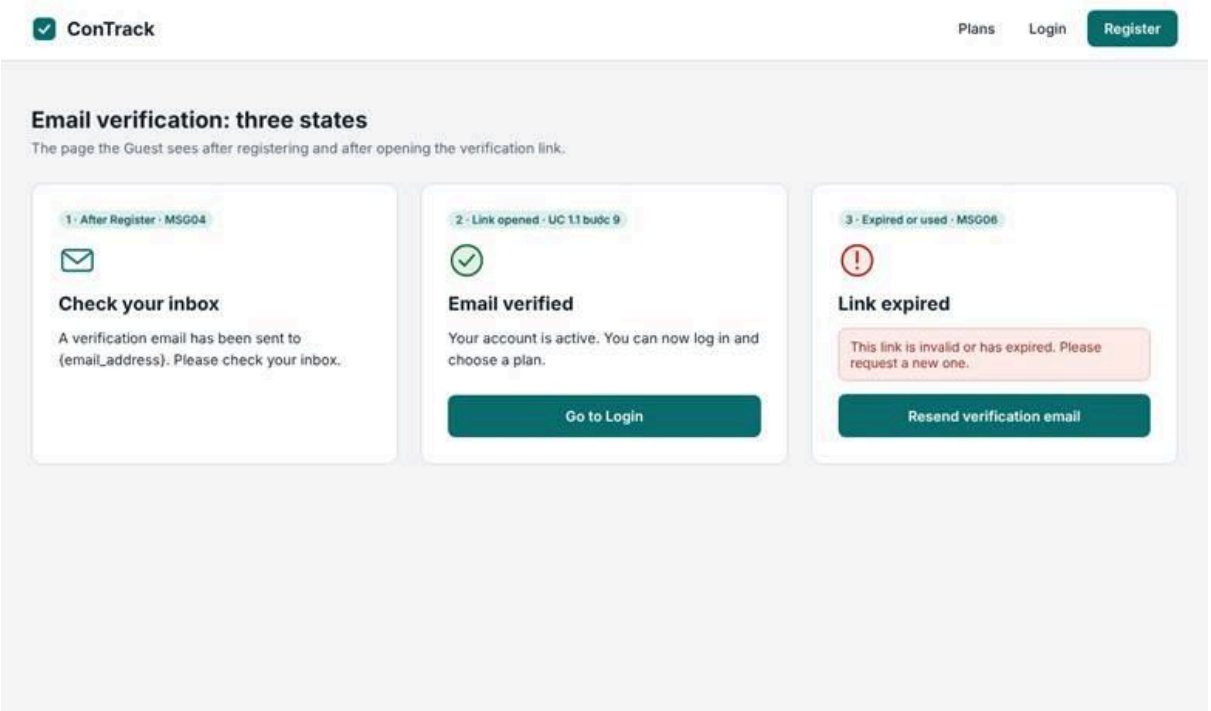
- Actions: Sign up with Google, Register

**Content #4: Permission and State**

- Permission: Guest only (Matrix: SaaS customer Register = Guest Full).

## 1.3 Verify Email

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: Shows the result of opening the verification link sent after registration (success, or invalid/expired link).
- Mapped use case: UC 1.1
- Actors: Guest
- Entities: customer_contacts (email verification)
- Business rules and messages: MSG06

**Content #3: Field and Component Descriptions**

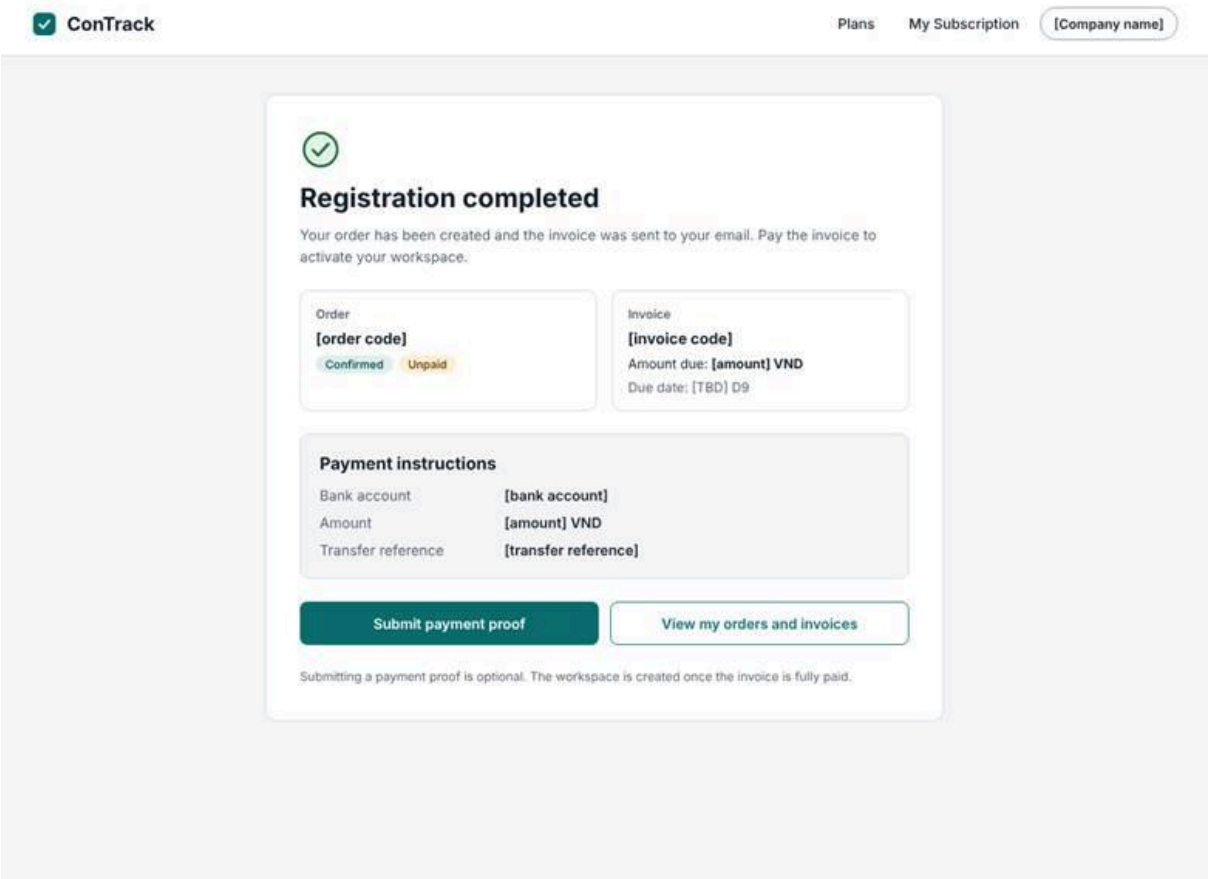
- Actions: Resend verification email

**Content #4: Permission and State**

- Permission: Open to Guest through the emailed link.
- State behavior: Success and invalid/expired link variants.

## 1.4 Registration Completed

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: Confirms that registration and the subscription order were submitted and tells the customer how to pay the invoice.
- Mapped use case: UC 2.1 steps 7–9
- Actors: SaaS Customer
- Entities: orders, invoices
- Business rules and messages:BR-05; MSG31

**Content #3: Field and Component Descriptions**

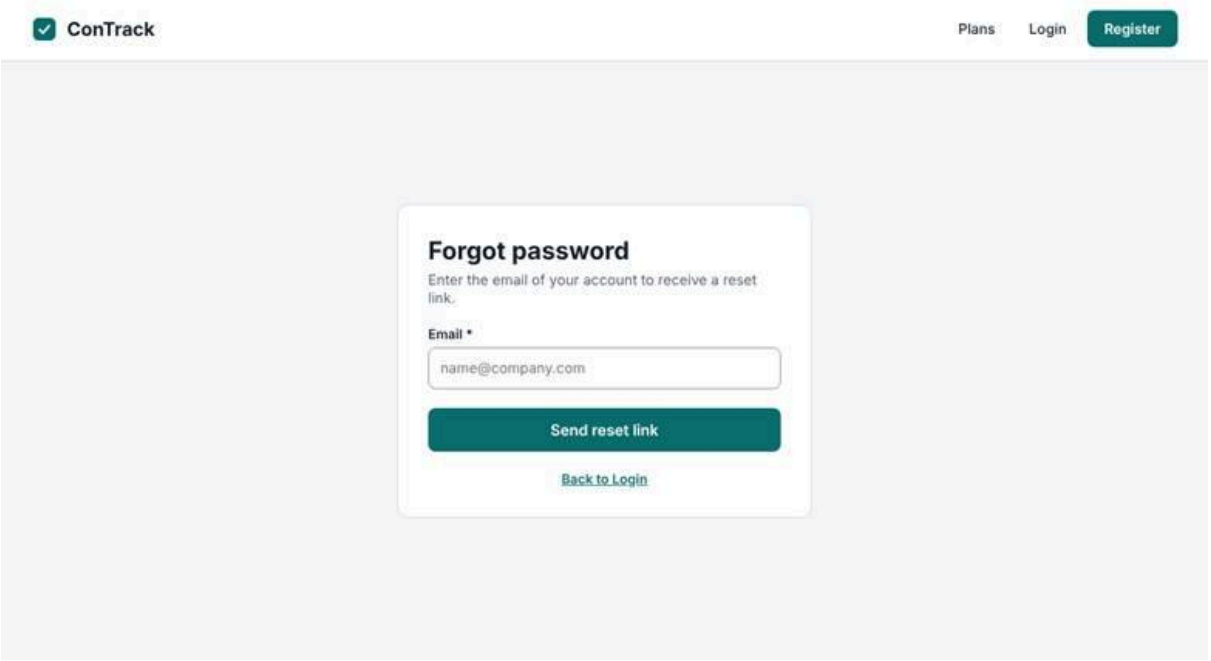
- Actions: Submit payment proof

**Content #4: Permission and State**

- Permission: SaaS Customer who has just submitted an order.

## 1.5 Forgot Password

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: Requests a password reset link by email without revealing whether the email exists.
- Mapped use case: UC 1.2
- Actors: All users
- Entities: users, password reset tokens
- Business rules and messages: BR-23; MSG01, MSG02

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|
| Email | Text input | Yes | Placeholder: name@company.com |

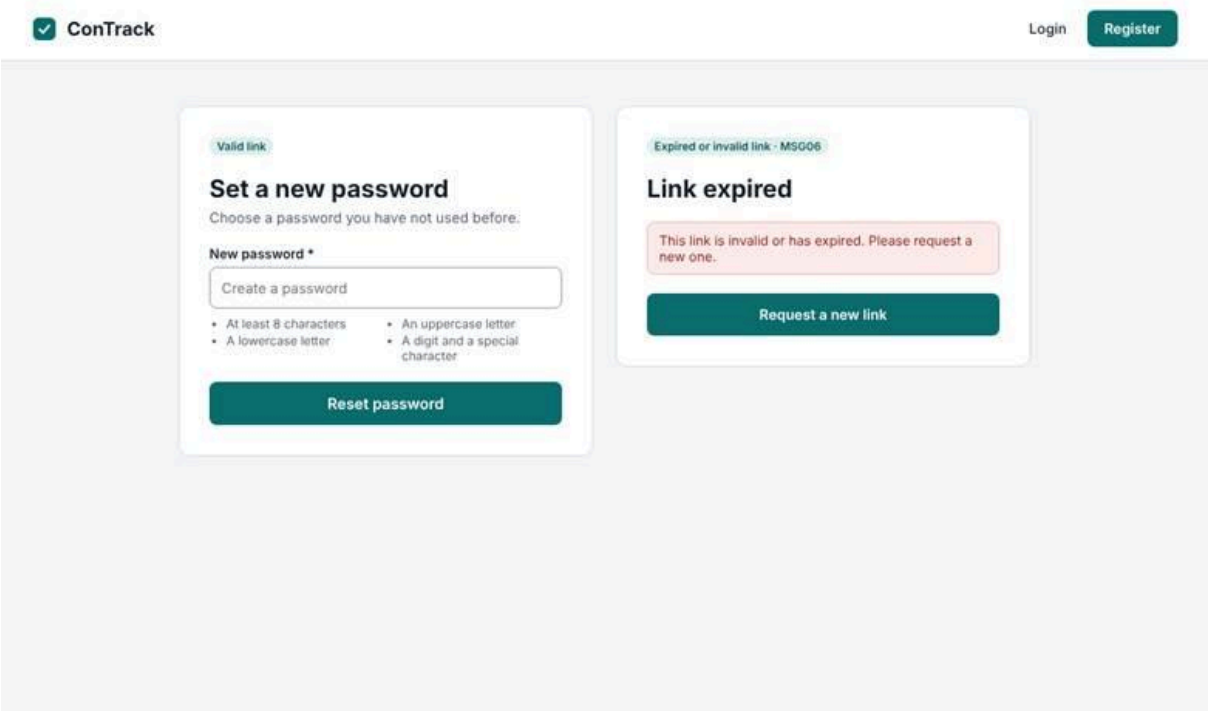
- Actions: Send reset link

**Content #4: Permission and State**

- Permission: Open to Guest.

## 1.6 Reset Password

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: Sets a new password from a reset link and applies the password policy.
- Mapped use case: UC 1.2
- Actors: All users
- Entities: users, password reset tokens
- Business rules and messages: BR-02; MSG01, MSG06

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|
| New password | Password | Yes | Placeholder: Create a password. Password policy (BR-02) |

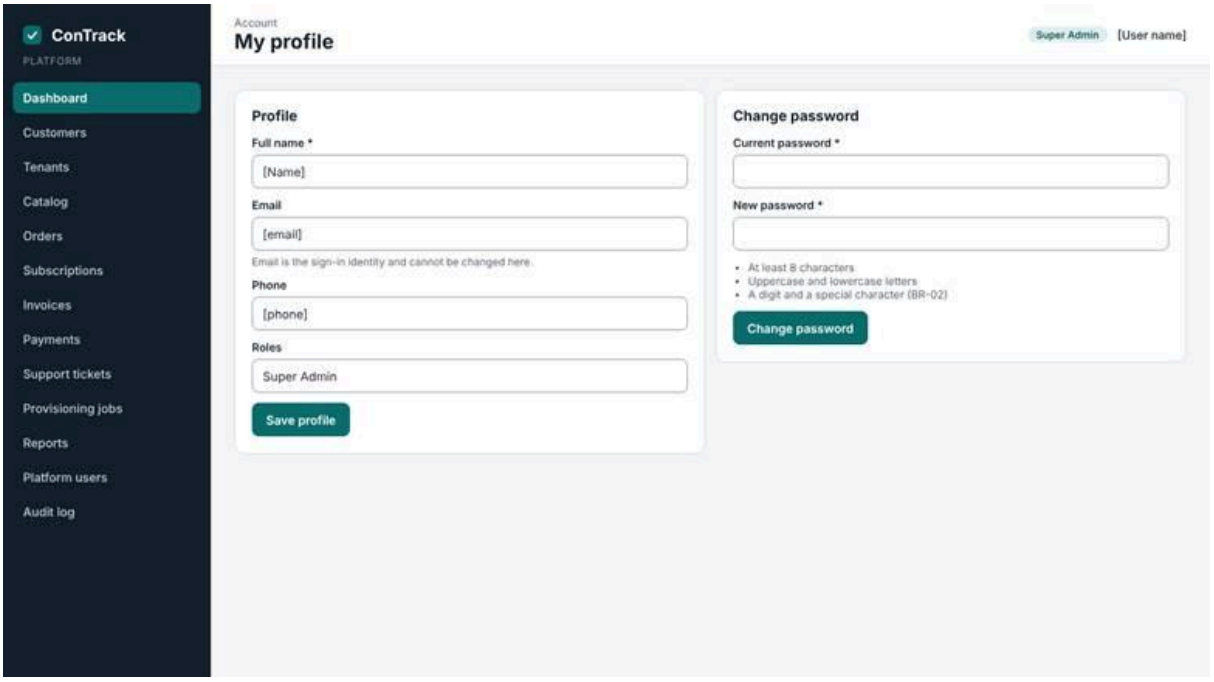
- Actions: Reset password

**Content #4: Permission and State**

- Permission: Open to Guest through the emailed link.
- State behavior: Invalid/expired link variant (MSG06).

## 1.7 My Profile and Change Password

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: Lets a signed-in user update personal details and change the password.
- Mapped use case: AU03, AU04
- Actors: Every signed-in user
- Entities: users
- Business rules and messages: BR-02; MSG01, MSG05

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|
| Full name | Text input | Yes | |
| Email | Text input | | Read-only |
| Phone | Text input | | |
| Roles | Text input | | Read-only |
| Current password | Password | Yes | |
| New password | Password | Yes | Password policy (BR-02) |

- Actions: Save profile, Change password

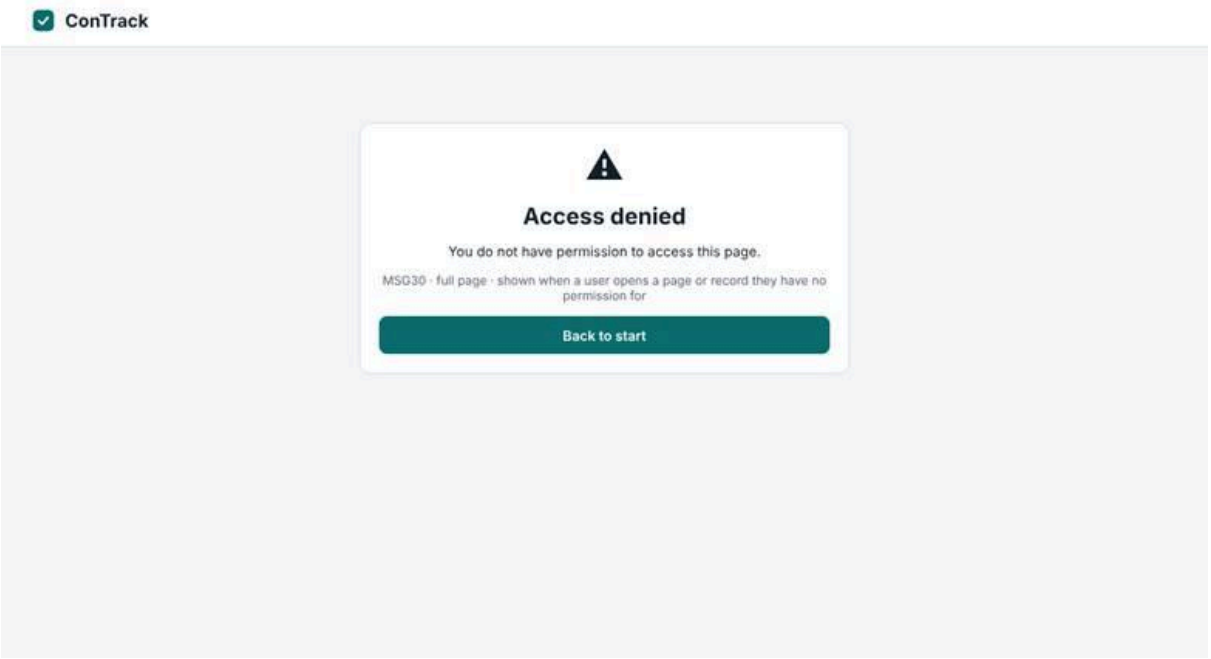
**Content #4: Permission and State**

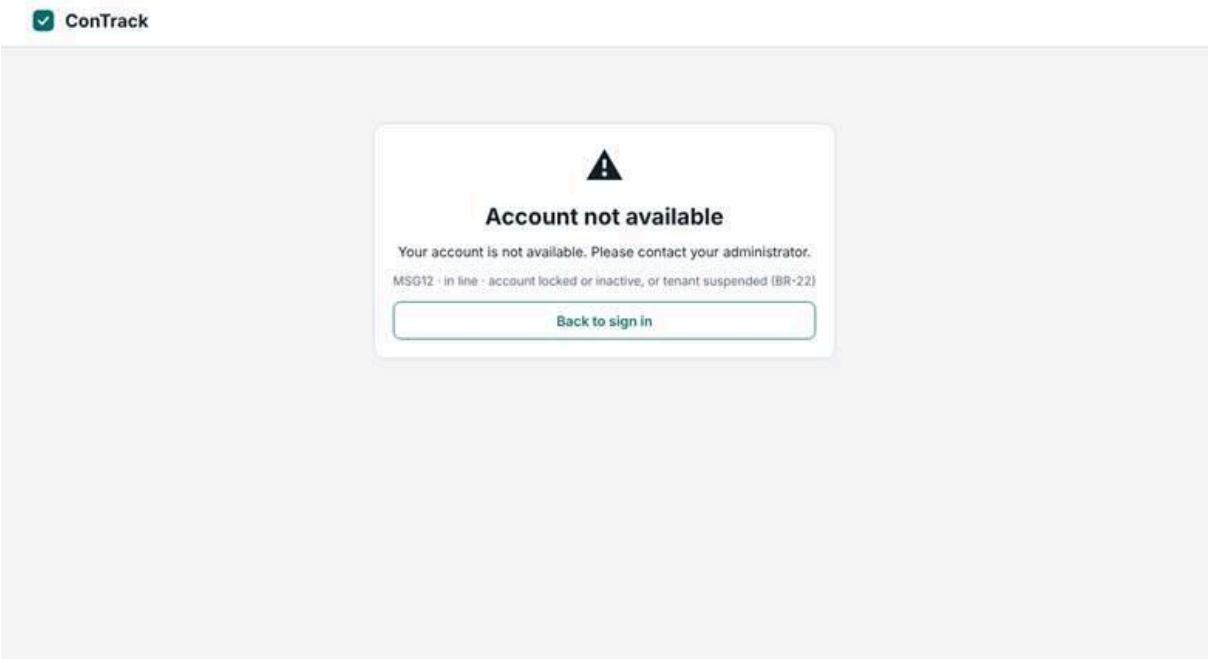
- Permission: Every signed-in user, own profile only.

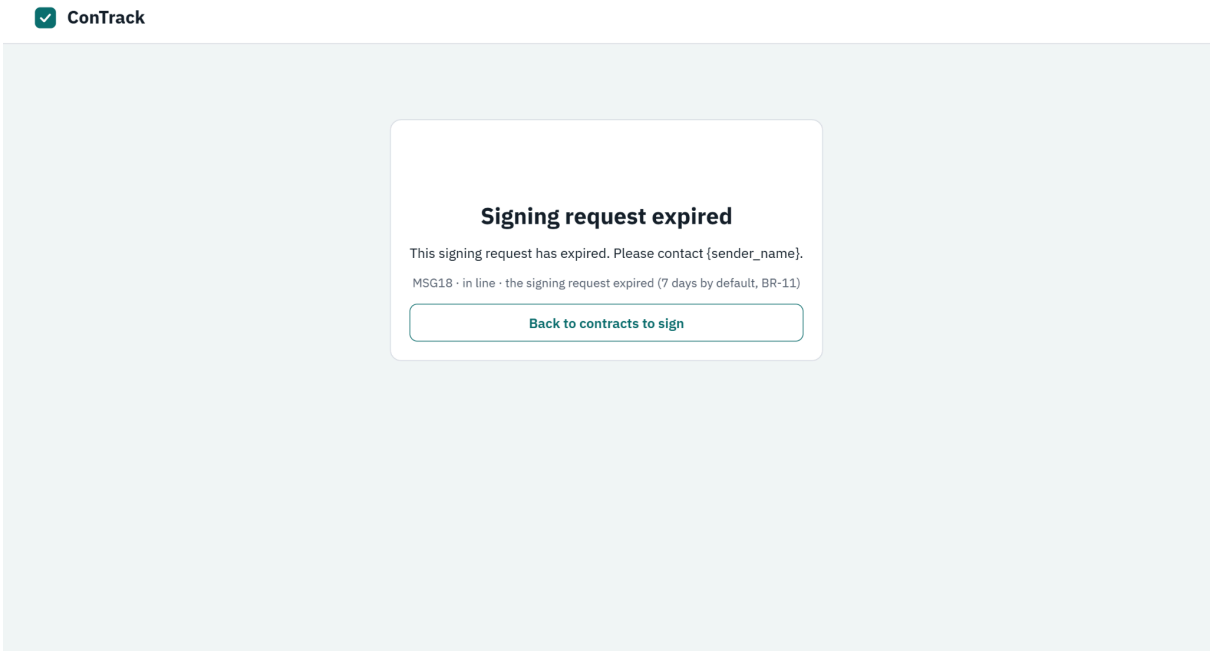
- Roles shown by the mockup tweak: Super Admin (SA), Product Admin Officer (PAO), Product Sales (PS), Product Technical Support (PTS)

## 1.8 Status pages (No Permission, Account Unavailable, Signing Expired)

**Content #1: UI Layout (Wireframe Mockup)**






**Content #2: Brief Description and Use Case Mapping**

- Description: Full-page messages for a record or page the user may not open (MSG30), an account or tenant that is unavailable (MSG12), and an expired signing request (MSG18).
- Mapped use case: All UCs
- Actors: All actors
- Entities: none
- Business rules and messages: BR-20, BR-21, BR-22; MSG12, MSG18, MSG30

**Content #3: Field and Component Descriptions**

- Actions: Back to start, Back to sign in, Back to contracts to sign

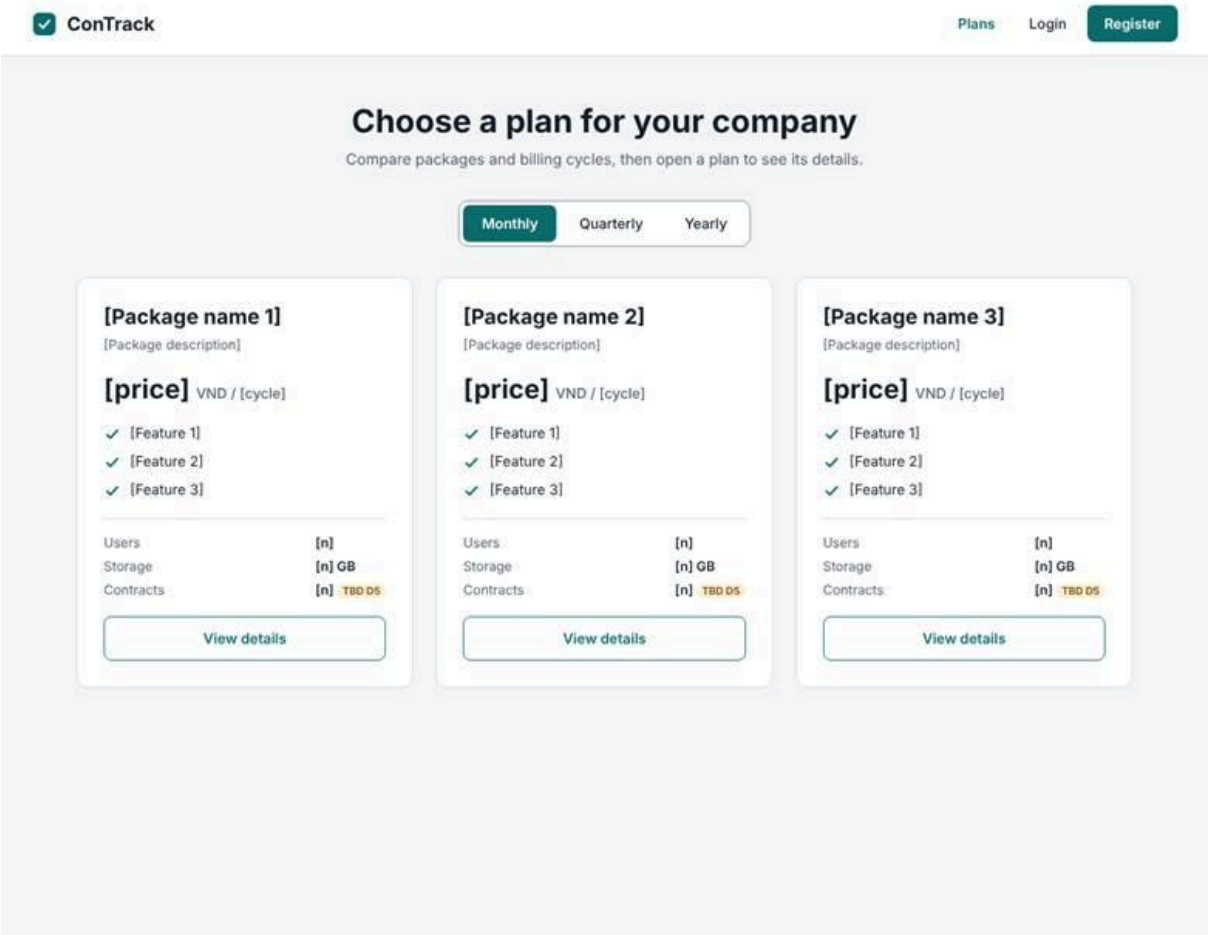
**Content #4: Permission and State**

- Permission: Shown to any role that opens a page or record whose Matrix entry is No.

# 2. SaaS Subscription and Platform Commerce

## 2.1 SaaS Package Selection

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: Public catalog of active SaaS packages with the billing-cycle switch so a visitor can compare and choose a plan.
- Mapped use case: UC 2.1
- Actors: Guest, SaaS Customer
- Entities: packages, package_prices, package_features
- Business rules and messages: none specific

**Content #3: Field and Component Descriptions**

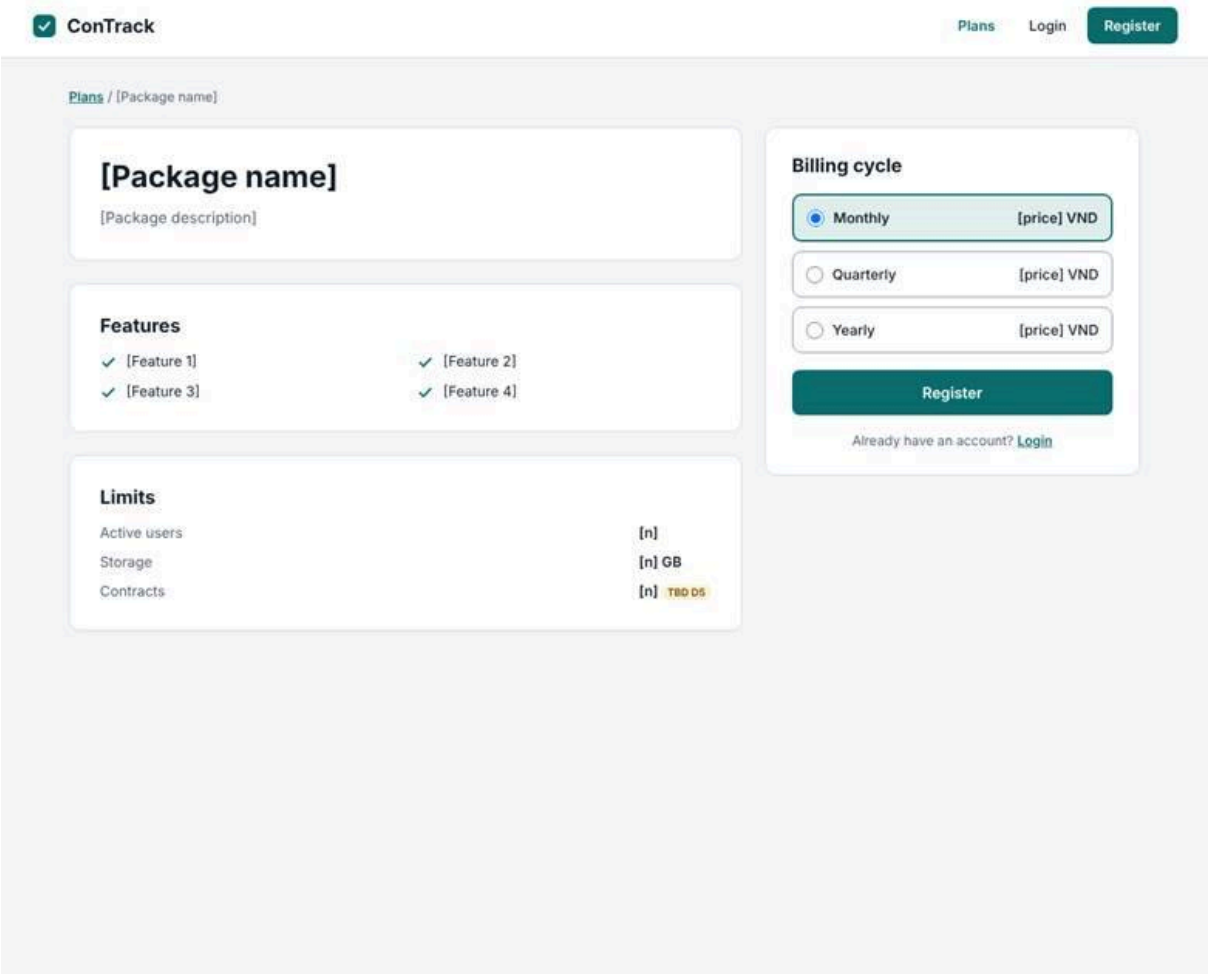
- Billing cycle switch: Monthly, Quarterly, Yearly
- Package cards: name, price for the selected cycle, key limits, link to Plan Detail

**Content #4: Permission and State**

- Permission: Catalog View: every role including Guest (active items only).

## 2.2 Plan Detail

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: Detail of one package: price per billing cycle, features and limits (users, storage, contracts).
- Mapped use case: UC 2.1
- Actors: Guest, SaaS Customer
- Entities: packages, package_prices, package_features
- Business rules and messages: BR-06

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|

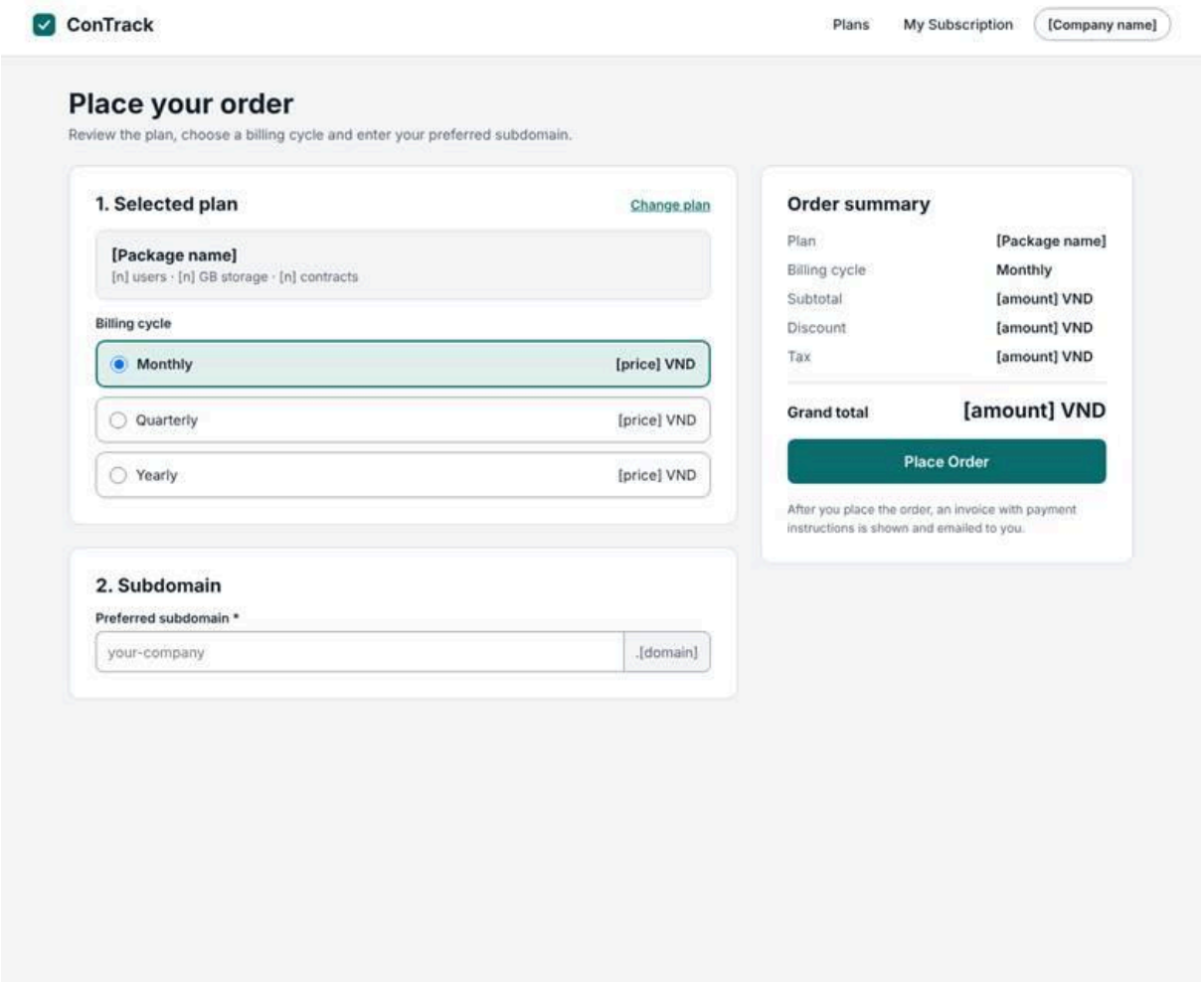
| | | | |
|---|---|---|---|
| Monthly price | Radio | | Selects the billing cycle |
| Quarterly price | Radio | | Selects the billing cycle |
| Yearly price | Radio | | Selects the billing cycle |

**Content #4: Permission and State**

- Permission: Catalog View: every role including Guest.
- State behavior: The call-to-action depends on the viewer (guest or customer).
- Mockup variants (viewer): guest, customer

## 2.3 SaaS Order / Subscription

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: Reviews the selected package and submits the SaaS order, including the desired tenant subdomain.

- Mapped use case: UC 2.1
- Actors: SaaS Customer
- Entities: orders, order_details, tenants (subdomain)
- Business rules and messages: BR-04, BR-17; MSG01, MSG03

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|
| Billing cycle (Monthly, Quarterly, Yearly) | Radio | | Price follows the selected cycle |
| Preferred subdomain | Text input | Yes | Placeholder: your-company. Unique (MSG03); suggestions are offered when taken |

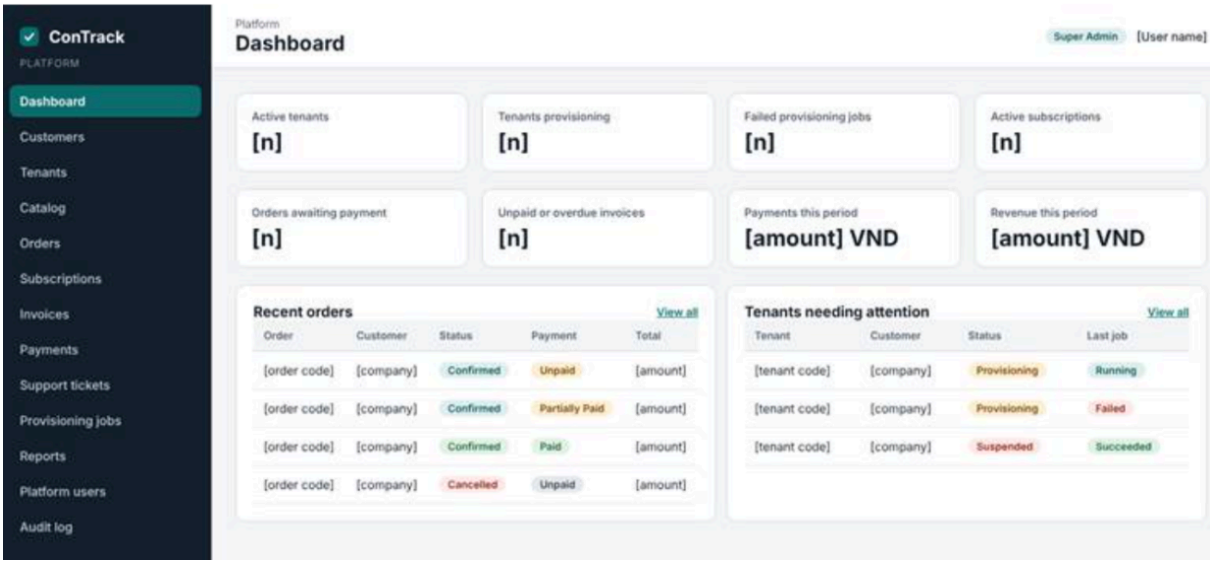
- Actions: Submit order, choose a suggested subdomain

**Content #4: Permission and State**

- Permission: SaaS Customer (Order Create = Full).
- State behavior: Subdomain-taken error variant.
- Mockup variants (showTaken): on, off

## 2.4 Platform Dashboard

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: Overview of platform activity for the Product Owner: customers, tenants, orders, invoices and tickets.
- Mapped use case: SA11, PS04
- Actors: SA, PAO, PS, PTS
- Entities: customers, tenants, orders, invoices, support_tickets
- Business rules and messages: none specific

**Content #3: Field and Component Descriptions**

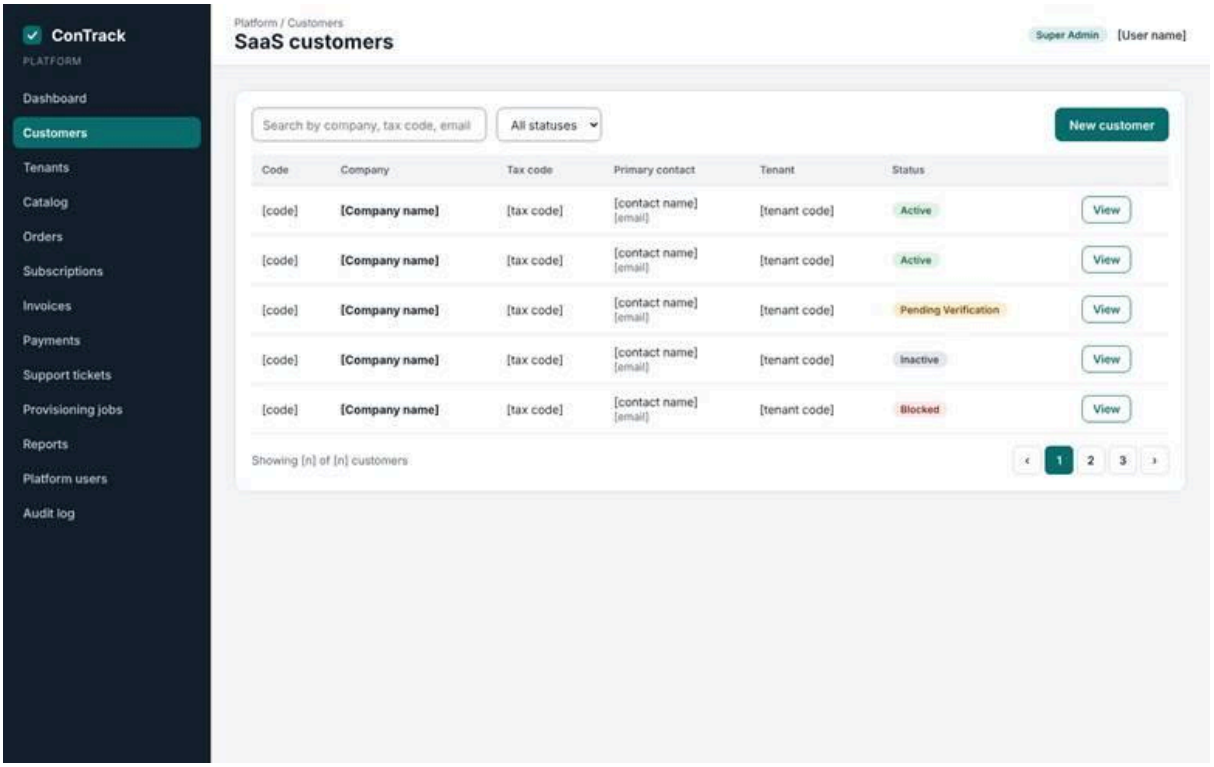
- Recent orders list columns: Order, Customer, Status, Payment, Total
- Tenants list columns: Tenant, Customer, Status, Last job

**Content #4: Permission and State**

- Permission: All four Product Owner roles can open it (SA = Super Admin, PAO = Product Admin Officer, PS = Product Sales, PTS = Product Technical Support). Widgets for Restricted roles follow their entity access.

## 2.5 SaaS Customer List

**Content #1: UI Layout (Wireframe Mockup)**



**Content #2: Brief Description and Use Case Mapping**

- Description: Searchable list of businesses registered for or using ConTrack.
- Mapped use case: PAS01
- Actors: SA, PAO, PS, PTS
- Entities: customers
- Business rules and messages: BR-24

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|
| Search | Search input | | By name, code or tax code |
| Status | Filter (dropdown) | | Options: All statuses, Active, Inactive, Blocked |

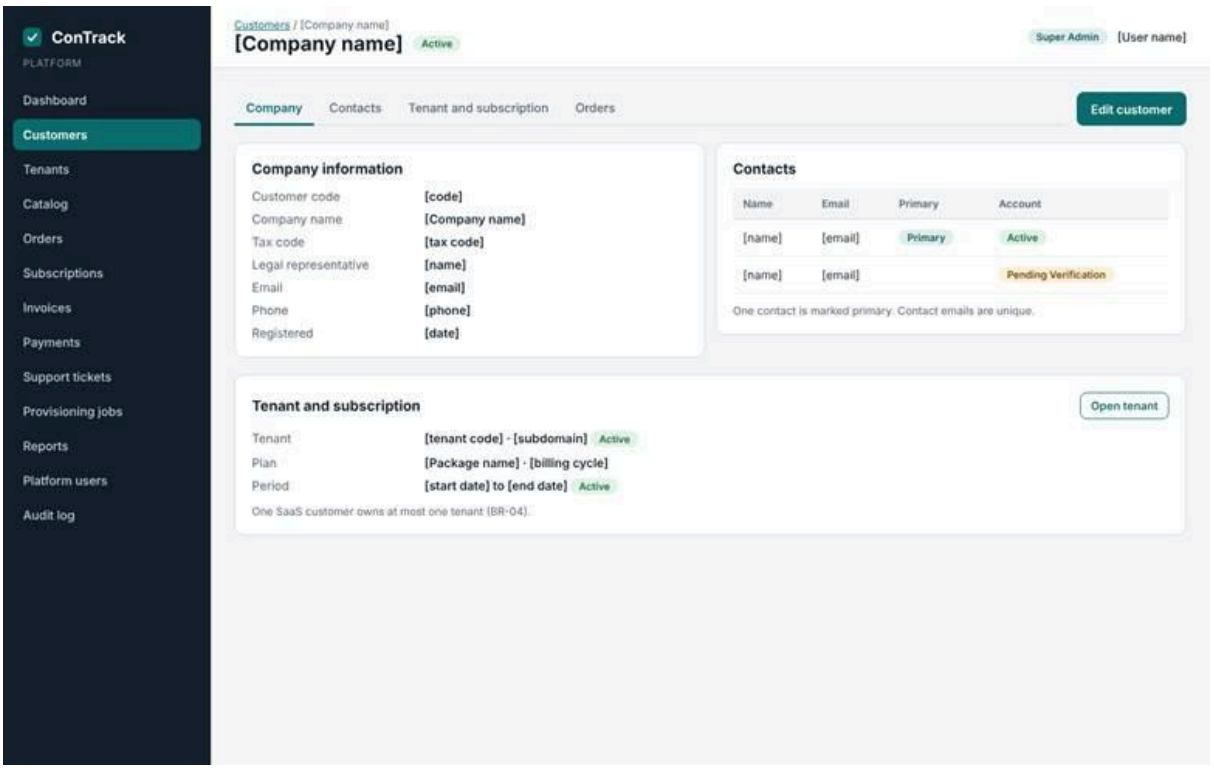
- List columns: Code, Company, Tax code, Primary contact, Tenant, Status
- Actions: New customer, View

**Content #4: Permission and State**

- Permission: SaaS customer View: SA, PAO, PTS Full; PS Restricted. Create/Edit: SA, PAO Full; PS Restricted; PTS No.
- Roles shown by the mockup tweak: Super Admin (SA), Product Admin Officer (PAO), Product Sales (PS), Product Technical Support (PTS)

## 2.6 SaaS Customer Detail

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: Company information, contacts, orders and the related tenant of a selected SaaS customer.
- Mapped use case: PAS01
- Actors: SA, PAO, PS, PTS
- Entities: customers, customer_contacts, tenants
- Business rules and messages: BR-01, BR-04, BR-24; MSG03

**Content #3: Field and Component Descriptions**

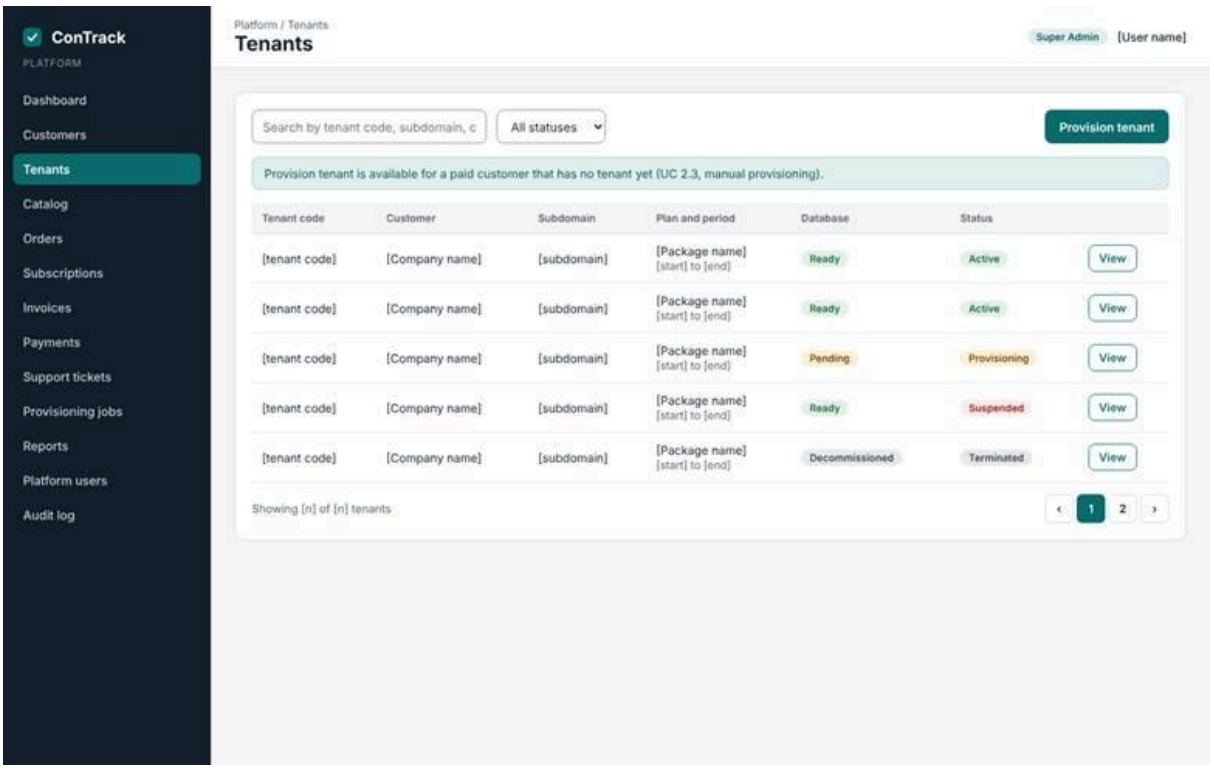
- Contacts list columns: Name, Email, Primary, Account
- Actions: Edit customer, Open tenant

**Content #4: Permission and State**

- Permission: View: SA, PAO, PTS Full; PS Restricted. Edit: SA, PAO Full; PS Restricted.
- Roles shown by the mockup tweak: Super Admin (SA), Product Admin Officer (PAO), Product Sales (PS), Product Technical Support (PTS)

## 2.7 Tenant List

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: Registered tenants with status, plan and subdomain.
- Mapped use case: UC 2.3
- Actors: SA, PAO, PTS
- Entities: tenants
- Business rules and messages: BR-04

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|
| Search | Search input | | By tenant code, customer or subdomain |
| Status | Filter (dropdown) | | Options: All statuses, Provisioning, Active, Suspended, Terminated |

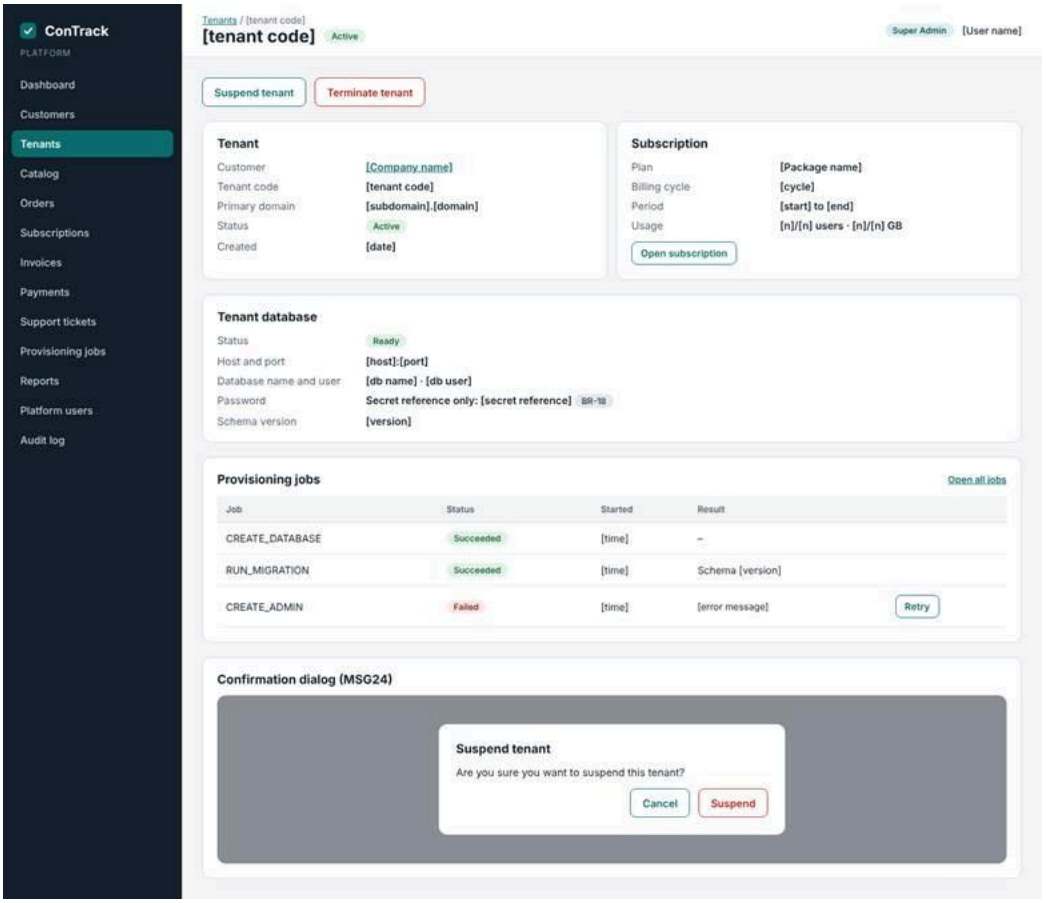
- List columns: Tenant code, Customer, Subdomain, Plan and period, Database, Status
- Actions: Provision tenant, View
- Notice: Provision tenant is available for a paid customer that has no tenant yet (UC 2.3, manual provisioning).

**Content #4: Permission and State**

- Permission: Tenant View: SA, PAO Full; PTS Restricted; PS No.
- Roles shown by the mockup tweak: Super Admin (SA), Product Admin Officer (PAO), Product Technical Support (PTS)

## 2.8 Tenant Detail

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: Tenant information, status, domain, subscription and tenant database status, with provisioning actions.
- Mapped use case: UC 2.3
- Actors: SA, PAO, PTS
- Entities: tenants, tenant_domains, tenant_databases, provisioning_jobs
- Business rules and messages: BR-04, BR-18; MSG14, MSG16, MSG24

**Content #3: Field and Component Descriptions**

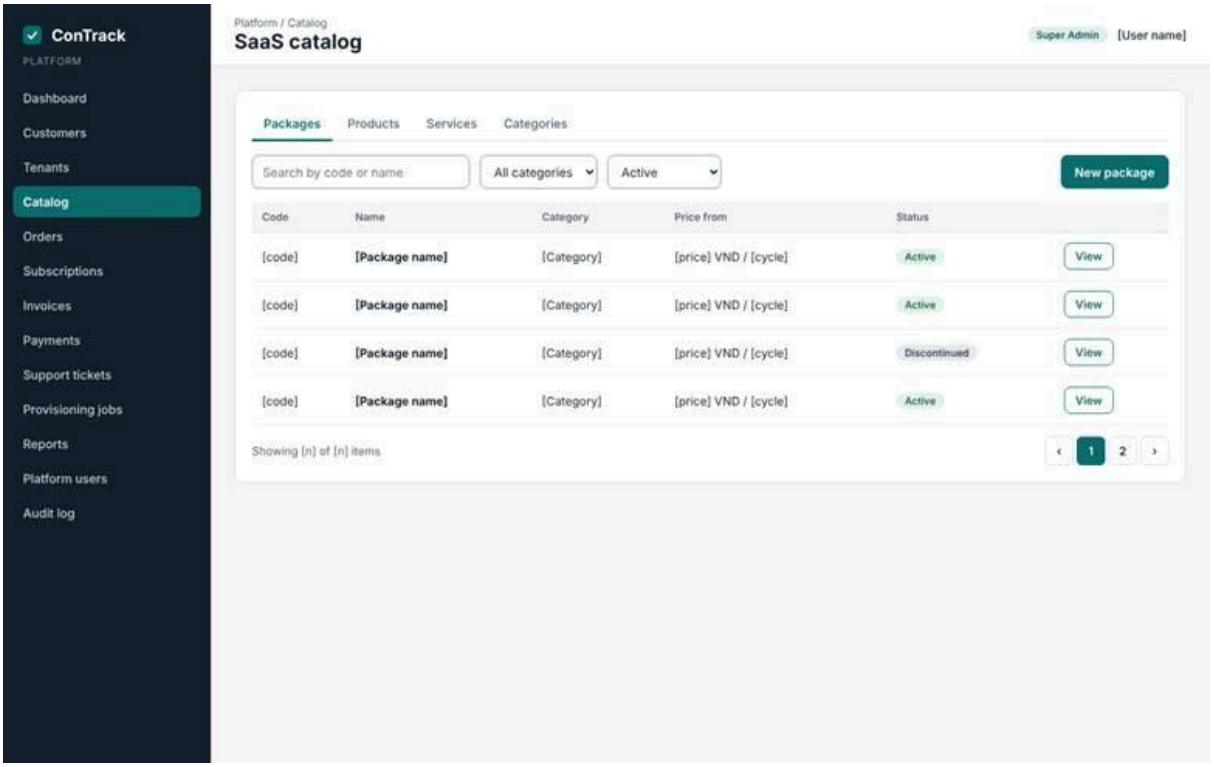
- Provisioning jobs list columns: Job, Status, Started, Result
- Actions: Suspend tenant, Activate tenant, Terminate tenant, Open subscription, Retry
- Messages: This action is not allowed while the tenant is {status}. (MSG16). Confirmation dialog: Are you sure you want to suspend this tenant? (MSG24)

**Content #4: Permission and State**

- Permission: View: SA, PAO Full; PTS Restricted. Provision, Activate, Suspend, Terminate: SA only.
- Roles shown by the mockup tweak: Super Admin (SA), Product Admin Officer (PAO), Product Technical Support (PTS)
- State behavior: Buttons depend on tenant status: Provisioning, Active, Suspended, Terminated.
- Mockup variants (tenantStatus): Provisioning, Active, Suspended, Terminated

## 2.9 SaaS Catalog List

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: List of packages, products and services offered by the platform.
- Mapped use case: SA04
- Actors: SA (edit), all roles (view)

- Entities: categories, packages, products, services
- Business rules and messages: BR-24

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|
| Search | Search input | | By code or name |
| Category | Filter (dropdown) | | Options: All categories, then each category |
| Status | Filter (dropdown) | | Options: Active, Discontinued |

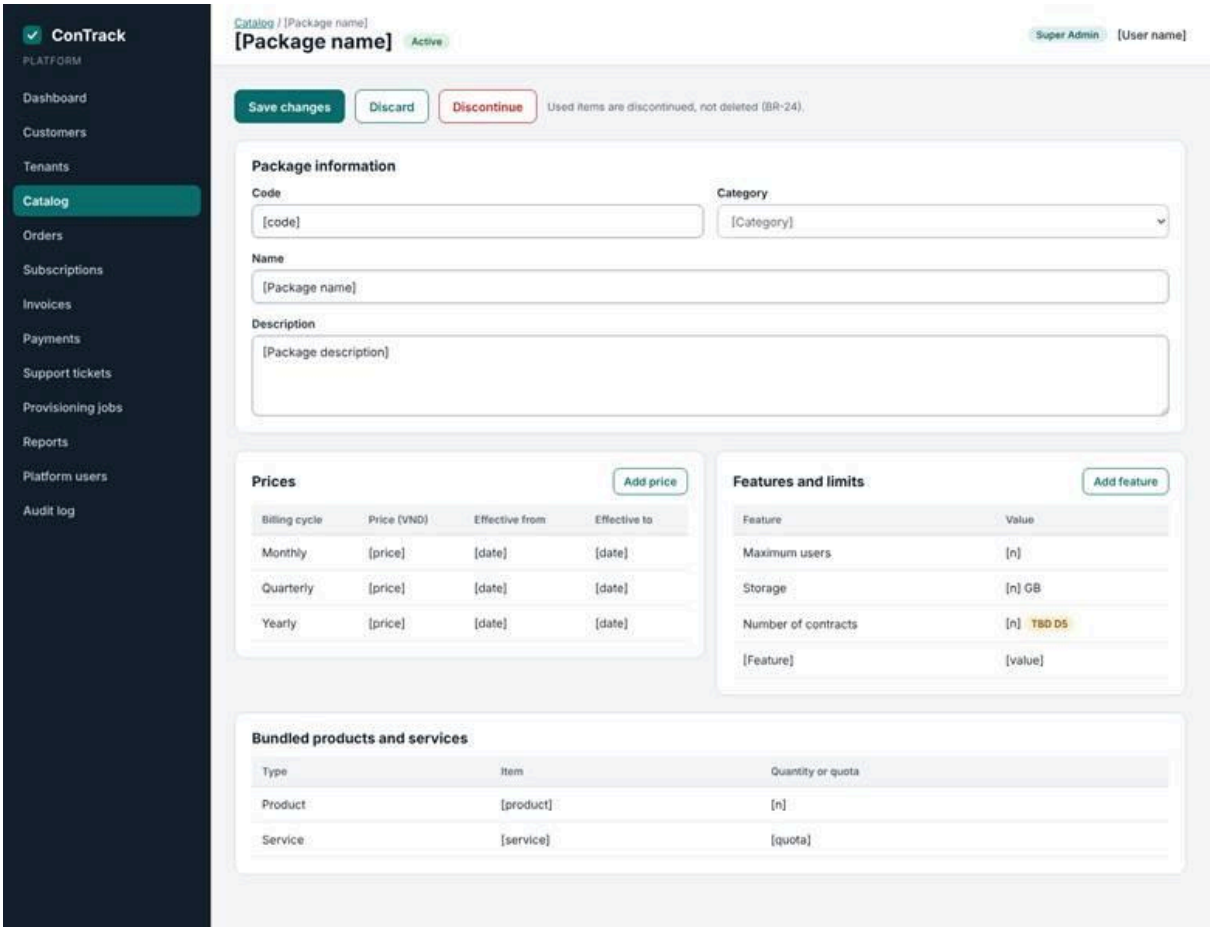
- List columns: Code, Name, Category, Price from, Status
- Actions: New package, View
- Notice: Only active items are listed. Catalog editing is available to Super Admin.

**Content #4: Permission and State**

- Permission: View (active items): all roles. Create/Edit/Discontinue: SA only.
- Roles shown by the mockup tweak: Super Admin (SA), Product Admin Officer (PAO), Product Sales (PS), Product Technical Support (PTS)

## 2.10 SaaS Catalog Detail

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: Detail of a package, product or service with prices and features; editable only by Super Admin.
- Mapped use case: SA04, SA05, SA06
- Actors: SA
- Entities: packages, package_prices, package_features
- Business rules and messages: BR-24; MSG01, MSG03, MSG05

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|
| Code | Text input | | Read-only |
| Category | Dropdown | | Category of the item |
| Name | Text input | | Unique (MSG03) |
| Description | Text area | | |

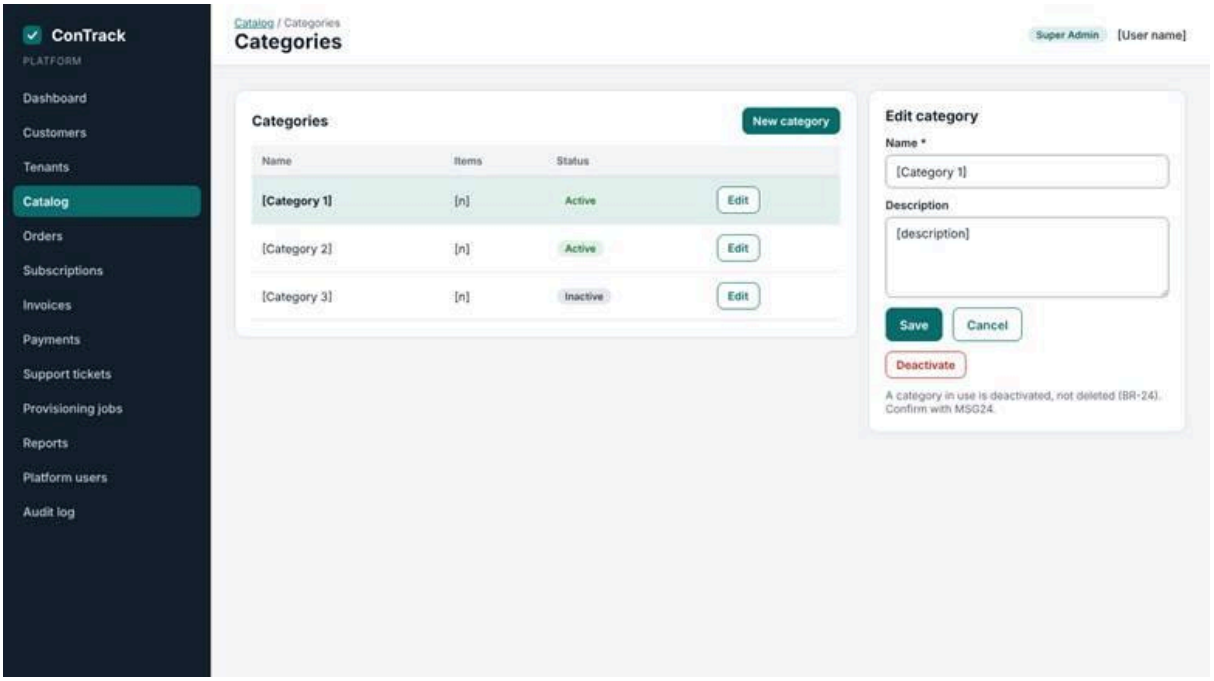
- Prices list columns: Billing cycle, Price (VND), Effective from, Effective to
- Features list columns: Feature, Value
- Contents list columns: Type, Item, Quantity or quota
- Actions: Save changes, Discard, Discontinue, Add price, Add feature
- Notice: You can view this item. Editing is available to Super Admin.

**Content #4: Permission and State**

- Permission: Plan price & features Create/Edit: SA only. Other roles see it read-only.
- Roles shown by the mockup tweak: Super Admin (SA), Product Admin Officer (PAO), Product Sales (PS), Product Technical Support (PTS)
- State behavior: Fields and actions are read-only for roles other than SA.

## 2.11 Category Management

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: Creates, edits and deactivates catalog categories.
- Mapped use case: SA03
- Actors: SA
- Entities: categories

- Business rules and messages: BR-24; MSG24

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|
| Name | Text input | Yes | |
| Description | Text area | | |

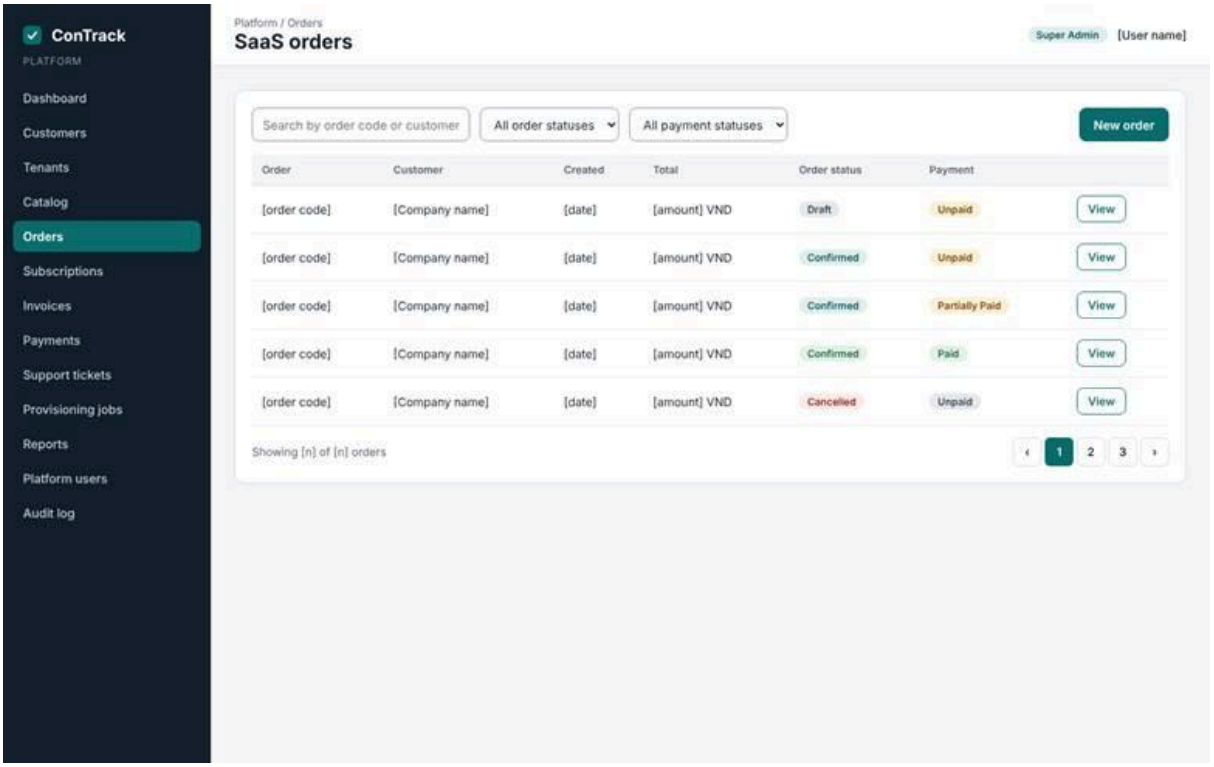
- List columns: Name, Items, Status
- Actions: New category, Edit, Save, Cancel, Deactivate

**Content #4: Permission and State**

- Permission: Category Create/Edit/Deactivate: SA only.

## 2.12 SaaS Order List

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: Orders placed by customers, with order and payment status.
- Mapped use case: UC 2.1, 2.2
- Actors: SA, PAO, PS

- Entities: orders
- Business rules and messages: BR-17

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|
| Search | Search input | | By order or customer |
| Order status | Filter (dropdown) | | Options: All order statuses, Draft, Confirmed, Cancelled |
| Payment status | Filter (dropdown) | | Options: All payment statuses, Unpaid, Partially Paid, Paid |

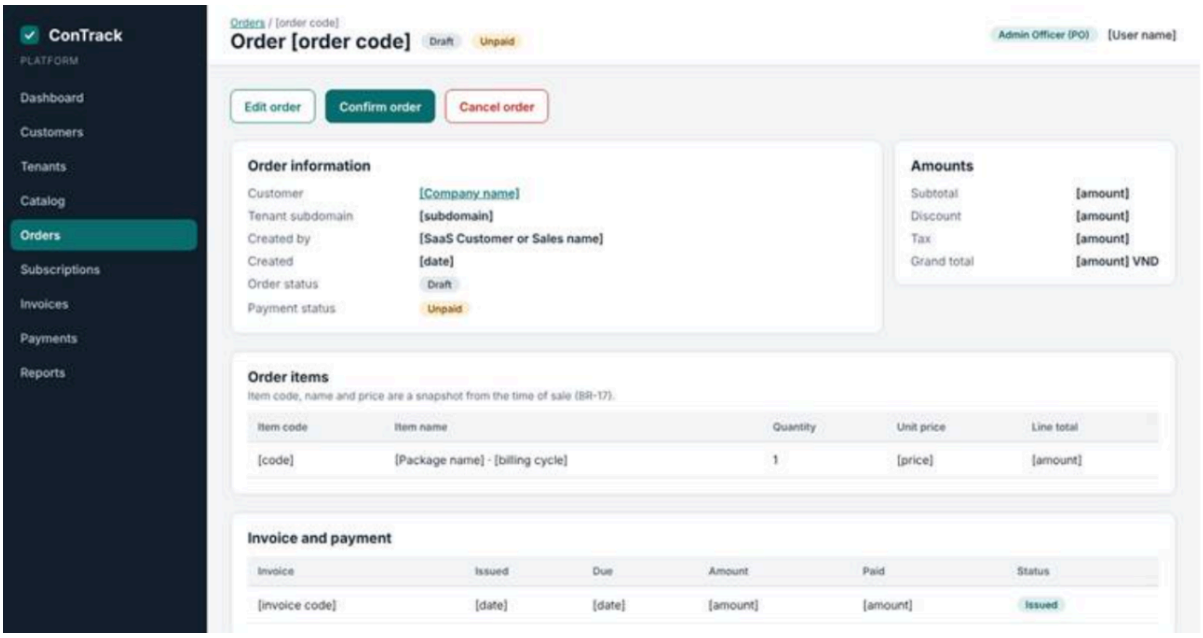
- List columns: Order, Customer, Created, Total, Order status, Payment
- Actions: New order, View

**Content #4: Permission and State**

- Permission: Order View: SA, PAO Full; PS Restricted; PTS No.
- Roles shown by the mockup tweak: Super Admin (SA), Product Admin Officer (PAO), Product Sales (PS)

## 2.13 SaaS Order Detail

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: Order information and items, with confirm and cancel actions.
- Mapped use case: UC 2.1, 2.2
- Actors: SA, PAO, PS
- Entities: orders, order_details
- Business rules and messages: BR-05, BR-17; MSG16, MSG24

**Content #3: Field and Component Descriptions**

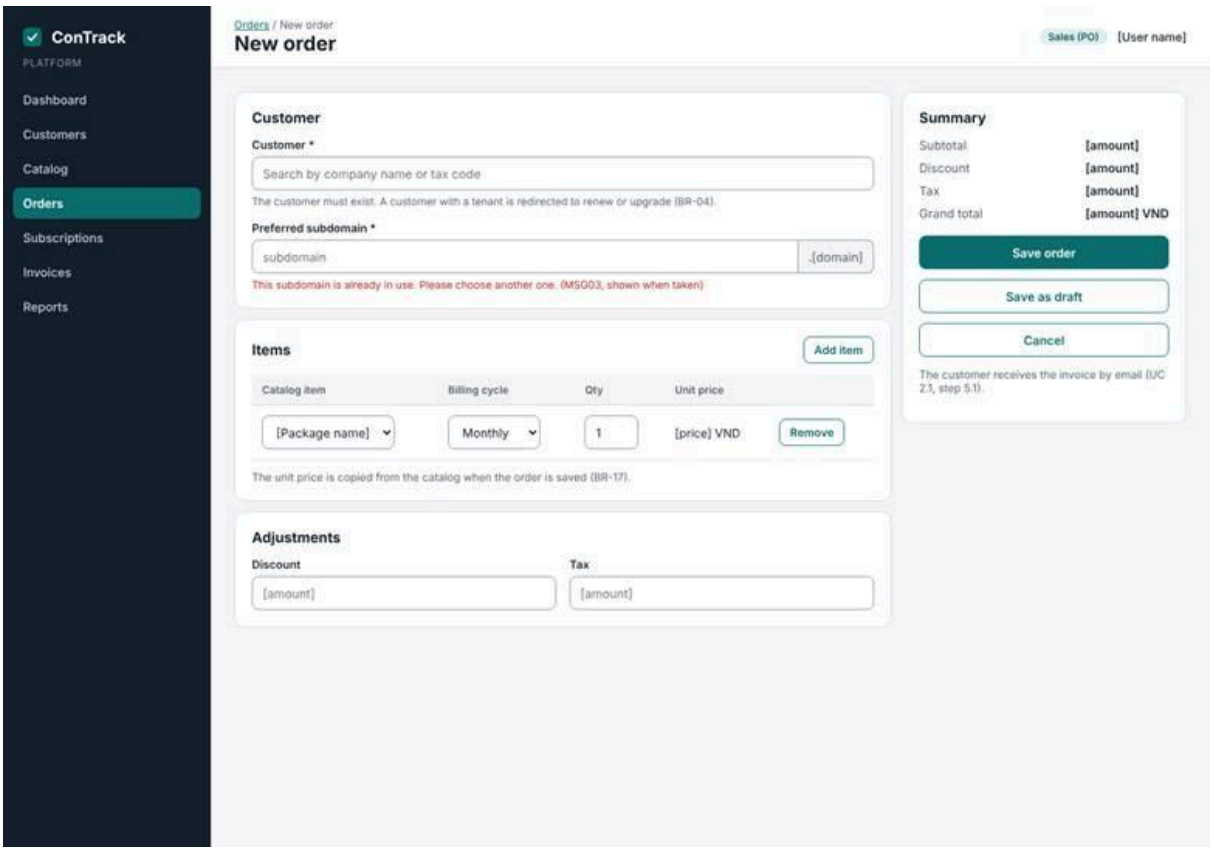
- Items list columns: Item code, Item name, Quantity, Unit price, Line total (a copy of code, name and price at the time of sale, BR-17)
- Invoices list columns: Invoice, Issued, Due, Amount, Paid, Status
- Actions: Edit order, Confirm order, Cancel order, Open invoice
- Message: This action is not allowed while the order is {status}. (MSG16)

**Content #4: Permission and State**

- Permission: Confirm/Cancel: SA, PAO only. Update (draft only): SA, PAO Full; PS Restricted.
- Roles shown by the mockup tweak: Super Admin (SA), Product Admin Officer (PAO), Product Sales (PS)
- State behavior: Buttons depend on order status (Draft, Confirmed, Cancelled) and payment status (Unpaid, Partially Paid, Paid).
- Mockup variants (orderStatus): Draft, Confirmed, Cancelled

## 2.14 Create or Edit Order

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: Creates or edits a draft order on behalf of a customer.
- Mapped use case: PS01, PS02
- Actors: SA, PAO, PS
- Entities: orders, order_details, catalog_items
- Business rules and messages: BR-17; MSG01, MSG03

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|
| Customer | Text input | Yes | Search by company name or tax code |
| Preferred subdomain | Text input | Yes | Unique: This subdomain is already in use. Please choose another one. (MSG03) |
| Discount | Text input | | Amount |
| Tax | Text input | | Amount |

- Items list columns: Catalog item, Billing cycle, Qty, Unit price

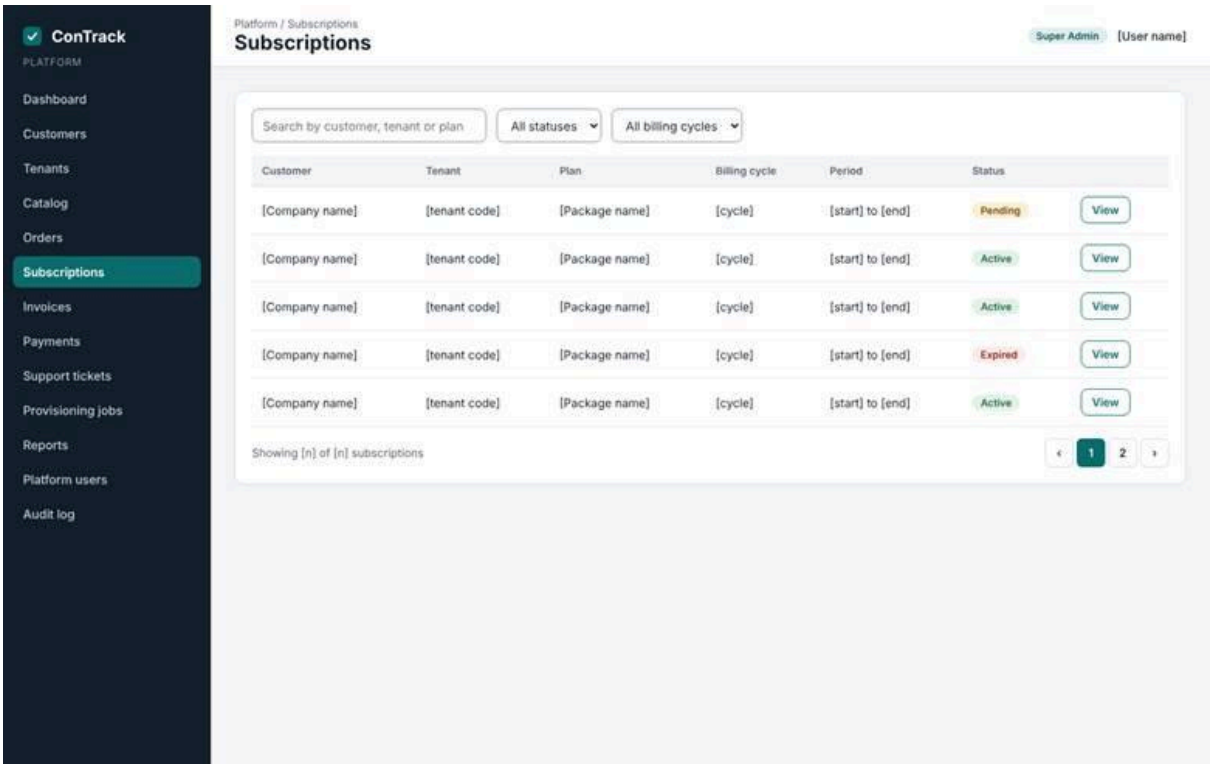
- Actions: Add item, Remove, Save order, Save as draft, Cancel

**Content #4: Permission and State**

- Permission: Order Create: SA, PAO, PS Full. Update (draft only): SA, PAO Full; PS Restricted.
- Roles shown by the mockup tweak: Super Admin (SA), Product Admin Officer (PAO), Product Sales (PS)

## 2.15 Subscription List

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: Active and historical subscriptions of tenants.
- Mapped use case: UC 2.2, 2.4
- Actors: SA, PAO, PS, PTS
- Entities: subscriptions
- Business rules and messages: none specific

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|

| | | | |
|---|---|---|---|
| Search | Search input | | By customer or tenant |
| Status | Filter (dropdown) | | Options: All statuses, Pending, Active, Expired |
| Billing cycle | Filter (dropdown) | | Options: All billing cycles, Monthly, Quarterly, Yearly |

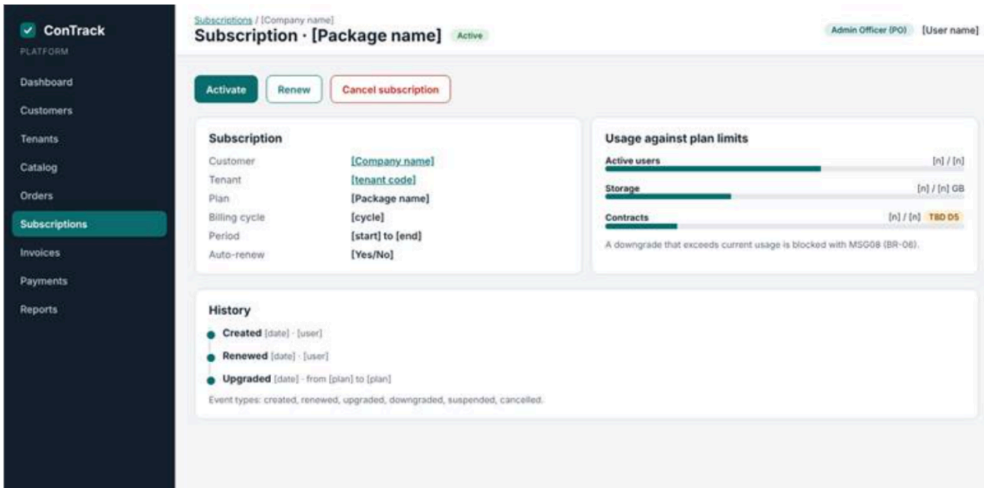
- List columns: Customer, Tenant, Plan, Billing cycle, Period, Status
- Actions: View

**Content #4: Permission and State**

- Permission: View: SA, PAO, PTS Full; PS Restricted.
- Roles shown by the mockup tweak: Super Admin (SA), Product Admin Officer (PAO), Product Sales (PS), Product Technical Support (PTS)

## 2.16 Subscription Detail

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: Package, billing cycle, validity period, status and history of one subscription, with renew and cancel actions.
- Mapped use case: UC 2.2, 2.4
- Actors: SA, PAO, PS, PTS
- Entities: subscriptions, subscription_history
- Business rules and messages: BR-05, BR-06; MSG16, MSG24

**Content #3: Field and Component Descriptions**

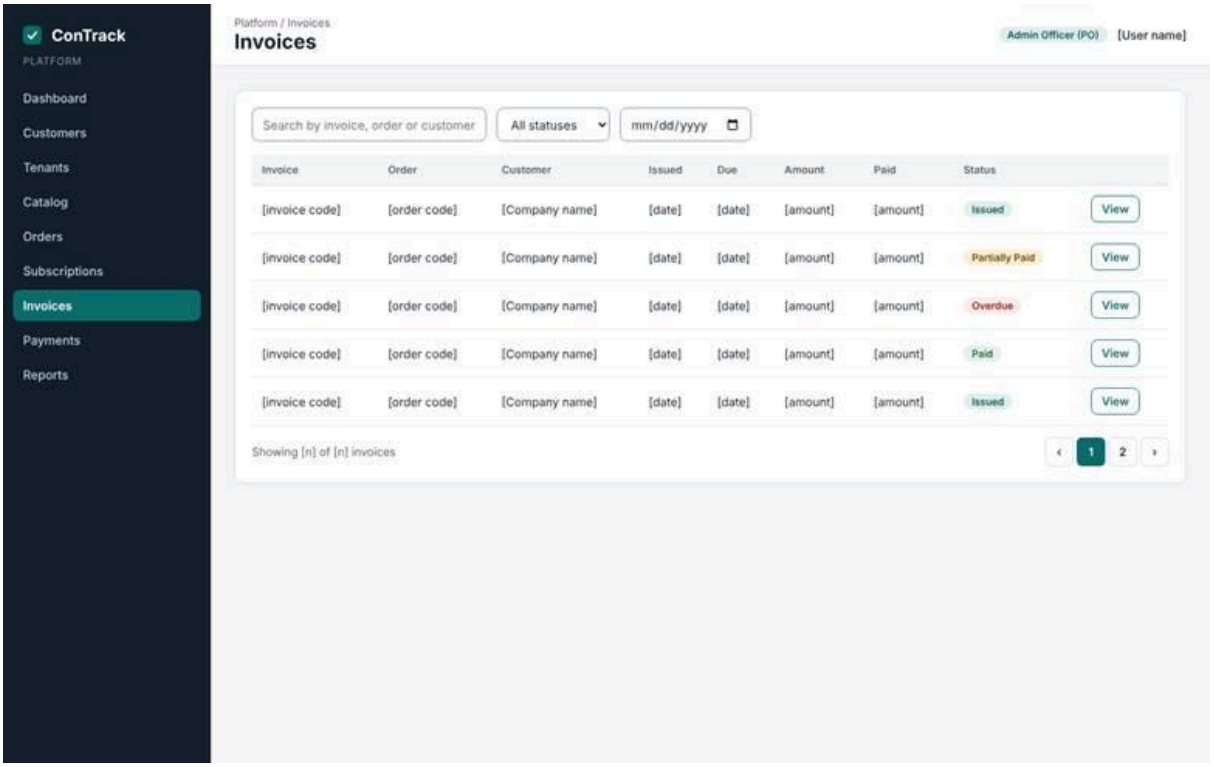
- Actions: Activate, Renew, Cancel subscription
- Notice: Activation starts after the invoice is fully paid (BR-05).

**Content #4: Permission and State**

- Permission: Activate/Renew/Cancel: SA, PAO only. View: SA, PAO, PTS Full; PS Restricted.
- Roles shown by the mockup tweak: Super Admin (SA), Product Admin Officer (PAO), Product Sales (PS), Product Technical Support (PTS)
- State behavior: Buttons depend on subscription status.
- Mockup variants (subStatus): Pending, Active, Expired

## 2.17 Invoice List

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: Invoices generated for orders and subscriptions.
- Mapped use case: UC 2.2
- Actors: SA, PAO, PS
- Entities: invoices

- Business rules and messages: none specific

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|
| Search | Search input | | By invoice, order or customer |
| Status | Filter (dropdown) | | Options: All statuses, Issued, Partially Paid, Paid, Overdue |
| Issued from | Date filter | | |

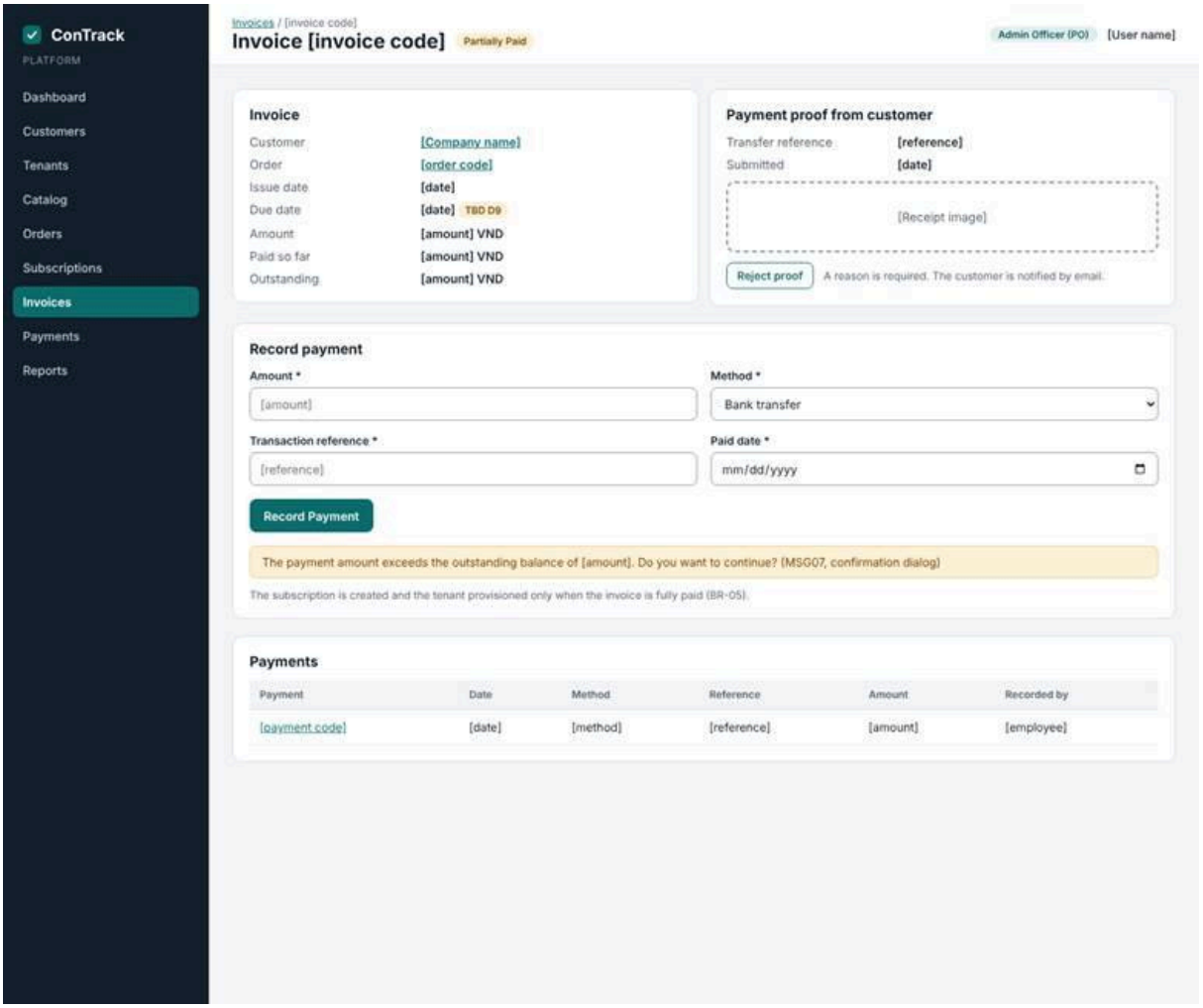
- List columns: Invoice, Order, Customer, Issued, Due, Amount, Paid, Status
- Actions: View

**Content #4: Permission and State**

- Permission: Invoice View: SA, PAO Full; PS Restricted; PTS No. Issue: PAO only.
- Roles shown by the mockup tweak: Super Admin (SA), Product Admin Officer (PAO), Product Sales (PS)

## 2.18 Invoice Detail

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: Invoice information, payments received and the form to record a payment.
- Mapped use case: UC 2.2
- Actors: SA, PAO, PS
- Entities: invoices, payments
- Business rules and messages: BR-05; MSG07

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|
| Amount | Text input | Yes | Overpayment shows a confirmation (MSG07) |
| Method | Dropdown | Yes | Options: Bank transfer, other methods |

| | | | |
|---|---|---|---|
| Transaction reference | Text input | Yes | |
| Paid date | Date | Yes | |
| Receipt image | File upload | | Drop zone |

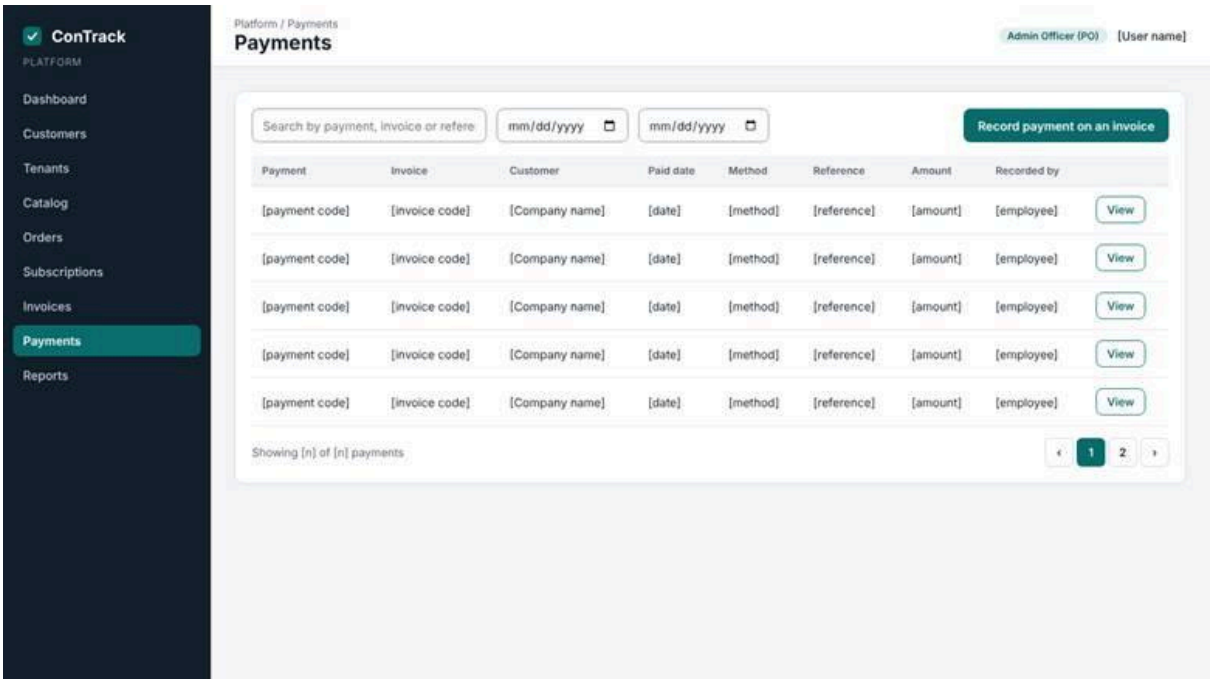
- Payments list columns: Payment, Date, Method, Reference, Amount, Recorded by
- Actions: Reject proof, Record Payment
- Messages: The payment amount exceeds the outstanding balance of {amount}. Do you want to continue? (MSG07, confirmation dialog). Recording payments is available to Admin Officer only.

**Content #4: Permission and State**

- Permission: Record/Confirm payment: PAO only. View: SA, PAO Full; PS Restricted.
- Roles shown by the mockup tweak: Super Admin (SA), Product Admin Officer (PAO), Product Sales (PS)
- State behavior: Record payment form with the overpayment confirmation (MSG07).

## 2.19 Payment List

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: Payment records tied to orders and invoices.
- Mapped use case: UC 2.2

- Actors: SA, PAO
- Entities: payments
- Business rules and messages: none specific

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|
| Search | Search input | | By payment, invoice or customer |
| Paid from | Date filter | | |
| Paid to | Date filter | | |

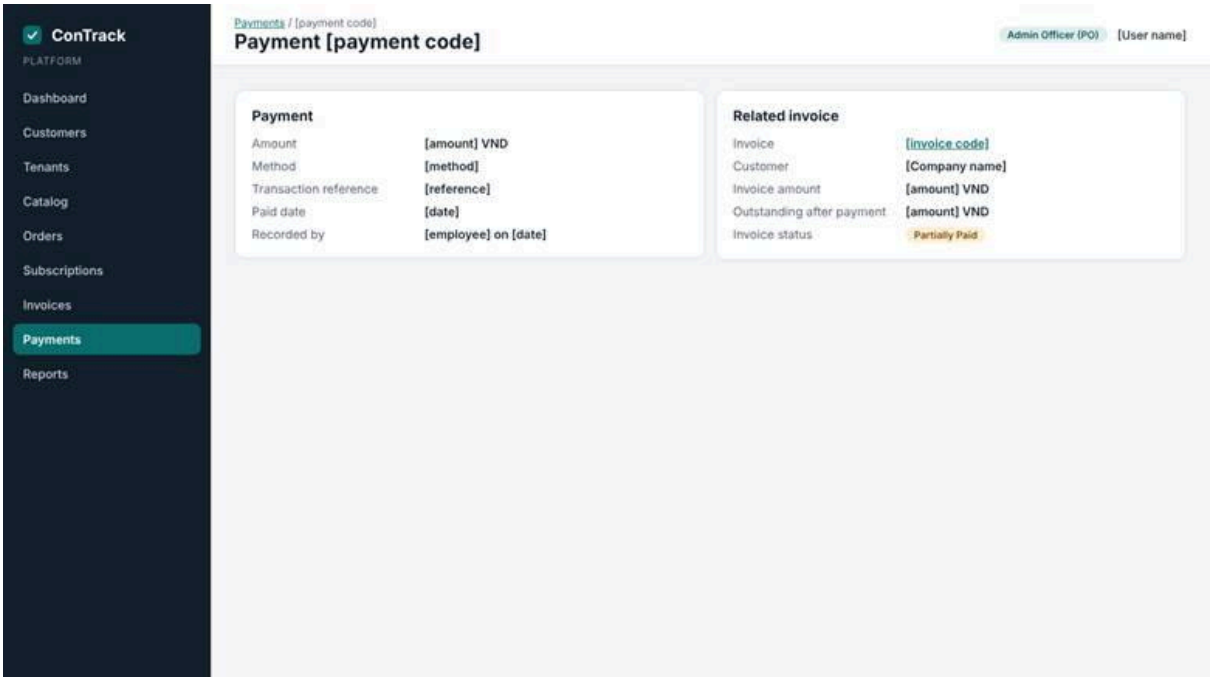
- List columns: Payment, Invoice, Customer, Paid date, Method, Reference, Amount, Recorded by
- Actions: Record payment on an invoice, View
- Notice: Super Admin can view payments. Recording is available to Admin Officer.

**Content #4: Permission and State**

- Permission: Payment Record/Confirm: PAO. SA view only.
- Roles shown by the mockup tweak: Super Admin (SA), Product Admin Officer (PAO)

## 2.20 Payment Detail

**Content #1: UI Layout (Wireframe Mockup)**



**Content #2: Brief Description and Use Case Mapping**

- Description: Details of one payment transaction and its proof.
- Mapped use case: UC 2.2
- Actors: SA, PAO
- Entities: payments, invoices
- Business rules and messages: MSG07

**Content #3: Field and Component Descriptions**

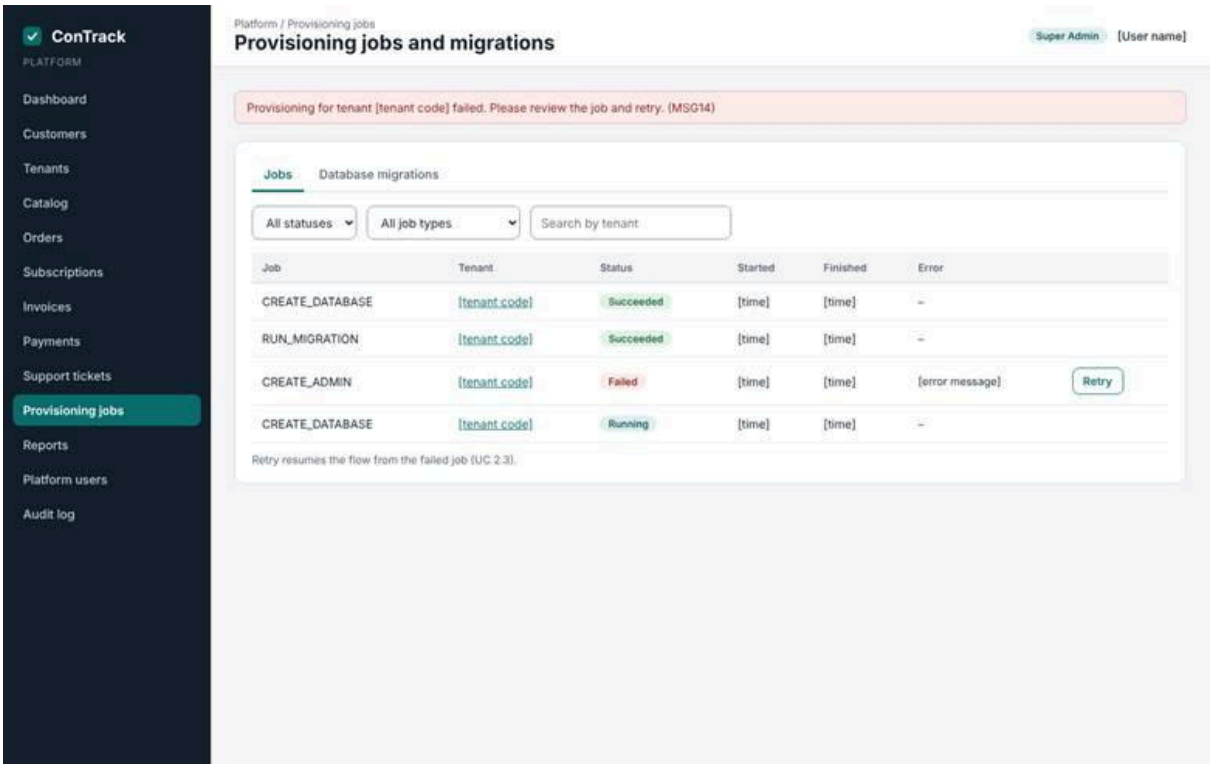
- Read-only display of the payment (amount, date, method, reference, recorded by), a link to the invoice and the receipt image.

**Content #4: Permission and State**

- Permission: Record/Confirm: PAO. SA view only.
- Roles shown by the mockup tweak: Super Admin (SA), Product Admin Officer (PAO)

## 2.21 Provisioning Jobs

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: Status of tenant provisioning and database migration jobs with retry.

- Mapped use case: UC 2.3
- Actors: SA, PTS
- Entities: provisioning_jobs, database_migrations
- Business rules and messages: BR-18; MSG14

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|
| Status | Filter (dropdown) | | Options: All statuses, Pending, Running, Succeeded, Failed, Cancelled |
| Job type | Filter (dropdown) | | Options: All job types, CREATE_DATABASE, RUN_MIGRATION, CREATE_ADMIN |
| Search tenant | Search input | | |

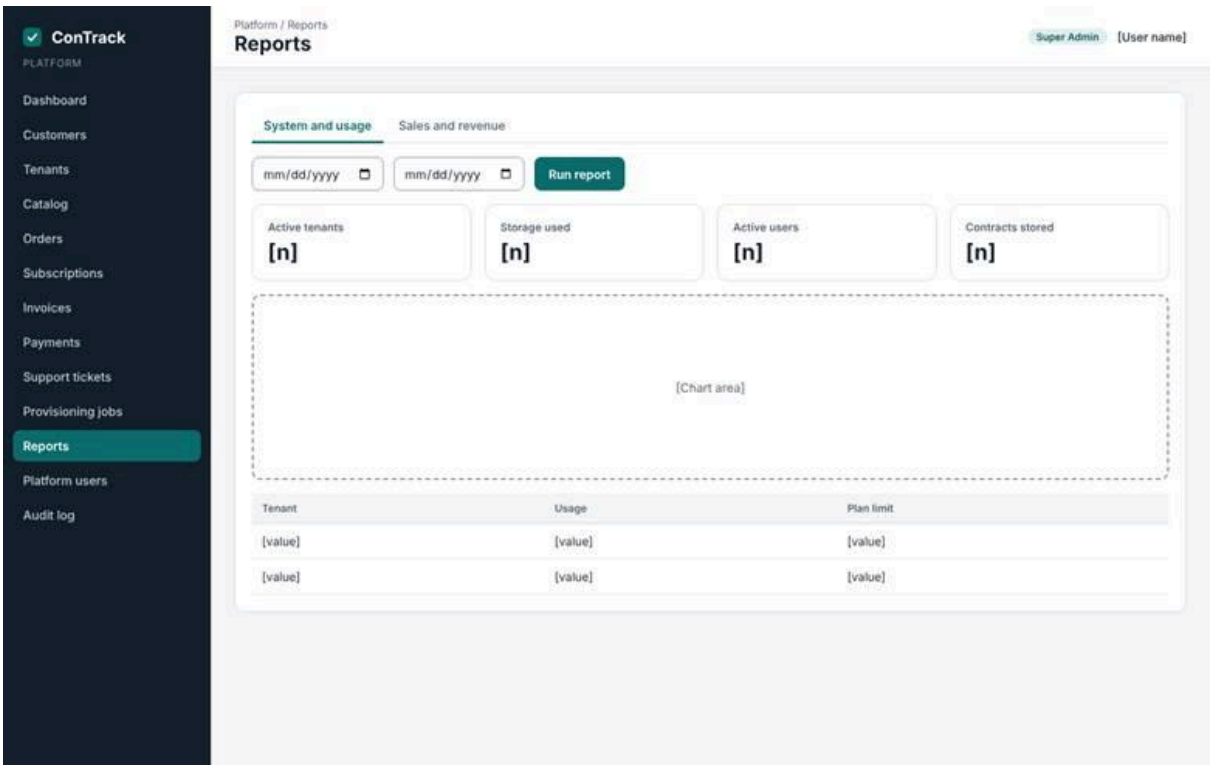
- List columns: Job, Tenant, Status, Started, Finished, Error
- Actions: Retry
- Message: Provisioning for tenant {tenant_code} failed. Please review the job and retry. (MSG14)

**Content #4: Permission and State**

- Permission: Provisioning job & migration View/Retry: SA Full; PTS Restricted.
- Roles shown by the mockup tweak: Super Admin (SA), Product Technical Support (PTS)
- State behavior: Job statuses: Pending, Running, Succeeded, Failed, Cancelled.

## 2.22 Platform Reports

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: System and usage reports and sales and revenue reports.
- Mapped use case: SA11, PS04
- Actors: SA, PAO, PS
- Entities: orders, invoices, tenants, subscriptions
- Business rules and messages: none specific

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|
| From | Date filter | | |
| To | Date filter | | |

- Chart area and result table (placeholders)
- Actions: Run report

**Content #4: Permission and State**

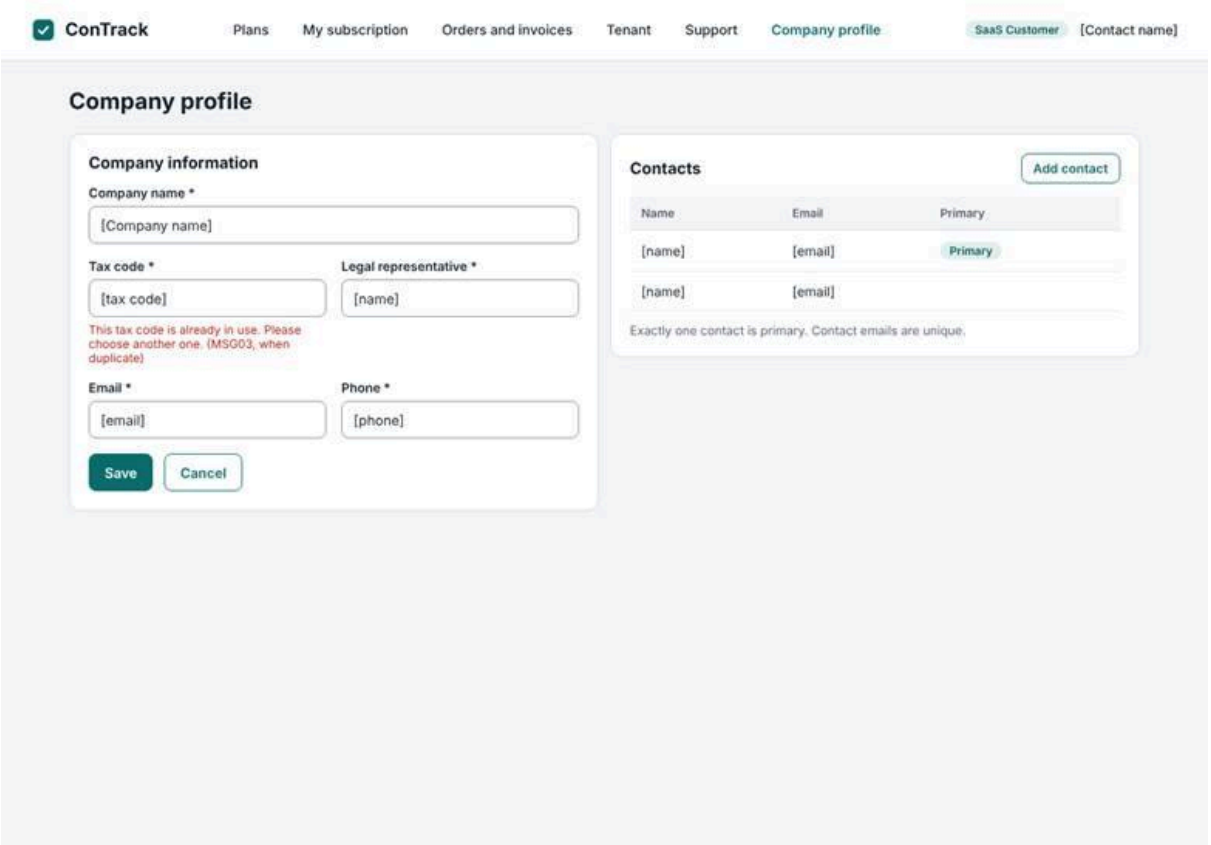
- Permission: System & usage report: SA only. Sales & revenue report: SA, PAO Full; PS Restricted.
- Roles shown by the mockup tweak: Super Admin (SA), Product Admin Officer (PAO), Product Sales (PS)

- State behavior: Report tabs depend on the role.

# 3. SaaS Customer Portal

## 3.1 Company Profile (SaaS Customer)

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: Lets the SaaS customer view and update company information and contacts.
- Mapped use case: SC01
- Actors: SaaS Customer
- Entities: customers, customer_contacts
- Business rules and messages: BR-01; MSG01, MSG03

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|

| | | | |
|---|---|---|---|
| Company name | Text input | Yes | |
| Tax code | Text input | Yes | Unique: This tax code is already in use. Please choose another one. (MSG03) |
| Legal representative | Text input | Yes | |
| Email | Text input | Yes | |
| Phone | Text input | Yes | |

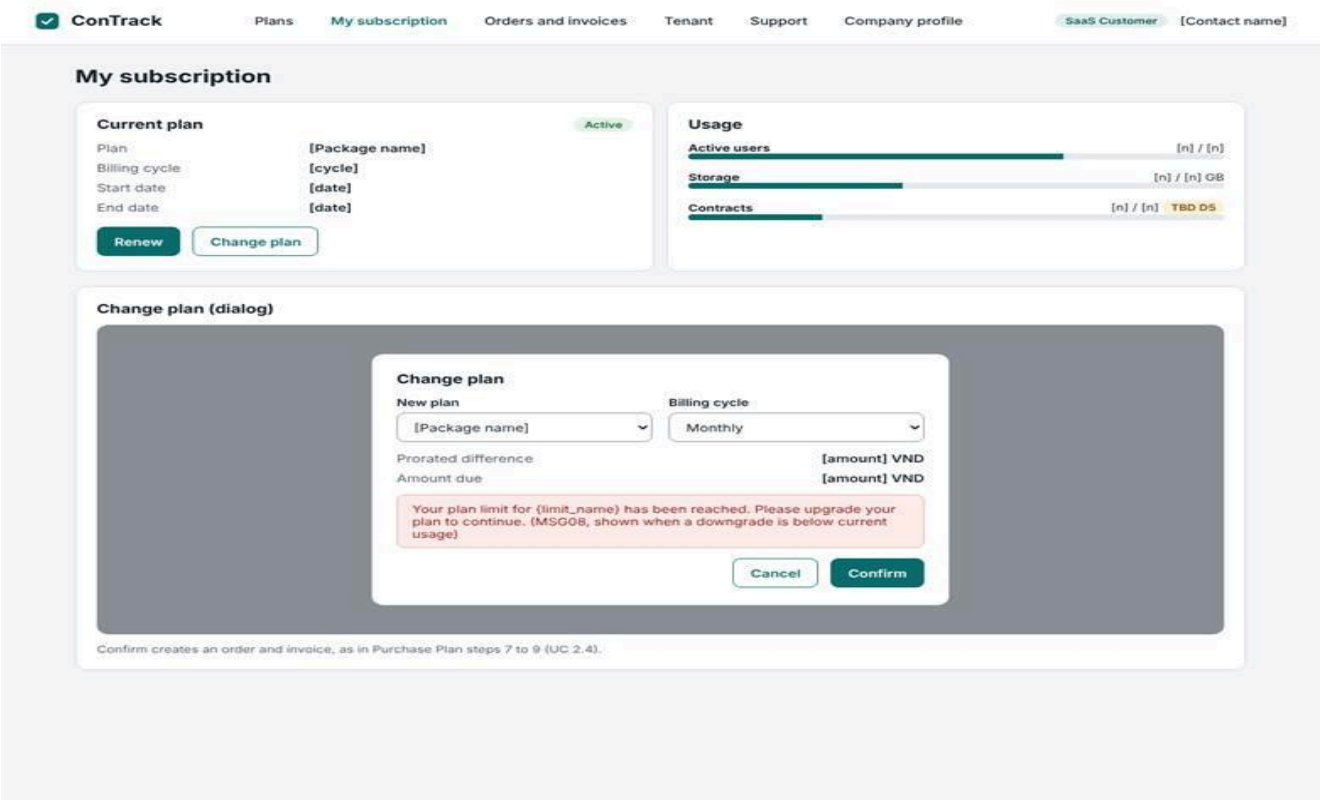
- Contacts list columns: Name, Email, Primary
- Actions: Save, Cancel, Add contact

**Content #4: Permission and State**

- Permission: SaaS customer Create/Edit: Restricted (own company only).

## 3.2 My Subscription

**Content #1: UI Layout (Wireframe Mockup)**



**Content #2: Brief Description and Use Case Mapping**

- Description: Shows the current plan, usage against limits and the renew or upgrade request.
- Mapped use case: UC 2.4; SC03–SC05
- Actors: SaaS Customer
- Entities: subscriptions, subscription_history, packages
- Business rules and messages: BR-06; MSG08

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|
| New plan | Dropdown | | Options: available packages |
| Billing cycle | Dropdown | | Options: Monthly, Quarterly, Yearly |

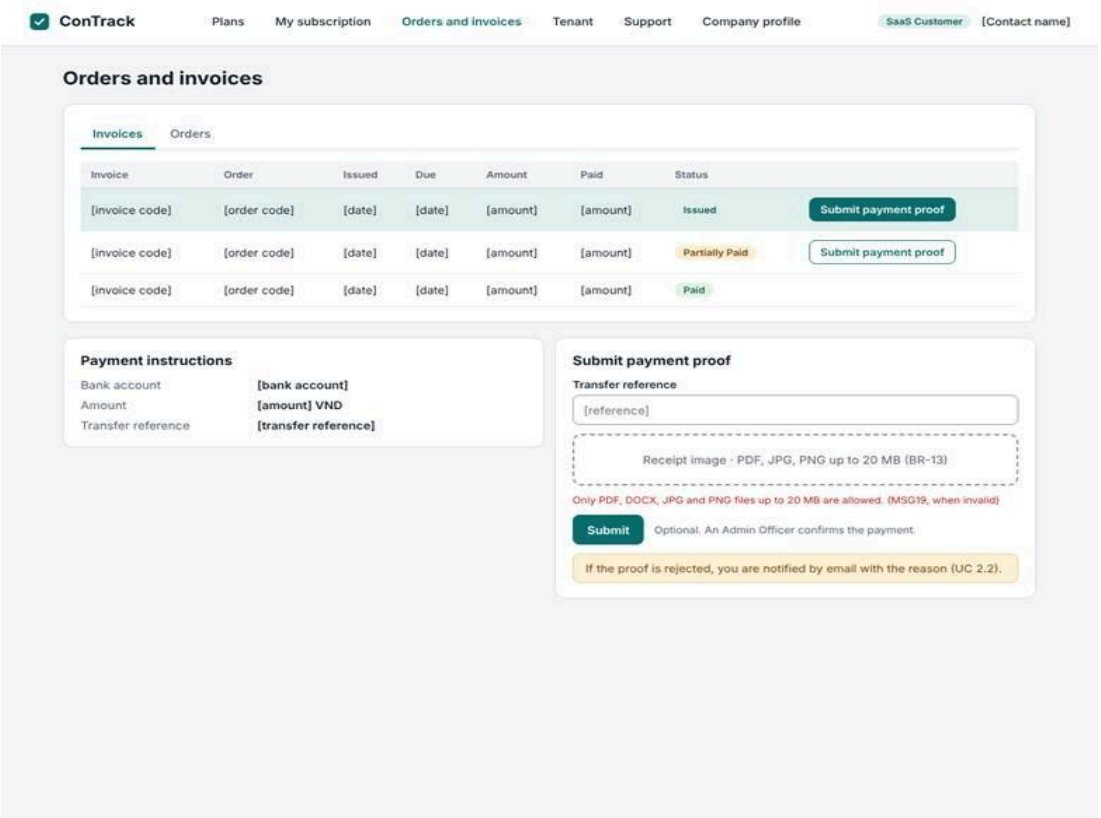
- Actions: Renew, Change plan, Cancel, Confirm
- Messages and notices:
  - Your subscription has expired. Renew to keep using your workspace. After the grace period the workspace is suspended.
  - Your order is waiting for payment. The subscription starts once the invoice is fully paid (BR-05).
  - Change plan dialog: new plan, billing cycle, prorated difference and amount due.
  - Your plan limit for {limit_name} has been reached. Please upgrade your plan to continue. (MSG08, shown when a downgrade is below current usage)

**Content #4: Permission and State**

- Permission: Subscription Purchase/Renew/Request upgrade: SC Restricted. View: SC Restricted (own).
- State behavior: The notices and actions depend on the subscription status.
- Mockup variants (subStatus): Pending, Active, Expired

## 3.3 Orders and Invoices

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: Lists the customer's orders and invoices and lets the customer submit payment proof.
- Mapped use case: UC 2.2; SC06, SC07
- Actors: SaaS Customer
- Entities: orders, invoices, payments
- Business rules and messages: BR-13, BR-17; MSG19

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|
| Transfer reference | Text input | | |
| Receipt image | File upload | | PDF, JPG, PNG up to 20 MB (BR-13). Invalid file shows MSG19 |

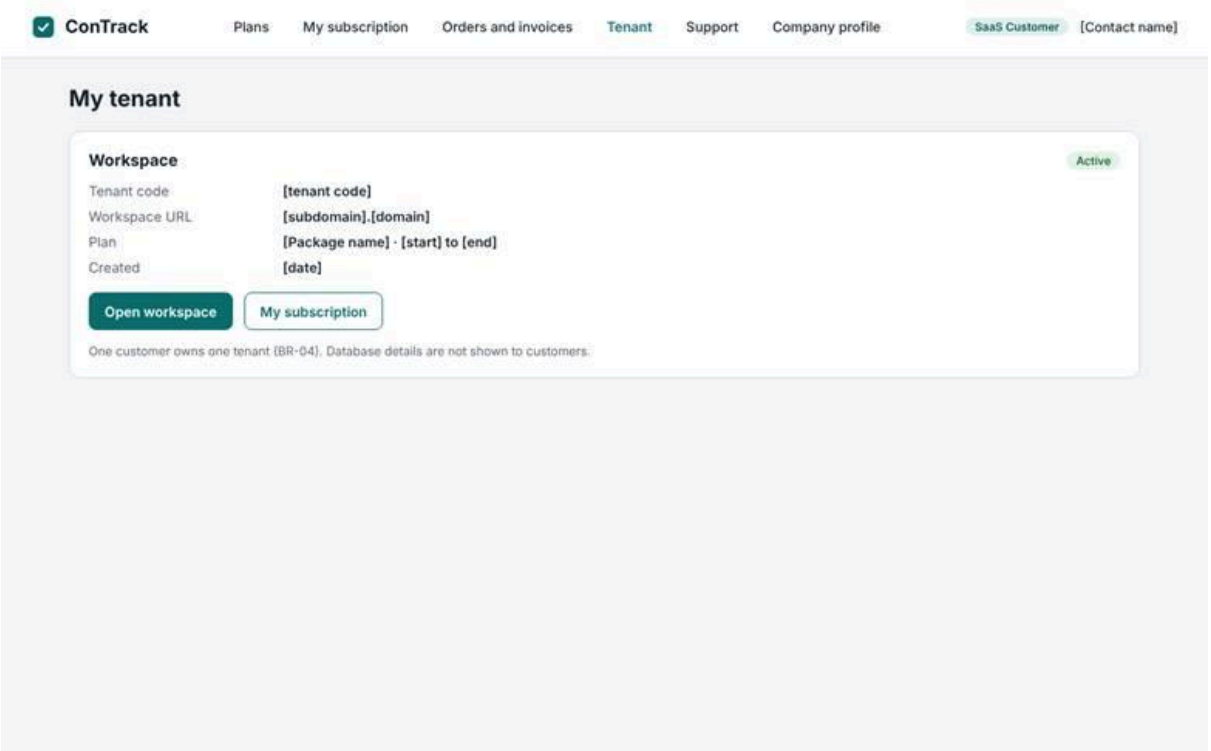
- Invoices list columns: Invoice, Order, Issued, Due, Amount, Paid, Status
- Actions: Submit payment proof, Submit
- Notice: If the proof is rejected, you are notified by email with the reason (UC 2.2).

**Content #4: Permission and State**

- Permission: Payment Submit proof: SC Restricted. Order and Invoice View: SC Restricted (own).

## 3.4 Tenant Information

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: Shows the customer's tenant: domain, status and access link.
- Mapped use case: SC08
- Actors: SaaS Customer
- Entities: tenants, tenant_domains
- Business rules and messages: BR-04, BR-18

**Content #3: Field and Component Descriptions**

- Actions: Open workspace, My subscription
- Messages and notices:
  - Your workspace is being prepared. You will receive an email with the workspace URL and an activation link when it is ready.
  - Your workspace is suspended. Renew your subscription to restore access.

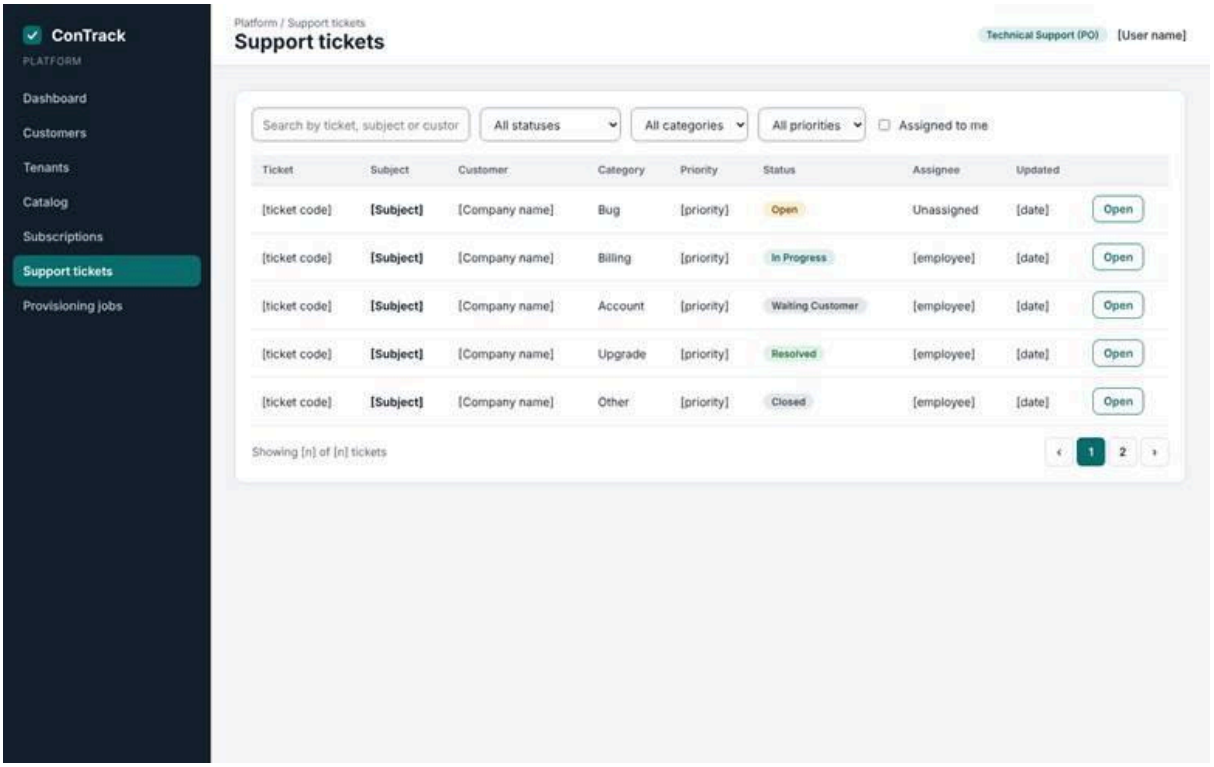
**Content #4: Permission and State**

- Permission: Tenant View: SC Restricted (own).
- State behavior: Notices and actions depend on tenant status.
- Mockup variants (tenantStatus): Provisioning, Active, Suspended, Terminated

# 4. Platform Support

## 4.1 Support Ticket List (Platform)

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: Queue of tickets submitted by SaaS customers, with assignment.
- Mapped use case: UC 3.1; PTS01, PTS02
- Actors: SA, PTS
- Entities: support_tickets, ticket_assignments
- Business rules and messages: BR-19

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|
| Assigned to me | Checkbox | | |
| Search | Search input | | By ticket, subject or customer |
| Status | Filter (dropdown) | | Options: All statuses, Open, In Progress, Waiting Customer, Resolved, Closed |
| Category | Filter (dropdown) | | Options: All categories, Bug, Account, Billing, Upgrade, Other |
| Priority | Filter (dropdown) | | Options: All priorities, then each priority |

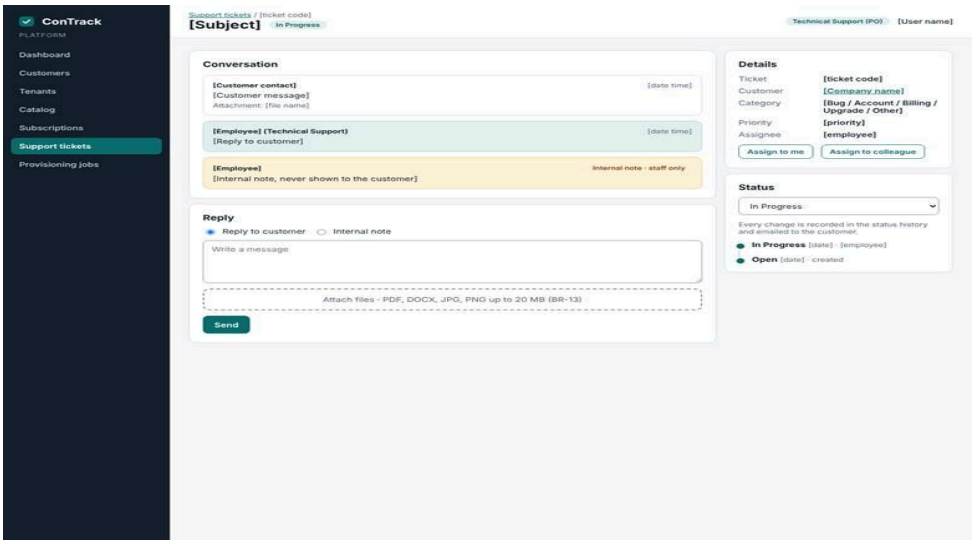
- List columns: Ticket, Subject, Customer, Category, Priority, Status, Assignee, Updated
- Actions: Open

**Content #4: Permission and State**

- Permission: Support ticket View/Assign: SA, PTS Full.
- Roles shown by the mockup tweak: Super Admin (SA), Product Technical Support (PTS)

## 4.2 Support Ticket Detail (Platform)

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: Ticket conversation, internal notes, assignment and status updates.
- Mapped use case: UC 3.1; PTS03–PTS05
- Actors: SA, PTS

- Entities: support_tickets, ticket_messages, ticket_attachments
- Business rules and messages: BR-13, BR-19; MSG16

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|
| Reply to customer | Radio | | Message visible to the customer |
| Internal note | Radio | | Never visible to the customer (BR-19) |
| Status | Dropdown | | Options: In Progress, Waiting Customer, Resolved, Closed |
| Attachments | File upload | | PDF, DOCX, JPG, PNG up to 20 MB (BR-13) |

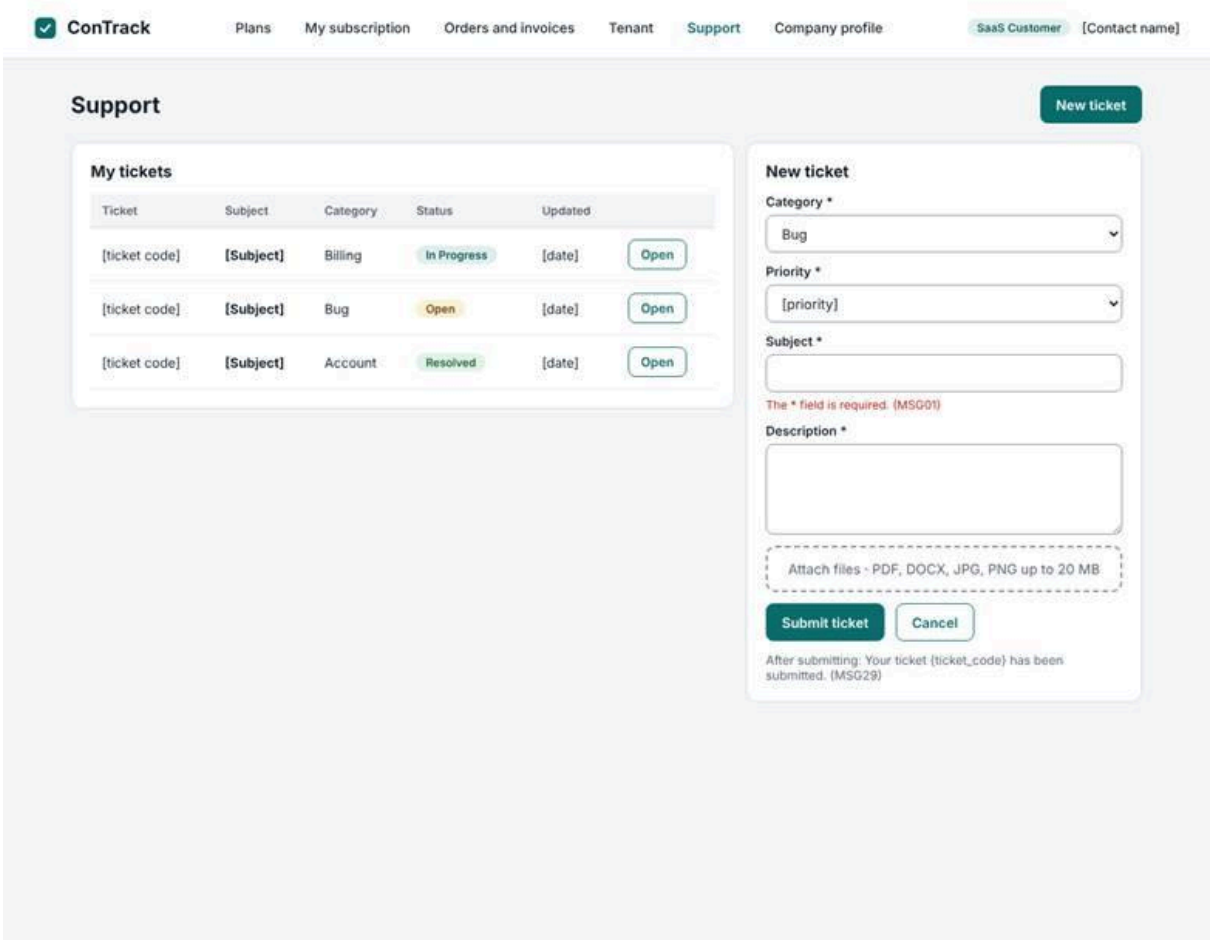
- Actions: Send, Assign to me, Assign to colleague

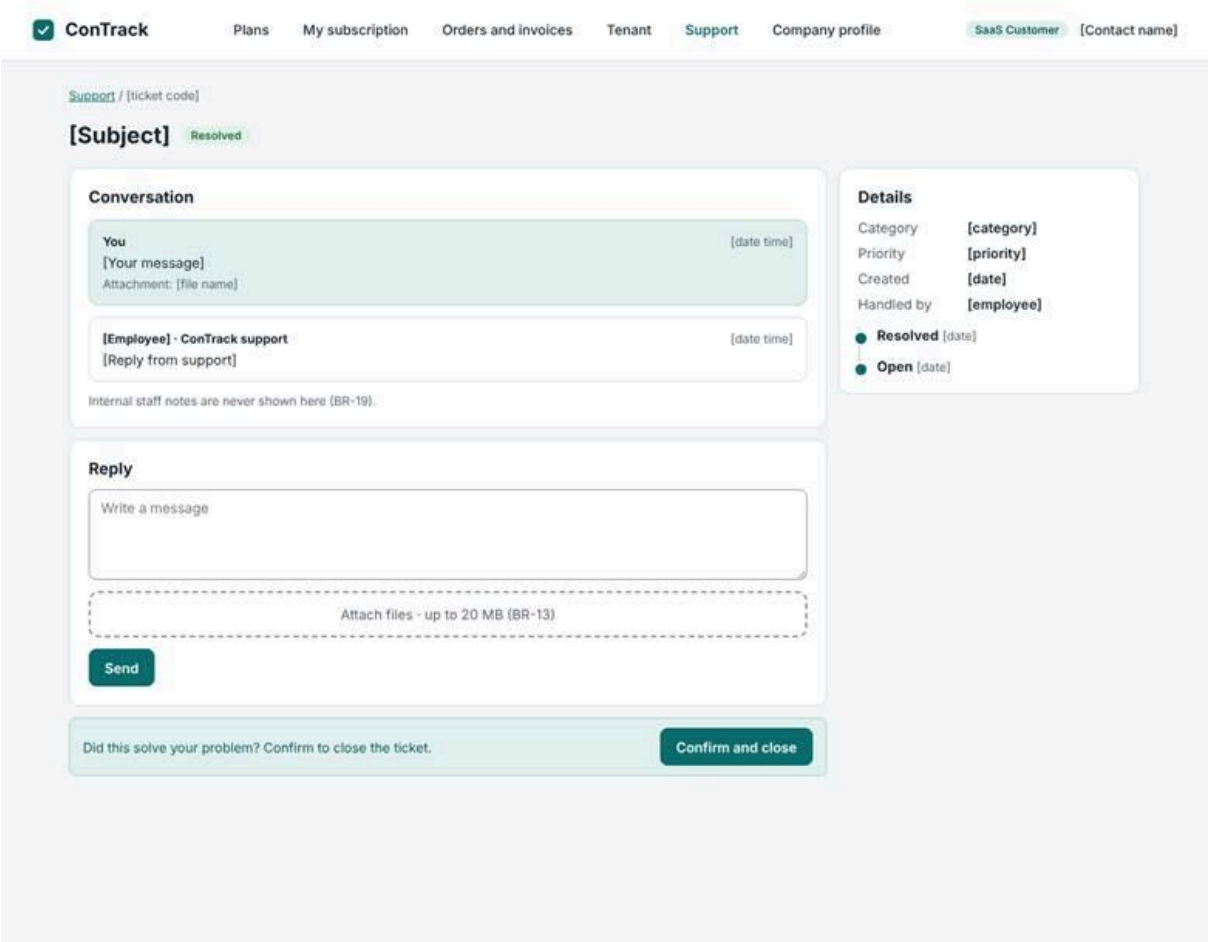
**Content #4: Permission and State**

- Permission: Reply/Update status: SA Full; PTS Restricted. Internal note: SA, PTS.
- Roles shown by the mockup tweak: Super Admin (SA), Product Technical Support (PTS)
- State behavior: Ticket status: Open, In Progress, Waiting Customer, Resolved, Closed.

## 4.3 Support Tickets (SaaS Customer)

**Content #1: UI Layout (Wireframe Mockup)**


ConTrack
Plans
My subscription
Orders and invoices
Tenant
Support
Company profile
SaaS Customer
[Contact name]
Support
New ticket
My tickets
Ticket
Subject
Category
Status
Updated
[ticket code]
[Subject]
Billing
In Progress
[date]
Open
[ticket code]
[Subject]
Bug
Open
[date]
Open
[ticket code]
[Subject]
Account
Resolved
[date]
Open
New ticket
Category *
Bug
Priority *
[priority]
Subject *
The * field is required. (MSG01)
Description *
Attach files - PDF, DOCX, JPG, PNG up to 20 MB
Submit ticket
Cancel
After submitting: Your ticket {ticket_code} has been submitted. (MSG29)




**Content #2: Brief Description and Use Case Mapping**

- Description: The customer creates a platform ticket and follows the replies.
- Mapped use case: UC 3.1; SC09, SC10
- Actors: SaaS Customer
- Entities: support_tickets, ticket_messages
- Business rules and messages: BR-13, BR-19; MSG01, MSG19, MSG29

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|
| Category | Dropdown | Yes | Options: Bug, Account, Billing, Upgrade, Other |
| Priority | Dropdown | Yes | |
| Subject | Text input | Yes | Error: The * field is required. (MSG01) |
| Description | Text area | Yes | |

| | | | |
|---|---|---|---|
| Attachments | File upload | | PDF, DOCX, JPG, PNG up to 20 MB (BR-13) |

- Tickets list columns: Ticket, Subject, Category, Status, Updated
- Actions: New ticket, Open, Submit ticket, Cancel, Send, Confirm and close
- Notice: Did this solve your problem? Confirm to close the ticket.

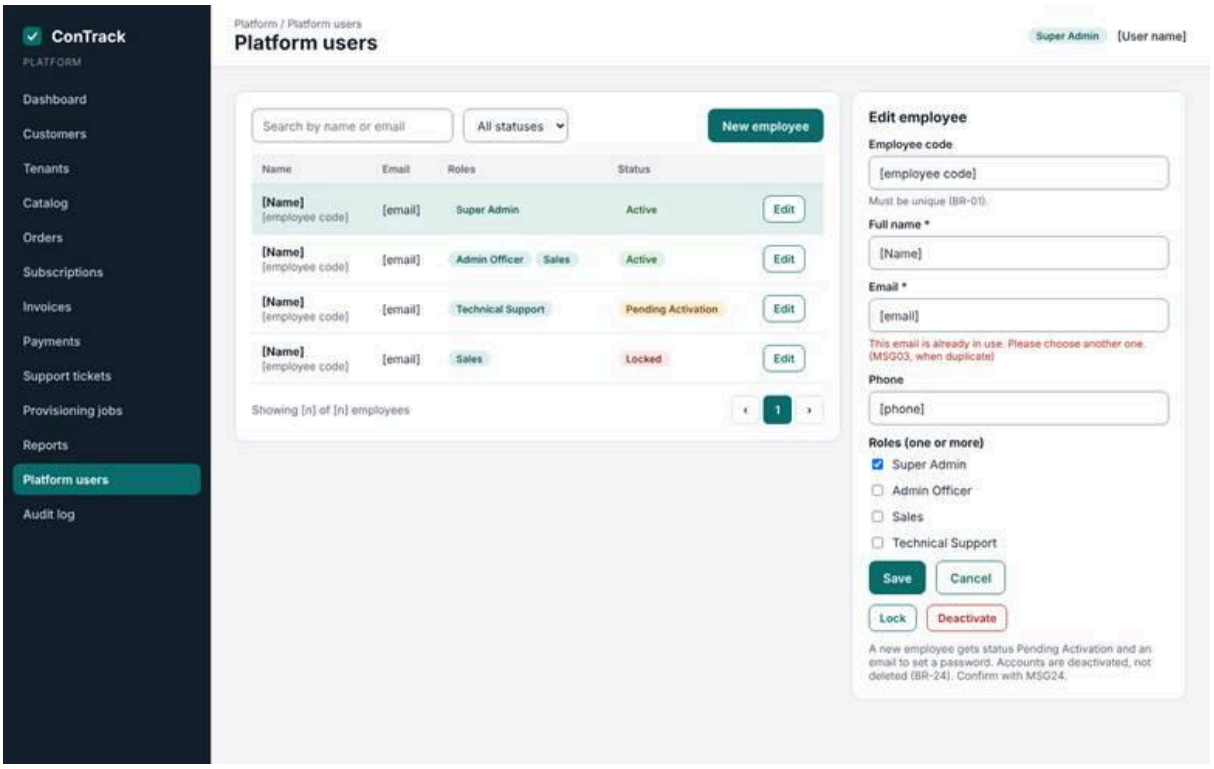
**Content #4: Permission and State**

- Permission: Ticket Create: SC Full. View and Reply: SC Restricted (own tickets). Internal notes are never shown (BR-19).
- State behavior: Resolved shows a confirm-close action; Closed locks replies.
- Mockup variants (ticketStatus): Open, In Progress, Waiting Customer, Resolved, Closed

# 5. Platform Administration

## 5.1 Platform User Management

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: Manages platform employees and assigns platform roles.

- Mapped use case: SA01, SA02
- Actors: SA
- Entities: employees, roles, employee_roles
- Business rules and messages: BR-01, BR-22; MSG03, MSG16, MSG24

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|
| Employee code | Text input | | Unique (BR-01) |
| Full name | Text input | Yes | |
| Email | Text input | Yes | Unique: This email is already in use. Please choose another one. (MSG03) |
| Phone | Text input | | |
| Roles | Checkbox group | | Super Admin, Admin Officer, Sales, Technical Support |
| Search | Search input | | By name, code or email |
| Status | Filter (dropdown) | | Options: All statuses, Active, Inactive, Locked |

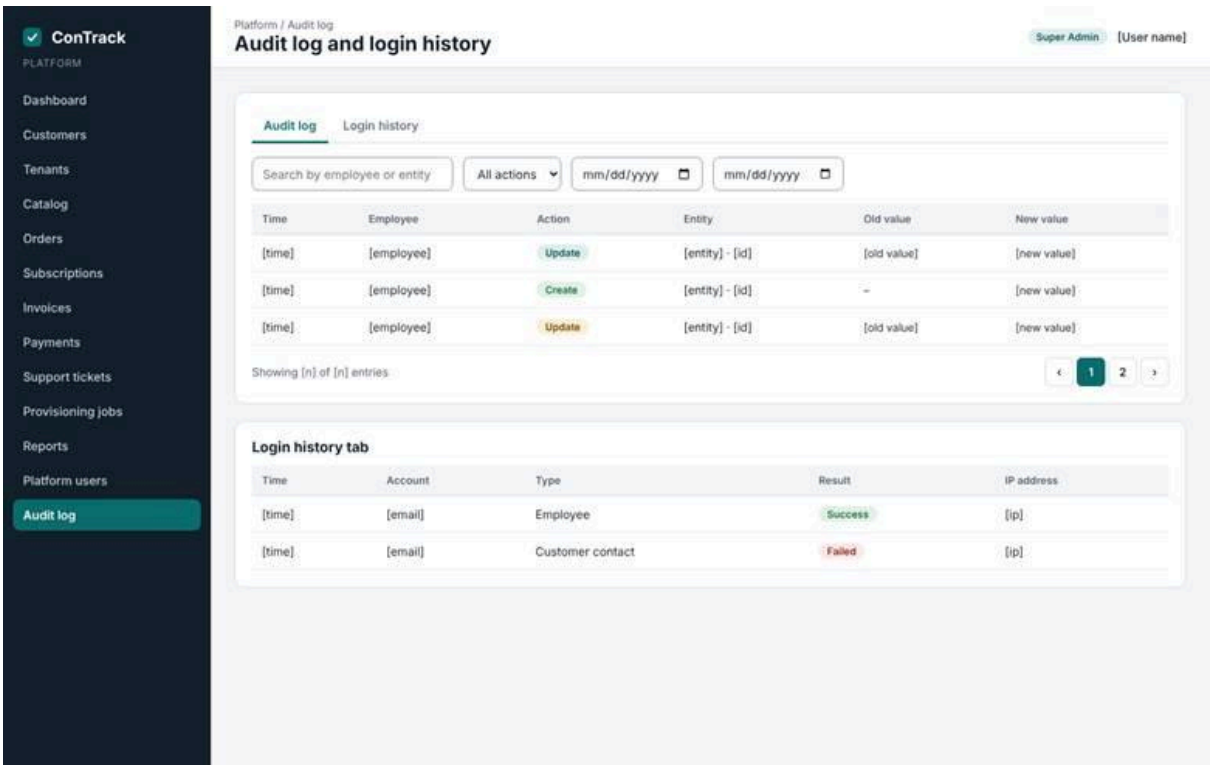
- List columns: Name, Email, Roles, Status
- Actions: New employee, Edit, Save, Cancel, Lock, Deactivate

**Content #4: Permission and State**

- Permission: Staff account Create/Edit/Lock/Deactivate/Assign roles: SA only.
- Roles shown by the mockup tweak: Super Admin (SA)

## 5.2 Platform Audit Log and Login History

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: Read-only trail of platform actions and sign-ins.
- Mapped use case: SA12
- Actors: SA
- Entities: audit_logs, login_histories
- Business rules and messages: BR-16

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|
| Search | Search input | | |
| Action | Filter (dropdown) | | Options: All actions, Create, Update, Delete |
| From | Date filter | | |
| To | Date filter | | |

- Audit list columns: Time, Employee, Action, Entity, Old value, New value
- Login history columns: Time, Account, Type, Result, IP address

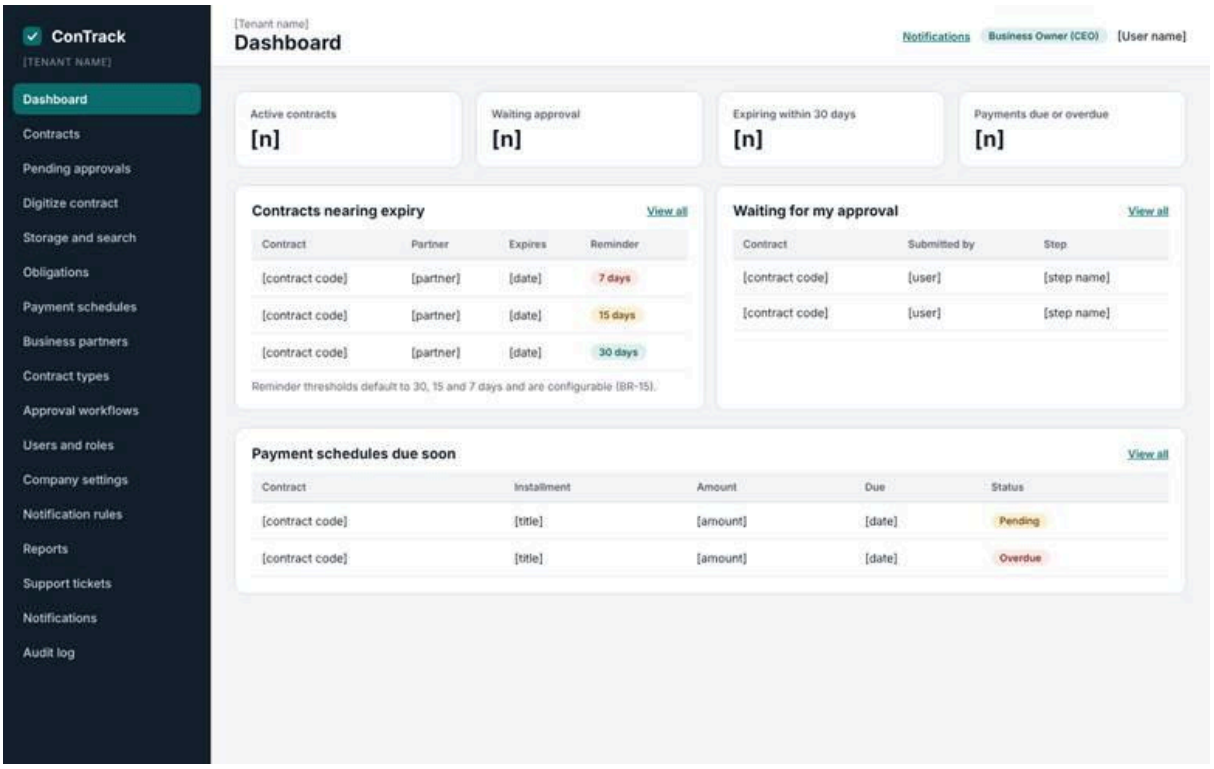
**Content #4: Permission and State**

- Permission: Audit log & login history View: SA only.
- Roles shown by the mockup tweak: Super Admin (SA)

# 6. TENANT ORGANIZATION AND USERS

## 6.1 Tenant Dashboard

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: Overview of contracts near expiry, approvals waiting for the user and upcoming payments.
- Mapped use case: CEO08
- Actors: CEO, BAO, BS
- Entities: contracts, contract_approval_instances, payment_schedules, notifications
- Business rules and messages: BR-15

**Content #3: Field and Component Descriptions**

- Summary tiles: Active contracts, Waiting approval, Expiring within 30 days, Payments due or overdue

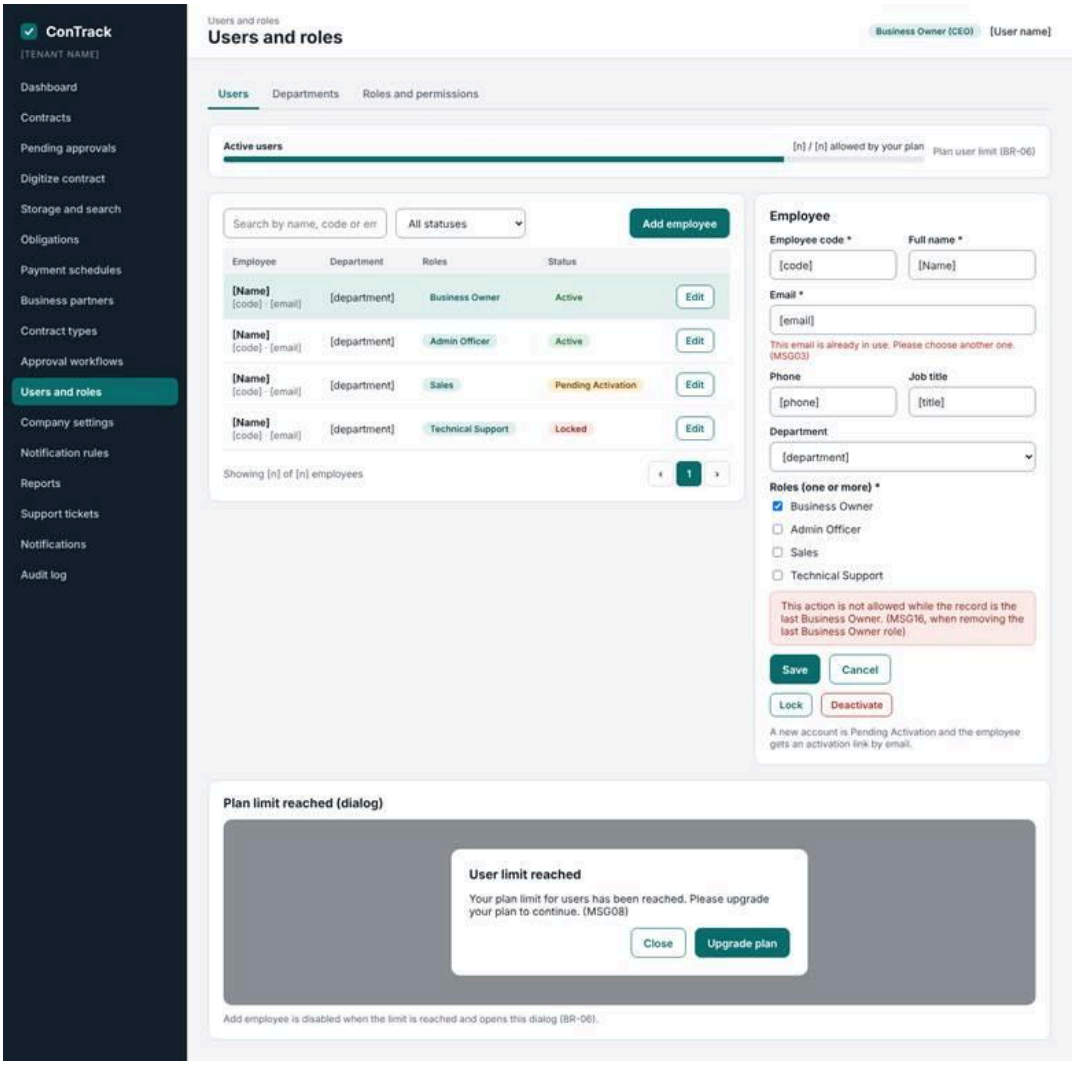
- Contracts nearing expiry list columns: Contract, Partner, Expires, Reminder
- Waiting for my approval list columns: Contract, Submitted by, Step
- Payment schedules due soon list columns: Contract, Installment, Amount, Due, Status
- Link: Notifications

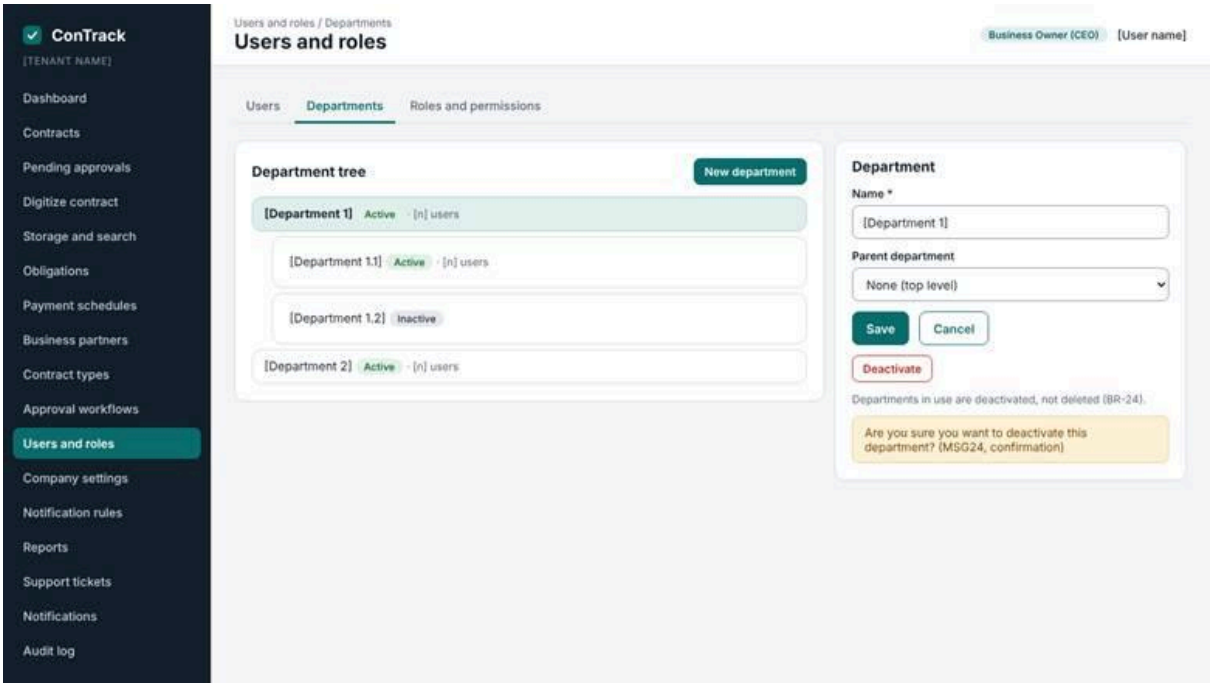
**Content #4: Permission and State**

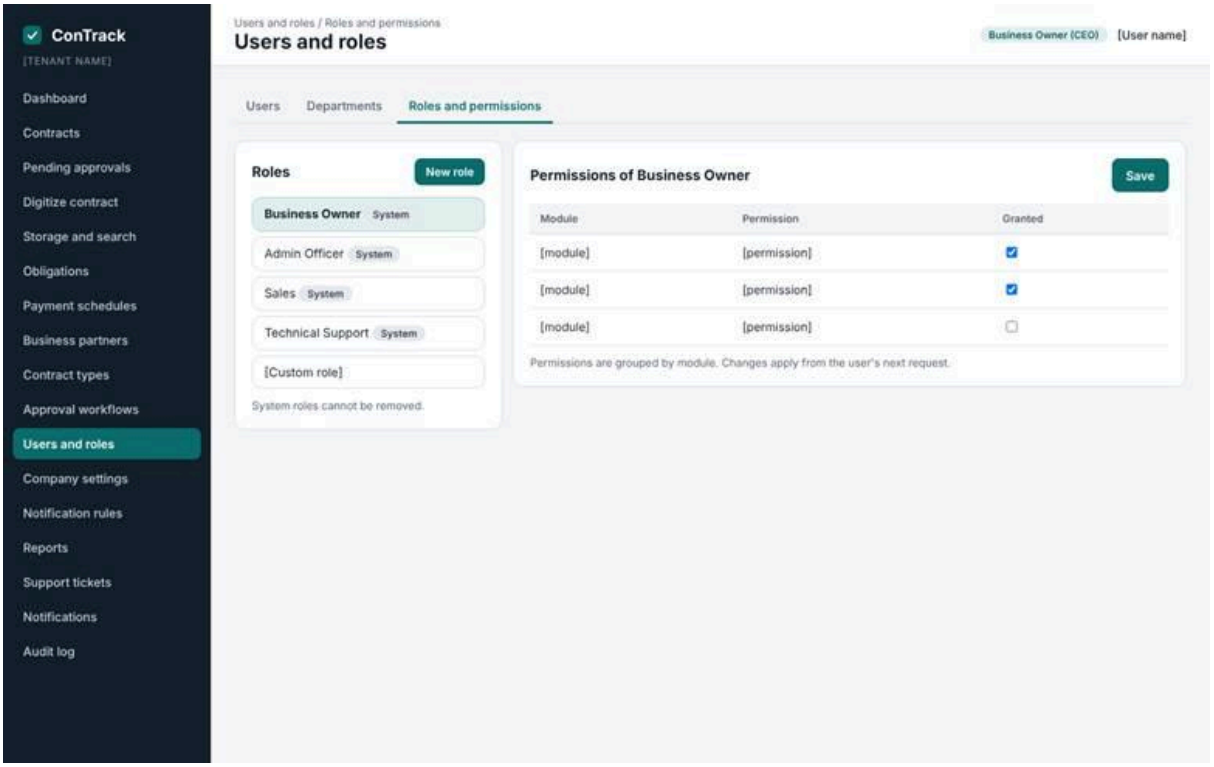
- Permission: Dashboard & report View: CEO Full; BAO, BS Restricted; BTS No.
- Roles shown by the mockup tweak: Business Owner (CEO), Admin Officer (BAO), Sales (BS)
- State behavior: Restricted roles see a notice that the visible widgets are limited.

## 6.2 User and Role Management

**Content #1: UI Layout (Wireframe Mockup)**






**Content #2: Brief Description and Use Case Mapping**

- Description: Three tabs: employees, department tree and roles with permissions.
- Mapped use case: UC 4.1; CEO02–CEO04
- Actors: CEO

- Entities: employees, departments, roles, permissions, user_roles
- Business rules and messages: BR-01, BR-06, BR-22, BR-24; MSG03, MSG08, MSG16, MSG24

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|
| Employee code | Text input | Yes | Unique (BR-01) |
| Full name | Text input | Yes | |
| Email | Text input | Yes | Unique: This email is already in use. Please choose another one. (MSG03) |
| Phone | Text input | | |
| Job title | Text input | | |
| Department | Dropdown | | Options: active departments |
| Roles | Checkbox group | Yes | Business Owner, Admin Officer, Sales, Technical Support (one or more) |
| Search | Search input | | By name, code or email |
| Status | Filter (dropdown) | | Options: All statuses, Active, Inactive, Locked, Pending Activation |
| Department name | Text input | Yes | |
| Parent department | Dropdown | | Options: None (top level), existing departments |

- Employees list columns: Employee, Department, Roles, Status
- Permissions list columns: Module, Permission, Granted
- Actions: Add employee, Edit, Save, Cancel, Lock, Deactivate, Upgrade plan, New department, New role
- Messages and notices:
  - This action is not allowed while the record is the last Business Owner. (MSG16, when removing the last Business Owner role)
  - Your plan limit for users has been reached. Please upgrade your plan to continue. (MSG08, dialog; Add employee is disabled at the limit)
  - Are you sure you want to deactivate this department? (MSG24, confirmation)

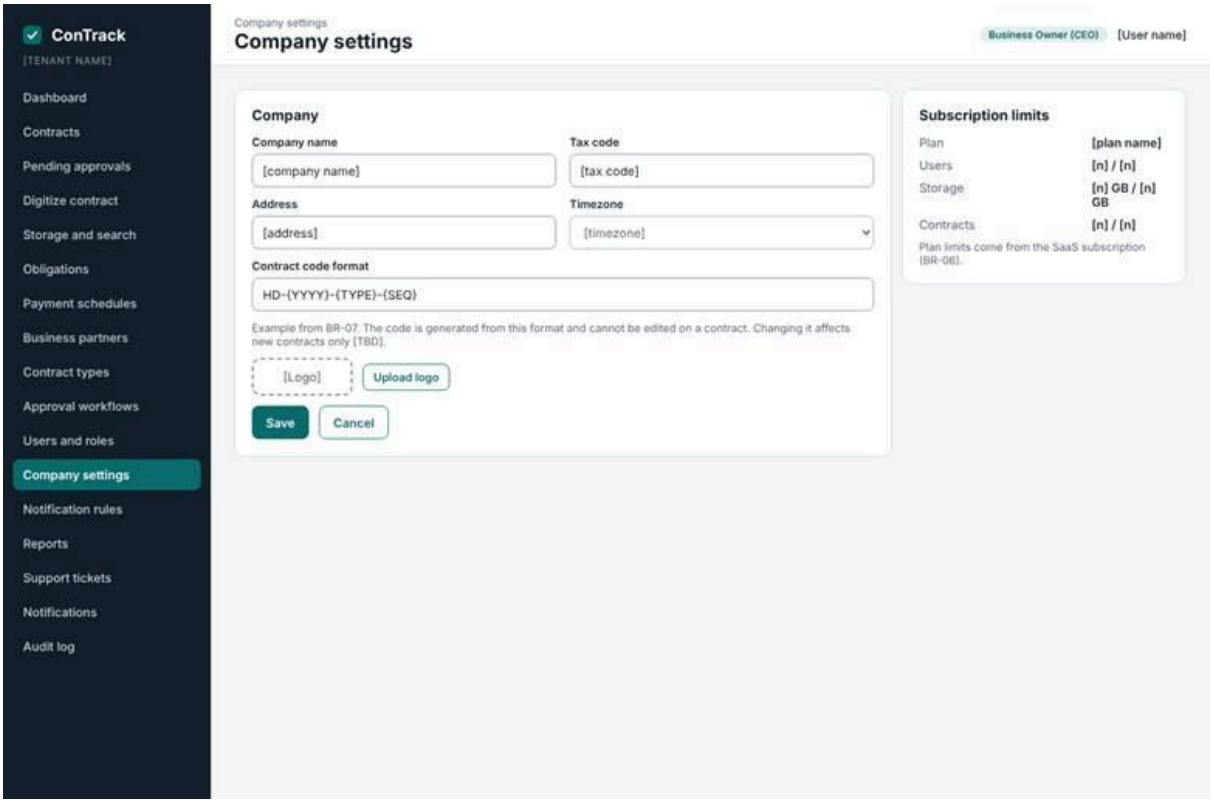
**Content #4: Permission and State**

- Permission: Department Create/Edit/Deactivate, Role & permission, Employee Create: CEO Full. Employee Edit/Lock/Deactivate: CEO Restricted. Other roles: No.
- Roles shown by the mockup tweak: Business Owner (CEO)

- State behavior: A new account is Pending Activation until the employee uses the activation link.

## 6.3 Company Settings

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: Company information, logo, timezone and the contract code format.
- Mapped use case: CEO01
- Actors: CEO (edit), BAO, BS, BTS (view)
- Entities: tenants (settings)
- Business rules and messages: BR-06, BR-07; MSG01, MSG05

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|
| Company name | Text input | | Editable by CEO only |
| Tax code | Text input | | Editable by CEO only |
| Address | Text input | | Editable by CEO only |

| | | | |
|---|---|---|---|
| Timezone | Dropdown | | Editable by CEO only |
| Contract code format | Text input | | Default HD-{YYYY}-{TYPE}-{SEQ} (BR-07).<br>Editable by CEO only |
| Logo | File upload | | Editable by CEO only |

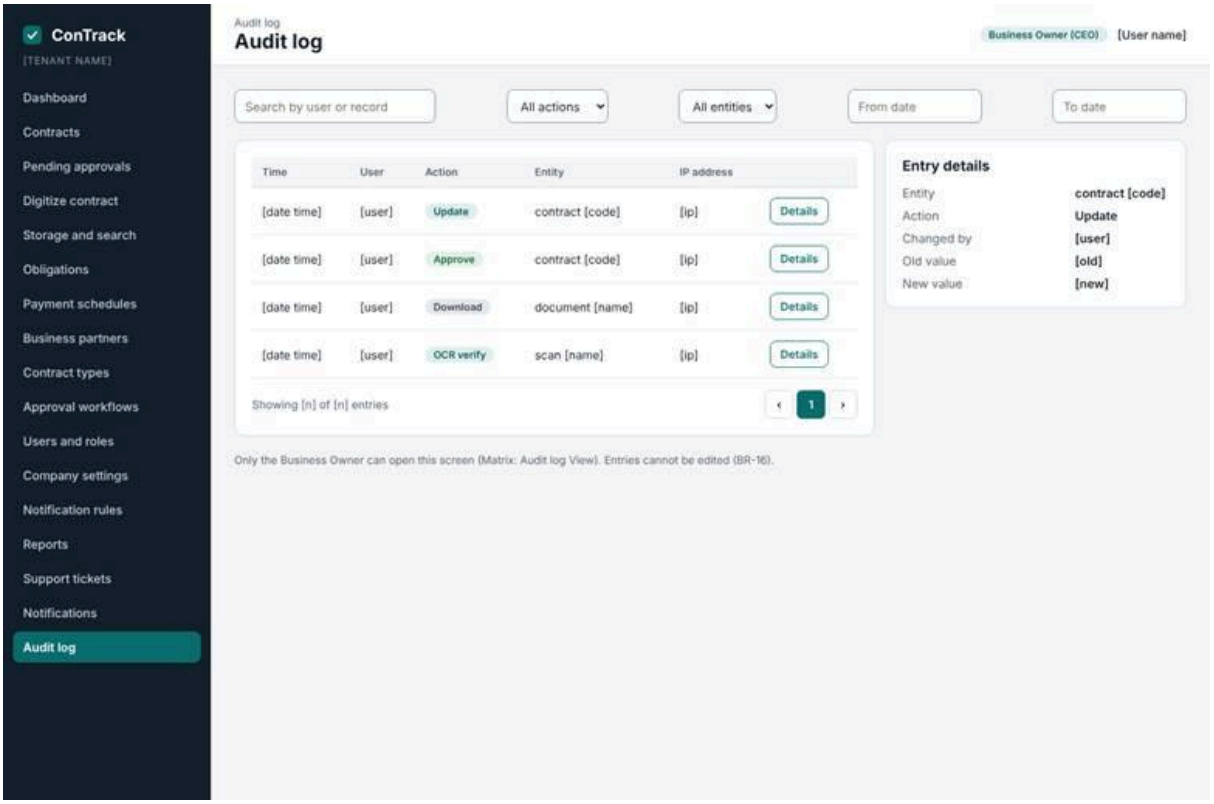
- Actions: Upload logo, Save, Cancel (CEO only)
- Subscription limits panel: plan, users, storage, contracts (BR-06)
- Notice: Read-only: only the Business Owner can edit company settings (Matrix).

**Content #4: Permission and State**

- Permission: View: CEO, BAO, BS, BTS. Edit: CEO only.
- Roles shown by the mockup tweak: Business Owner (CEO), Admin Officer (BAO), Sales (BS), Technical Support (BTS)
- State behavior: Fields are read-only for roles other than CEO.

## 6.4 Audit Log (Tenant)

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: Who created, changed, approved, viewed or downloaded contracts and files, with old and new values.
- Mapped use case: CEO10
- Actors: CEO
- Entities: audit_logs
- Business rules and messages: BR-16

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|
| Search | Search input | | By user or record |
| Action | Filter (dropdown) | | Create, Update, Delete, Approve, Sign, Download, OCR verify |
| From | Date filter | | |
| To | Date filter | | |

- List columns: Time, User, Action, Entity, IP address
- Actions: Details (old value, new value, changed by)

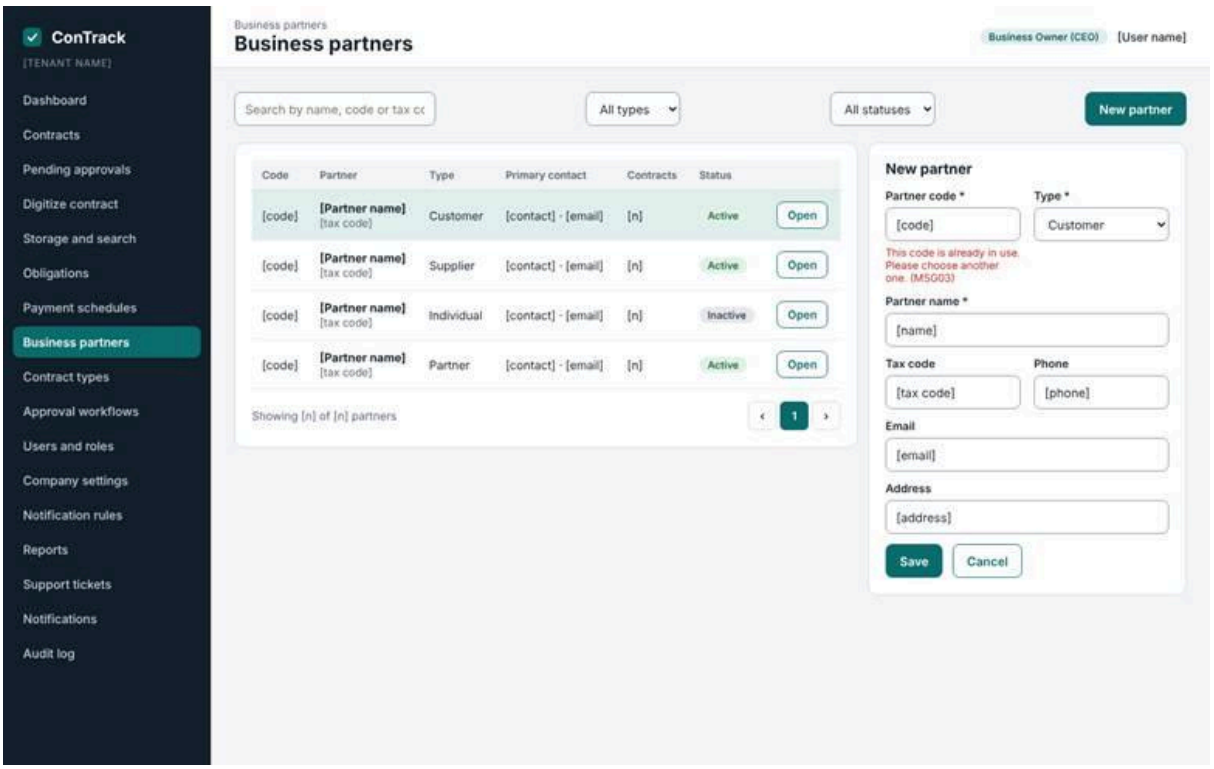
**Content #4: Permission and State**

- Permission: Audit log View: CEO only. Entries cannot be edited (BR-16).
- Roles shown by the mockup tweak: Business Owner (CEO)

# 7. PARTNER MANAGEMENT

## 7.1 Business Partner List

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: Customers, suppliers, partners and individuals managed by the tenant.
- Mapped use case: BS01
- Actors: CEO, BAO, BS, BTS
- Entities: business_partners
- Business rules and messages: BR-01, BR-24; MSG01–MSG03

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|
| Partner code | Text input | Yes | Unique: This code is already in use. Please choose another one. (MSG03) |
| Type | Dropdown | Yes | Options: Customer, Supplier, Partner, Individual |
| Partner name | Text input | Yes | |
| Tax code | Text input | | |
| Phone | Text input | | |
| Email | Text input | | |
| Address | Text input | | |

| | | | |
|---|---|---|---|
| Search | Search input | | By name, code or tax code |

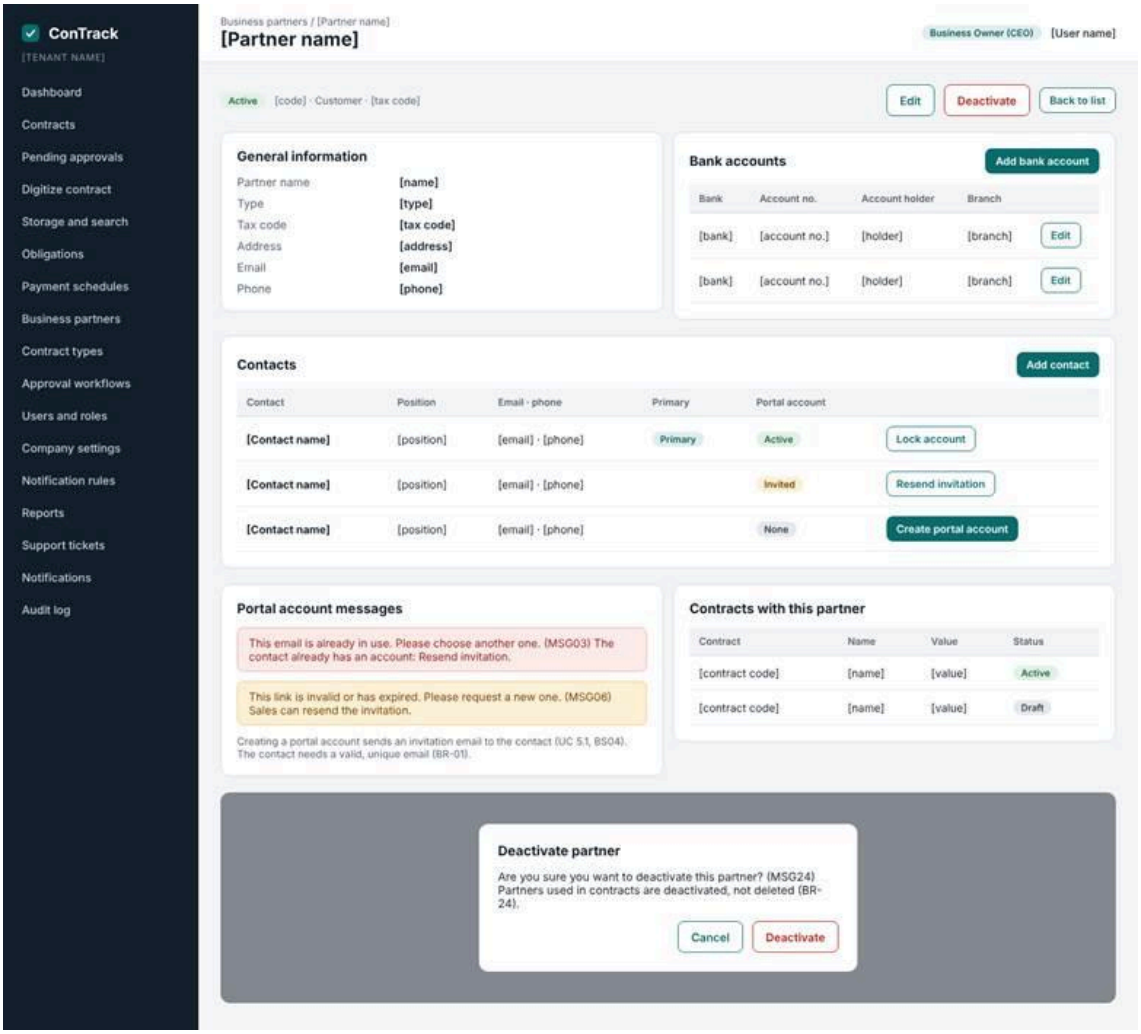
- List columns: Code, Partner, Type, Primary contact, Contracts, Status
- Actions: New partner, Open, Save, Cancel
- Notice: Read-only for this role: Technical Support and Admin Officer can view partners but not create or edit them (Matrix).

**Content #4: Permission and State**

- Permission: View: CEO, BAO, BS, BTS. Create/Edit: CEO, BS. Deactivate: CEO Full; BS Restricted.
- Roles shown by the mockup tweak: Business Owner (CEO), Sales (BS), Admin Officer (BAO), Technical Support (BTS)
- State behavior: New partner and the form are hidden for BAO and BTS.

## 7.2 Business Partner Detail

**Content #1: UI Layout (Wireframe Mockup)**



**Content #2: Brief Description and Use Case Mapping**

- Description: Partner information, contacts, bank accounts, portal accounts and related contracts.
- Mapped use case: BS01–BS04; UC 5.1
- Actors: CEO, BS (edit); BAO, BTS (view)
- Entities: business_partners, partner_contacts, partner_bank_accounts, customer_contacts
- Business rules and messages: BR-01, BR-24; MSG03, MSG06, MSG24

**Content #3: Field and Component Descriptions**

- Bank accounts list columns: Bank, Account no., Account holder, Branch
- Contacts list columns: Contact, Position, Email and phone, Primary, Portal account
- Contracts list columns: Contract, Name, Value, Status
- Actions: Edit, Deactivate, Back to list, Add bank account, Add contact, Lock account, Resend invitation, Create portal account
- Messages and notices:
  - This email is already in use. Please choose another one. (MSG03) The contact already has an account: Resend invitation.
  - This link is invalid or has expired. Please request a new one. (MSG06) Sales can resend the invitation.
  - Are you sure you want to deactivate this partner? (MSG24) Partners used in contracts are deactivated, not deleted (BR-24).

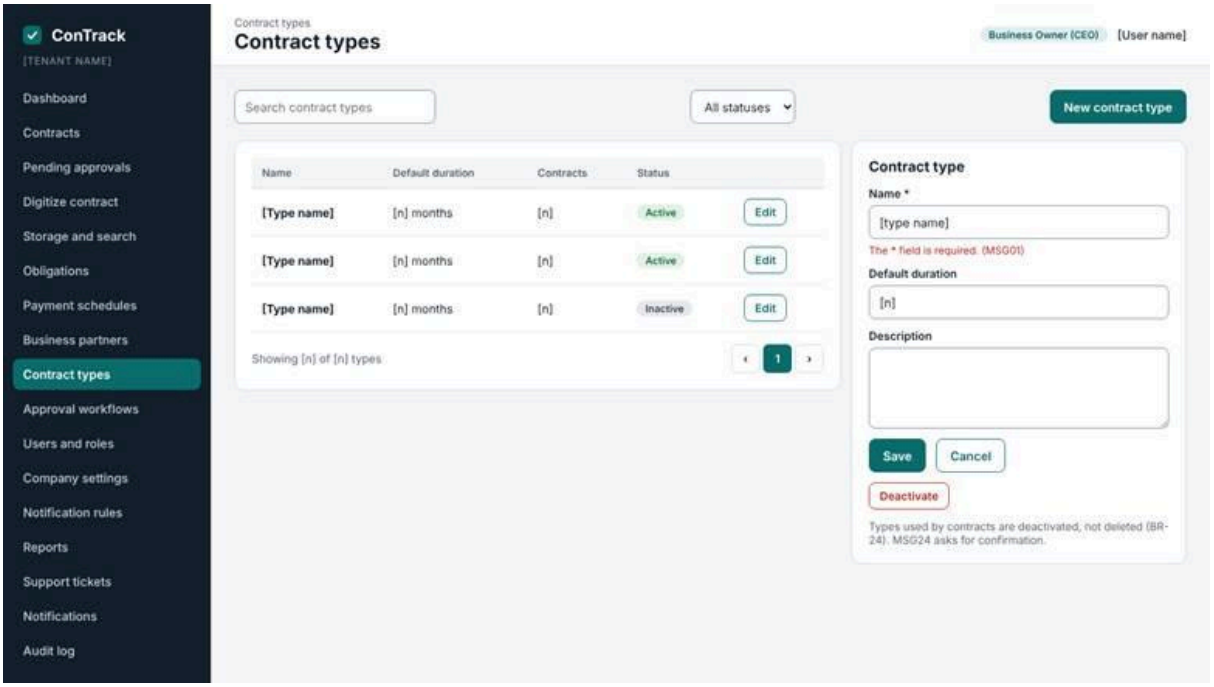
**Content #4: Permission and State**

- Permission: Partner contact and bank account Create/Edit, portal account Create/Lock: CEO, BS. Deactivate: BS Restricted.
- Roles shown by the mockup tweak: Business Owner (CEO), Sales (BS), Admin Officer (BAO), Technical Support (BTS)
- State behavior: Portal account per contact: None, Invited, Active.

# 8. CONTRACT MANAGEMENT AND APPROVAL

## 8.1 Contract Type Management

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: Contract types with default duration used to classify contracts.
- Mapped use case: BAS01
- Actors: BAO, CEO
- Entities: contract_types
- Business rules and messages: BR-07, BR-24; MSG01, MSG24

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|
| Name | Text input | Yes | Error: The * field is required. (MSG01) |
| Default duration | Text input | | |
| Description | Text area | | |
| Search | Search input | | |

- List columns: Name, Default duration, Contracts, Status
- Actions: New contract type, Edit, Save, Cancel, Deactivate

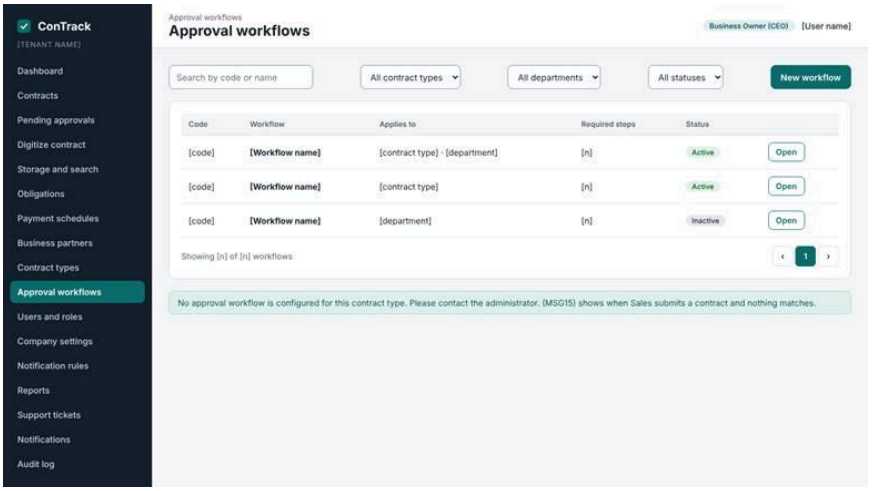
**Content #4: Permission and State**

- Permission: Contract type Create/Edit/Deactivate: CEO, BAO Full.

- Roles shown by the mockup tweak: Business Owner (CEO), Admin Officer (BAO)

## 8.2 Approval Workflow List

## Content #1: UI Layout (Wireframe Mockup)




**Content #2: Brief Description and Use Case Mapping**

- Description: Configured approval workflows by contract type and department.
- Mapped use case: CEO05; UC 4.2
- Actors: CEO
- Entities: approval_workflows
- Business rules and messages: BR-09; MSG15

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|
| Search | Search input | | |

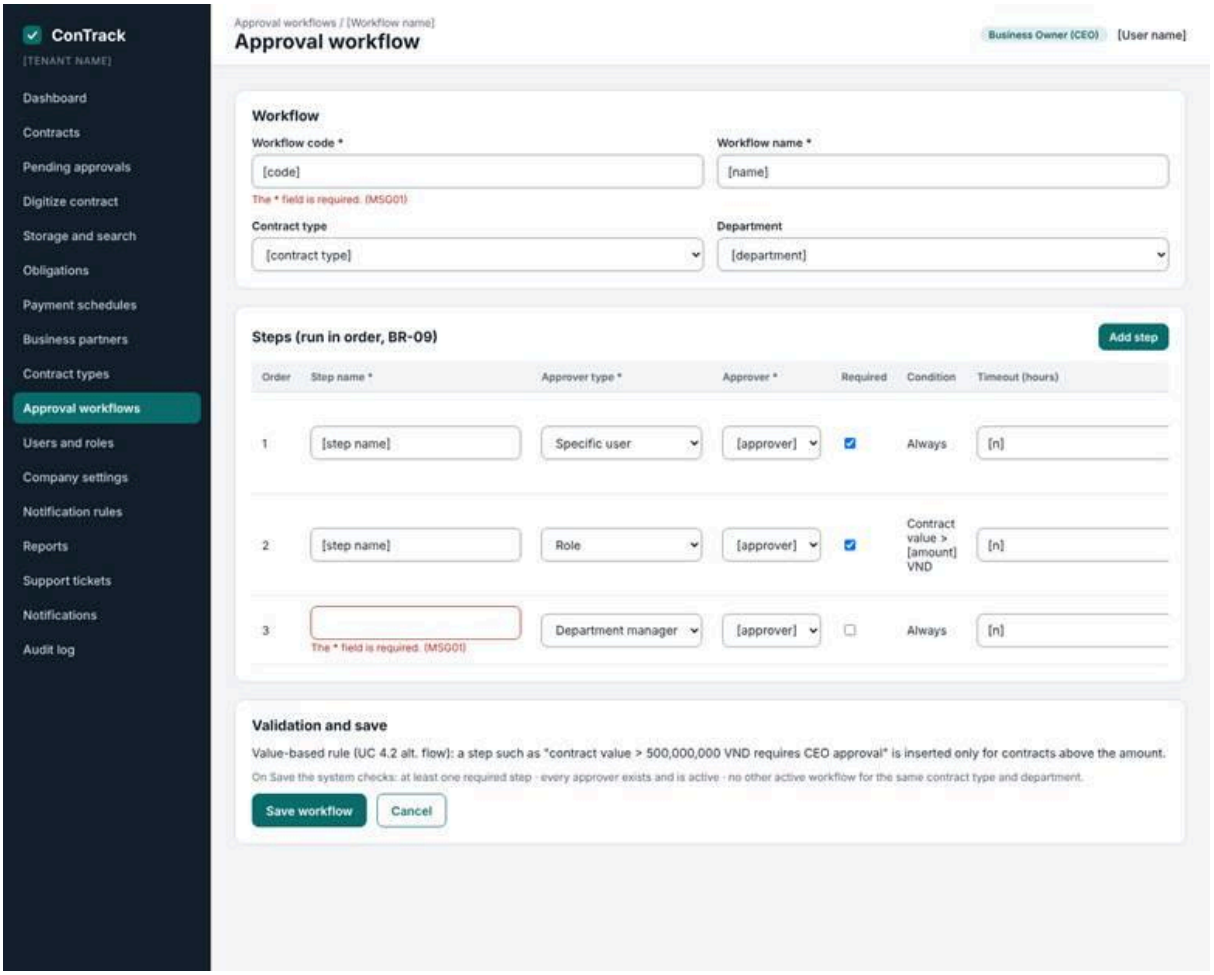
- List columns: Code, Workflow, Applies to, Required steps, Status
- Actions: New workflow, Open
- Messages and notices:
  - No approval workflow is configured for this contract type. Please contact the administrator. (MSG15) Shown when Sales submits a contract and no workflow matches.

**Content #4: Permission and State**

- Permission: Approval workflow Configure: CEO only.
- Roles shown by the mockup tweak: Business Owner (CEO)

## 8.3 Approval Workflow Detail

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: Defines ordered steps, approver type, required flag, value-based conditions and timeout.
- Mapped use case: CEO05; UC 4.2
- Actors: CEO
- Entities: approval_workflows, approval_workflow_steps
- Business rules and messages: BR-09; MSG01, MSG02, MSG05

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|
| Workflow code | Text input | Yes | Error: The * field is required. (MSG01) |
| Workflow name | Text input | Yes | |
| Contract type | Dropdown | | Options: a contract type, All types |
| Department | Dropdown | | Options: a department, All departments |

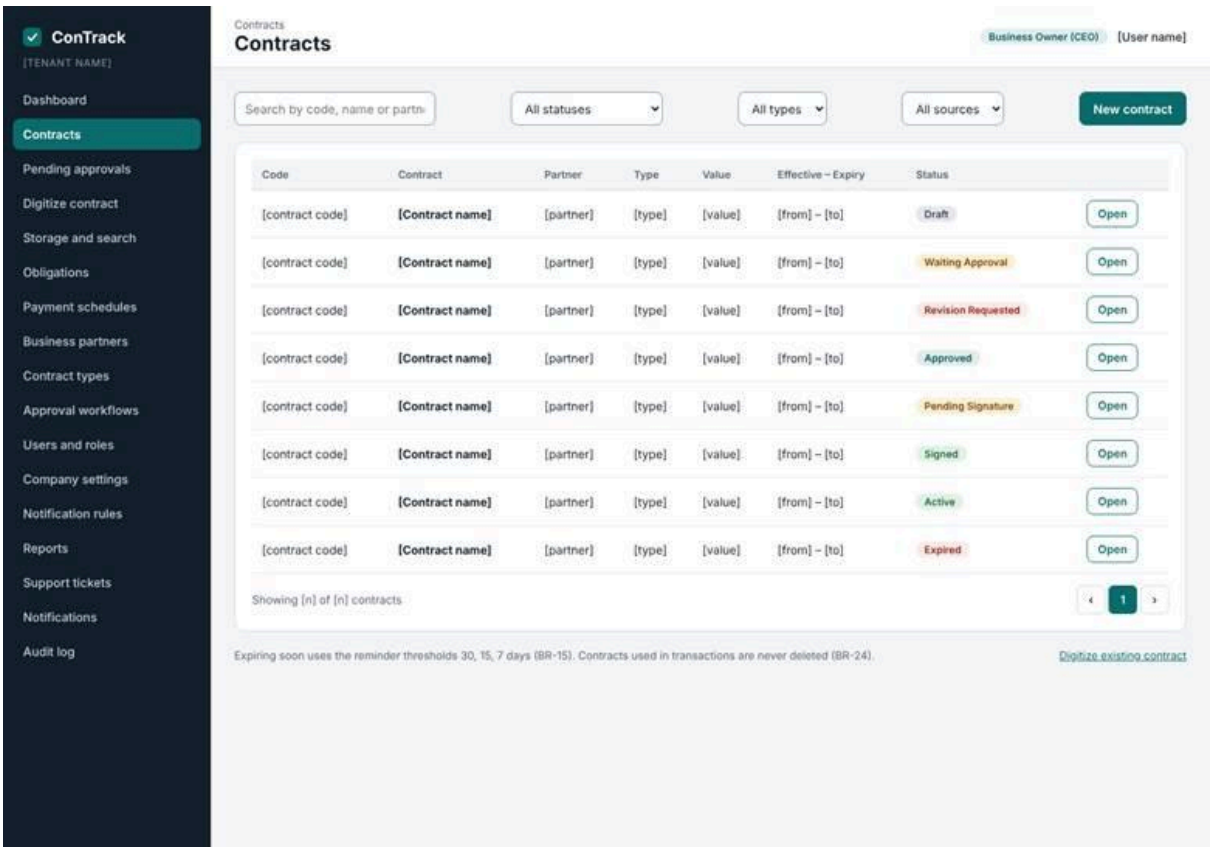
- Steps list columns: Order, Step name (required), Approver type (required), Approver (required), Required, Condition, Timeout (hours)
- Actions: Add step, Up, Down, Remove, Save workflow, Cancel

**Content #4: Permission and State**

- Permission: Approval workflow Configure: CEO only.
- Roles shown by the mockup tweak: Business Owner (CEO)

## 8.4 Contract List

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: Contracts of the tenant with search, filters and entry to contract operations.
- Mapped use case: BS14, BAS08
- Actors: CEO, BAO, BS, BTS
- Entities: contracts, business_partners, contract_types
- Business rules and messages: BR-15, BR-24

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|
| Search | Search input | | |

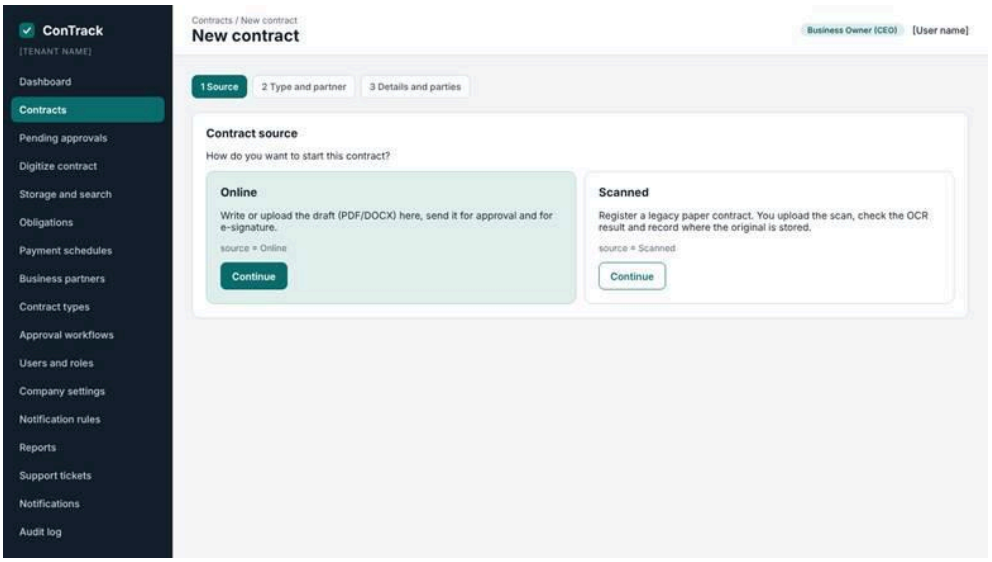
- List columns: Code, Contract, Partner, Type, Value, Effective – Expiry, Status
- Actions: New contract, Open

**Content #4: Permission and State**

- Permission: Contract View: CEO, BAO Full; BS, BTS Restricted. Create: CEO, BS.
- Roles shown by the mockup tweak: Business Owner (CEO), Sales (BS), Admin Officer (BAO), Technical Support (BTS)
- State behavior: New contract is hidden for BAO and BTS.

## 8.5 Create Contract

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: Step 1 of contract creation: choose Online or Scanned source.
- Mapped use case: BS05; UC 6.1 steps 1–2
- Actors: BS, CEO
- Entities: contracts
- Business rules and messages: BR-06; MSG08

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|
| Source | Radio | Yes | Options: Online, Scanned |

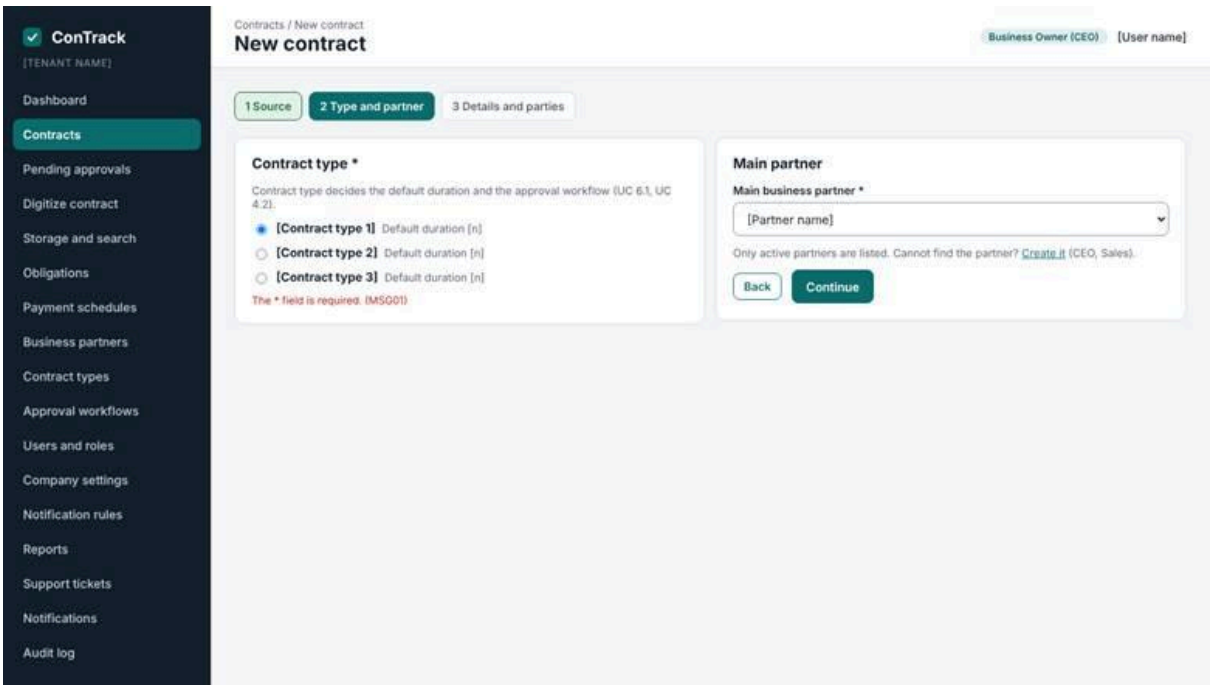
- Actions: Continue, Close, Upgrade plan
- Messages and notices:
  - Contract limit dialog: Your plan limit for contracts has been reached. Please upgrade your plan to continue. (MSG08) Buttons: Close, Upgrade plan. Continue is blocked when the limit is reached (BR-06).

**Content #4: Permission and State**

- Permission: Contract Create: CEO, BS Full.
- Roles shown by the mockup tweak: Business Owner (CEO), Sales (BS)

## 8.6 Select Contract Type

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: Step 2: choose the contract type and the main business partner.
- Mapped use case: BS05; UC 6.1 step 2
- Actors: BS, CEO
- Entities: contract_types, business_partners
- Business rules and messages: BR-24; MSG01

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|
| Contract type | Radio list | Yes | One option per active contract type, with its default duration |
| Main business partner | Dropdown | Yes | Options: active business partners |

- Actions: Back, Continue
- Messages and notices:
  - The * field is required. (MSG01)

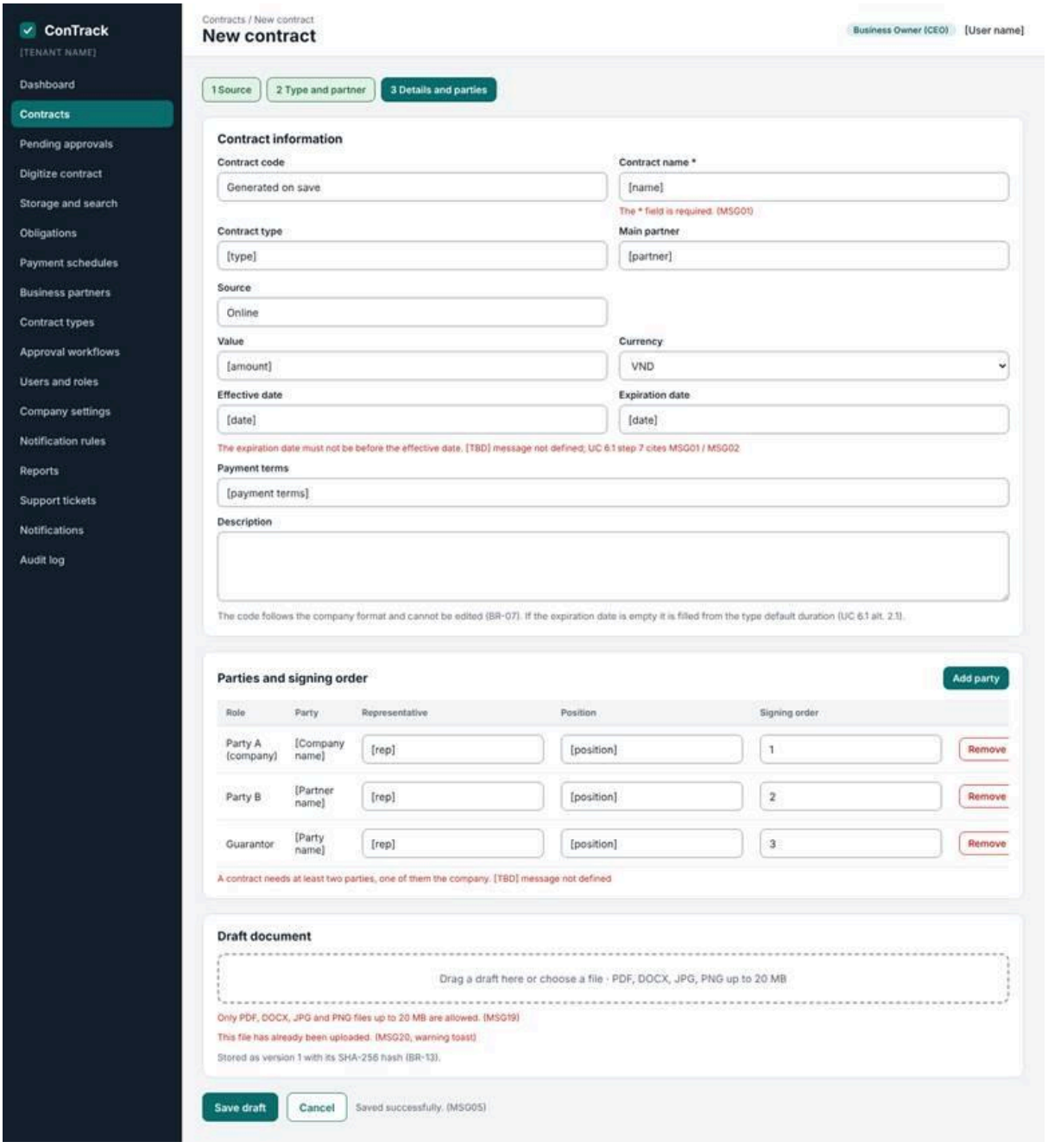
**Content #4: Permission and State**

- Permission: Contract Create: CEO, BS Full.

- Roles shown by the mockup tweak: Business Owner (CEO), Sales (BS)

## 8.7 Contract Form

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: Step 3: contract information, parties with signing order and the draft document.
- Mapped use case: BS05, BS06; UC 6.1 steps 3–10

- Actors: BS, CEO
- Entities: contracts, contract_parties, contract_documents
- Business rules and messages: BR-07, BR-12, BR-13, BR-16; MSG01, MSG02, MSG05, MSG19, MSG20

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|
| Contract code | Text (read-only) | | Generated on save |
| Contract name | Text input | Yes | Error: The * field is required. (MSG01) |
| Contract type | Text (read-only) | | Taken from the previous step |
| Main partner | Text (read-only) | | Taken from the previous step |
| Source | Text (read-only) | | Online |
| Value | Text input | | |
| Currency | Dropdown | | Options: VND, other currencies |
| Effective date | Date input | | |
| Expiration date | Date input | | |
| Payment terms | Text input | | |
| Description | Text area | | |
| Draft document | File upload | | Drop zone. PDF, DOCX, JPG, PNG up to 20 MB |

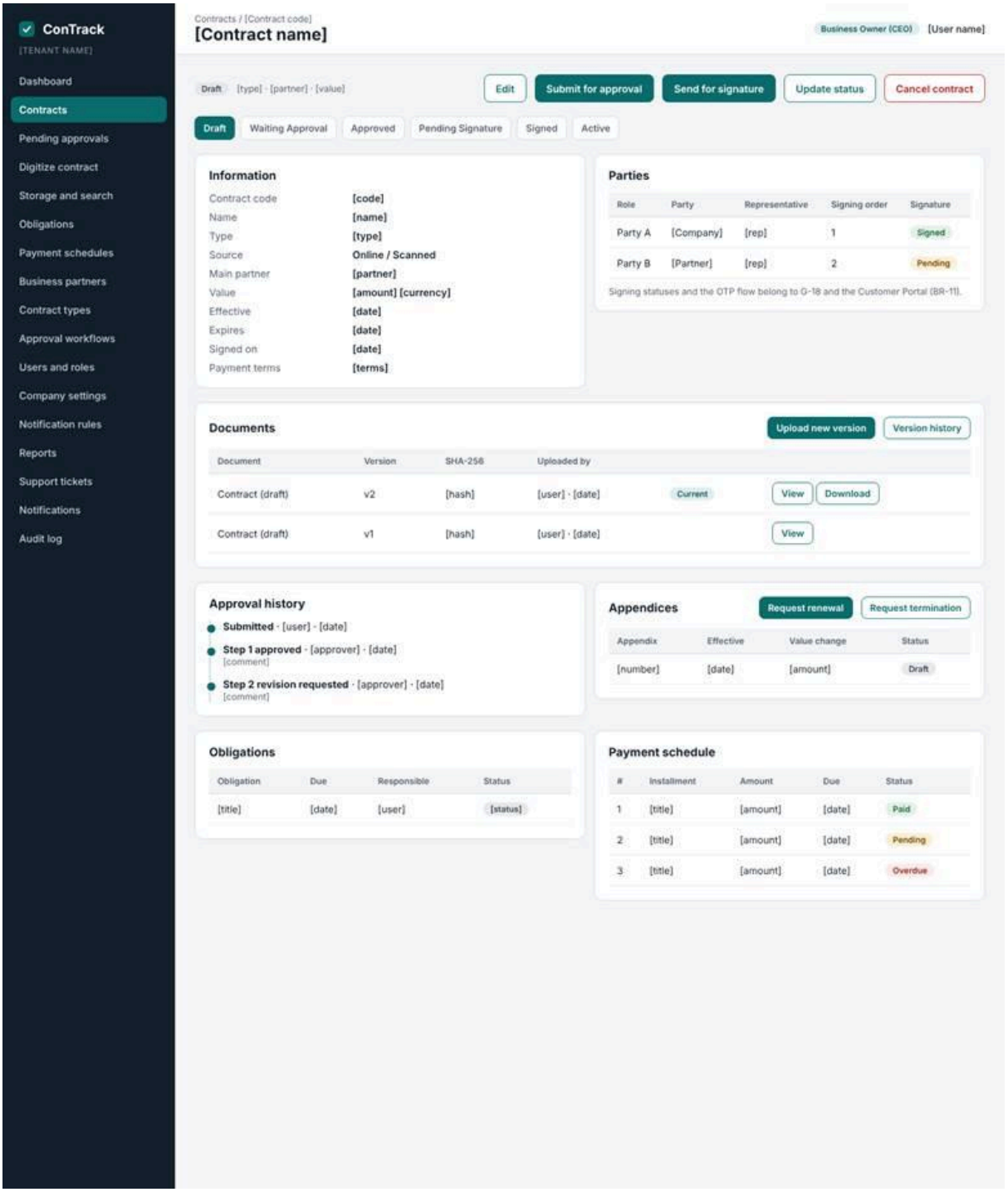
- Parties list columns: Role, Party, Representative, Position, Signing order
- Actions: Add party, Remove, Save draft, Cancel
- Messages and notices:
  - Only PDF, DOCX, JPG and PNG files up to 20 MB are allowed. (MSG19)
  - This file has already been uploaded. (MSG20, warning toast)
  - Source is Scanned: the document step is replaced by Upload Scan after the record is saved (UC 7.2).

**Content #4: Permission and State**

- Permission: Contract Create: CEO, BS Full. Edit: CEO Full; BS Restricted.
- Roles shown by the mockup tweak: Business Owner (CEO), Sales (BS)

## 8.8 Contract Detail

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: Contract information, parties, documents, approval history, appendices, obligations and payment schedule with status-driven actions.

- Mapped use case: BS05–BS08, BS14; UC 6.2
- Actors: CEO, BS, BAO, BTS
- Entities: contracts, contract_parties, contract_documents, approval history, appendices, payment_schedules
- Business rules and messages: BR-07, BR-08, BR-10, BR-12, BR-20; MSG16, MSG27

**Content #3: Field and Component Descriptions**

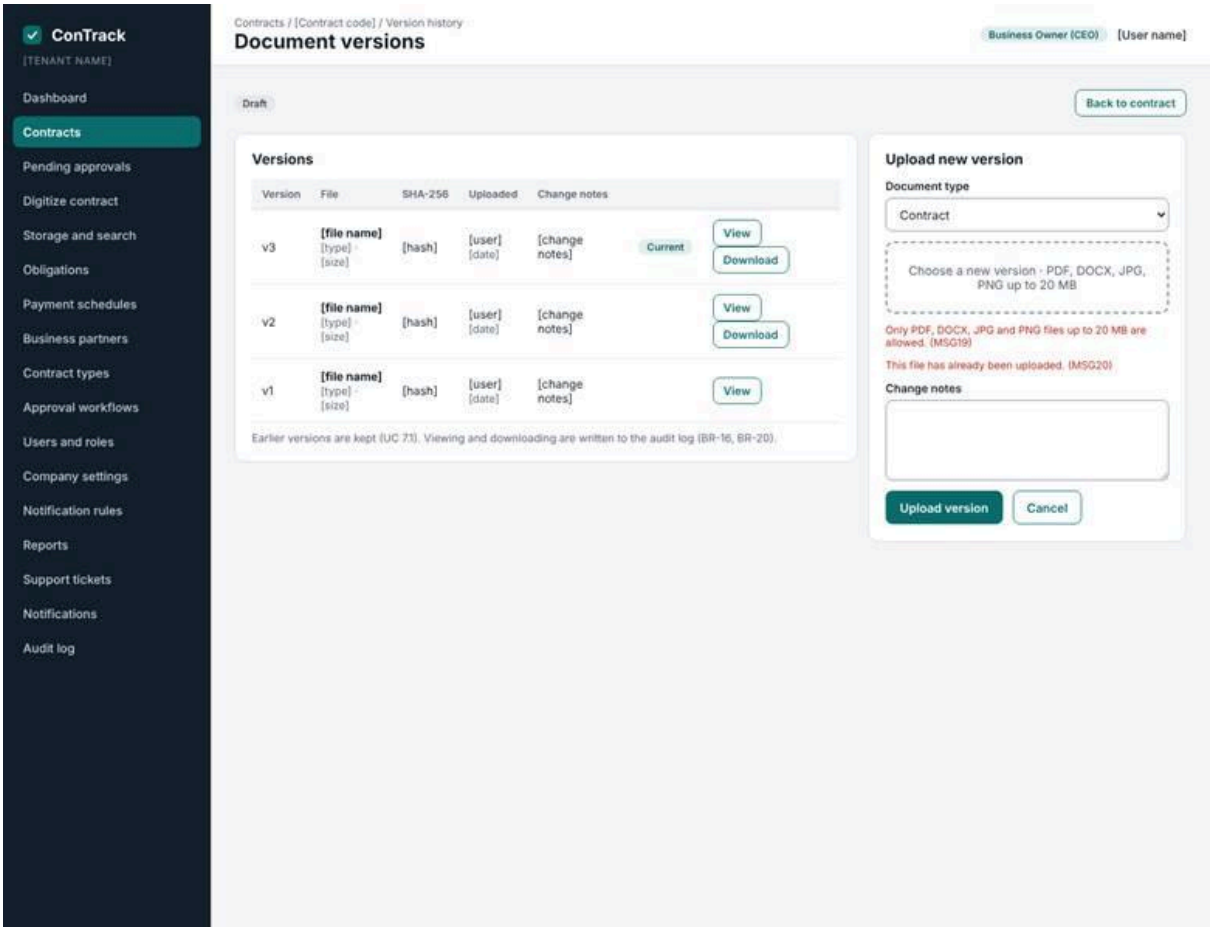
- Parties list columns: Role, Party, Representative, Signing order, Signature
- Documents list columns: Document, Version, SHA-256, Uploaded by
- Appendices list columns: Appendix, Effective, Value change, Status
- Obligations list columns: Obligation, Due, Responsible, Status
- Payment schedule list columns: #, Installment, Amount, Due, Status
- Actions: Edit, Submit for approval, Send for signature, Update status, Cancel contract, Upload new version, Version history, View, Download, Request renewal, Request termination
- Messages and notices:
  - This action is not allowed while the contract is [status]. (MSG16) Editing and submitting are possible only in Draft or Revision Requested (BR-08).
  - A signed contract cannot be edited or deleted. Any change must be made through an appendix. (BR-12)
  - Revision requested: [reviewer comment]. Correct the contract and submit again (BR-10).
  - Waiting for approval at step [n] of [n]: [approver]. The approver gets the contract in Pending approvals.

**Content #4: Permission and State**

- Permission: Contract View: CEO, BAO Full; BS, BTS Restricted. Edit/Submit: CEO Full; BS Restricted. Documents tab: hidden for BTS (Contract document View = No).
- Roles shown by the mockup tweak: Business Owner (CEO), Sales (BS), Admin Officer (BAO), Technical Support (BTS)
- State behavior: Edit and Submit are enabled only in Draft or Revision Requested (BR-08); Send for signature only in Approved; after Signed a notice explains that changes go through an appendix (BR-12).
- Mockup variants (status): Draft, Waiting Approval, Revision Requested, Approved, Pending Signature, Signed, Active, Expired, Terminated, Liquidated

## 8.9 Contract Version History

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: All versions of the contract documents with SHA-256 hash, and upload of a new version.
- Mapped use case: BAS02, BAS03; UC 7.1
- Actors: BAO, CEO, BS
- Entities: contract_documents
- Business rules and messages: BR-12, BR-13, BR-16, BR-20; MSG19, MSG20

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|
| Document type | Dropdown | | Options: Contract, Appendix, Acceptance Record, Other |
| Change notes | Text area | | |

| | | | |
|---|---|---|---|
| New version file | File upload | | Drop zone. PDF, DOCX, JPG, PNG up to 20 MB |

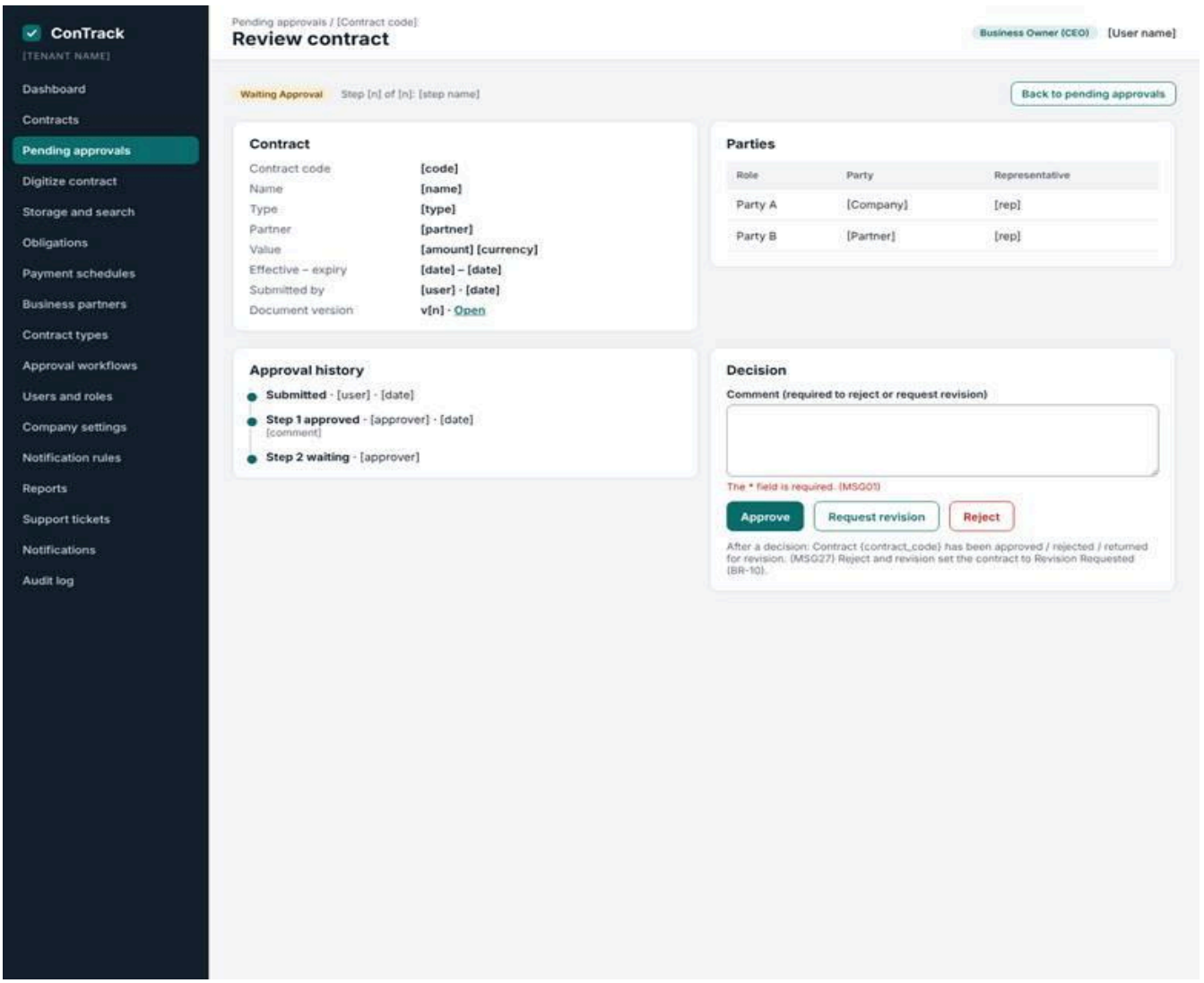
- Versions list columns: Version, File, SHA-256, Uploaded, Change notes
- Actions: Back to contract, View, Download, Upload version, Cancel
- Messages and notices:
  - Only PDF, DOCX, JPG and PNG files up to 20 MB are allowed. (MSG19)
  - This file has already been uploaded. (MSG20)
  - The contract is signed: only Appendix, Acceptance Record or Other documents can be added (BR-12, UC 7.1).

**Content #4: Permission and State**

- Permission: Contract document Upload/New version: CEO, BAO Full; BS Restricted. View/Download: CEO, BAO Full; BS Restricted.
- Roles shown by the mockup tweak: Business Owner (CEO), Admin Officer (BAO), Sales (BS)
- State behavior: On a signed contract only Appendix, Acceptance Record or Other documents can be added.
- Mockup variants (status): Draft, Waiting Approval, Revision Requested, Approved, Pending Signature, Signed, Active, Expired, Terminated, Liquidated

## 8.10 Approval Review

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: An approver reviews the submitted contract and approves, rejects or requests revision with a comment.
- Mapped use case: CEO06; UC 6.2
- Actors: Approver of the current step (default CEO)
- Entities: contract_approval_instances, contract_approval_history
- Business rules and messages: BR-08, BR-09, BR-10; MSG01, MSG27

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|

| | | | |
|---|---|---|---|
| Comment | Text area | Conditional | Required to reject or request revision. Error: The * field is required. (MSG01) |

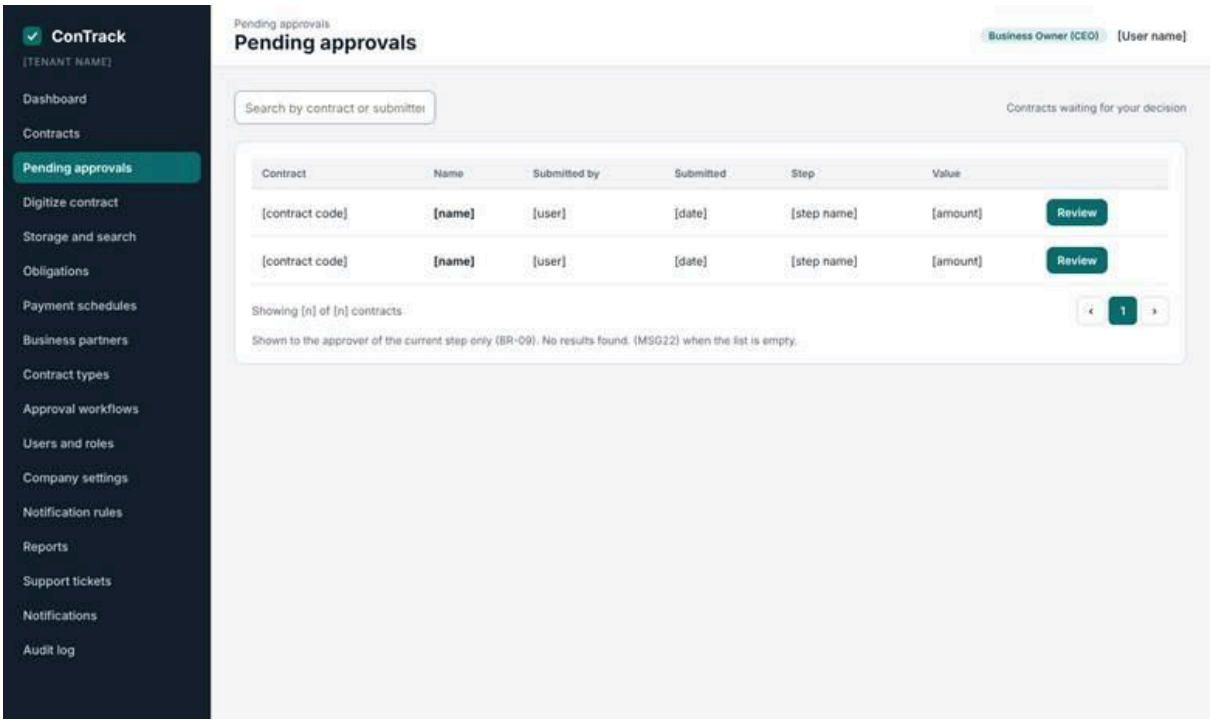
- Parties list columns: Role, Party, Representative
- Actions: Back to pending approvals, Approve, Request revision, Reject
- Messages and notices:
  - This contract is not waiting for your decision. Only the approver of the current step can approve, reject or request revision (BR-09).

**Content #4: Permission and State**

- Permission: Approve/Reject/Request revision: CEO Full; BAO, BS, BTS Restricted (only when they are the current approver).
- Roles shown by the mockup tweak: Business Owner (CEO), Admin Officer (BAO), Sales (BS), Technical Support (BTS)
- State behavior: Approve, Reject and Request revision are shown only to the approver of the current step.
- Mockup variants (approver): current, other

## 8.11 Pending Approvals

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: Contracts waiting for the signed-in user's decision.
- Mapped use case: CEO06; UC 6.2 step 5
- Actors: Approvers
- Entities: contract_approval_instances
- Business rules and messages: BR-09; MSG22

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|
| Search | Search input | | |

- List columns: Contract, Name, Submitted by, Submitted, Step, Value
- Actions: Review

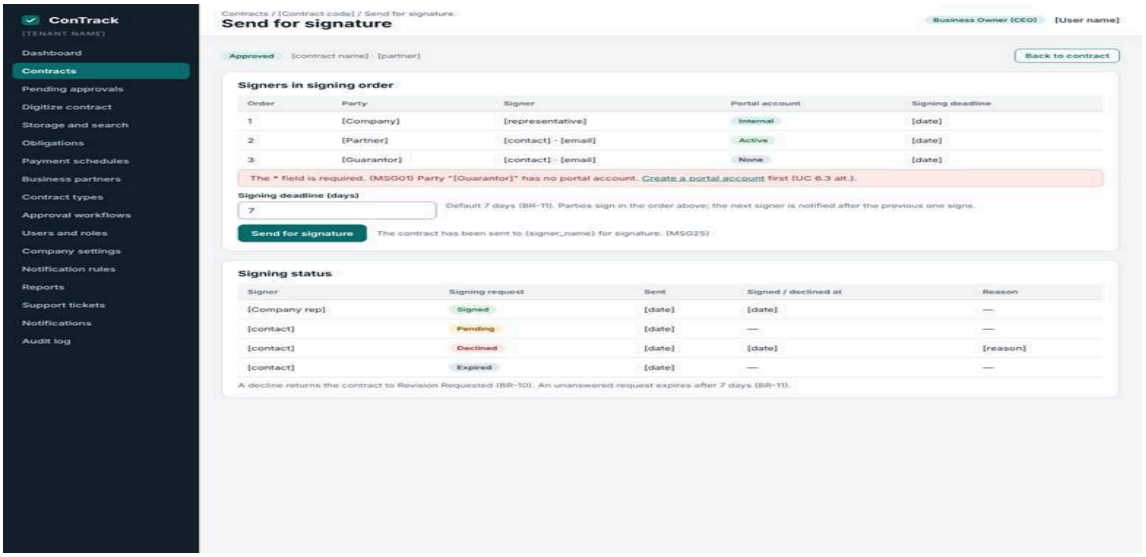
**Content #4: Permission and State**

- Permission: CEO, BAO, BS, BTS (Restricted roles see only their own steps).
- Roles shown by the mockup tweak: Business Owner (CEO), Admin Officer (BAO), Sales (BS), Technical Support (BTS)

# 9. E-SIGNATURE

## 9.1 Send for Signature

**Content #1: UI Layout (Wireframe Mockup)**



**Content #2: Brief Description and Use Case Mapping**

- Description: Confirms signers in signing order and the deadline, sends signing requests and tracks their status.
- Mapped use case: UC 6.3 (no Inventory entry;Mapped use case: UC 6.3 )
- Actors: CEO, BS (send); BAO (view status)
- Entities: signing_requests, contract_parties
- Business rules and messages: BR-10, BR-11, BR-12, BR-15; MSG01, MSG16, MSG25

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|
| Signing deadline (days) | Text input | | Deadline applied to every signing request |

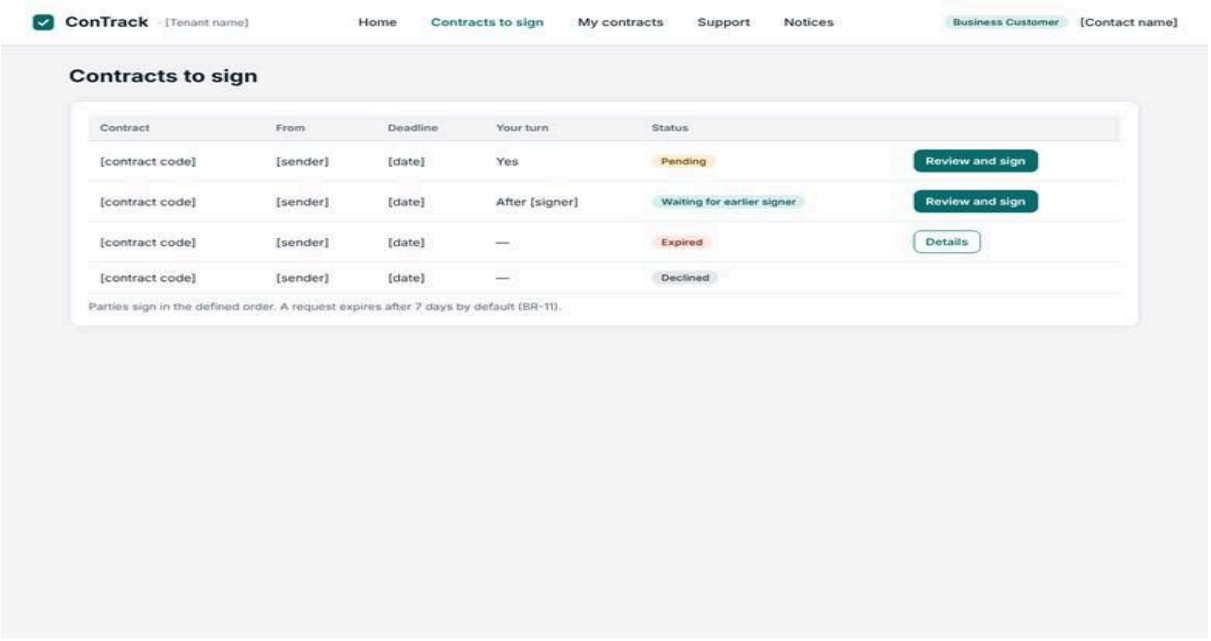
- Signers list columns: Order, Party, Signer, Portal account, Signing deadline
- Signing status list columns: Signer, Signing request, Sent, Signed / declined at, Reason
- Actions: Back to contract, Send for signature
- Messages and notices:
  - This action is not allowed while the contract is [status]. (MSG16) A contract can be sent only when it is Approved.
  - Party "[Guarantor]" has no portal account. Create a portal account first (UC 6.3 alternative flow). The required-field message (MSG01) applies to the missing signer.

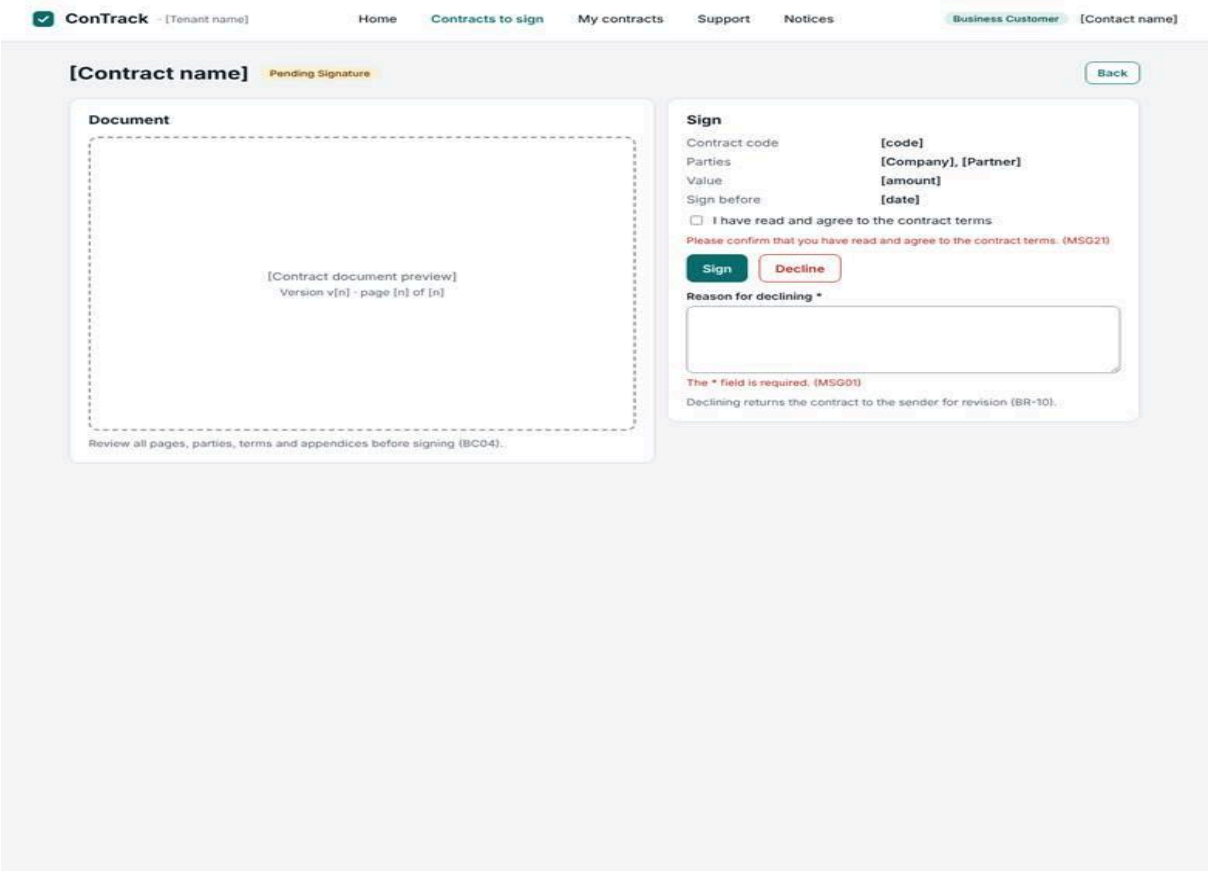
**Content #4: Permission and State**

- Permission: Send for e-signature: CEO Full; BS Restricted; others No. View signing status: CEO, BAO Full; BS Restricted.
- Roles shown by the mockup tweak: Business Owner (CEO), Sales (BS), Admin Officer (BAO)
- State behavior: Allowed only when the contract is Approved; shows the Pending Signature and Signed states.
- Mockup variants (status): Approved, Pending Signature, Signed

## 9.2 Contracts to Sign and E-sign

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: The customer lists signing requests, reviews the document, accepts the terms, verifies by OTP and signs or declines.

- Mapped use case: BC04–BC07; UC 6.4
- Actors: Business Customer
- Entities: signing_requests, contract_documents
- Business rules and messages: BR-10, BR-11, BR-12, BR-16; MSG01, MSG17, MSG21, MSG26, MSG28

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|
| Agree to terms | Checkbox | Yes | Label: I have read and agree to the contract terms. Error: MSG21 |
| Reason for declining | Text area | Yes | Error: The * field is required. (MSG01) |
| Verification code | Text input | Yes | 6 digits |
| Document preview | Viewer | | Shows the document version and page [n] of [n] |

- Signing requests list columns: Contract, From, Deadline, Your turn, Status
- Actions: Review and sign, Details, Back, Sign, Decline, Verify and sign, Resend code, Back to my contracts
- Messages and notices:
  - Please confirm that you have read and agree to the contract terms. (MSG21)
  - The OTP is incorrect or has expired. You have {n} attempt(s) left. (MSG17)
  - Signing is blocked for 15 minutes after 3 incorrect attempts. The exact message is not in the message list (BR-11).
  - You have signed contract [contract code] successfully. (MSG26)

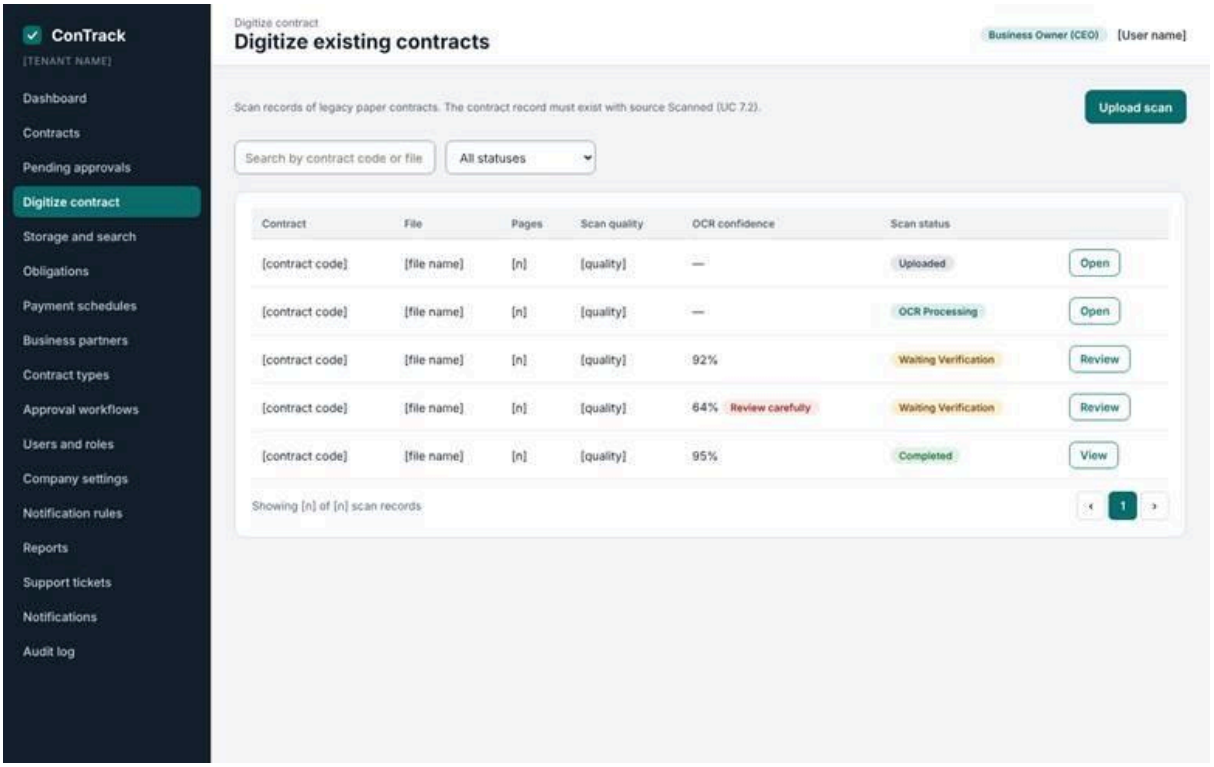
**Content #4: Permission and State**

- Permission: E-sign and Decline signing: Business Customer Restricted (only when it is their turn and the request is not expired).
- State behavior: Stages: review, OTP, locked after 3 wrong attempts for 15 minutes, done.
- Mockup variants (stage): review, otp, locked, done

# 10. Document and Digitization

## 10.1 Digitize Existing Contract

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: Scan records of paper contracts with OCR status and confidence.
- Mapped use case: BAS04, BAS05; UC 7.2
- Actors: BAO, CEO
- Entities: contract_documents (scan records), document_ocr_results
- Business rules and messages: BR-13, BR-14

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|
| Search | Search input | | |

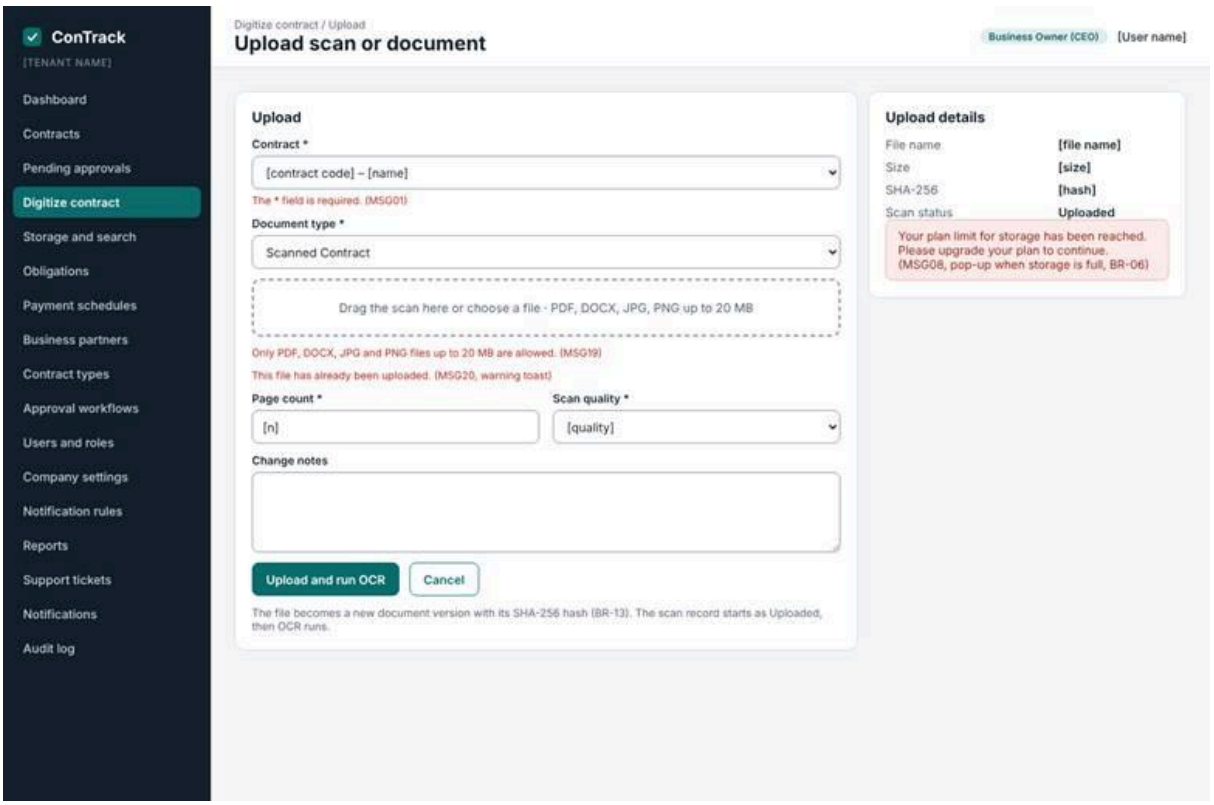
- List columns: Contract, File, Pages, Scan quality, OCR confidence, Scan status
- Actions: Upload scan, Open, Review, View

**Content #4: Permission and State**

- Permission: Scan/Upload scan and Verify OCR: CEO, BAO Full; BS, BTS No.
- Roles shown by the mockup tweak: Business Owner (CEO), Admin Officer (BAO)
- State behavior: Scan status: Uploaded, OCR Processing, Waiting Verification, Completed.

## 10.2 Upload Document

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: Uploads a scan or document to a contract with page count, quality and notes.
- Mapped use case: BAS02, BAS04; UC 7.1, 7.2
- Actors: BAO, CEO
- Entities: contract_documents
- Business rules and messages: BR-06, BR-13; MSG01, MSG08, MSG19, MSG20

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|

| | | | |
|---|---|---|---|
| Contract | Dropdown | Yes | Options: contract code and name. Error: The * field is required. (MSG01) |
| Document type | Dropdown | Yes | Options: Scanned Contract, Appendix, Acceptance Record, Other |
| Page count | Text input | Yes | |
| Scan quality | Dropdown | Yes | Values: Not specified in the SRS |
| Change notes | Text area | | |
| File | File upload | Yes | Drop zone. PDF, DOCX, JPG, PNG up to 20 MB |

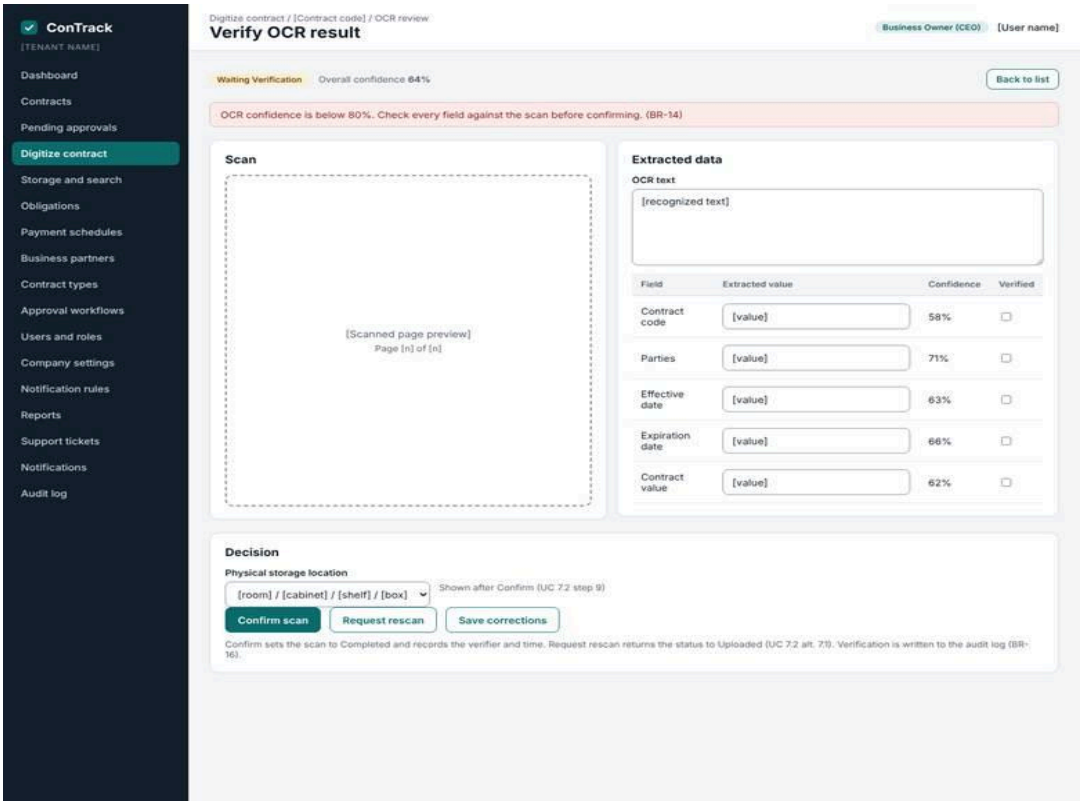
- Actions: Upload and run OCR, Cancel
- Messages and notices:
  - Only PDF, DOCX, JPG and PNG files up to 20 MB are allowed. (MSG19)
  - This file has already been uploaded. (MSG20, warning toast)
  - Your plan limit for storage has been reached. Please upgrade your plan to continue. (MSG08, pop-up when storage is full, BR-06)

**Content #4: Permission and State**

- Permission: Upload scan: CEO, BAO Full.
- Roles shown by the mockup tweak: Business Owner (CEO), Admin Officer (BAO)

## 10.3 OCR / Review Extracted Data

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: Compares the scan with OCR text, corrects extracted fields, confirms or requests a rescan, then assigns the physical location.
- Mapped use case: BAS05, BAS07; UC 7.2 steps 6–10
- Actors: BAO, CEO
- Entities: document_ocr_results, contract_extracted_fields, physical_storage_locations
- Business rules and messages: BR-14, BR-16; MSG05

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|
| Scanned page preview | Viewer | | Page [n] of [n] |
| OCR text | Text area | | Editable text recognized from the scan |
| Physical storage location | Dropdown | | Options: room / cabinet / shelf / box |

- Extracted fields list columns: Field, Extracted value, Confidence, Verified
- Actions: Back to list, Confirm scan, Request rescan, Save corrections

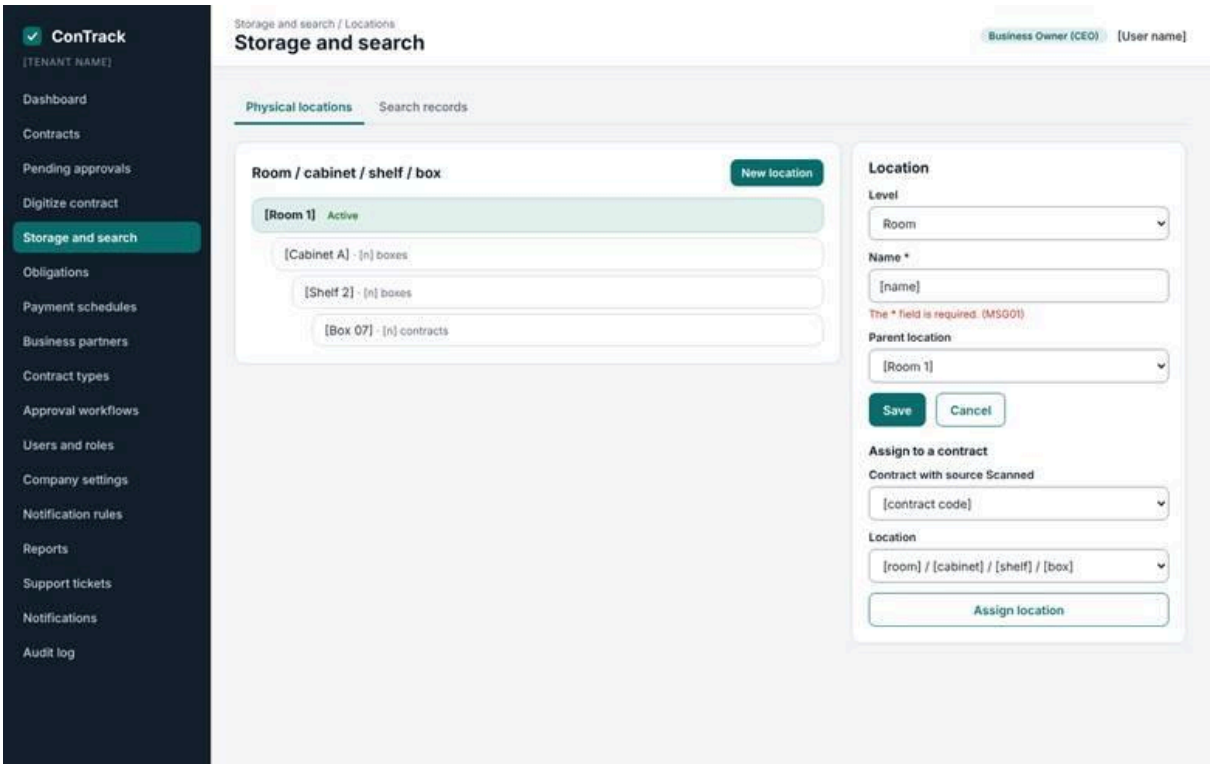
- Messages and notices:
  - OCR confidence is below 80%. Check every field against the scan before confirming. (BR-14)

**Content #4: Permission and State**

- Permission: Verify OCR/Confirm: CEO, BAO Full.
- Roles shown by the mockup tweak: Business Owner (CEO), Admin Officer (BAO)
- State behavior: Confidence below 80% raises a review warning (BR-14).
- Mockup variants (confidence): low, high

## 10.4 Storage Locations

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: Manages room, cabinet, shelf and box and assigns the paper original of a contract.
- Mapped use case: BAS06, BAS07
- Actors: BAO, CEO
- Entities: physical_storage_locations

- Business rules and messages: BR-24; MSG01, MSG05

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|
| Level | Dropdown | | Options: Room, Cabinet, Shelf, Box |
| Name | Text input | Yes | Error: The * field is required. (MSG01) |
| Parent location | Dropdown | | Options: locations of the level above |
| Contract (source Scanned) | Dropdown | | Options: scanned contracts |
| Location | Dropdown | | Options: room / cabinet / shelf / box |

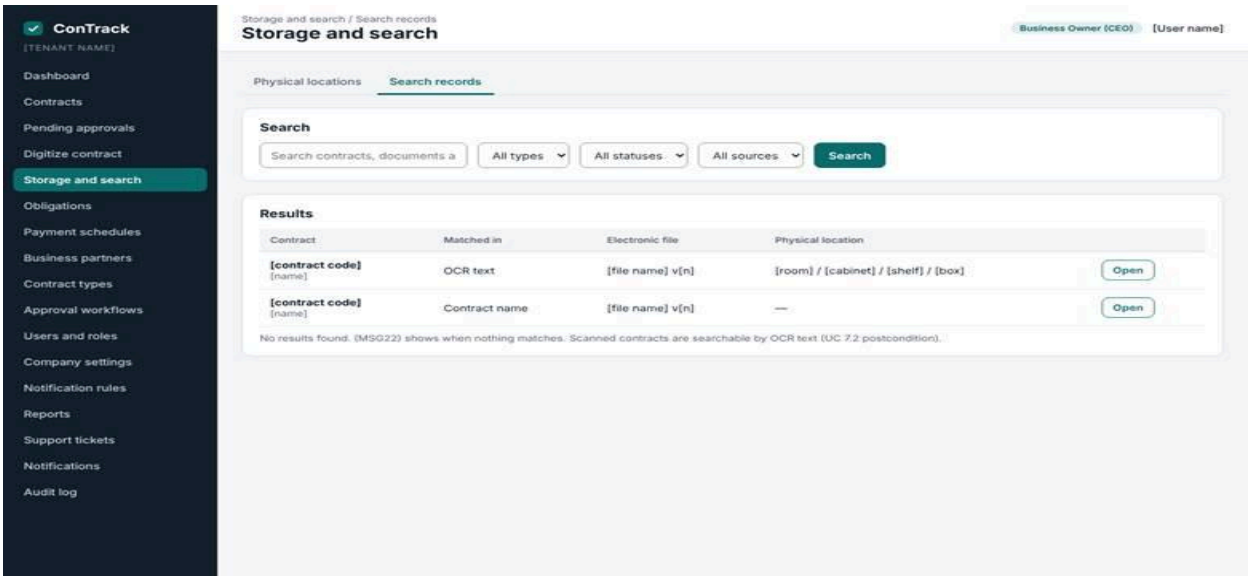
- Actions: New location, Save, Cancel, Assign location

**Content #4: Permission and State**

- Permission: Physical storage location Create/Edit/Assign: CEO, BAO Full.
- Roles shown by the mockup tweak: Business Owner (CEO), Admin Officer (BAO)

## 10.5 Search Records

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: Searches contracts and documents and shows both the electronic file and the paper location.
- Mapped use case: BAS08

- Actors: BAO, CEO
- Entities: contracts, contract_documents, document_ocr_results
- Business rules and messages: BR-20; MSG22

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|
| Search | Search input | | |

- Results list columns: Contract, Matched in, Electronic file, Physical location
- Actions: Search, Open

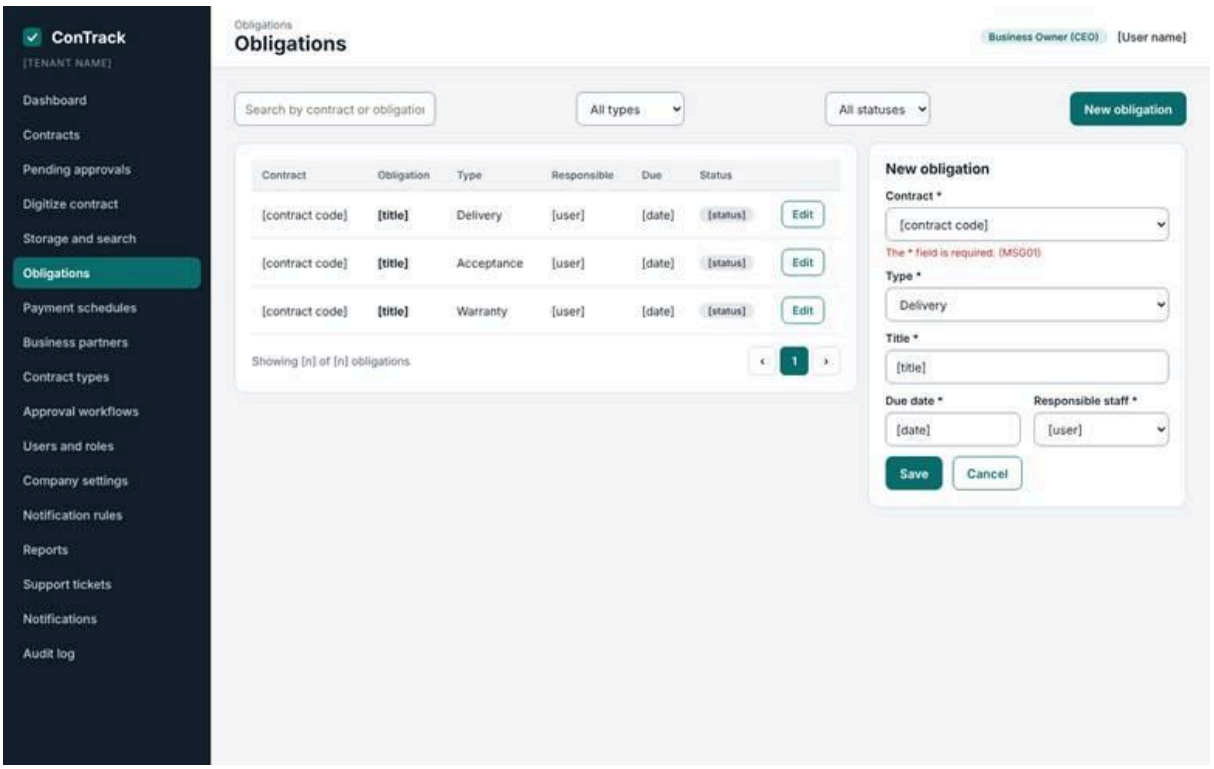
**Content #4: Permission and State**

- Permission: Contract View: CEO, BAO Full.
- Roles shown by the mockup tweak: Business Owner (CEO), Admin Officer (BAO)

# 11. CONTRACT EXECUTION

## 11.1 Contract Obligation List

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: Obligations (delivery, acceptance, warranty) with due date and responsible staff.
- Mapped use case: BS10
- Actors: CEO, BS
- Entities: contract obligations (no entity in the ERD)
- Business rules and messages: BR-15; MSG01

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|
| Contract | Dropdown | Yes | Options: contract codes. Error: The * field is required. (MSG01) |
| Type | Dropdown | Yes | Options: Delivery, Acceptance, Warranty |
| Title | Text input | Yes | |
| Due date | Date input | Yes | |
| Responsible staff | Dropdown | Yes | Options: tenant users |
| Search | Search input | | |

- List columns: Contract, Obligation, Type, Responsible, Due, Status

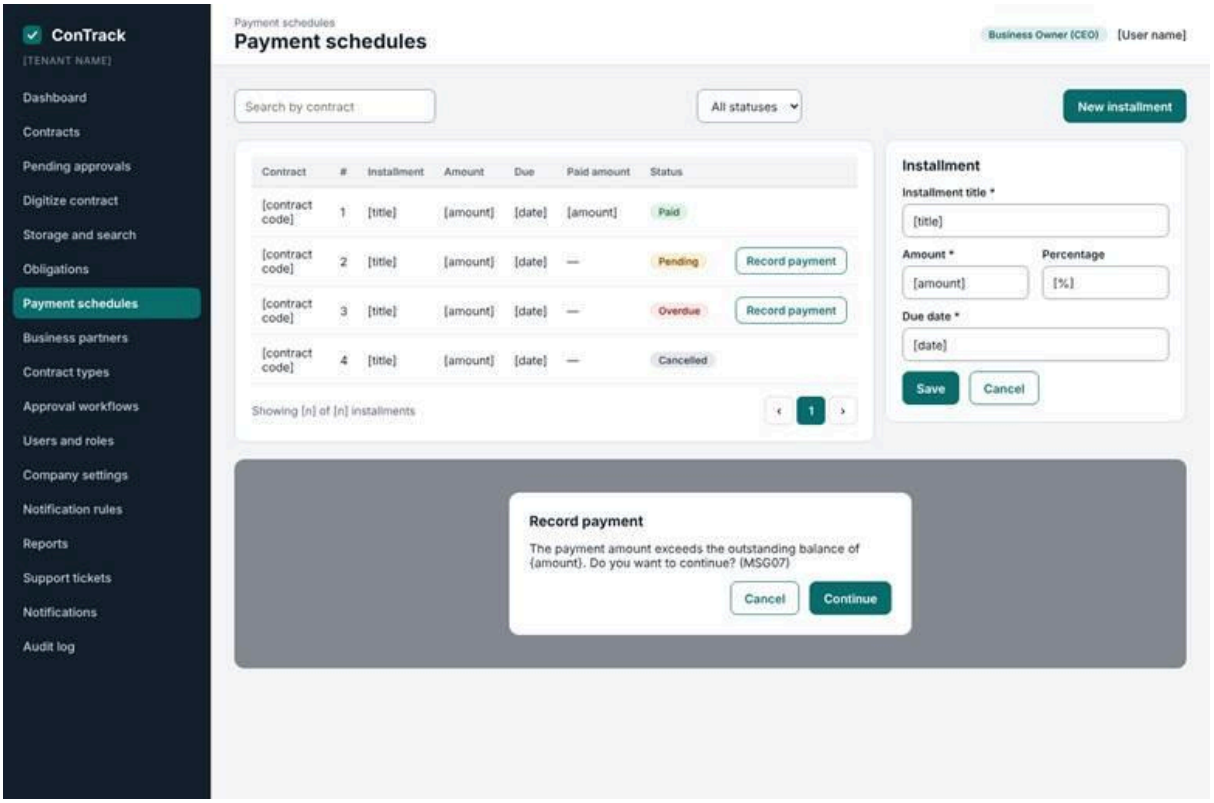
- Actions: New obligation, Edit, Save, Cancel

**Content #4: Permission and State**

- Permission: Appendix, obligation, payment schedule Create/Edit: CEO Full; BS Restricted.
- Roles shown by the mockup tweak: Business Owner (CEO), Sales (BS)

## 11.2 Payment Schedule List

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: Installments with amount, due date and status; records actual payments.
- Mapped use case: BS11
- Actors: CEO, BS
- Entities: contract_payment_schedules
- Business rules and messages: BR-15; MSG07

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|

| | | | |
|---|---|---|---|
| Installment title | Text input | Yes | |
| Amount | Text input | Yes | |
| Percentage | Text input | | |
| Due date | Date input | Yes | |
| Search | Search input | | |

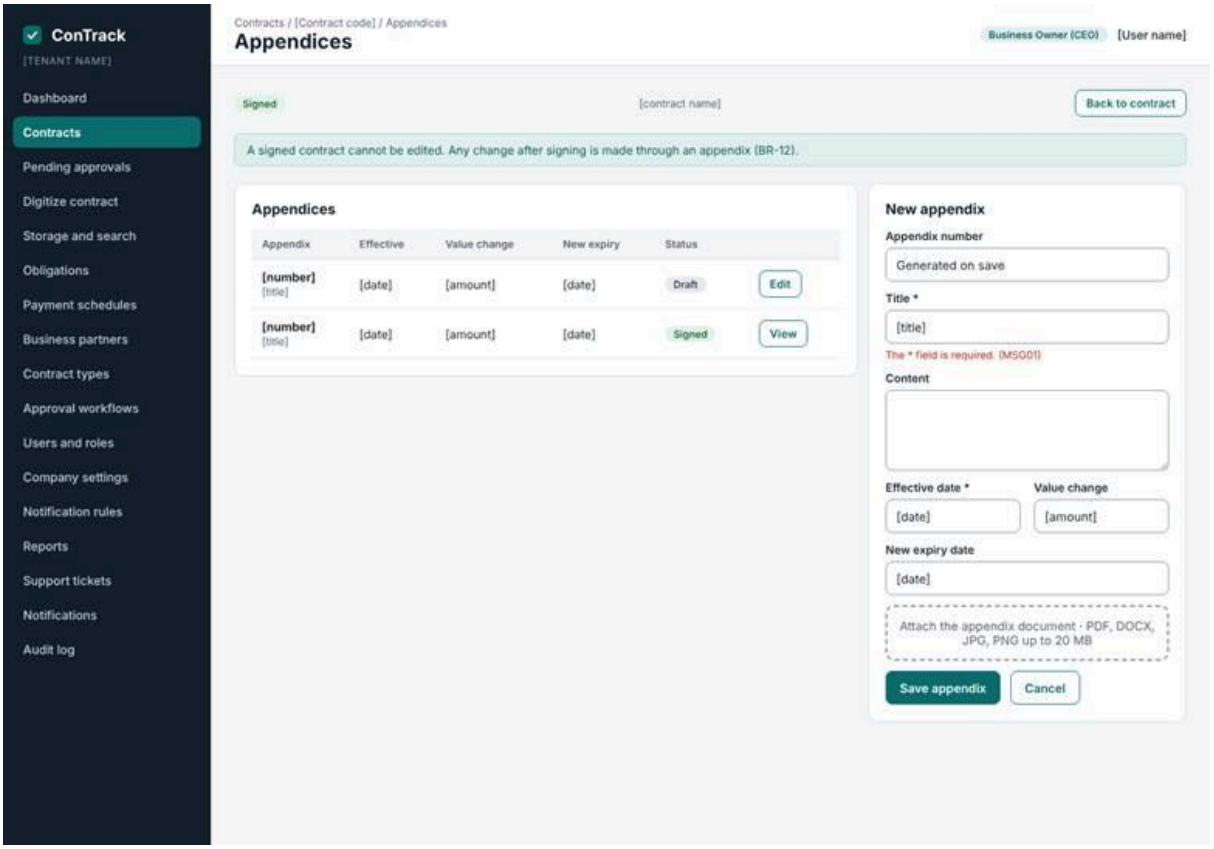
- List columns: Contract, #, Installment, Amount, Due, Paid amount, Status
- Actions: New installment, Record payment, Save, Cancel, Continue
- Messages and notices:
  - Record payment dialog: The payment amount exceeds the outstanding balance of {amount}. Do you want to continue? (MSG07) Buttons: Cancel, Continue.

**Content #4: Permission and State**

- Permission: Create/Edit: CEO Full; BS Restricted.
- Roles shown by the mockup tweak: Business Owner (CEO), Sales (BS)
- State behavior: Statuses: Pending, Paid, Overdue, Cancelled.

## 11.3 Contract Appendices

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: Adds appendices that change value or expiry date, the only way to change a signed contract.
- Mapped use case: BS09; UC 8.1
- Actors: CEO, BS
- Entities: contract_appendices
- Business rules and messages: BR-12, BR-13; MSG01

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|
| Appendix number | Text (read-only) | | Generated on save |
| Title | Text input | Yes | Error: The * field is required. (MSG01) |
| Content | Text area | | |
| Effective date | Date input | Yes | |
| Value change | Text input | | |

| | | | |
|---|---|---|---|
| New expiry date | Date input | | |
| Appendix document | File upload | | Drop zone. PDF, DOCX, JPG, PNG up to 20 MB |

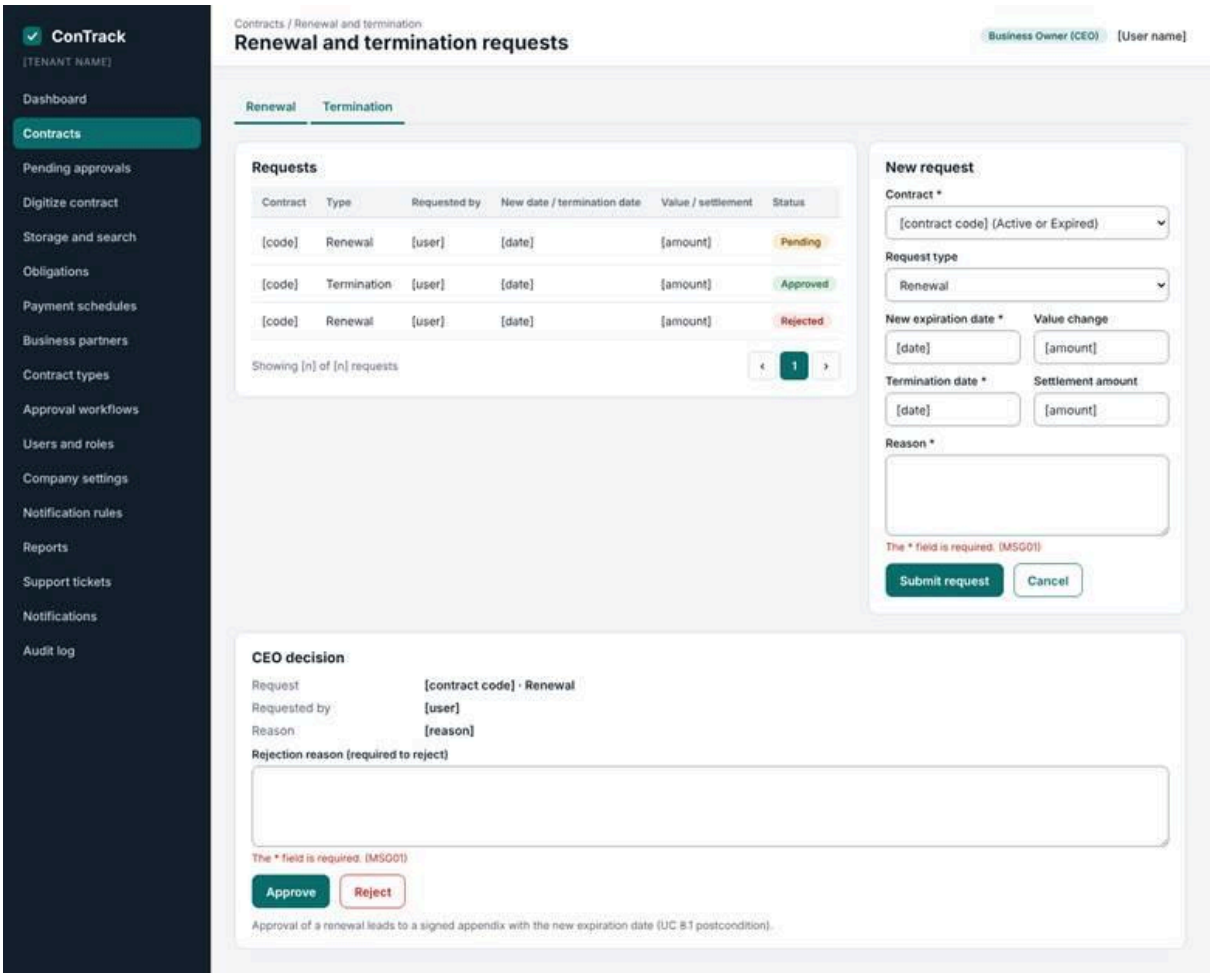
- List columns: Appendix, Effective, Value change, New expiry, Status
- Actions: Back to contract, Edit, View, Save appendix, Cancel
- Messages and notices:
  - A signed contract cannot be edited. Any change after signing is made through an appendix (BR-12).

**Content #4: Permission and State**

- Permission: Create/Edit: CEO Full; BS Restricted.
- Roles shown by the mockup tweak: Business Owner (CEO), Sales (BS)

## 11.4 Renewal and Termination

**Content #1: UI Layout (Wireframe Mockup)**



**Content #2: Brief Description and Use Case Mapping**

- Description: Requests, approves or rejects contract renewal and early termination.
- Mapped use case: BS12, BS13; UC 8.1
- Actors: BS (request), CEO (request and approve)
- Entities: renewal requests, contract_appendices
- Business rules and messages: BR-12; MSG01, MSG24

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|
| Contract | Dropdown | Yes | Options: contracts that are Active or Expired |
| Request type | Dropdown | | Options: Renewal, Termination |
| New expiration date | Date input | Yes | Renewal only |
| Value change | Text input | | Renewal only |
| Termination date | Date input | Yes | Termination only |
| Settlement amount | Text input | | Termination only |
| Reason | Text area | Yes | Error: The * field is required. (MSG01) |
| Rejection reason | Text area | Conditional | Required to reject. Error: The * field is required. (MSG01) |

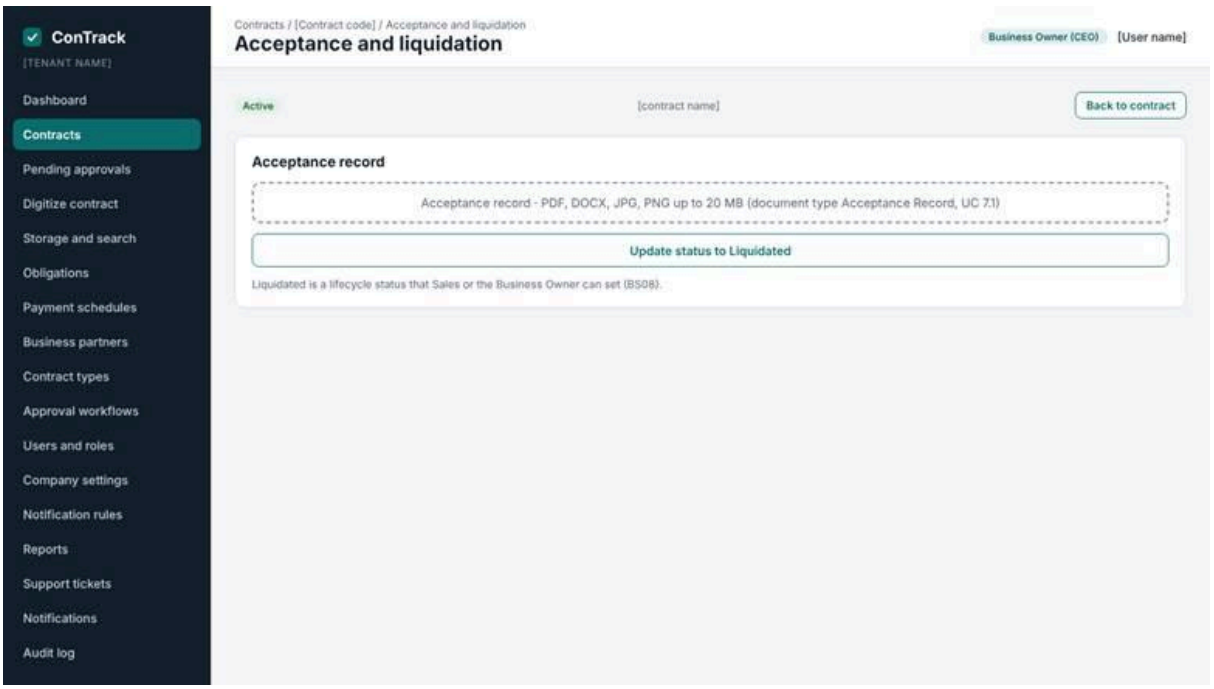
- List columns: Contract, Type, Requested by, New date / termination date, Value / settlement, Status
- Actions: Review, Submit request, Cancel, Approve, Reject

**Content #4: Permission and State**

- Permission: Request: CEO Full; BS Restricted. Approve: CEO only.
- Roles shown by the mockup tweak: Business Owner (CEO), Sales (BS)
- State behavior: Only the CEO sees the decision panel; the contract must be Active or Expired.

## 11.5 Acceptance and Liquidation

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: Placeholder for recording the acceptance record and marking a contract Liquidated.
- Mapped use case: BS08, BAS02
- Actors: CEO, BS
- Entities: contract_documents (Acceptance Record), contracts
- Business rules and messages: BR-12

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|
| Acceptance record | File upload | | Document type Acceptance Record. Drop zone. PDF, DOCX, JPG, PNG up to 20 MB |

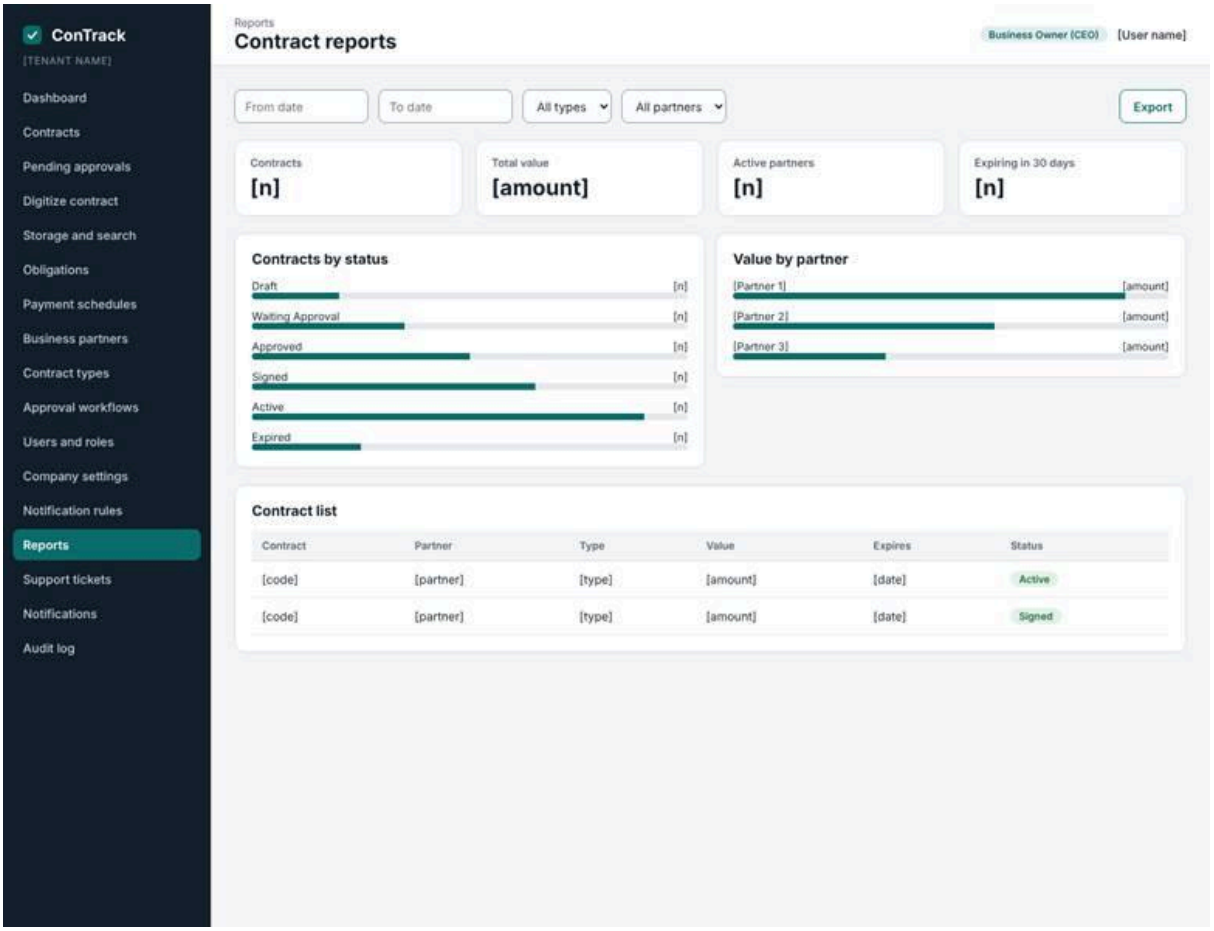
- Actions: Back to contract, Update status to Liquidated

**Content #4: Permission and State**

- Permission: Update lifecycle status: CEO Full; BS Restricted.
- Roles shown by the mockup tweak: Business Owner (CEO), Sales (BS)

## 11.6 Contract Reports

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: Reports on contracts, partners and contract value with export.
- Mapped use case: CEO09
- Actors: CEO
- Entities: contracts, business_partners
- Business rules and messages: none

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|
| From | Date input | | Report period start |
| To | Date input | | Report period end |

- List columns: Contract, Partner, Type, Value, Expires, Status
- Actions: Export

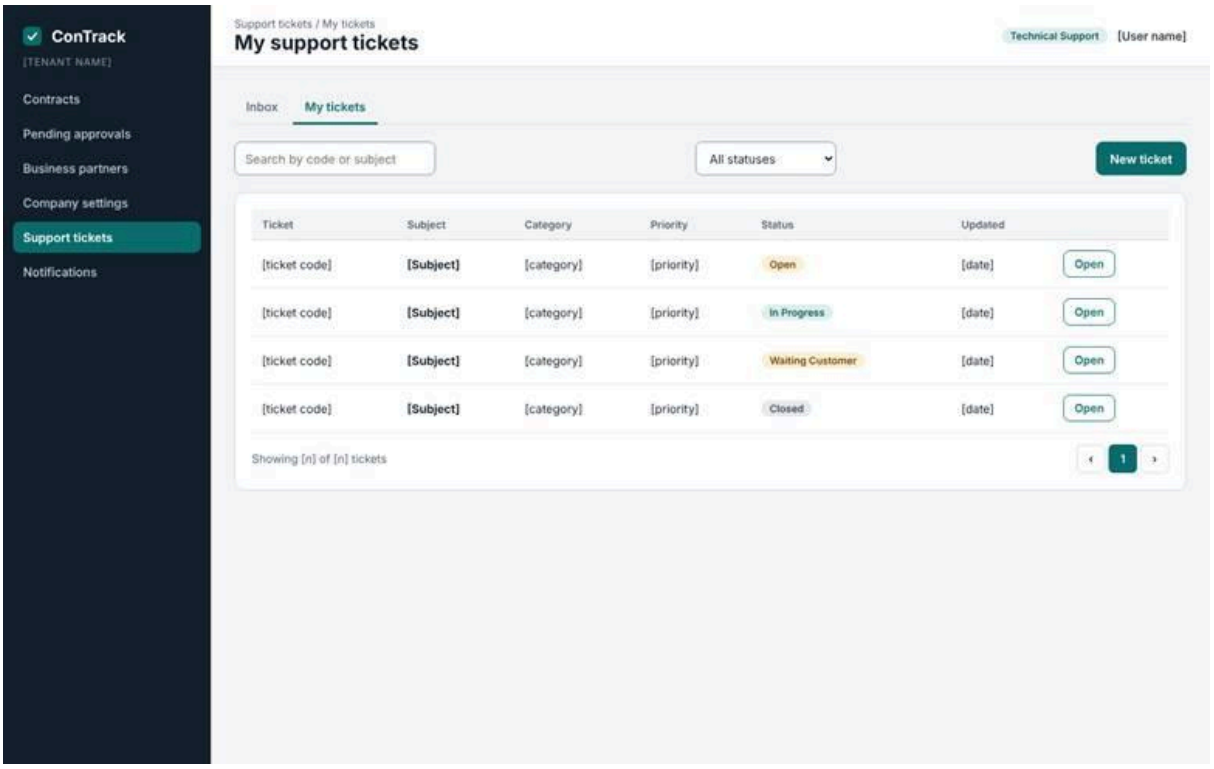
**Content #4: Permission and State**

- Permission: Dashboard and report View: CEO Full; BAO, BS Restricted.
- Roles shown by the mockup tweak: Business Owner (CEO), Admin Officer (BAO), Sales (BS)

# 12. CUSTOMER SUPPORT (TENANT)

## 12.1 My Support Tickets (Tenant)

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: Tickets raised by Technical Support and their status.
- Mapped use case: BTS01
- Actors: BTS, CEO
- Entities: support_tickets
- Business rules and messages: MSG29

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|
| Search | Search input | | |

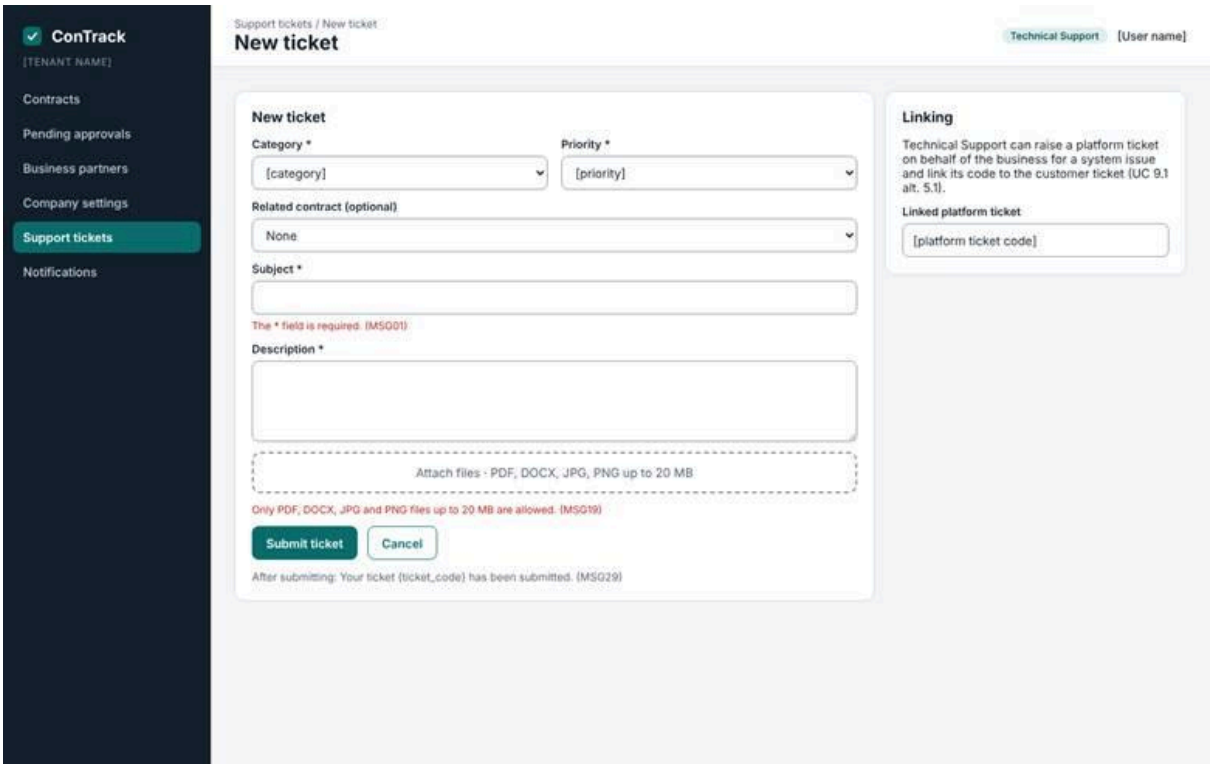
- List columns: Ticket, Subject, Category, Priority, Status, Updated
- Actions: New ticket, Open

**Content #4: Permission and State**

- Permission: Ticket Create: BTS. View: BTS, CEO Full.
- Roles shown by the mockup tweak: Technical Support (BTS), Business Owner (CEO)

## 12.2 Create Support Ticket (Tenant)

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: Submits a ticket with category, priority, related contract and attachments.
- Mapped use case: BTS; UC 9.1
- Actors: CEO
- Entities: support_tickets, ticket_categories

- Business rules and messages: BR-13; MSG01, MSG19, MSG29

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|
| Category | Dropdown | Yes | Options: ticket categories |
| Priority | Dropdown | Yes | Options: Not specified in the SRS |
| Related contract | Dropdown | | Optional. Options: None, contract codes |
| Subject | Text input | Yes | Error: The * field is required. (MSG01) |
| Description | Text area | Yes | |
| Linked platform ticket | Text input | | |
| Attachments | File upload | | Drop zone. PDF, DOCX, JPG, PNG up to 20 MB |

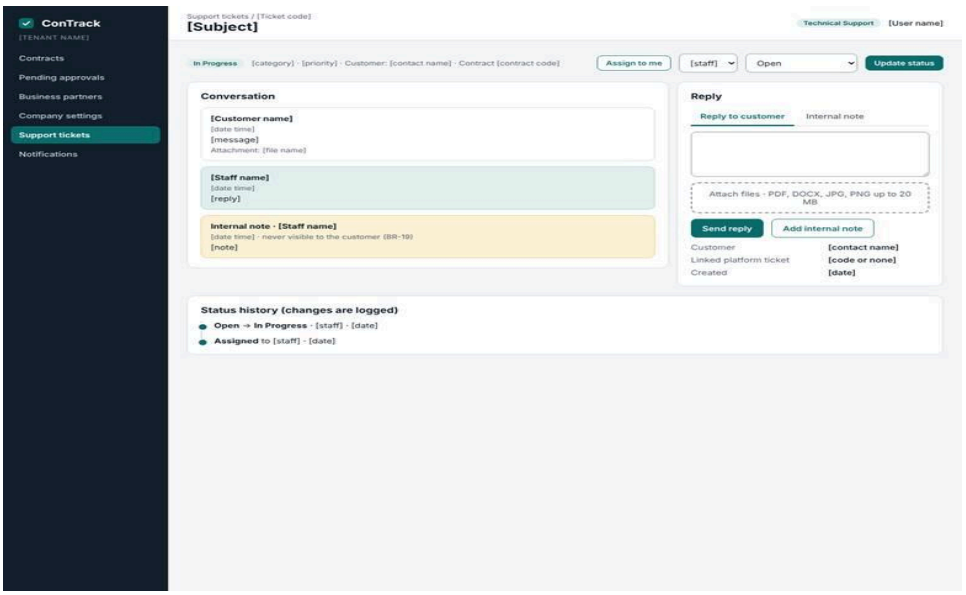
- Actions: Submit ticket, Cancel
- Messages and notices:
  - Only PDF, DOCX, JPG and PNG files up to 20 MB are allowed. (MSG19)

**Content #4: Permission and State**

- Permission: Create: BTS Full.
- Roles shown by the mockup tweak: Technical Support (BTS)

## 12.3 Support Ticket Detail (Tenant)

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: Conversation, internal notes, assignment and status updates for a customer or tenant ticket.
- Mapped use case: BTS02–BTS05; UC 9.1
- Actors: BTS, CEO
- Entities: ticket_messages, ticket status history
- Business rules and messages: BR-13, BR-19; MSG16, MSG29

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|
| Attachments | File upload | | Drop zone. PDF, DOCX, JPG, PNG up to 20 MB |

- Actions: Assign to me, Update status, Send reply, Add internal note
- Messages and notices:
  - This action is not allowed while the ticket is Closed. (MSG16) The customer can reopen it (UC 9.1 alternative flow).

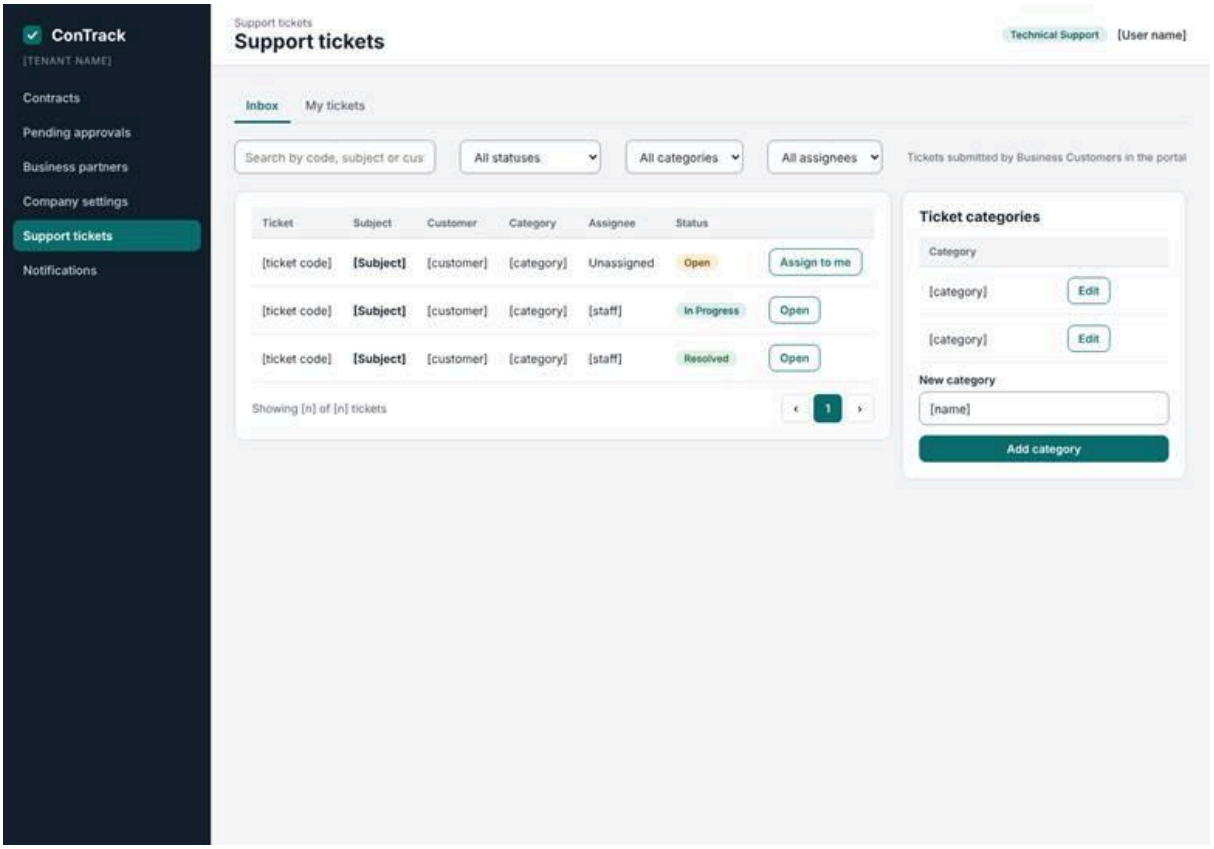
**Content #4: Permission and State**

- Permission: Reply/Update status: CEO Full; BTS Restricted. Internal note: CEO, BTS.
- Roles shown by the mockup tweak: Technical Support (BTS), Business Owner (CEO)
- State behavior: Closed locks actions (MSG16).

- Mockup variants (status): Open, In Progress, Waiting Customer, Resolved, Closed

## 12.4 Customer Ticket Inbox

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: Inbox of tickets submitted by Business Customers with assignment and ticket categories.
- Mapped use case: BTS01, BTS02, BTS06; UC 9.1
- Actors: BTS, CEO
- Entities: support_tickets, ticket_categories
- Business rules and messages: BR-19

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|
| New category | Text input | | |

| | | | |
|---|---|---|---|
| Search | Search input | | |

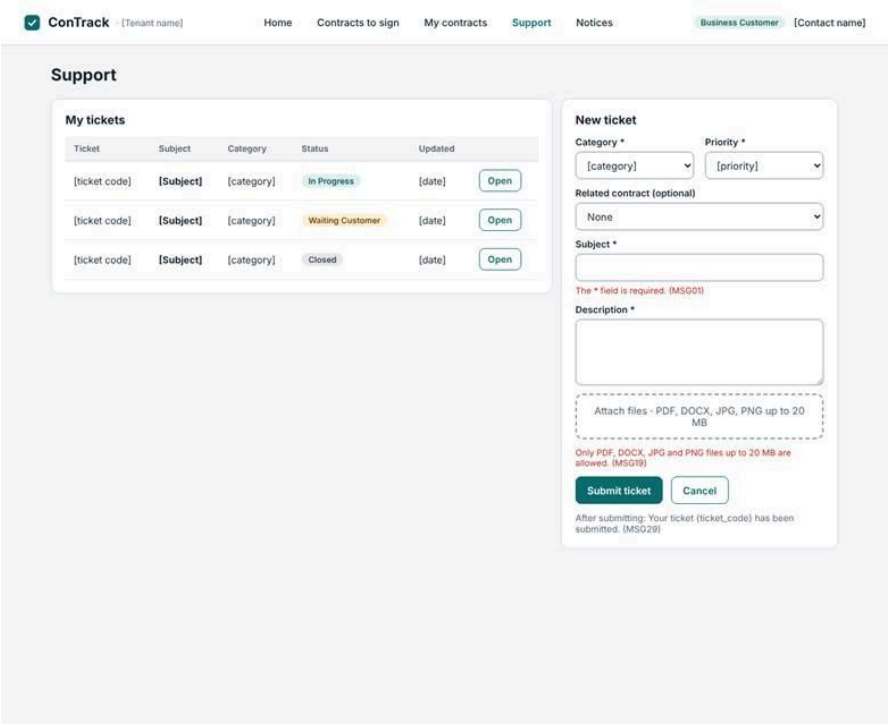
- Tickets list columns: Ticket, Subject, Customer, Category, Assignee, Status
- Categories list columns: Category
- Actions: Assign to me, Open, Edit, Add category

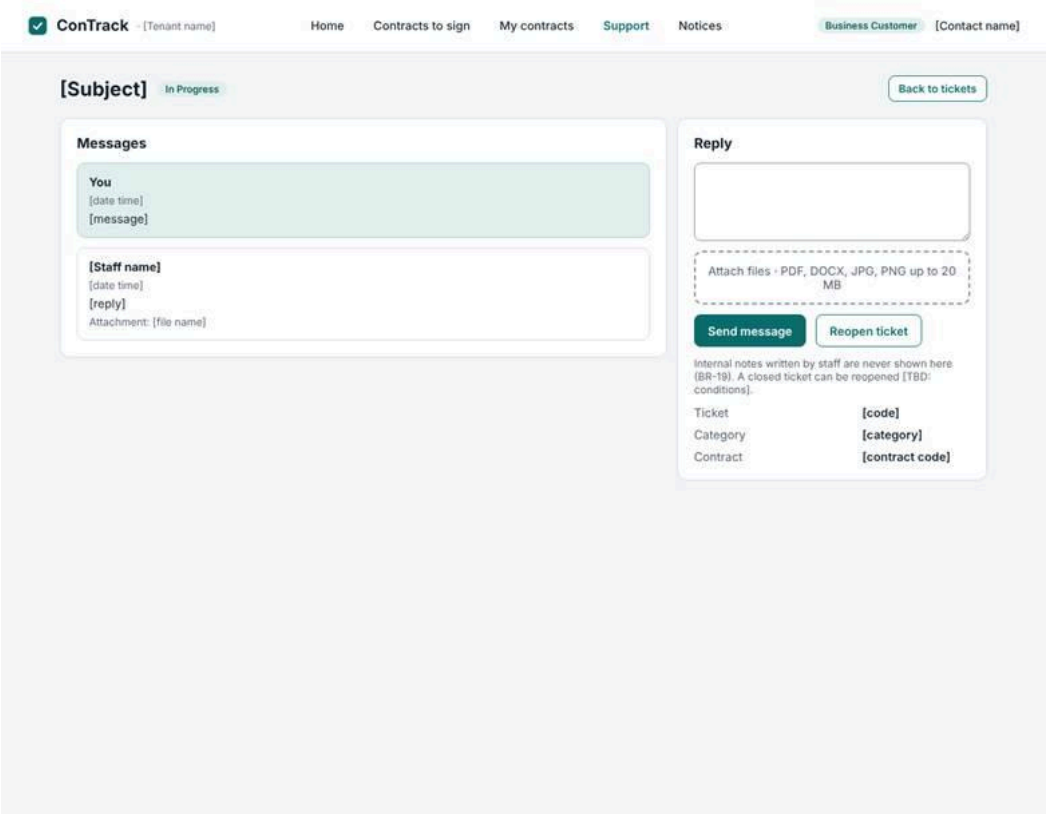
**Content #4: Permission and State**

- Permission: Ticket View/Assign and Category Create/Edit: CEO, BTS Full; BAO, BS No.
- Roles shown by the mockup tweak: Technical Support (BTS), Business Owner (CEO)

## 12.5 Portal Support (Business Customer)

**Content #1: UI Layout (Wireframe Mockup)**






**Content #2: Brief Description and Use Case Mapping**

- Description: The customer creates a ticket linked to a contract and follows the replies.
- Mapped use case: BC08, BC09; UC 9.1
- Actors: Business Customer
- Entities: support_tickets, ticket_messages
- Business rules and messages: BR-13, BR-19; MSG01, MSG19, MSG29

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|
| Category | Dropdown | Yes | Options: ticket categories |
| Priority | Dropdown | Yes | Options: Not specified in the SRS |
| Related contract | Dropdown | | Optional. Options: None, contracts of the customer |
| Subject | Text input | Yes | Error: The * field is required. (MSG01) |
| Description | Text area | Yes | |
| Attachments | File upload | | Drop zone. PDF, DOCX, JPG, PNG up to 20 MB |

- List columns: Ticket, Subject, Category, Status, Updated
- Actions: Open, Submit ticket, Cancel, Back to tickets, Send message, Reopen ticket
- Messages and notices:
  - Only PDF, DOCX, JPG and PNG files up to 20 MB are allowed. (MSG19)

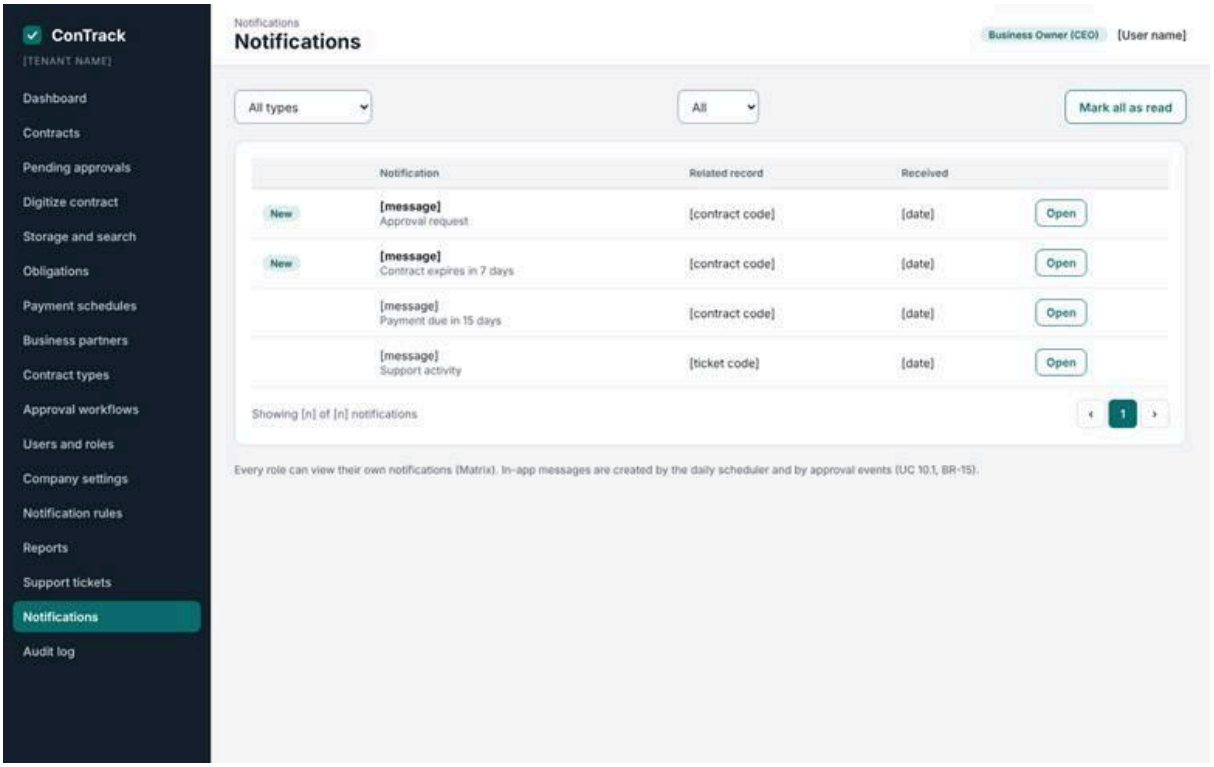
**Content #4: Permission and State**

- Permission: Create: Business Customer Full. View and reply: Business Customer Restricted (own tickets).

# 13. Notification

## 13.1 Notification Center

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: In-app notifications about approvals, expiries, payments, signing and support.
- Mapped use case: UC 10.1
- Actors: CEO, BAO, BS, BTS
- Entities: notifications

- Business rules and messages: BR-15

**Content #3: Field and Component Descriptions**

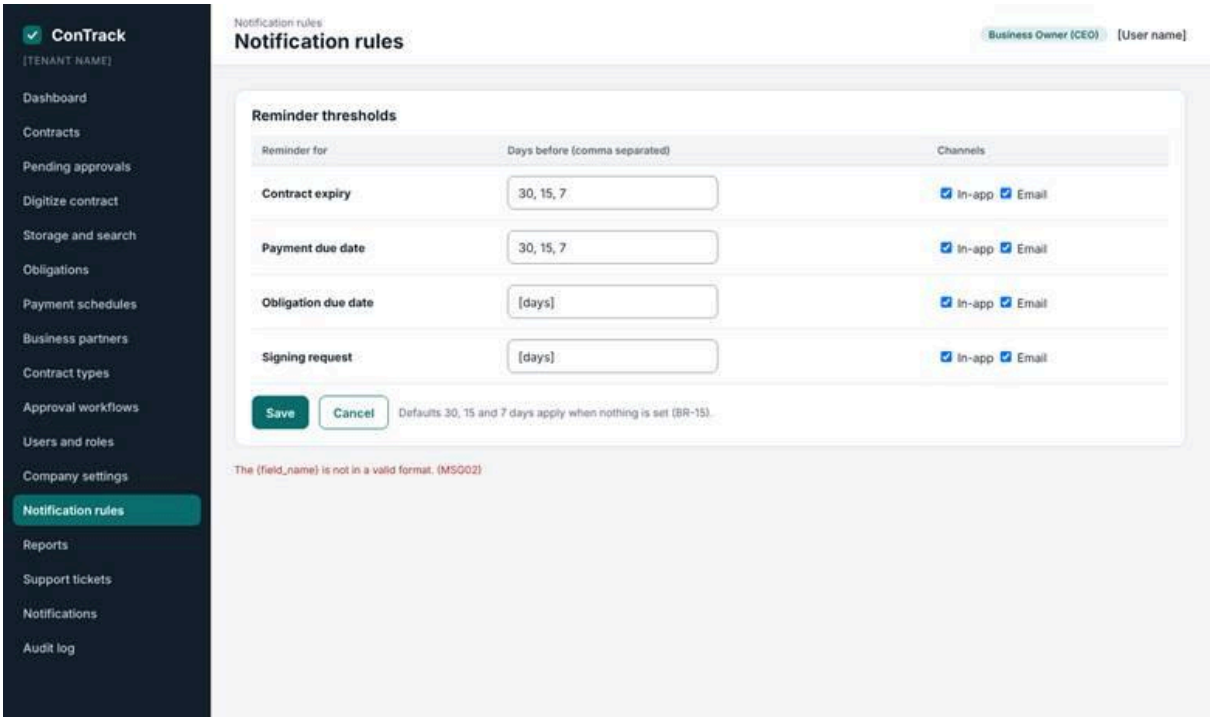
- Notifications list columns: Notification, Related record, Received
- Actions: Mark all as read, Open

**Content #4: Permission and State**

- Permission: Notification View own: all tenant roles Full.
- Roles shown by the mockup tweak: Business Owner (CEO), Admin Officer (BAO), Sales (BS), Technical Support (BTS)

## 13.2 Notification Rules

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: Reminder thresholds (default 30, 15, 7 days) per reminder type.
- Mapped use case: CEO07; UC 10.1
- Actors: CEO
- Entities: notification rules
- Business rules and messages: BR-15; MSG02, MSG05

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|
| Days before | Text input | | Comma-separated list, default 30, 15, 7 (BR-15). Error: The {field_name} is not in a valid format. (MSG02) |

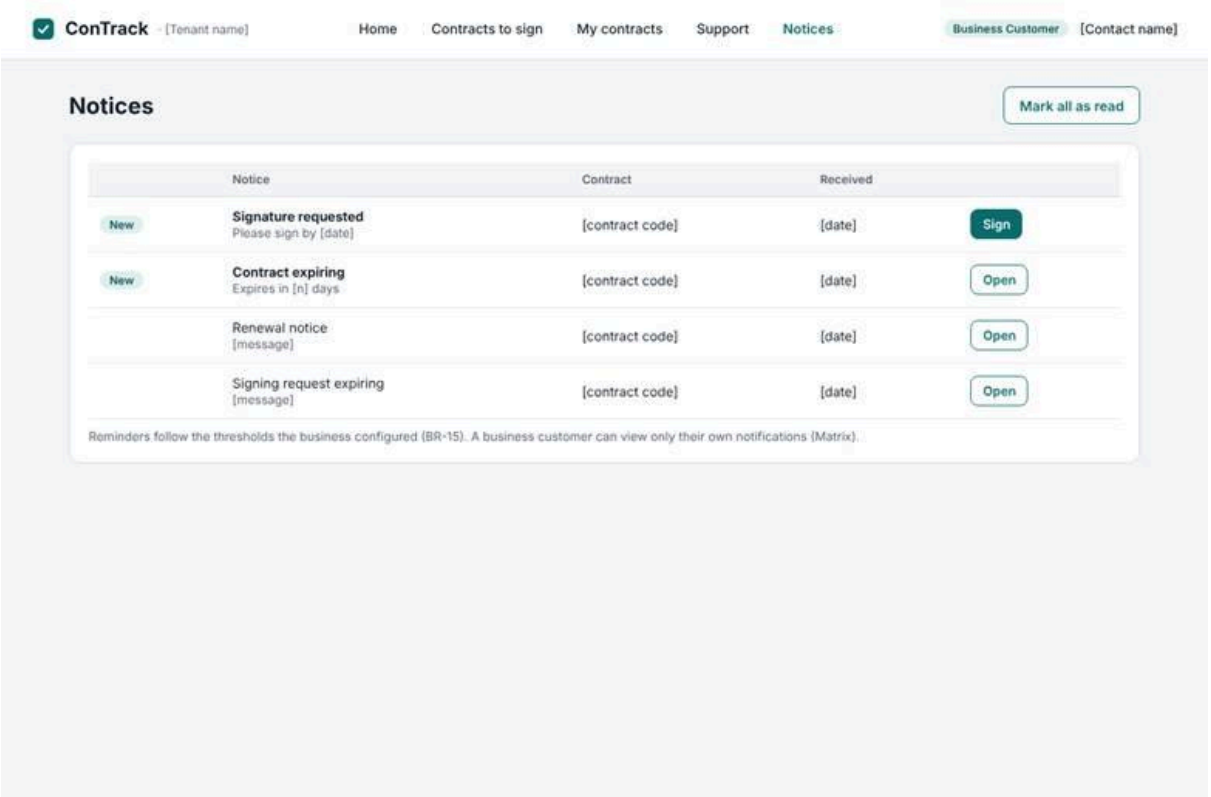
- Rules list columns: Reminder for, Days before, Channels
- Actions: Save, Cancel

**Content #4: Permission and State**

- Permission: Notification rule Configure: CEO only.
- Roles shown by the mockup tweak: Business Owner (CEO)

## 13.3 Portal Notices (Business Customer)

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: Notices about contract expiry, renewal and pending signatures.
- Mapped use case: BC10; UC 10.1

- Actors: Business Customer
- Entities: notifications
- Business rules and messages: BR-15

**Content #3: Field and Component Descriptions**

- Notices list columns: Notice, Contract, Received
- Actions: Mark all as read, Sign, Open

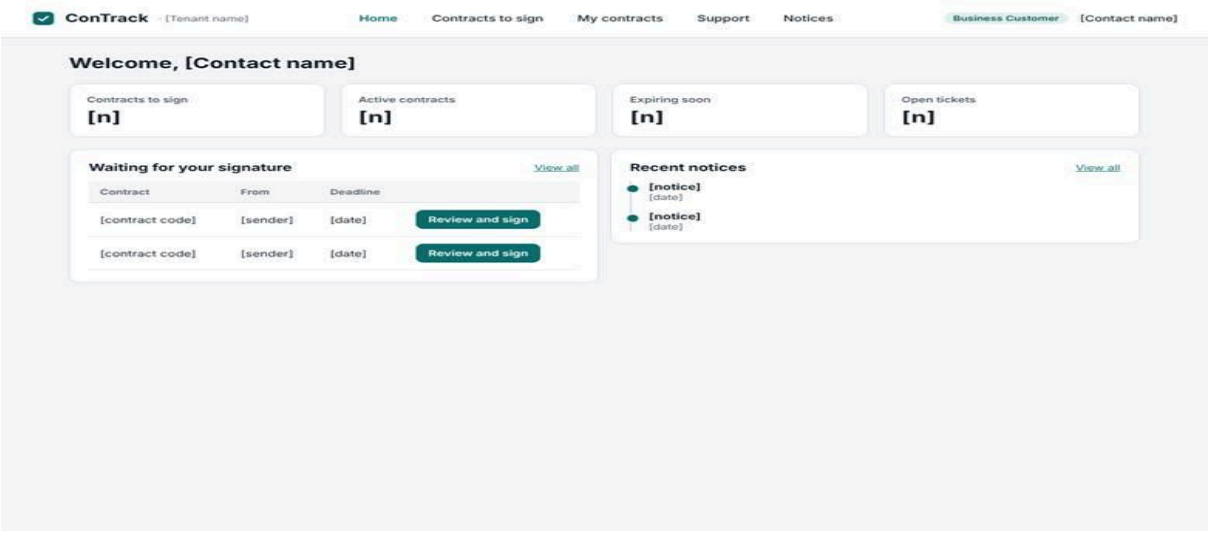
**Content #4: Permission and State**

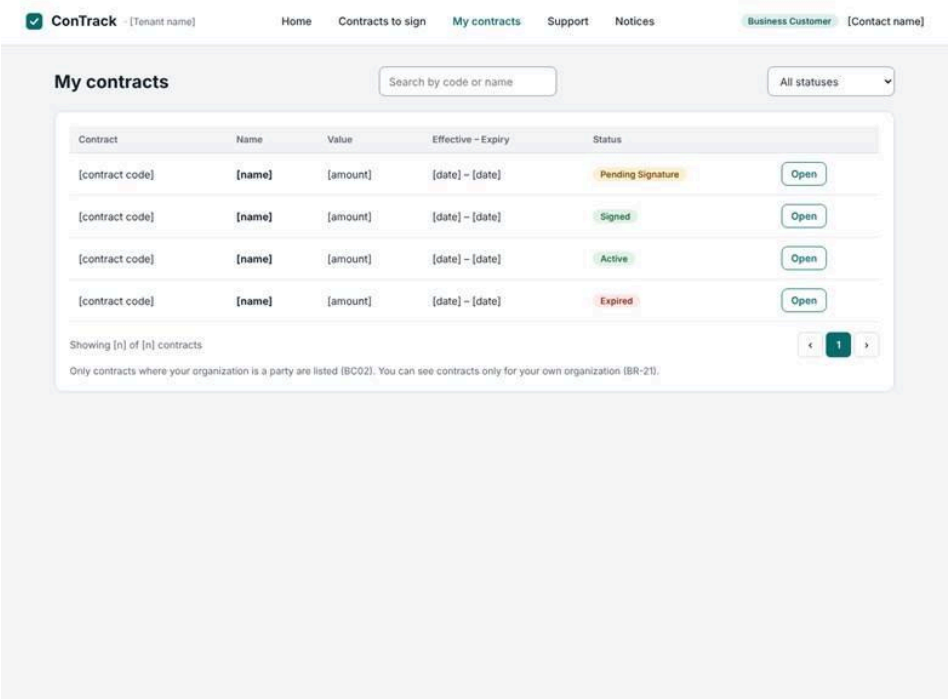
- Permission: Notification View own: Business Customer Full.

# 14. Business Customer Portal

## 14.1 Portal Home and My Contracts

**Content #1: UI Layout (Wireframe Mockup)**






**Content #2: Brief Description and Use Case Mapping**

- Description: Landing page and list of contracts where the customer's organization is a party.
- Mapped use case: BC01, BC02
- Actors: Business Customer
- Entities: contracts, contract_parties, signing_requests
- Business rules and messages: BR-20, BR-21

**Content #3: Field and Component Descriptions**

| Field / Component | Type | Required | Description / Rule |
|---|---|---|---|
| Search | Search input | | |

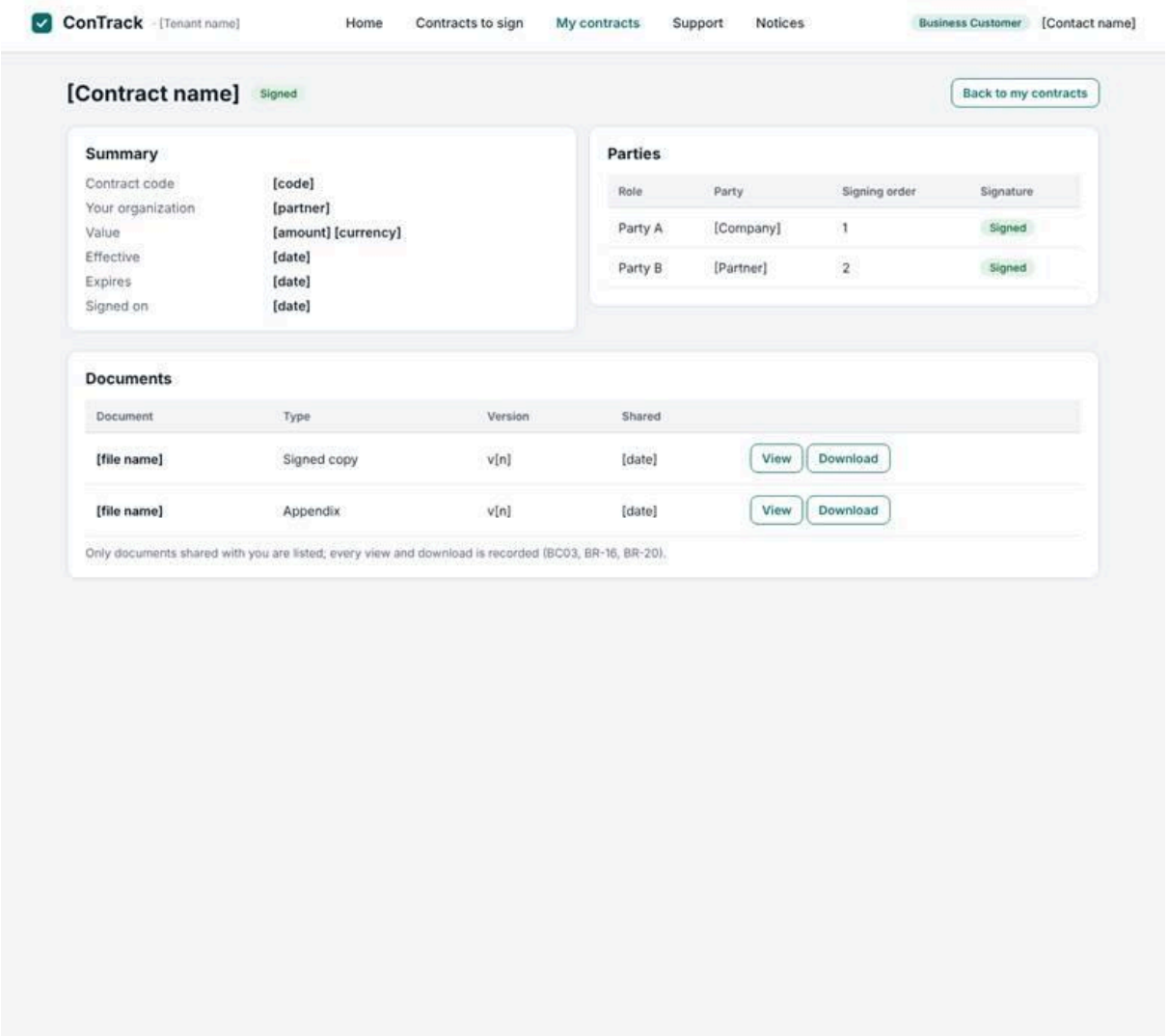
- Waiting for my signature list columns: Contract, From, Deadline
- My contracts list columns: Contract, Name, Value, Effective – Expiry, Status
- Actions: Review and sign, Open

**Content #4: Permission and State**

- Permission: Contract View: Business Customer Restricted (own organization).

## 14.2 Contract Detail and Documents (Business Customer)

**Content #1: UI Layout (Wireframe Mockup)**




**Content #2: Brief Description and Use Case Mapping**

- Description: Contract summary, parties and the documents shared with the customer.
- Mapped use case: BC03, BC04
- Actors: Business Customer
- Entities: contracts, contract_documents
- Business rules and messages: BR-16, BR-20

**Content #3: Field and Component Descriptions**

- Parties list columns: Role, Party, Signing order, Signature
- Documents list columns: Document, Type, Version, Shared

- Actions: Back to my contracts, View, Download, Review and sign

**Content #4: Permission and State**

- Permission: Contract document View/Download: Business Customer Restricted.
- State behavior: A contract waiting for signature shows an action card.
- Mockup variants (status): Pending Signature, Signed

# 15. Background Job Inventory

Background jobs have no screen. They are listed here because they produce the notifications, status changes and reminders shown on the screens above. Job names come from Inventory 5.3 and UC 10.1; anything the SRS does not state is marked as not specified in the SRS.

| # | Job | Purpose | Related screens | Related rules and use cases | Schedule |
|---|---|---|---|---|---|
| 1 | Contract Expiration Reminder | Sends reminders before a contract expires, using the configured thresholds (default 30, 15 and 7 days) | 6.1 Tenant Dashboard, 13.1 Notification Center, 13.2 Notification Rules, 13.3 Portal Notices | BR-15; UC 10.1; CEO07 | Not specified in the SRS: run frequency |
| 2 | Contract Status Update | Updates contract status as dates pass (for example Active to Expired) | 8.4 Contract List, 8.8 Contract Detail | BR-08, BR-12 | Not specified in the SRS: run frequency and exact transitions |
| 3 | Support Ticket Reminder | Reminds staff about tickets that need attention | 12.1 My Support Tickets, 12.4 Customer Ticket Inbox | BR-19; UC 9.1 | Not specified in the SRS: run frequency and reminder rule |
| 4 | Signing Request Expiry | Marks a signing request Expired when its deadline passes and informs the sender | 9.1 Send for Signature, 9.2 Contracts to Sign and E-sign | BR-10, BR-15; UC 6.3, UC 10.1 | Not specified in the SRS: run frequency |

# 16. EXTERNAL API INVENTORY

| # | External service | Used for | Related screens | Related rules and use cases | Notes |
|---|---|---|---|---|---|
| 1 | Google Sign-In API | Signing in with a Google account | 1.1 Sign In | UC 1.1 | Not specified in the SRS: which user types may use Google sign-in and how an account is linked to an existing user |
| 2 | Google Service for email | Sending email: activation links, password reset, signing invitations and reminders | 1.1 to 1.6 Authentication, 9.1 Send for Signature, 13 Notification | UC 1.x, UC 6.3, UC 10.1; MSG06 | Failed emails are retried up to 3 times (UC 10.1); not specified in the SRS: sender address and templates |
| 3 | Payment gateway | Paying subscription orders; sends authenticated payment notifications (callbacks) | Registration Completed, Orders and Invoices, Invoice Detail, Payment Detail | Context Diagram; BF-01 swimlane 2.1; Section IV.1; UC 2.1, UC 2.2; SC07 | A callback must be verified and duplicates prevented before payment success is recorded, a subscription activated or provisioning started (IV.1). See ASM-02 for how the gateway and the bank-transfer fallback fit together |

| | | | | | |
|---|---|---|---|---|---|
| 4 | Document storage service | Uploading contract documents, scanned files and support attachments through a secured backend integration | 8.7 Contract Form, 8.9 Contract Version History, 10 Document and Digitization | BR-13, BR-16; Section IV.1 | Not specified in the SRS: provider and access rules beyond the authorization check named in IV.1 |

# IV. Non-Functional Requirements

## 1. External Interfaces

ConTrack is a web-based, multi-tenant SaaS system that exchanges information with users and external services through secure interfaces. External-service failures shall not invalidate completed business transactions.

- **User Interface:** A responsive web interface supports Guests, SaaS Customers, Product Owner employees, Business Owner employees, and Business Customers. Forms provide inline validation, clear status feedback, and navigation appropriate to each user's permissions.
- **Authentication Service:** ConTrack integrates with Google Sign-In for supported registration and authentication workflows. Successful external authentication remains subject to account status, tenant membership, and access permission checks.
- **Payment via QR Code:** ConTrack displays a payment QR code and transfer instructions for the customer. Payments remain pending until an authorized Sales employee checks the received transfer and confirms the payment in the system. Displaying or scanning the QR code shall not be treated as proof of successful payment.
- **Email Service:** ConTrack sends account invitations, password-reset links, signing OTPs, approval requests, and contract and support notifications. Delivery failures shall be recorded and support an applicable retry flow without reversing completed business operations.
- **Document Storage Service:** ConTrack uploads contract documents, scanned files, and support attachments through a secured backend integration. Stored files shall retain their tenant and business-record associations, and access shall require appropriate authorization.
- **OCR Service:** ConTrack submits documents to the configured OCR engine and associates processing results with the correct contract and document version. Failed processing shall not be treated as completed digitization.
- **Electronic Signing:** OTP-based electronic signing is implemented within ConTrack. Email delivery supports OTP verification. Integration with external digital-signature providers and Certificate Authorities is excluded from the current version.

## 2. Quality Attributes

### 2.1 Usability

ConTrack shall provide a consistent, learnable interface for platform administrators, tenant employees, and occasional customer portal users.

- Users shall be able to identify their current workspace and locate the main functions permitted by their roles.

- Navigation labels, contract statuses, button placement, confirmation patterns, and validation behavior shall remain consistent across modules.

- Required fields and invalid values shall be identified next to the relevant input. Valid entered information shall be preserved when corrections are required.

- Long-running operations, including OCR, payment processing, document signing, and tenant provisioning, shall display a clear pending, processing, successful, or failed status.

- Contract approval status, signing progress, payment obligations, subscription validity, and service quotas shall be clearly presented to authorized users.

- Destructive or significant actions, such as cancelling a contract, suspending a tenant, or deactivating an account, shall require explicit confirmation.

- Core workflows shall support Vietnamese Unicode content, keyboard navigation, readable contrast, and responsive layouts on common desktop and mobile browsers.

### 2.2 Performance

The following targets apply under normal workload on the documented test deployment. External-provider processing and file-transfer times shall be measured separately from application response times.

- At least 95% of ordinary page loads, list searches, and read-only API requests shall complete within 3 seconds. Standard create or update requests shall acknowledge success or validation failure within 2 seconds.

- The initial deployment shall support at least 100 concurrent active users while meeting the ordinary-response targets.

- Standard contract, subscription, and system-statistics dashboards shall render within 5 seconds for the agreed test dataset and normal filter range.

- OCR, signing, and provisioning requests shall display a processing state within 2 seconds of acceptance. Payment records shall clearly display their current verification status while awaiting Sales review. The system shall prevent accidental duplicate submissions and apply configurable timeouts to external requests.

- Contract digitization uploads shall support PDF, JPEG, and PNG files up to 20 MB per file. Unsupported files and files exceeding the limit shall be rejected before permanent storage.

- Background processing, including OCR, tenant provisioning, and notification delivery, shall not block the interactive request until all processing is complete.

- Search results and large record lists shall use pagination to avoid loading the entire dataset into a single response.

### 2.3 Security and Privacy

ConTrack shall protect authentication information, tenant data, contract documents, signing evidence, and payment records through access control and secure data handling.

- Network communication containing credentials, personal information, contract documents, signing data, or payment information shall be encrypted in transit.

- Passwords shall be stored using a salted, one-way password hash. Raw passwords, OTPs, reset tokens, signing credentials, and provider secrets shall not appear in application logs or user-facing error messages.

- Role-based, tenant-scoped, and record-level authorization shall be enforced on the server for every protected request.

- A tenant user shall not access another tenant's records by changing a URL, record identifier, request parameter, or workspace subdomain. Platform support access shall not automatically grant permission to view tenant contract contents.

- Contract documents and support attachments shall be accessible only to authorized users. Knowing a file identifier or storage location shall not be sufficient to access protected files.

- Signing OTPs shall be time-limited, single-use, and associated with the intended signer and signing transaction. Expired or previously used OTPs shall be rejected.

-Only authorized Sales employees shall confirm payments after checking the received transfer against the corresponding payment record. Each confirmation shall record the employee, timestamp, amount, and payment reference. Repeated confirmation attempts shall not create duplicate payments or trigger the same downstream operation more than once.

- Internal support notes shall remain accessible only to authorized employees and shall not appear in customer-facing responses or notifications.

- Security-sensitive actions, permission changes, payment outcomes, contract approvals, and signing events shall be audit logged with the actor, affected record, timestamp, and outcome.

- Locked, suspended, or inactive accounts shall be denied protected access. Session expiration and failed sign-in restrictions shall follow a consistent policy across authentication functions and background jobs.

## 2.4 Reliability and Availability

ConTrack shall preserve consistent business records when users submit concurrent requests or external services become slow, unavailable, or return duplicate responses.

- Completed account, order, contract, and support operations shall remain valid when email delivery fails. Failed notifications shall be recorded and remain eligible for retry.

- Payment records shall remain pending until an authorized Sales employee confirms receipt of the transfer. For subscription payments, subscription activation and tenant provisioning shall proceed only after payment confirmation. Provisioning shall use recoverable processing states, and a tenant shall not be marked ready until all required steps have completed successfully.

- Retrying a payment confirmation or background job shall not duplicate payment records, subscription activations, provisioning requests, or other completed business effects.

- Failed OCR processing shall preserve the original document. Extracted information shall remain subject to authorized review before being confirmed as contract data.

- Previous document versions shall remain available to authorized users. Signed contract files and their signing evidence shall be protected against overwriting or deletion in accordance with the document retention rules.

- Concurrent edits shall not silently overwrite another user's changes. Conflicting updates shall require the user to review the latest record.

- The initial deployment shall target at least 99% monthly availability, excluding announced maintenance. External-provider failures shall be tracked separately, and unrelated functions shall remain available where possible.

- The Master Database, tenant databases, and contract files shall be backed up at least daily, with a target recovery point of 24 hours and a target recovery time of 4 hours.

- Recovery verification shall confirm that restored records preserve tenant ownership, document versions, file references, and signing evidence.

### 2.5 Maintainability and Scalability

ConTrack shall remain maintainable as tenant numbers, contract volumes, and external-service integrations grow.

- Authentication, tenant administration, subscription and payment processing, contract management, document digitization, and technical support shall be organized into clearly separated modules with documented interfaces.

- Provider endpoints, credentials, service quotas, notification schedules, file limits, timeouts, and retry settings shall be configurable rather than hard-coded in user-interface logic.

- Application logs and health information shall support diagnosis of failed requests, payment processing, OCR jobs, provisioning operations, and notification delivery without exposing confidential information.

- Database changes shall use versioned migrations, with migration status and failures traceable for each affected tenant.

- Routine log cleanup shall preserve signed-contract evidence and signature audit records required by the retention rules.

- External and internal service interfaces shall use documented contracts and controlled versioning to prevent updates from silently breaking existing workflows.

- Request processing and background workers shall support capacity expansion as usage increases beyond the initial 100-concurrent-user target, while preserving tenant isolation and transaction consistency.

## V. Requirement Appendix

### 1. Business Rules

| BR-ID | Rule Statement | Applies To | Rationale | Enforced By |
|---|---|---|---|---|
| BR-01 | Email addresses must be unique within their scope (platform, or each tenant). Employee codes, customer codes, partner codes and contract codes must also be unique. | Registration, employee accounts, portal accounts, partners, contracts | Prevents duplicate identities and ambiguous records. | The system automatically on every create and update |
| BR-02 | Passwords must have at least 8 characters, including uppercase, lowercase, a digit and a special character, and are stored only as a hash (bcrypt / Argon2id). | All accounts | Protects accounts against weak passwords and database leaks. | The system automatically when a password is set or changed |
| BR-03 | After 5 consecutive failed sign-in attempts, the account is locked for 30 minutes. | Sign-in | Protects accounts against brute-force attacks. | The system automatically on every failed sign-in |
| BR-04 | Each SaaS customer can own only one tenant in version 1. | Tenant provisioning | Keeps billing and data ownership simple in the first release. | The system automatically when an order or tenant is created |

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| BR-05 | A subscription is created and the tenant provisioned only when the invoice is fully paid. | Payments, subscriptions | Ensures service is delivered only after payment. | The system automatically when a payment is recorded |
| BR-06 | The number of active users and the storage used must not exceed the limits of the tenant's current plan. | Employee accounts, document upload, plan downgrade | Enforces the commercial terms of each plan. | The system automatically when a user is created or a file is uploaded |
| BR-07 | Contract codes are generated from the company's contract code format (for example HD-{YYYY}-{TYPE}-{SEQ}) and cannot be edited manually. | Contract creation | Guarantees consistent, unique and traceable contract numbers. | The system automatically when a contract is saved for the first time |
| BR-08 | A contract can be edited or submitted for approval only while its status is Draft or Revision Requested. | Contract editing, submission | Prevents changes to a contract that is being approved or has been approved. | The system automatically on every contract update |
| BR-09 | Approval steps run in their defined order. A contract is approved only when every required step has been approved. | Contract approval | Ensures every required approver reviews the contract. | The system automatically on every approval action |
| BR-10 | A rejection or revision request at any approval step, or a customer declining to sign, returns the contract to Revision Requested. | Contract approval, e-signature | Keeps one clear path for correcting a contract. | The system automatically on reject, revision request or decline |
| BR-11 | Parties sign in their defined signing order. An OTP is valid for 5 minutes and allows 3 attempts, after which signing is blocked for 15 minutes. A signing request expires after 7 days by default. | E-signature | Verifies the signer's identity and keeps signing requests time-bound. | The system automatically on every signing attempt |
| BR-12 | A signed contract cannot be edited or deleted. Any change after signing must be made through an appendix. | Signed contracts and their documents | Preserves the legal integrity of signed agreements. | The system automatically on every update or delete request |
| BR-13 | Only PDF, DOCX, JPG and PNG files up to 20 MB each may be uploaded. A SHA-256 hash is stored for every file. | All file uploads | Limits storage abuse and allows integrity checks and duplicate detection. | The system automatically on every upload |

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| BR-14 | OCR results with confidence below 80% are flagged for careful verification before a scan can be confirmed. | Contract digitization | Reduces errors when extracting data from paper contracts. | The system automatically when OCR finishes |
| BR-15 | Reminders are sent at the days-before values defined in the notification rules (default 30, 15 and 7 days). | Contract expiry, payment schedules, obligations, signing requests | Prevents missed deadlines. | The System Scheduler daily |
| BR-16 | Every create, update, approve, sign, download and delete action on contracts, documents and accounts is written to the audit log, with old and new values. | Contracts, documents, accounts | Provides a complete audit trail of who did what and when. | The system automatically on every audited action |
| BR-17 | Order details store a copy of the item code, item name and price at the time of sale. | Orders | Keeps historical orders correct when catalog names or prices change. | The system automatically when an order is created |
| BR-18 | Tenant database passwords are never stored in the Master DB; only a secret reference is kept. | Tenant databases | Prevents credential leaks if the Master DB is exposed. | The system automatically when a tenant database is provisioned |
| BR-19 | Internal notes on support tickets are never visible to customers, in any view or notification. | Support tickets (platform and tenant) | Allows staff to discuss issues candidly. | The system automatically on every page and data request |
| BR-20 | Contract documents may only be viewed or downloaded by users with permission for the related contract. Access is checked on every request. | All document access | Protects confidential contract data. | The system automatically on every file access request |
| BR-21 | A user can access only the data of the tenant they belong to. Data from one tenant is never visible to another tenant. | All tenant data | Ensures data isolation between businesses on a shared platform. | The system automatically on every request (tenant resolved by domain and account) |
| BR-22 | A locked, inactive or deactivated account cannot sign in, regardless of whether the password is correct. Users | Sign-in | Ensures removed users and unpaid tenants lose access immediately. | The system automatically on every sign-in attempt |

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| | of a suspended tenant cannot sign in. | | | |
| BR-23 | When a sign-in fails, the message must not reveal whether the email exists or the password was wrong. | Sign-in | Prevents attackers from discovering valid accounts. | The system automatically on every failed sign-in |
| BR-24 | Records that have been used in transactions (customers, partners, packages, contracts, tenants with subscriptions) are deactivated by status instead of being deleted. | Master data | Preserves history and referential integrity. | The system automatically on every delete request |

**2. System Messages**

| # | Message code | Message Type | Context | Content |
|---|---|---|---|---|
| 1 | MSG01 | In red, under the text box | A required field is empty | The * field is required. |
| 2 | MSG02 | In red, under the text box | An input value has an invalid format | The {field_name} is not in a valid format. |
| 3 | MSG03 | In red, under the text box | A unique value is already in use (email, tax code, subdomain, code) | This {field_name} is already in use. Please choose another one. |
| 4 | MSG04 | Toast message | Registration submitted successfully | A verification email has been sent to {email_address}. Please check your inbox. |
| 5 | MSG05 | Toast message | Data saved successfully | Saved successfully. |
| 6 | MSG06 | In line | A verification, invitation or reset link is invalid or expired | This link is invalid or has expired. Please request a new one. |
| 7 | MSG07 | Confirmation dialog | The payment amount exceeds the outstanding balance | The payment amount exceeds the outstanding balance of {amount}. Do you want to continue? |
| 8 | MSG08 | Pop-up message | A plan limit (users, storage, contracts) has been reached | Your plan limit for {limit_name} has been reached. Please upgrade your plan to continue. |
| 9 | MSG09 | In line | Email or password is not correct when signing in | Incorrect email or password. Please check again. |

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| 10 | MSG10 | In red, under the text box | Email and/or password is empty when signing in | Please enter your email and password. |
| 11 | MSG11 | In line | The email has not been verified when signing in | Your email has not been verified. Please check your inbox. |
| 12 | MSG12 | In line | The account is locked, inactive, or the tenant is suspended | Your account is not available. Please contact your administrator. |
| 13 | MSG13 | In line | 5 consecutive failed sign-in attempts | Too many failed attempts. Your account is locked for 30 minutes. |
| 14 | MSG14 | Toast message (error) | A tenant provisioning job fails | Provisioning for tenant {tenant_code} failed. Please review the job and retry. |
| 15 | MSG15 | Pop-up message | No approval workflow matches the contract when submitting | No approval workflow is configured for this contract type. Please contact the administrator. |
| 16 | MSG16 | Toast message (error) | An action is not allowed in the record's current status | This action is not allowed while the {record} is {status}. |
| 17 | MSG17 | In red, under the text box | The OTP is incorrect or expired when signing | The OTP is incorrect or has expired. You have {n} attempt(s) left. |
| 18 | MSG18 | In line | The signing request has expired | This signing request has expired. Please contact {sender_name}. |
| 19 | MSG19 | In red, under the upload box | An uploaded file has an unsupported type or exceeds the size limit | Only PDF, DOCX, JPG and PNG files up to 20 MB are allowed. |
| 20 | MSG20 | Toast message (warning) | An uploaded file is identical to an existing version | This file has already been uploaded. |
| 21 | MSG21 | In red, under the checkbox | The customer clicks Sign without accepting the terms | Please confirm that you have read and agree to the contract terms. |
| 22 | MSG22 | In line | There is no search result | No results found. |
| 23 | MSG23 | In red, under the text box | An input value is longer than the maximum length | Exceeds the maximum length of {max_length} characters. |
| 24 | MSG24 | Confirmation dialog | The user deactivates or cancels a record | Are you sure you want to {action} this {record}? |
| 25 | MSG25 | Toast message | A contract is sent for signature successfully | The contract has been sent to {signer_name} for signature. |

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| 26 | MSG26 | Toast message | A contract is signed successfully | You have signed contract {contract_code} successfully. |
| 27 | MSG27 | Toast message | A contract is approved, rejected or returned for revision | Contract {contract_code} has been {action}. |
| 28 | MSG28 | Toast message | An OTP is sent to the signer | A verification code has been sent to {email_address}. |
| 29 | MSG29 | Toast message | A support ticket is created successfully | Your ticket {ticket_code} has been submitted. |
| 30 | MSG30 | Full page | The user opens a page or record they have no permission for | You do not have permission to access this page. |

## 3. Other Requirements